

Print

Two-component dispensing machine with rotary table and material preparation (PPS)

B5200

Translation of the original Operating Instructions
5453

1.0-en

Table of contents

1	Notices for use	1 - 1
1.1	Service legal notices	1 - 1
1.2	CE documents.....	1 - 2
1.3	Guide.....	1 - 3
1.4	Intended use	1 - 5
1.5	Foreseeable misuse Dispensing machines.....	1 - 7
1.6	Operation of the machine	1 - 9
1.7	Duties of the personnel	1 - 10
1.8	Personnel deployment	1 - 12
1.9	Special dangers	1 - 13
1.10	Handling hazardous materials.....	1 - 13
1.11	Transport.....	1 - 16
1.11.1	Set-up or installation of the component, system part(s), system	1 - 16
1.11.2	Disconnecting the machine from the power supply.....	1 - 18
1.11.3	Removing and making the machine or system available	1 - 18
1.11.4	Moving the machine or component.....	1 - 20
1.12	Storing the machine	1 - 23
1.13	Taking out of operation/disposal	1 - 24
2	Structure and functionality	2 - 1
2.1	Protective equipment for the machine	2 - 1
2.2	Technical data.....	2 - 1
2.3	Technical description	2 - 2
2.3.1	Axis system module	2 - 3
2.3.2	Operating and display elements.....	2 - 3
2.3.2.1	Main switch	2 - 3
2.3.2.2	Enabling switch	2 - 3
2.3.2.3	Emergency stop command device	2 - 4
2.3.2.4	Signal column.....	2 - 5
2.3.2.5	Membrane keypad move axis	2 - 6
2.3.2.6	Membrane keypad general operation	2 - 8
2.3.2.7	Key switch for Setup/Automatic	2 - 10
2.3.2.8	Captron-Sensortaste	2 - 12
2.3.3	Mixing technology	2 - 12
2.3.3.1	Duplex+	2 - 12
2.3.4	Air preparation.....	2 - 14
2.3.4.1	Service unit	2 - 14
2.3.4.2	Manual start-up unit	2 - 15
2.3.5	Sensors	2 - 15
2.3.5.1	Wika pressure sensor	2 - 16
2.3.6	Safety elements	2 - 18
2.3.6.1	Safety light barrier arrangement for round tables.....	2 - 18
2.3.6.2	Door safety switch.....	2 - 18
2.3.6.3	Safety switches	2 - 19
2.3.7	Machine positions	2 - 20
2.3.7.1	Blind shot position with cup wiper and load cell evaluation.....	2 - 22
2.3.7.2	Weighing position.....	2 - 24

2.3.7.3	Needle check	2 - 25
2.3.7.4	WIPER 2 belt cleaner.....	2 - 26
2.3.8	Switchable remote access	2 - 27
2.3.9	Material preparation technology	2 - 27
2.3.9.1	PPS – Configuration and function	2 - 27
2.4	Function of static mixing principle	2 - 36
3	Operation	3 - 1
3.1	Danger notes.....	3 - 1
3.2	Commissioning.....	3 - 3
3.2.1	Preparation for the operation of a system	3 - 3
3.2.2	Connecting the power supply	3 - 3
3.2.3	Info - Logging in to the machine	3 - 5
3.3	Switching on the machine	3 - 6
3.4	Changing over languages	3 - 8
3.5	Starting material preparation	3 - 9
3.6	Operation modes.....	3 - 10
3.6.1	Activating PROCESS MONITORING operation mode.....	3 - 11
3.6.2	Activating MANUAL operation mode.....	3 - 11
3.6.3	Activating AUTOMATIC operation mode	3 - 12
3.6.4	Activating STANDBY operation mode.....	3 - 13
3.6.5	Activating SETUP operation mode.....	3 - 13
3.6.6	Activating SAFETY operation mode.....	3 - 14
3.7	Commissioning activities	3 - 15
3.7.1	Referencing a multi-turn drive	3 - 15
3.7.2	Automatic golden calibration	3 - 20
3.7.3	Automatic recalibration.....	3 - 21
3.7.4	Creating a new dispensing program	3 - 23
3.7.5	Teaching in the scale	3 - 24
3.7.6	Save values of weighing diagram in form	3 - 25
3.7.7	Carry out initial commissioning of pail.....	3 - 26
3.7.8	Calibration of the follower plate valve	3 - 28
3.7.8.1	Plate valve calibration – Start manually	3 - 29
3.7.8.2	Plate valve check – Start manually	3 - 31
3.8	Manual functions .. Please read!.....	3 - 32
3.8.1	Overview of the drive programs	3 - 33
3.8.1.1	Creating a drive program	3 - 33
3.8.1.2	Jump command "@121"	3 - 35
3.8.1.3	Jump command "@124"	3 - 35
3.8.1.4	Multiple branching "@111".....	3 - 36
3.8.1.5	Unconditional jump "@100"	3 - 36
3.8.1.6	"M110" command.....	3 - 37
3.8.1.7	Activating absolute dimension details	3 - 37
3.8.1.8	Activating relative dimension details	3 - 38
3.8.1.9	Moving axis in rapid traverse	3 - 38
3.8.1.10	Travelling a straight line at feed speed	3 - 39
3.8.1.11	Travel in a circle in clockwise direction (Mpt).....	3 - 39
3.8.1.12	Travel in a circle in anticlockwise direction (Mpt).....	3 - 40
3.8.1.13	Travel in a circle in clockwise direction (Radius).....	3 - 41
3.8.1.14	Travel in a circle in anticlockwise direction (radius)	3 - 41
3.8.1.15	Dwell time.....	3 - 42

3.8.1.16	Exit drive program	3 - 42
3.8.1.17	Example drive program structure	3 - 42
3.8.1.18	Notes on how to change or create a drive program	3 - 43
3.8.1.19	Loading a drive program	3 - 44
3.8.1.20	Starting the drive program manually	3 - 44
3.8.1.21	Inserting a drive program in the program selection list	3 - 44
3.8.1.22	Assigning drive program to a recipe	3 - 45
3.8.1.23	Renaming a drive program	3 - 45
3.8.1.24	Deleting a drive program	3 - 45
3.8.2	Pail change with optional bar codes	3 - 46
3.8.3	Relieving pail	3 - 48
3.8.4	Calibration of Mettler scales, Type 'WKC'	3 - 49
3.8.4.1	Internal calibration of the Mettler scales Type WKC	3 - 49
3.8.4.2	External calibration of the Mettler scales Type WKC	3 - 49
3.8.5	Weighing in	3 - 50
3.8.5.1	Separate weighing via static mixing tube (automatic scale)	3 - 51
3.8.6	Weight check according to the component simulation method	3 - 62
3.8.7	Manual functions, general	3 - 65
3.8.8	Manual functions Dispensing system Processes	3 - 76
3.8.8.1	Function key bar 1C Static 2C	3 - 76
3.8.9	Needle Check - Single-head System	3 - 79
3.8.9.1	Set up 'Golden Needle'	3 - 79
3.8.9.2	Needle check procedure	3 - 80
3.8.9.3	Needle replacement procedure	3 - 82
3.8.10	Parameterisation of a blindshot bin/bucket on load cell	3 - 83
3.9	Dispensing machine symbols	3 - 83
3.10	PPS symbols	3 - 86
3.11	Menu structure	3 - 88
3.11.1	Login	3 - 89
3.11.2	User	3 - 90
3.11.3	Changing over languages	3 - 93
3.11.4	Graphic operation mode changeover - dispensing	3 - 94
3.11.5	Graphic operation mode changeover - material preparation	3 - 95
3.11.6	Dispensing	3 - 95
3.11.6.1	Starting screen	3 - 95
3.11.6.2	Head	3 - 101
3.11.6.3	Pump A without valve	3 - 112
3.11.6.4	Pump B without valve	3 - 120
3.11.6.5	Parameterisation of blind shot container on load cell	3 - 129
3.11.6.6	Behaviour after Mat NOK	3 - 132
3.11.6.7	Dispensing programs	3 - 134
3.11.7	Weighing screen	3 - 136
3.11.7.1	Weighing cup with cup weight detection	3 - 139
3.11.7.2	Weighing cup – Wiping off	3 - 140
3.11.7.3	Automatic weighing with part simulation	3 - 141
3.11.7.4	Scale - Weighing limits	3 - 144
3.11.7.5	Scale - Actual values - Evaluations - Result	3 - 146
3.11.7.6	Scales parameters	3 - 148
3.11.7.7	Scale - Part weighing simulation	3 - 149
3.11.7.8	Scale - Weighing programs	3 - 150
3.11.8	Needle check	3 - 151
3.11.8.1	Master positions	3 - 151

3.11.8.2	Deviation dialogue.....	3 - 153
3.11.8.3	Settings	3 - 155
3.11.9	Cleaning station with belt	3 - 156
3.11.10	Basic settings	3 - 158
3.11.10.1	Calibration A with scale	3 - 158
3.11.10.2	Calibration B with scale	3 - 162
3.11.10.3	Basic settings – Turn-back determination	3 - 166
3.11.10.4	Machine settings	3 - 170
3.11.11	Material preparation PPS	3 - 179
3.11.11.1	Start page 1 x 2K	3 - 179
3.11.11.2	Component A - Pressure.....	3 - 179
3.11.11.3	Component A - Drive.....	3 - 186
3.11.11.4	Component A - PID controller	3 - 191
3.11.11.5	Component A - fill level	3 - 196
3.11.11.6	Component A - general	3 - 198
3.11.11.7	Component A - Valves - Pail	3 - 201
3.11.11.8	Component A - Pressure relief via material pump	3 - 204
3.11.11.9	Component A evacuation - vacuum pump	3 - 205
3.11.11.10	Component A evacuation - Current setting Affects	3 - 207
3.11.11.11	Configuration.....	3 - 209
3.11.11.12	“Barcode scan” dialogue	3 - 210
3.11.12	Transport.....	3 - 221
3.11.12.1	Rotary table.....	3 - 221
3.11.12.2	Workplace, left	3 - 222
3.11.13	Axis dialogues	3 - 224
3.11.13.1	Management of servo axis positions.....	3 - 230
3.11.13.2	Manual operation of the axes	3 - 232
3.11.13.3	Drive program editor	3 - 234
3.11.13.4	Drive program management	3 - 236
3.11.14	Recipe dialogue	3 - 238
3.11.14.1	Mixing tube change indicator	3 - 241
3.11.14.2	Part counter.....	3 - 242
3.11.14.3	Maintenance dialogue - Pump change.....	3 - 243
3.11.15	Active messages	3 - 245
3.11.16	Log file settings	3 - 246
3.12	Production flow.....	3 - 247
3.13	Shutting down the dispensing machine.....	3 - 248
4	Maintenance and service	4 - 1
4.1	BDT folder backup	4 - 8
4.1.1	Backup group field	4 - 8
4.1.2	Manual backup of the “_BDT” folder on drive N of the bdtronic LAN	4 - 10
4.1.3	Manual backup of the “_BDT” folder on a random medium	4 - 12
4.1.4	Automatic backup of the “_BDT” folder	4 - 13
4.2	Venting of material circuits Dispensing units.....	4 - 14
4.2.1	Venting a material line.....	4 - 15
4.2.2	Venting a progressive cavity pump	4 - 16
4.2.3	Venting a Duplex-Plus mixing head	4 - 18
4.3	Machine maintenance before production start.....	4 - 20
4.4	Mixing tube maintenance requirement	4 - 20
4.5	Dispensing pump maintenance requirements.....	4 - 21
4.6	Maintenance requirement of emptying the condensate container	4 - 21

4.7	Maintenance request - Blind shot - Replace or empty vessel	4 - 22
4.8	Maintenance requirement - replace weighing container	4 - 22
4.9	PPS maintenance requirement	4 - 23
4.10	Maintenance prompt from EMERGENCY STOP command devices	4 - 23
4.11	Maintenance requirements for the activation switch	4 - 23
4.12	Maintenance requirements for the door safety switch.....	4 - 24
4.13	Light curtain maintenance requirement.....	4 - 25
4.14	Maintenance requirement - Festo cylinder	4 - 25
4.15	Maintenance requirement - vacuum shut-off valve of follower plate	4 - 25
4.16	Maintenance prompt – Internal calibration of Mettler scales.....	4 - 26
4.17	Maintenance of mixing and dispensing components	4 - 27
4.17.1	Maintenance work on the static Duplex mixing head	4 - 27
4.17.1.1	Analysis of the material flow after the dispensing	4 - 28
4.17.1.2	Bleeding operation before the dispensing pump.....	4 - 29
4.17.1.3	Bleeding operation at the pump holder	4 - 31
4.17.1.4	Dismantling the mixer holder.....	4 - 33
4.17.1.5	Dismantle the eccentric worm pump from the mixing head	4 - 34
4.17.1.6	Replacing a rotor/stator on an eccentric worm pump.....	4 - 37
4.17.1.7	Mounting a bdtronic eccentric worm pump	4 - 39
4.17.1.8	Replacing the Wittenstein motor	4 - 45
4.17.2	Maintenance work on bdtronic progressing cavity pump PPS	4 - 47
4.17.2.1	Changing the wiping ring.....	4 - 47
4.17.2.2	Changing the stator.....	4 - 50
4.17.2.3	Changing the rotor	4 - 54
4.18	Maintenance of special components.....	4 - 66
4.18.1	Oil change vacuum pump	4 - 66
4.18.2	Oil level vacuum pump.....	4 - 67
4.19	Lubricating LineTech axles	4 - 67
4.20	How to read the maintenance table	4 - 83
4.21	Maintenance table.....	4 - 84
5	Overview of faults	5 - 1
5.1	Faults on compressed air-driven units and components.....	5 - 3
5.1.1	Pneumatic cylinder shut-off valve SRU.....	5 - 3
5.1.2	Pneumatic cylinder end position inlet.....	5 - 3
5.1.3	Pneumatic cylinder end position	5 - 4
5.1.4	Power supply - Compressed air	5 - 4
5.2	Faults on sensor units and components	5 - 4
5.2.1	Sensors - scanner - read error	5 - 4
5.2.2	Sensors - balance - cup missing	5 - 4
5.2.3	Sensors - balance - cup full.....	5 - 5
5.2.4	Sensors - balance - communication error	5 - 5
5.2.5	Sensors - balance - taring	5 - 5
5.2.6	Sensors - needle check - outside of tolerance	5 - 6
5.2.7	Sensors - needle check needle check error.....	5 - 6
5.2.8	Sensors - needle check position calculation incorrect	5 - 6
5.2.9	Sensors - needle check sensor incorrect.....	5 - 7
5.2.10	Sensors – Light barrier.....	5 - 7
5.2.11	Fault - Drive motor - SEW	5 - 17
5.3	Faults on transport and movement components.....	5 - 33

5.3.1	Transport - lift-turn unit - turning working position NOK	5 - 33
5.3.2	Transport - lift-turn unit - turning home position NOK	5 - 33
5.3.3	Transport - lift-turn unit - home position NOK	5 - 33
5.3.4	Transport - WC.....	5 - 34
5.3.5	Transport - pallet holder not open	5 - 34
5.3.6	Drive technology - error in axis system	5 - 34
5.3.7	Drive technology - lift-turn unit - mechanical system.....	5 - 34
5.3.8	Drive technology - lift-turn unit - working position	5 - 35
5.3.9	Drive technology - lift-turn unit - home position.....	5 - 35
5.3.10	Drive technology - Reference run	5 - 35
5.3.11	Drive technology - Transport belt	5 - 35
5.4	Faults caused by processes.....	5 - 36
5.4.1	Dispensing technology - filling prohibited	5 - 36
5.4.2	Dispensing technology - process dispensing pressure	5 - 36
5.4.3	Dispensing technology - blindshot container.....	5 - 37
5.4.4	Dispensing technology - blindshot container full	5 - 37
5.4.5	Dispensing technology - dispensing pumps - pressure low	5 - 37
5.4.6	Dispensing technology - material preparation - material NOK	5 - 38
5.4.7	Dispensing technology - pump speed above max.	5 - 38
5.4.8	Dispensing technology - pump speed under min.	5 - 38
5.4.9	Process - dispense - chamber not ready	5 - 38
5.4.10	Process - dispense - chamber error after cleaning	5 - 39
5.4.11	Process - dispense - undef. Chamber status after recipe loading	5 - 39
5.4.12	Software - Drive program	5 - 39
5.4.13	System - vacuum - not filled	5 - 39
5.4.14	System - vacuum error.....	5 - 40
5.4.15	System - vacuum - vacuum pump.....	5 - 40
5.4.16	System - vacuum - vacuum value	5 - 40
5.4.17	System - vacuum chamber - not filled	5 - 41
5.4.18	System - vacuum chamber - process still active	5 - 41
5.4.19	System - vacuum - vacuum not reached.....	5 - 41
5.5	Material-conditioned faults	5 - 41
5.5.1	Material preparation - container - fill level max.	5 - 41
5.5.2	Material preparation - container - fill level min.	5 - 42
5.5.3	Material preparation - container - stirrer	5 - 42
5.5.4	Material preparation - container - sensors - safety switch.....	5 - 42
5.5.5	Material preparation - container - prepressure low	5 - 43
5.5.6	Material preparation - container - prepressure max.	5 - 43
5.5.7	Material preparation - ball cock - end position	5 - 43
5.5.8	Material preparation - ball cock - message box	5 - 44
5.6	Safety faults	5 - 44
5.6.1	Safety - E-STOP pressed.....	5 - 44
5.6.2	Safety - Door not locked.....	5 - 44
5.7	Fault due to pump wear	5 - 44
6	Layout	6 - 1
7	Setting data	7 - 1
8	Machine log	8 - 1
9	Replacement part documents.....	9 - 1
10	Electrical schematic	10 - 1
11	Supplier documentation.....	11 - 1

1 Notices for use

1.1 Service legal notices

Contacting the bdtronic Service team

These operating instructions are used to generally describe the function and operation of the machine.

For special problem solutions and the performance of repairs, service work and further tasks, which are not described in these operating instructions due to space limitations, the expert bdtronic GmbH Service team is available on weekdays during normal working hours, from 7:00 a.m. to 5:00 p.m., and optionally in the time agreed upon in the maintenance contract. Prolonged operational interruptions (e.g. Christmas, turn of the year) are, of course, bridged with on-call service support. In these periods, employees of the Service team continue to provide support during the stated times.

For questions about quotations, order processing or spare parts sales, please contact the central office or the responsible sales employees for inside or outside sales.

We can be reached at this address:

- bdtronic GmbH
- Lindenstraße 3
- 97990 Weikersheim (Germany)

We can also be reached by telephone:

- Central switchboard: +49 7934 104-0
- Service hotline: -140
- Spare parts sales and order processing: -210
- Fax: +49 7934 104 371
- E-mail: info@bdtronic.de
- E-mail for Service: service@bdtronic.de

1.2 CE documents

The declaration of conformity and installation declaration with assembly instructions are included in these instructions.

1.3 Guide

Guide to these operating instructions

Foreword

These operating instructions are part of the machine documentation from bdtronic GmbH.

They include all the information that is required for the intended use of the machine.

Their contents correspond to the state of construction of the machine at the time these operating instructions were created. Changes to the construction and the technical data are subject to constant enhancement and customer-specific design requirements.

Therefore, no claims can be derived from the contents of these operating instructions (details, graphics, drawings, descriptions, etc.). Errors excepted!

These operating instructions should make it easy to get to know the machine and to use it as intended, in a goal-oriented, safe fashion.

If any print errors, incomprehensible or mistaken information is/are noticed when reading these operating instructions, please communicate this to bdtronic GmbH immediately. The editorial team will check this information, make adjustments or changes, and make the edited information available in data form or as a printout.

Structure

The operating instructions contain important notes for operating the machine safely properly and economically, for avoiding dangers, for minimising repair costs and downtimes and for increasing the reliability and service life of the machine.

The operating instructions consist of 10 chapters and an appendix with supplier documentation.

The header line indicates the current chapter. The footer line contains the machine designation (in the middle of the line) and the chapter number/page number (at the right).

Symbols and pictograms

Warnings are indicated by a warning triangle with hazard symbol and warn of dangers that can result in property damage and personal injury.

Prohibitions are marked with circles with a red slash through the danger symbol and prohibit an activity or the carrying along of particular objects.

	NOTICE Smoking prohibited! Smoking is prohibited here.
---	---

	NOTICE Access prohibited! No access for unauthorised persons.
---	--

Requirements are marked with circles with mandatory sign or rectangles with instructive text and require an activity or the use of particular objects.

	NOTICE Wear eye protection! Wear suitable eye protection for every dangerous situation, e.g. laser, infrared, etc.
---	---

	NOTICE Wear protective gloves! Wear suitable protective gloves for every dangerous situation.
---	--

	NOTICE Wear ear protection Wear suitable ear protection for every dangerous situation.
--	---

	NOTICE Dispose of properly Work materials and cleaning fluids must be disposed of properly so that the requirements of local and especially regional disposal regulations are fulfilled completely.
---	--

Notes are marked with an information pictogram and include recommendations or additional information.

Example:

	NOTICE Connection to the compressed air system All work on system parts subject to compressed air may only be performed by trained personnel.
---	--

For detailed instructions on chemical testing procedures, please contact your supplier.

1.4 Intended use

Use of the dispensing machine

bdtronic GmbH produces among other things dispensing solutions which, depending on the customer's requirements, cover the range from purely externally controlled modules to controlled dispensing assemblies, machines as purely manual workstations, without further automation, as well as partially automated table-top systems for one operator or as partially or fully automated dispensing cells for one operator or without operator. In each case, the process task - mixing and dispensing - is implemented on the machine using bdtronic's own components such as dispensing pumps, dispensing valves, mixing heads, etc. so that single- or multi-component material systems such as polyurethanes, epoxy resins, silicones, etc. can be prepared, mixed and dispensed. Optional sensors support the machine's quality assurance process. Additional automation solutions such as product carrier transport systems, heat treatment systems, etc. optimise the production flow.

Main areas of application:

1. Gluing of components such as oil filters, housing covers, etc.
2. Sealing of component housings such as control units for the automotive supply industry
3. Manufacturing of seals for small housings
4. Covering of sensitive electronics for protection against environmental influences
5. Application of heat-conducting pastes to electronic circuit assemblies that generate power losses

Machine requirements

The machine is constructed and set according to:

1. The agreements stipulated in the contract, such as the process materials, model, selection of machine components and accessories.
2. The agreed parameters.
3. User-specific needs.
4. The state of technology and the recognised technical safety rules.

Non-adherence to the operating instructions for the machine

Any other use or additional use of the machine than described in these operating instructions is considered to be improper use.

In case of unintended use

The operator alone is liable for all consequences of unintended use.

In case of impermissible operation modes

1. Operation of the machine is especially not permitted in case of:
2. performance of work for which the machine is not designed.

3. work that does not comply with the procedures for assembly, operation, and servicing that is not described in these operating instructions or is not recommended by bdtronic.
4. unauthorised alterations and/or repairs.
5. disassembly and/or bypassing of safety devices.
6. the use of components, tools or additional equipment that is not approved or recommended by bdtronic.
7. the use of spare parts that are not bdtronic original spare parts or spare parts from a supplier recommended by bdtronic.
8. the use of auxiliary components and operating materials that are not approved or recommended by bdtronic.

In case of unauthorised changes to the machine

The manufacturer is not liable for any consequences of unauthorised changes to the machine and/or the materials used. The risk of compromising the life and limb of the user and third parties as well as damage to the machine and other associated property is borne solely by the operating firm.



NOTICE

Changeover to other potting materials

Dispensing, impregnation and material processing systems are usually only designed and delivered for a specific material system. The type approval pertains to these materials.

The changeover of systems to other potting materials, whether for laboratory operations or for the normal production system, may require a new risk assessment by the system manufacturer. Particularly when fluid class 1 material systems are used instead of fluid class 2, goes out immediately so that the operating approval for the system, originally approved for fluid class 2, is voided as further measures such as extraction, safety interlocks, etc. must be taken into account! We recommend contacting both the material manufacturer and bdtronic, as the system manufacturer, **before a material change is made** in order to discuss any further measures in good time. **This shall not apply to systems which, at the time of delivery, were already configured and built to handle various specific material systems!**

About the permissible operation modes

The machine is only intended for the tasks specified in the order confirmation or agreed upon contractually at a later point in time. Operation of the machine is only permissible:

1. if these operating instructions have been read and understood completely.
2. In accordance with the procedures described in these operating instructions.

3. if the process materials intended for this machine whose components and optional cleaning products have been approved by bdtronic.

Recommended extraction system

Pollutants that occur during the processing of solid and liquid materials (e.g. resin casting, etc.), must be collected at the point of origin and sent for disposal, especially if they endanger or lead to endangerment of humans or the environment due to their concentration or material properties.

bdtronic GmbH recommends installing an extraction system for effective removal of the emissions at the site of operation for all substances that are classified as pollutants according to the GHS STANDARD as of 2014 and that are used in their systems. In addition, an extraction system is mandatory for certain products that are classified as fluid class 1 or 2 (e.g. isocyanates). That is to say, commissioning or production, maintenance or troubleshooting, or the filling or refilling of containers may not be carried out without an effective extraction system. The time of the onset of the extraction measure starts at the latest with the loading or reloading of the system or machine component(s) with this material.

In individual cases, bdtronic GmbH already calculates the relevant minimum extraction power(s) for the extraction connections, e.g. on the working cell or the material room, etc., during the machine design by the specialist departments. These values are entered in the internal and the external documentation.

Depending on the order, a special sensor monitors the function of the extraction of the machine or the machine components in such a way that it is not possible to switch on or switch on again or to carry out production if there is insufficient extraction.



NOTICE

Further safety measures

Effective extraction does not replace any specifications additionally required by the safety datasheets for a product for the safety of the workplace or the person who handles the product with respect to protective clothing, training, etc.



NOTICE

Fresh air supply

Sufficient fresh air supply must be provided to the workplaces connected with the system or to the workspace surrounding the system, especially when the extraction system is running.

1.5 Foreseeable misuse | Dispensing machines

Operation of the machine is especially not permissible:

- If it is possible to reach into a danger zone during the ongoing production process due to missing barriers, etc.

- If set-up work is carried out on open material circuits other than for the purpose of checking supply bottlenecks or leaks.
- If a workpiece carrier or a part variant has become jammed in the ongoing process and manual intervention on the part of the operator is required.
- If there is a possibility of treatment for other than the specified parts and their variants when using a kiln.
- If, during the use of a plasma device, there is a possibility of treatment for other components and their variants than those specified.
- If an optional transport system (rotary plates, transport belts, etc.) is used for the transport of components and their variants other than those specified.
- If maintenance work is implemented on pressurised components which, despite stopping of the machine, are still under pressure (material carriers for dynamic systems!).
- If maintenance work is carried out on an automated movement systems (axes, cylinders, etc.) if these machine components are still live or under pressure.
- If, when refilling material containers, the specified materials are not filled and the flows and safety measures specified in the operating instructions are not adhered to.
- If, in case of disruptions in the production flow, impermissible tools or objects are used to restart either a product carrier flow or the further processing of a pallet
- For manual functions – especially for the movement of linear systems (axes) and rotation systems (rotary tables, etc.) – that are not performed in the correct sequence (info: the person performing the functions is also responsible for the result if particular combinations of functions are not blocked one against the other. For manual functions, there is a duty to train).
- If, during manual evacuation of a material vessel, the correct sequence for initiating the evacuation process is not adhered to (info: first depressurise the material container, then close all inlets and outlets, open the evacuation circuit).
- If manual ball valves requiring a particular position for the process have been set to the wrong position.
- If safety devices such as door safety switches, guard plates, protective tunnels on transport lines, enabling switches, etc. are de-installed or deactivated.

1.6 Operation of the machine

	NOTICE About the details in the operating instructions All details in these operating instructions refer only to the machine whose design and settings are listed in the "Setting data" for the machine or that are contractually agreed upon with bdtronic GmbH at a later point in time.
	NOTICE About the work clothing to be worn when working on the machine Danger of injury due to getting caught or being pulled in. • Do not wear long hair loose, loose clothing or jewellery (bracelets, chains, rings, etc.)
	NOTICE Correct workplace lighting Workplace lighting must be designed according to "ASR 7/3" of the workplace regulations of 1st October, 1993.
	NOTICE About the operating instructions, instructions and regulations The operating instructions, instructions and regulations are a component of the machine and must always be easily available, legible and complete for all people working with and on the machine.
	NOTICE About the operator's operating instructions §20 GefStoffV obligates the operator to create operating instructions for the operating personnel on how to handle hazardous materials and to train them.
	NOTICE About the adjustment, maintenance and inspection work! Adjustment, maintenance and inspection work must be performed according to the instructions and on time.

	NOTICE About the protective and safety devices The protective and safety devices, locks and couplings, etc. must be inspected by an expert at least once a year on the basis of DGUV regulation 3. These inspection results must be logged in an inspection certification and kept.
---	--

1.7 Duties of the personnel

	NOTICE About the safety devices in the workplace • The safety devices must be properly attached and functional. • Safety and danger markings on the machine must be clean and legible. • Never remove, cover, work around or bypass the safety equipment and markings on the machine.
---	---

	NOTICE In case the machine stops • Switch off the machine if it is not being used, before maintenance work, repairs or if a malfunction is determined or suspected. • Report malfunctions or damage to the responsible supervisor immediately. • Make sure that the work area is properly safeguarded. • Discontinue further production until the damage has been repaired.
--	--

	NOTICE Eating and drinking are forbidden in the work area. Eating, drinking and smoking, as well as the storage of food or foodstuffs in the work area are strictly forbidden.
---	---

	NOTICE About checking before starting work on the machine • Before starting work, check the proper state of the machine. Pay special attention to loose connections, leaks, abrasion, cuts, cracks, bubbles or kinked and crushing points. • Report such damage to your supervisor immediately. • Secure the work area properly and forbid the continuation of production until the damage has been repaired.
---	---

	NOTICE About working on the machine • Who is responsible for what work on the machine must be regulated. • Only qualified, trained personnel who have understood the descriptions in the instructions are allowed to perform work.
	NOTICE About general safety Find out exactly where the machine documents, escape routes, fire extinguishers and first aid supplies are located.
	NOTICE About cleanliness in the workplace • Keep your workplace clean and easily accessible. • Do not use the machine table as storage, for sitting on or as a climbing aid.
	NOTICE Ensure sufficient fresh air supply at the workstation Workplaces for fluid class 1 and fluid class 2 machines that process isocyanate must always be well ventilated. The operator must ensure a sufficient supply of fresh air
	NOTICE About the use of personal protective equipment Personal protective equipment must be available, in proper condition and used. In particular for fluid class 1 materials and fluid class 2 materials such as isocyanates, a protective mask is prescribed (fluid class 1) or urgently recommended (fluid class 2) during manual refilling of the material vessel. The protective mask should fulfil at least the requirements of an ABE1 filter.

1.8 Personnel deployment

	NOTICE Performing work on the machine The machine is classified in the category "fuel-driven tools with dangerous movements". • Therefore, only personnel that has been trained for such work or is familiar with how to perform it safely and independently and is commissioned by the operator with this work may perform any work on or with the machine. • People in training and people under 18 years of age may only work on or with the machine under the supervision of a trained or instructed person.
	NOTICE About the training duty of the operator For all work with or on a machine or device where there is a danger to the health and/or life of people, the operator has an obligation to provide training and instruction.
	NOTICE About the safety training of personnel People who work with and on the machine must confirm with their signature that they have read and understood these operating instructions, especially the safety instructions, and that they have been trained in the handling of the machine as well as in the handling of hazardous materials.
	NOTICE In case of work due on the machine The connection, set-up, maintenance and repair work may only be performed by trained personnel.
	NOTICE About the responsibility of personnel • The operator of the machine must clearly specify the responsibilities of personnel for particular task areas such as assembly, commissioning, operation and maintenance. • If there are questions or a lack of clarity with and on the machine, the responsible personnel must be consulted.

	NOTICE About ambiguities on the machine If ambiguities or questions arise when working with or on the machine, they must be clarified with the responsible personnel.
---	--

1.9 Special dangers

	DANGER Use of high-voltage devices When using high-voltage devices or special assemblies, it must be ensured that no conductive parts are fed into or protrude out of the area of the high-voltage discharge zone. These must always be earthed! This can cause danger to life. Furthermore, it can cause irreversible damage to the device so that safety can no longer be guaranteed.
---	--

	WARNING Warning about burns! When using high-voltage devices and special components (e.g. plasma devices), care must be taken because great quantities of heat are also released in the area of the high-voltage discharge zone; severe burns can result if these areas are touched. Before performing maintenance and repair work on components and parts in this danger zone, first wait after switching off the machine until a temperature of under 40 °C has been reached. In any case, the wearing of sufficient protective clothing is recommended.
--	---

	NOTICE Danger point due to incorporation into a production line! Danger points that arise due to the incorporation of the machine into a production line or through combination with upstream or downstream machines, systems or components must be determined by the operator and secured.
---	--

1.10 Handling hazardous materials

1. If a replenishing pail is not sufficient: Repeat the replenishing process from the point "Taking the old pail out of the storage vessel"



WARNING

Disposal of the old pail

Immediately eliminate spilled materials with the suitable cleaner. Return the old pail together with soiled cleaning material to your regional or national waste disposal company. Pay attention here to the hazardous material classification of the old pail and the cleaning materials soiled with cleaner residues.

- Immediately eliminate spilled substances with the cleaner prescribed for this and ensure the proper storage and disposal of waste.
- Smoking, naked flames and lights in the workspace and when handling hazardous materials are strictly forbidden!
- Do not use any substances that are unknown or that have been changed. Never mix substances which can result in unknown reactions. Pay attention to information regarding statutory regulations.
- Avoid any direct contact with hazardous materials.
- Carefully read the safety data sheets for the materials used when working with and on the machine, and adhere to the details and instructions they contain.
- Machines that process materials of fluid class 1 or fluid class 2, in particular isocyanates, are designed for one operator, i.e. other persons should avoid this workplace because of the possible escape of vapours or outgasings!
- According to the bdtronic company standard "**Instructions for use of isocyanates or CMR materials**" a minimum suction capacity of 167 NL/min (= approx. 10 m³/h) must be connected when using a bdtronic flushing agent recirculation, and an MPS 500 NL/min when using a vacuum pump on the hardener side. (= approx. 30 m³/h). When using isocyanates, an extraction capacity of at least 100 m³/h must always be connected for the entire system!

	DANGER Hazardous handling of caustic materials! Severe burns and irritations of breathing passages and skin, on through to death. • Any person who handles caustic materials must wear protective equipment. • The protective equipment must be available and in proper condition. • Avoid touching the skin. • After contact with the material, clean the point immediately with the prescribed agent.
	NOTICE About the handling of hazardous materials For the handling of hazardous materials, the safety regulations for handling these materials must be heeded: Ask your supplier for the current safety data sheets for all materials used for all work with and on the machine. Among other things, the safety data sheets include the following binding details: Physical and chemical material properties, possible dangers, storage/transport, fire/explosion protection, disposal/environmental protection, first aid, personal protective equipment, reaction with other materials.



NOTICE

About the duty of training for hazardous materials

§20 GefStoffV obligates the operator to create operating instructions for operating personnel on the handling of hazardous materials and to train them.

1.11 Transport

Machines and systems are relocated after a factory acceptance or as a result of a change of location. It often requires special knowledge to disconnect the machine or system from the power supply, to dismantle it into sub assemblies, to pack, ship and reinstall it. Due to the number of existing machine concepts, these processes can only be described in general terms.

Even the onward movement of larger units with forklifts, cranes, etc. also requires the experience of vehicle drivers who are totally familiar with the object or local circumstances and can handle the goods without any danger.

1.11.1 Set-up or installation of the component, system part(s), system

- ✓ The scope of delivery was checked positively for completeness and correctness
 - ✓ The component, system part(s), system has/have been delivered undamaged
 - ✓ The component, system part(s), system is/are directly at the intended destination (without transport pallet)
 - ✓ The installation location is freely accessible, clean, free of dust, level
 - ✓ The installation location is a closed room whose temperature remains constant at between 15 °C and 25 °C with a constant absolute humidity of up to 40 %
 - ✓ Energy supply (electrical, pneumatics, etc.) is available at the installation location
 - ✓ Access to an exhaust system is available if the material or system necessitate this
1. Free the component, system part(s), system from packaging materials (e.g. protective film)
 2. Remove all transport tools (locking of optional axes, fixations of optional doors, etc.)



NOTICE

Disposal instructions

All transport tools that accumulate from unpacking the component, system part(s), system (protective film, transport belts, etc.) must be disposed of according to the regional or national disposal regulations.

3. Connect the component, system part(s), system to an operational exhaust system if the material or system necessitate this
4. Connect electrical separating points so that the plug and associated bushing fit together according to designation (see also the electrical wiring plan in the switch cabinet)
5. Connect pneumatic separating points so that the plug and associated bushing or the associated hose ends fit together via coupling pieces according to designation (see also the pneumatic plan in the operating instructions.)
6. Install loose parts or components (such as covers, balance, etc.) mechanically so that later commissioning is guaranteed
7. Integrate process modules such as the hot riveting head and plasma head according to assembly instructions
8. Integrate components such as dispensing modules, etc. according to assembly instructions
9. Align and fix optional signal column(s)
10. In dispensing systems, material preparation units: Pull the material line up to the connection point if necessary, remove the blind plugs from the line ends; immediately connect the material lines to the assigned connection point (see the material line plan)
11. In systems with flow detection: Place the flow detection at the point intended for this purpose, and wire it if disconnected
12. In separate transport belts: Grind the separating point and join together with belt welding equipment; pull in the belt



NOTICE

Running direction

Heed the running direction of the belt.

13. Remove the transport strut(s)
14. Connect all disconnected protective conductors



WARNING

Trained electrician

This work may only be carried out by a trained electrician.

15. Screw all plates, panels, etc. disassembled for transport purposes together with the protective conductors again at the original position
16. In swivel arm structures: Remove the transport locks so that the arm which swings freely when unpacking cannot cause any injury (e.g. by covering the outlet nozzle, limitation of the arm movement)
17. If installed, remove the blind plugs on the ruby red gel filter



NOTICE

Switching on

During cold weather or at extreme temperatures, let the component, the system part(s), system adjust slowly to room temperature before being placed in operation. In the case of condensation, the component, the system part(s), system may only be switched on after a waiting time of at least 12 hours.

1.11.2 Disconnecting the machine from the power supply

- ✓ The machine has already been shut down (see also chapter "Shutting down the machine")
- 1. Remove the power supply cable from the mains socket
- 2. Close the hand switch-on valve of the main maintenance unit
- 3. Disconnect the compressed air supply hose from the building compressed air network
- 4. Release the pressure on the optional material vessel with ventilation screw of the maintenance unit
- 5. Disconnect optional devices with their own power supply from the operational power supply

1.11.3 Removing and making the machine or system available

- ✓ The power supply has already been disconnected
- 1. if agreed with the customer: Dismantle the machine into smaller components
- 2. Fix moving parts such as the axle system, drawers



NOTICE

Centre of gravity

Lock the optional axle system so the machine centre of gravity is over the middle of the table.

1. Seal all switch cabinet doors
2. Secure the front sliding door against accidental opening using straps
3. Remove the optional balance and pack separately
4. For plasma or hot riveting machines, pack the head separately once again.
5. Remove the stamps from the hot riveting heads and pack in the containers provided
6. Transfer and secure the signal columns
7. Disconnect the material lines from the dispensing or impregnation machines and seal with screw plugs

8. Remove optional solvent recycling units and pack separately with cleanser
9. Seal off the open ends of tempering lines with screw plugs
10. Secure the lifter arrangement of a PPS against lowering
11. For pump over units: Removal of the barrel, locking of the lifting cylinder, securing of the cover; tying down the barrel pump with frame
12. Screens remain on the frame: Pack additionally
13. Disconnect optional flow detection if the drive-through or transport height is exceeded or fasten laterally on the frame
14. Remove workpiece pallets and pack separately
15. Transport sections fitted directly into cells remain there
16. Remove external transport systems on supports and pack separately
17. Where possible, lift all components onto suitable pallets and tie down or screw onto them
18. Fasten roller-supported machine onto the pallet with special fixtures
19. Machines or machine parts, their dimensions and weight distribution that endanger stability on the pallet (top heavy) must be specially marked
20. Vessel racks mostly remain on the dispensing machine
21. Protruding parts that project beyond the surface that must be raised must be additionally secured
22. Insert transport struts on the front side of table systems or cells
23. Individual components, such as dispensing pumps, dispensing valves, etc., are packed in clean conditions and in certain cases such as eccentric worm pumps usually with sealing medium, delivered in special packaging.
24. Protective conductors and plug connectors are separated if necessary
25. Remove any manipulators of safety devices that were fitted for test purposes
26. Back up software versions of the machine
27. Divide pneumatic hoses if necessary: provide both associated ends with a unique marking
28. Seal the ruby red gel filter with open circuits on both sides with blind plugs
29. Divide conveyors if necessary.
30. Fix optional swivel arm structures in such a way that the swivel arm cannot make any dangerous movements after packing.



NOTICE

Protect against dust and water

Seal components, machine parts, machines with tear-proof protective film against dust and penetration of water.

	NOTICE Packaging routes Heed special regulations of specific packaging routes (air freight, sea freight, etc.).
	NOTICE Vibrations and impacts Despite the robust construction, the installed components are sensitive to strong vibrations and impacts. Protect the device during transport against large mechanical loads. The original packaging should be used for specific components (balance, etc.).
	NOTICE Plug connectors The plug connectors are often coded. Separating points are marked on both sides according to the schematic.

1.11.4 Moving the machine or component

- ✓ Module, machine part(s) or machine have been packed according to instructions
 - ✓ Special hazard points, e.g. due to top-heaviness, have been identified and marked
 - ✓ The assembly, machine part(s) or machine is/are capable of being transported (e.g. by pallet, transport lugs, etc.)
1. Pick up the component, machine part(s) or machine with suitable load lifting devices (e.g. forklift, crane, electric lifting truck, etc.) in accordance with the adjacent graphic on the points marked with arrows

	WARNING Hoisting capacity It is essential to avoid any swinging back and forth of the suspended load when picking up by crane! Moving under the supported load is forbidden!
	WARNING Transport route Select the transport route and velocity with regard to weight, weight distribution and sensitivity of the transported goods so there is no hazard when loading and unloading and the transported goods reach the loading or unloading position undamaged.



WARNING

Condensation

When transporting in cold weather, or if the assembly or the machine is exposed to extreme temperature fluctuations, make sure that no moisture condenses.

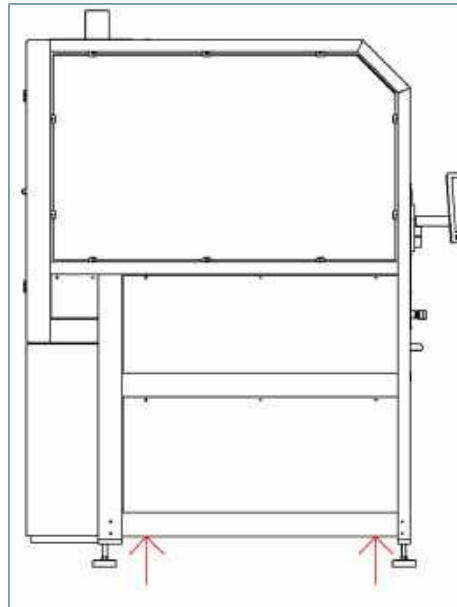


Illustration 1: B5200

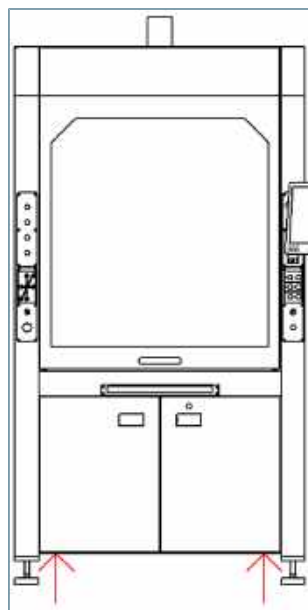


Illustration 2: B5200

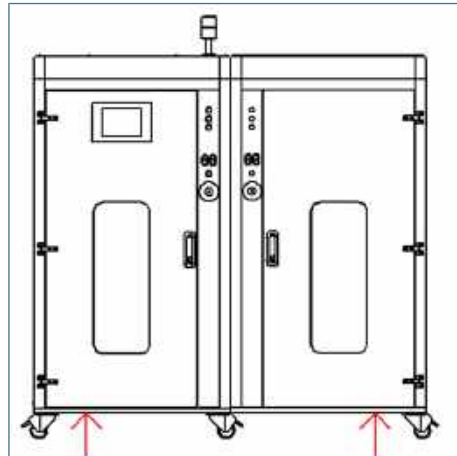


Illustration 3: PPS



NOTICE

About the exceeding of lifting and carrying capacity!

If the reasonable lifting and carrying forces are exceeded during set-up or work on the machine, suitable auxiliary equipment must be present and used.

Here, special attention must be paid to the safety instructions for load lifting devices.

1.12 Storing the machine

- The system must be shut down properly.
- The energy supplies (voltage, compressed air) must be disconnected from the machine and the energy storage systems must be drained.
- The machine and machine assemblies must be stored in a protected area (dry, humidity under 40%, dust-free, at temperatures between 15 and 40 °C).
- Sensitive assembly entry points (e.g. axes or sensors) must be protected against mechanical impacts by additional covers and locks to ensure that the machine can be taken back into operation properly.
- Remove any fluids (for example: transmission fluid, temperature control media, etc.) from the machine or the machine assemblies, to the extent possible.



NOTICE

Missing operating fluids

Resupply any missing operating fluids before taking the machine back into operation.

- Remove adherent material remnants (dust, resins, etc., arising from the process steps) completely.

1.13 Taking out of operation/disposal

- The system must be shut down properly.
- The energy supplies (voltage, compressed air, water circuits) must be disconnected from the machine.
- Optional material containers must be vented.
- All assemblies that contain materials (for example: containers, material lines, ball valves, pressure sensors) and that are difficult to clean must be disposed of according to local disposal regulations.
- All assemblies that contain materials (for example: dispensing pumps, dispensing valves, mixing heads) and for which there is a specific cleaning instruction in the operating manual must be disposed of according to local disposal regulations.
- The (one-component) remnant containers (resin, hardener, cleaner) are supplied to the certified supply company according to the EAK key, which is mentioned in the "Disposal" chapter of the respective safety data sheet.
- Optional material mixtures consisting of resin and hardener can be disposed of as household waste once they have hardened.
- Optional material mixtures consisting of resin and hardener must be disposed of separately as hazardous waste.
- All other modules (power supply units, controls, screens, transport equipment etc.) are disposed of in accordance with regional regulations.



NOTICE

Regional disposal regulations!

Follow the regional disposal regulations in your area when disposing of special type of waste.

2 Structure and functionality

2.1 Protective equipment for the machine

To protect the operating personnel and the environment, the machine is equipped with the following protective equipment:

- Acoustic malfunction signals.
- Identification of the material lines (lines that transport the isocyanates are marked specially on both ends), compressed air connections and the electrical connections.
- Switch cabinet with mechanically-locking main switch.
- Cover panels for voltage-conducting parts.
- Warning signs on danger points.
- Visual malfunction indicators as signal column or as individual lamps, e.g. for access control.
- Automatic controller for the work processes with automatic interruption of operation in the event of malfunctions.
- EMERGENCY STOP command device with latching button.

2.2 Technical data

The following table lists the main dimensions and connection features of the machine described in these operating instructions in such a way that only standard machines are described. The actual dimensions are the topic of the enclosed layout.

Dimensions, space requirement (details in mm)	
Width	1190
Depth	1600
Height (maximum)	2200
Optional drive paths (details in mm)	
Y-direction	870
X-direction	645
Z-direction	200
Weight (details in kg)	
Without additional components	600
Electrical connection	
Voltage supply (V)	400
Fuse protection (A)	See type plate
Frequency (Hz)	50
Compressed air connection	
Supply pressure (bar)	6
Flow rate (l/min)	500
Compressed air quality	DIN ISO
Optional air cooling rate (l/min)	----

Extraction system	
Extraction rate (NI/min)	----
Noise level	
Noise measurement (dB(A))	73

**NOTICE****Other technical data**

Additional technical data can be found in the function plans (electrical, pneumatic, etc.). They can be found in the appendix to these operating instructions.

2.3 Technical description

Structure of the machine

The main components of the machine are described below: The most important features of the component that are necessary for understanding the role it plays in the entire system are explained in an easy-to-understand manner in the points "**Function**", "**Application**" and "**Structure**". In addition, the manufacturer's supplier documentation which is included with the operating instructions can be applied for additional understanding of details.

**NOTICE****Organisation and completeness**

The organisation of the description modules is arbitrary and does not reflect the structural layout of the machine. Only such components that belong to the basic understanding of the function of the machine for operation, maintenance, troubleshooting are described. According to the contract, constructive details are included in the drawings of the machine in the appendix. In particular, the colour and signal programming of the machine lights is highly customer-specific and is described in detail and in the specifications underlying the system and optional operating standards.

2.3.1 Axis system module

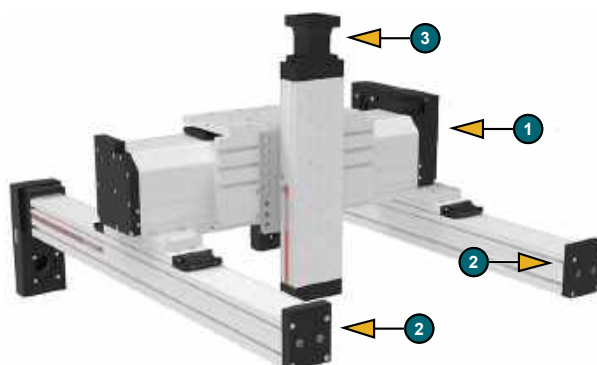


Illustration 4: Exemplary illustration (without drive units)

1 X-axis	2 Y-axis: here designed as a portal
3 Z-axis	

Application

3-axis system for tool movements (mixing head, etc.). The travel path lengths of the axes can be found in chapter "Technical data".

2.3.2 Operating and display elements

2.3.2.1 Main switch

Application

The main switch is available in different colours: red-yellow and black-grey. It has 2 switching positions: "ON" or "I" or "OFF" or "O":

"OFF": The voltage was disconnected.

"ON": The voltage was forwarded to the electrical components.

Type, red-yellow	Type, black-grey
Disconnects the voltage supply of an entire machine	Disconnects the voltage supply of the function component, etc. in the assigned machine part only
Also as an E-STOP in machines/ devices without an additional E-STOP command device	-----

2.3.2.2 Enabling switch

Application:

The enabling switch is used if machine personnel (set-up technician, service technician, etc.) intervene in ongoing operation in order to perform special work such as setting up the position exactly, aligning the sensors, etc. when the access doors are open. The SETUP operation mode is necessary for operation.

Function:

The enabling switch recognises three settings:

1. Not pressed: SAFETY operation mode is active, axes are stopped
2. Semi-pressed: SETUP operation mode is active, the axes can be moved at a particular velocity and to a particular position.
3. Fully pressed: SAFETY operation mode is active; axes are stopped



Illustration 5: Wenglor enabling switch

1	Enabling switch	2	Confirmation button
---	-----------------	---	---------------------

2.3.2.3 Emergency stop command device

Application:

The EMERGENCY STOP command device is circuited directly to the PLC safety elements, e.g. safety PLC, in order to de-energise all electrical and pneumatic components of the machine immediately when pressed regardless of the operation mode and user log-in. It is a safety element that can work networked with several additional elements in the system network.

Function:

When pressed, the unlocked button will be locked: The control and load voltage are disconnected and an optional pressure accumulator emptied. EMERGENCY STOP operation mode is displayed.

**NOTICE****Actuation of EMERGENCY STOP command device**

The EMERGENCY STOP command device should only be used to protect people in dangerous situations, i.e. from the effects of the system or system components. Except for the purpose of testing the command device itself, it may not be used to switch off systems or system parts simply to test their function!



Illustration 6: Figure: EMERGENCY STOP command device

2.3.2.4 Signal column

Application

The signal column indicates particular operating states of the machine so that personnel responsible for the machine and production can react as quickly as possible to changes (e.g. errors, lack of materials, etc.) without having to be directly in the working area of the machine.

Function

Depending on the operating state of the machine, one or several of the different coloured signal lamps is activated so that either a steady light or a flashing signal can be perceived until the operating states have changed again. An additional horn produces an intermittent audible signal. The programming and arrangement of the lamp colours is specified by bdtronic and can naturally be adjusted on customer request.

Standard design



NOTICE

Machine configuration

The following description of this module shows only one possible application configuration. The illustration may differ from your customer-specific machine or module. This means that instead of a ruby gel filter, a diaphragm dryer can also be installed, and there are different versions of flow rate valves (FRV), etc. The basic function remains the same!

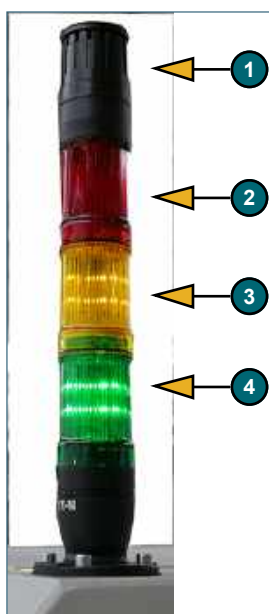


Illustration 7: Signal column red-yellow-green horn

1	Acoustic signal (e.g. potlife)	2	Error (e.g. print error, etc.)
3	Warning (e.g. lack of material, etc.)	4	Operating (operational readiness of AUTOMATIC as steady light, PROCESS MONITORING as flashing signal)

2.3.2.5 Membrane keypad move axis

Application:

Machines can also be controlled manually through interventions by the machine operator, set-up or service personnel. This requires elements that must be operable from outside. In particular, important functions such as operating states such as 'Machine on', change of operation mode, etc. are switched and sometimes also visualised via signal columns and display fields on a menu-guided operator interface.

Function:

By pressing the membrane keys, e.g. < Machine on >, a function is switched on or sometimes switched over (only if the keys are double programmed). The function remains active until another intervention is made.

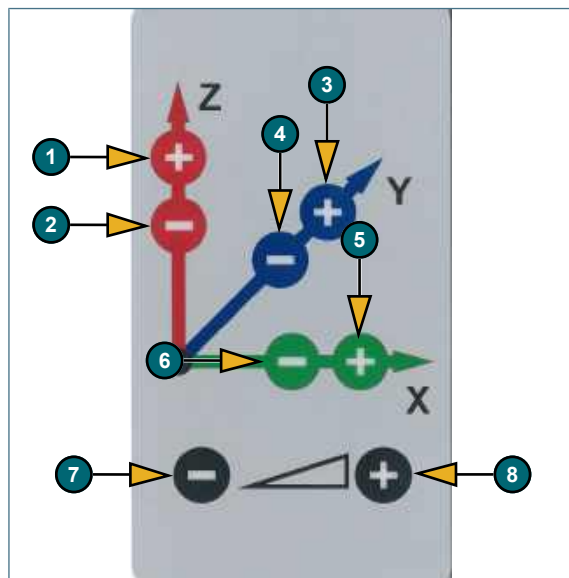


Illustration 8: Axis membrane

1 < Move Z+ axis > membrane key	2 < Move Z- axis > membrane key
3 < Move Y+ axis > membrane key	4 < Move Y- axis > membrane key
5 < Move X+ axis > membrane key	6 < Move X- axis > membrane key
7 < Axis speed - > membrane key	8 < Axis speed + > membrane key

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

< Move Z+ axis > membrane key (1):

Pressing this button moves the Z-axis in the positive direction.

< Move Z- axis > membrane key (2):

Pressing this button moves the Z-axis in the negative direction.

< Move Y+ axis > membrane key (3):

Pressing this button moves the Y-axis in the positive direction.

< Move Y- axis > membrane key (4):

Pressing this button moves the Y-axis in the negative direction.

< Move X+ axis > membrane key (5):

Pressing this button moves the X-axis in the positive direction.

< Move X- axis > membrane key (6):

Pressing this button moves the X-axis in the negative direction.

< Axis speed - > membrane key (7):

Pressing this key reduces the speed default for the axes.

< Axis speed + > membrane key (8):

Pressing this key increases the speed default for the axes.

2.3.2.6 Membrane keypad general operation

Application:

Machines can also be controlled manually through interventions by the machine operator, set-up or service personnel. This requires elements that must be operable from outside. In particular, important functions such as operating states such as 'Machine on', change of operation mode, etc. are switched and sometimes also visualised via signal columns and display fields on a menu-guided operator interface.

Function:

By pressing the membrane keys, e.g. < Machine on >, a function is switched on or sometimes switched over (only if the keys are double programmed). The function remains active until another intervention is made.

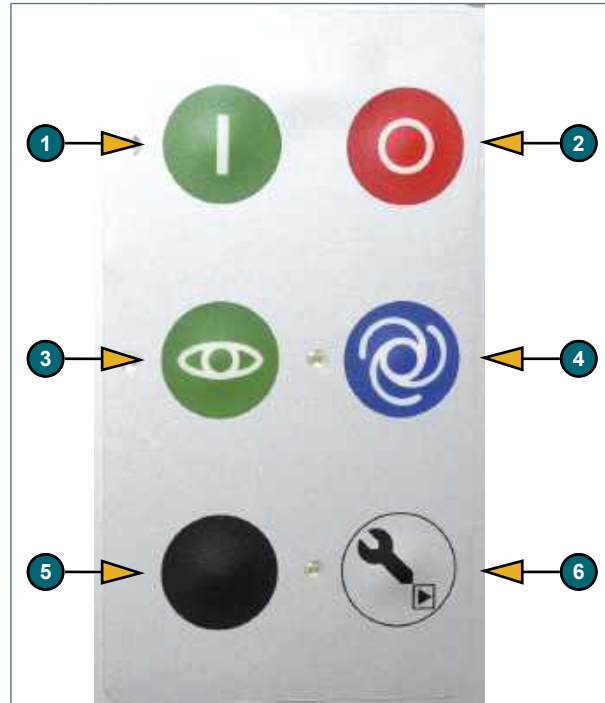


Illustration 9: Operation mode buttons

1 < Machine on > membrane key	2 < Machine off > membrane key
3 < Process monitoring on > membrane key	4 < Automatic on / off > membrane key
5 < Stop > membrane key	6 < Move to set-up position > membrane key

< Machine on > membrane key (1):

Pressing the key connects the control and load voltage to the components:

1. Safety systems are active
2. MANUAL operation mode is active

< Machine off > membrane key (2):

Pressing the key disconnects the control and load voltage: electrical consumers are no longer active; the compressed air supply is also disconnected.

< Process monitoring on > membrane key (3):

PROCESS MONITORING operation mode is switched on.

**NOTICE****Key switch**

The function of this key must only be viewed in association with the position of the key switch: Only in the "Automatic" position does it switch to PROCESS MONITORING operation mode.

< Automatic on / off > membrane key (4):

With the first key press, the machine switches into AUTOMATIC operation mode; with the next, to PROCESS MONITORING operation mode.

**NOTICE****Key switch**

The function of this key must only be viewed in association with the position of the key switch: Only in the "Automatic" position does it switch to PROCESS MONITORING operation mode.

< Stop > membrane key (5):

Pressing the key stops the processing step immediately: MANUAL operation mode is active.

< Move to set-up position > membrane key (6):

Pressing the key moves the dispensing unit to the set-up position.

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

2.3.2.7 Key switch for Setup/Automatic

Application

Key switches have several switching positions (here: two). They can be used to put a machine into different operating states. In our application, it is possible to use the key switch to switch between the SETUP ("Setup" position) and MANUAL (position: "Automatic") operation modes

**NOTICE****Homonymy**

The "Automatic" switch position designation is not the AUTOMATIC operation mode itself. This is only switched on or preselected by further operating buttons or via a symbol on the status bar of the visualisation system.

"Setup" position: Safety devices have been bridged: To move axes when doors are open, the activation switch must be pushed halfway in. The key switch cannot be removed.



Illustration 10: "Setup" key switch position

"Automatic" position: Safety devices are all active: To move the axes or an optional pneumatic actuator, the access doors must be closed and the optional light barriers in the "non interrupted" status. The key switch can be removed.

**NOTICE****Operation modes**

In the "Automatic" position, the access doors are always locked in the AUTOMATIC and PROCESS MONITORING operation modes.



Illustration 11: "Automatic" key switch position



NOTICE

Key switch variants

Naturally, as for each product, there are also modified construction designs with slightly different functional characteristics. The key switch is available where the key can be removed in both switch positions. At the customer's request, the positions can also be labelled "Setup off" instead of "Automatic", or "Setup on" instead of just "Setup". The differences in designation do not change anything with regard to the functionality represented.

2.3.2.8 Captron-Sensortaste



Illustration 12: Captron sensor button

Application

This button reacts to contact. Its status is indicated by a play of colours on the LEDs.

LED-Color	Activity	Description
Green	Permanent light	Command has been saved
Green	Blinking	Requirement for operator
Red	Blinking	Error state

2.3.3 Mixing technology

2.3.3.1 Duplex+

Application:

In case of a filling or dispensing command, a mixing tube blends two material streams (resin and hardener) via a mixing tube so that a reactive mixture is available at its end.

Mode of operation:

The materials (resin and hardener) are introduced separately into the single-use mixing tube via 2 progressing cavity pumps. Here, a dead space-free arrangement of the feed channels ensures optimal intake into the mixing tube head. The special arrangement of the mixing elements in the mixing tube brings the resin and hardener into contact step by step. The material quantity added per second determines the mixing ratio.

At the end of the production cycle, the mixing tube is simply replaced with a new one: thus, no cleaning work is required.

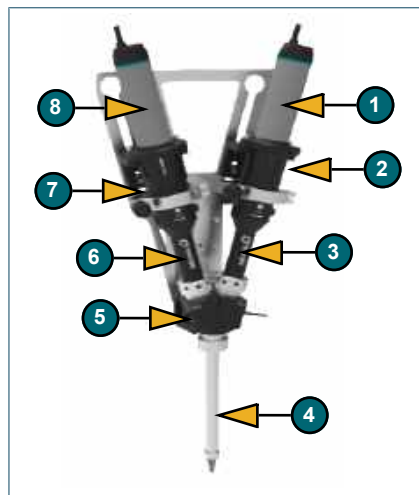


Illustration 13: Duplex+ plus pump design

1	Pump drive B	2	Material line B
3	Progressing cavity pump B	4	Single-use mixing tube
5	Static mixing head	6	Progressing cavity pump A
7	Material line A	8	Pump drive A

2.3.4 Air preparation

2.3.4.1 Service unit

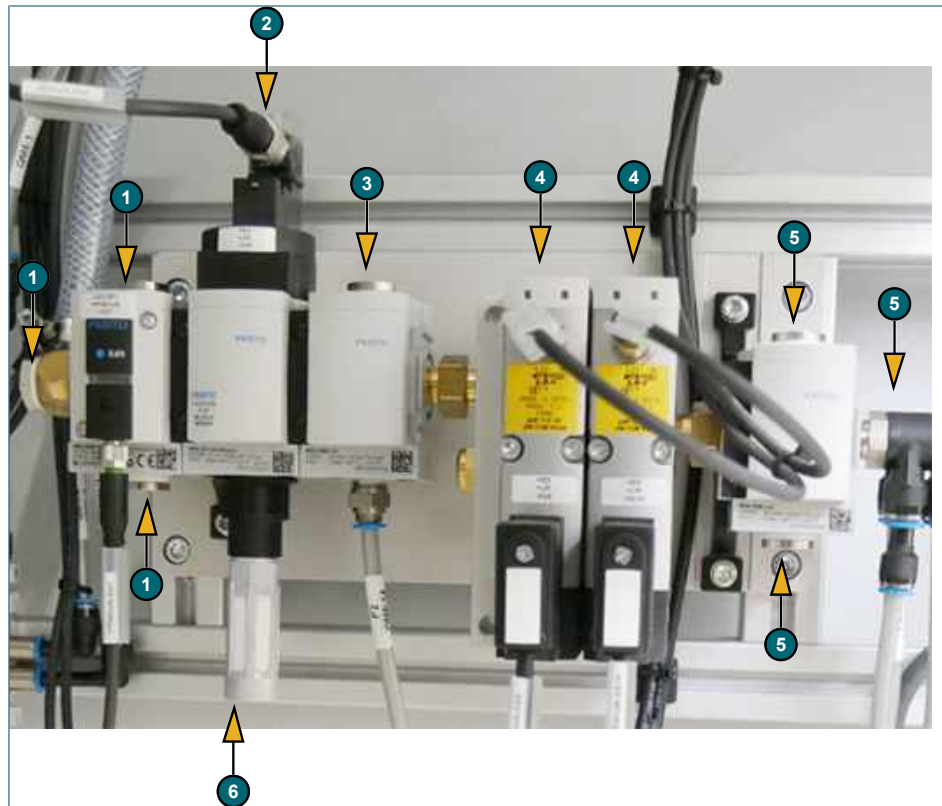


Illustration 14: Maintenance unit | Double safety circuit

1	Compressed air inlets	2	Electric safety valve (Emergency STOP)
3	Distributor	4	Electric safety valves (door safety)
5	Distributor	6	Silencer

Application

Control and enabling of the compressed air circuits.

2.3.4.2 Manual start-up unit

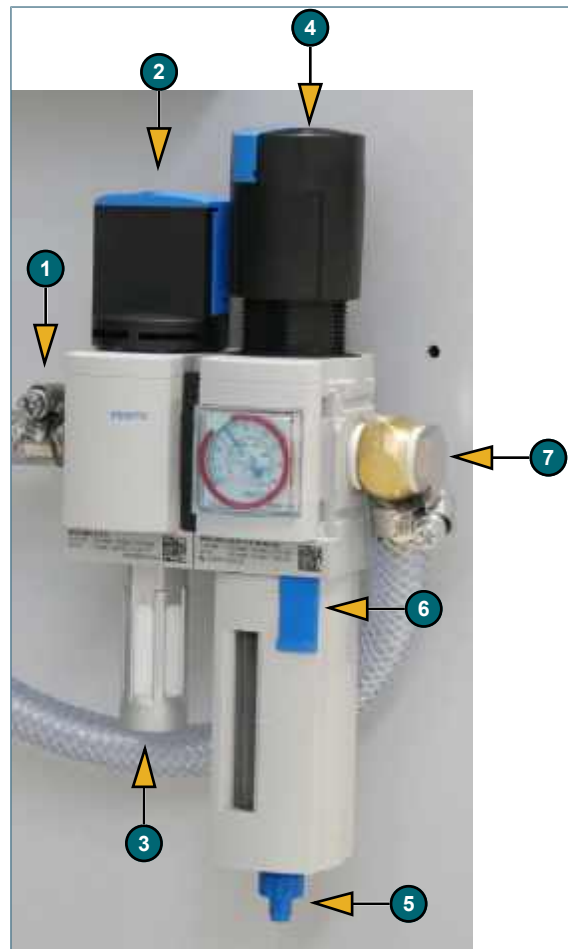


Illustration 15: Manual start-up unit | Manual start-up valve | Pressure regulator with condensate drain plug

1	Compressed air inlet with shut-off function	2	Manual start-up valve
3	Silencer	4	Pressure regulator with shut-off function
5	Condensate drain plug	6	Condensate trap unlocking slide

Application

Supply of a variable flow of compressed air to the service unit.

2.3.5 Sensors

Application

Sensors detect physical values such as distances, weights, material properties, etc. and transform them into numbers or signals so that an evaluation unit, e.g. PLC, can process them further. To do this, the values that are required for decision-making in algorithms are screened out. This can be implemented using input fields that specify the limits (e.g. pressure limits, etc.) on the system visualisation.

The incorporation of the sensors into the system environment is a topic for the electrical schematic. There, the individual sensors can be identified and the function components assigned. It is possible to query the values in real time via the linking with the appropriate variables in the TWINCAT SYSTEMMANAGER.

Typical sensors that are used in bdtronic systems are:

Pressure sensors

Fill level sensors

Optical sensors for component and component position detection

Distance sensors

etc.

Mode of operation

The detailed description of the use and mode of operation is a topic for the included manufacturer documentation. There, depending on the sensor type and complexity, information can be found about processes such as connection, optional teaching, usage limits, etc.

2.3.5.1 Wika pressure sensor

Application:

A dispensing circuit is only stable if the pressures defined at the marked points are stable and move only within specified limits. The control unit queries the installed pressure sensors cyclically in order to decide about further interventions. Important is the recording of the supply pressure of the dispensing pump, of the pressure during the filling process, the outlet pressure of the dispensing pump. Furthermore, through suitable specification of limits on the linked screens of the operator interface, it is possible to detect an error situation in timely fashion or also to work effectively to counteract the start of wear, e.g. of the dispensing pump.

Mode of operation:

The sensors are incorporated at the point to be monitored and linked to the control core as part of an input component. As soon as the evaluation unit has power, it provides current values. The PLC visualises these in real time and also provides error messages in case of limit over- or underruns.

**NOTICE****Machine configuration**

The following description of this module shows only one possible application configuration. The illustration may differ from your customer-specific machine or module. This means that instead of a ruby gel filter, a diaphragm dryer can also be installed, and there are different versions of flow rate valves (FRV), etc. The basic function remains the same!

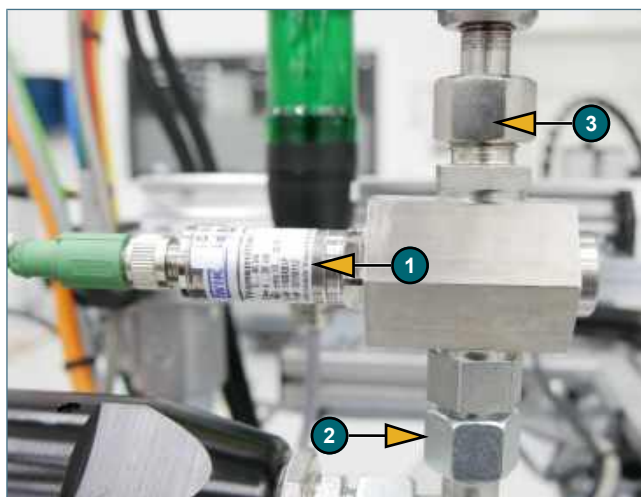
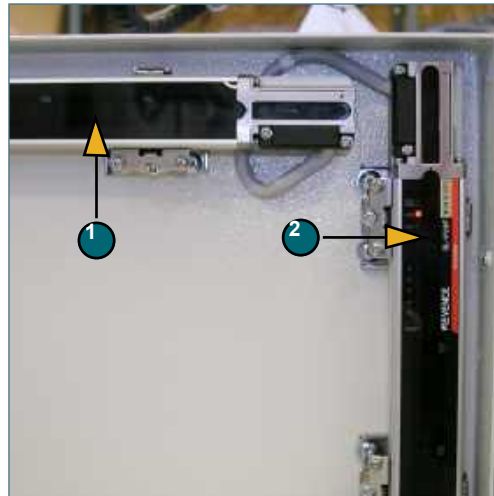


Illustration 16: Wika pressure sensor

1	Pressure sensor	2	Material supply
3	Material removal		

2.3.6 Safety elements

2.3.6.1 Safety light barrier arrangement for round tables



1	Safety light barrier for horizontal scanning Drives and ongoing processes in automatic operation such as plasma treatment, hot riveting, etc. are stopped or de-energised immediately in case of an intervention.	2	Safety light barrier for vertical scanning Drives and ongoing processes in automatic operation such as plasma treatment, hot riveting, etc. are stopped or de-energised immediately in case of an intervention.
---	---	---	---

2.3.6.2 Door safety switch

Application:

A door safety switch protects the operating personnel of a system against interventions in ongoing production operation.

Based on the operating principle, a distinction is made between two types:

- With interlock: Doors are mechanically locked during the automatic sequence
- Without interlock: Doors can be opened during the automatic sequence.

Function:

In ongoing operation (AUTOMATIC or PROCESS MONITORING operation mode is active) with switches without interlock, if the door safety circuit is interrupted, SAFETY operation mode active: Operator intervention is possible without danger. With door safety switches with interlock, the secured access doors can only be unlocked by switching to MANUAL or SETUP operation mode.



Illustration 17: Door safety with interlock



Illustration 18: Door safety without interlock

2.3.6.3 Safety switches

Application:

A door safety switch protects the operating personnel of a system against interventions in ongoing production operation.

Based on the operating principle, a distinction is made between two types:

- With interlock: Doors are mechanically locked during the automatic sequence
- Without interlock: Doors can be opened during the automatic sequence.

Function:

In ongoing operation (AUTOMATIC or PROCESS MONITORING operation mode is active) with switches without interlock, if the door safety circuit is interrupted, SAFETY operation mode active: Operator intervention is possible without danger. With door safety switches with interlock, the secured access doors can only be unlocked by switching to MANUAL or SETUP operation mode.



NOTICE

Machine configuration

The following description of this module shows only one possible application configuration. The illustration may differ from your customer-specific machine or module. The basic function remains the same!



Illustration 19: Door safety switch

2.3.7 Machine positions

Application

Depending on the process, on bdtronic systems, there are a number of fixed positions for different functions. In general, it is the case that they are within the axle drive range and are moved to using an NC program that is started manually or automatically. They are made known uniquely to the system via the axle dialog and can then be called up via a keypress or programmed flows.

Weighing, filling, blind shooting, parking, starting, needle checking, set-up, basic position, etc. are examples of machine positions that can also occur twice with double head systems.

Function

Weighing position:

A precision weigher with container accepts the set number of weighing shots for further evaluation from there.

Fill position:

The dispensing unit (mixing head or dispensing valve or dispensing pump) is filled once in this position before the actual dispensing: components or mixtures flow into a container or via a tube in a drop ratio until "Chamber full" appears.

Blindshot position:

After expiry of the potlife, dispensing unit moves into the blindshot position in order to process a number of set dispensing shots there. These are introduced into a container or via a tube into a catch basin.

Park position:

Materials can be sensitive to humidity or other environmental conditions. Therefore, dispensing units - largely single-component ones - must be protected in production downtimes so that the material at the outlet does not harden and create a pressure disturbance during the next dispensing. For this, leave the outlet nozzle of the dispensing unit plunge into a container full of suitable liquid (e.g. silicone, etc.). This liquid is specially chosen for this.

Start position:

The start position is a coordinate combination XYZ which reflects the start point of the drive program on the component to be processed.

Needle check position:

The first time the needle check is set up, a reference needle has passed its measurement test within specified tolerances. This is the original position. Any additional needle position that was specified with a new needle is only a correction of the original position. The measurement provides information about the alignment of the vertical. Dispensing needles that do not pass the test must be straightened or replaced.

Set-up position:

This position is placed so that machine operators can access the tool, e.g. riveting head, dispensing unit, etc. relatively easily.

Basic position:

For static and single-component mixing and dispensing machines, a basic position is realized. The dispensing unit drives there in order to declare the system as "chamber empty" and end the pot processing cycle.

2.3.7.1 Blind shot position with cup wiper and load cell evaluation

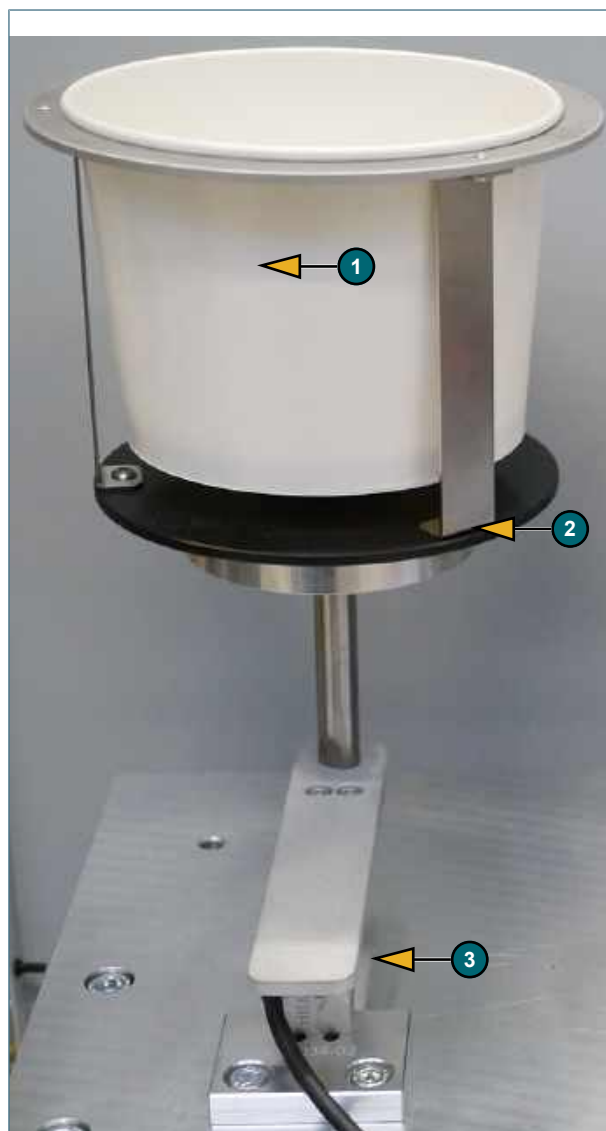


Illustration 20: Blind shot cup with cup tilting device on load cell

1 Weighing cup. The size is determined during the machine configuration.	2 Cup holder: The design is modified according to the customer's requirements and the cup dimensions. The cup tilts by approx. 10° on contact with the needle. In this way, the dispensing needle can be moved along the inside of the cup towards the edge bead and any material residues adhering to it can be wiped off. The next time material is wiped off, wiping is started at a different point on the circumference of the cup.
---	---

3	Load cell: This measuring element records and evaluates the weights of the blind shot. Via a dialogue in the visualisation system, the cup detection is taught so that an empty cup and a completely filled cup are recognised.		
---	---	--	--

Application

This station is a process safety component, as a blind shot can always be dispensed after the potlife counter has expired and as long as the maximum filling weight of the cup has not been reached. The system therefore does not harden out.

2.3.7.2 Weighing position



Illustration 21: Weighing position with cup tilting device

1	Weighing cup: project-specific determination of the size	2	Cup holder: The design is modified according to the customer's requirements and the cup dimensions. The cup tilts by approx. 10° on contact with the needle. In this way, the dispensing needle can be moved along the inside of the cup towards the edge bead and any material residues adhering to it can be wiped off. The next time material is wiped off, wiping is started at a different point on the circumference of the cup.
---	--	---	--

3 Weighing module: project-specific determination of the maximum weight. This module weighs and tares. The measured values are transmitted directly to the requesting PLC. An internal calibration can be started via a visualisation system element.		
--	--	--

2.3.7.3 Needle check



Illustration 22: Needle check 60 x 60

1 Needle check: Block with diagonally arranged, perpendicularly intersecting light barriers and an evaluation unit. The commissioning engineer can set the check sequence via a dialogue in the visualisation system. The module can perform two functions: one to measure a needle and the second to calculate a new needle position after a needle change. The machine operator later replaces the damaged needle and confirms a new needle position.	2 Light barrier 1: A PLC algorithm guides the object to be measured into the measuring area. By alternately interrupting light barriers 1 and 2 in combination with the parameter settings, the needle is measured during the needle check and recognised as good or bad, or a new position for a new needle or mixing tube is calculated and transmitted to the active drive program as an offset.
3 Light barrier 2	

Application

Optical measurement of the alignment of an undamaged vertical dispensing needle or mixing tube. Bent dispensing needles or mixing tubes can be reliably detected. These can then be replaced by the operator.

2.3.7.4 WIPER 2 belt cleaner

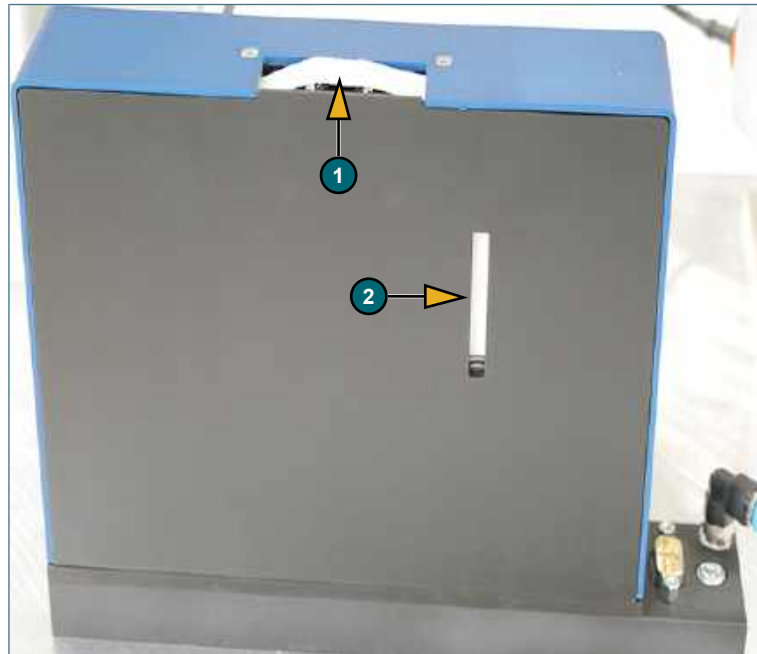


Illustration 23: Wiper 2

1 Cleaning belt for wiping material residues off the metering needle or mixing tube outlet	2 Sight glass for checking the belt stock. Additional info: During each cleaning process, the cleaning belt length is wound onto another reel and is therefore missing here in the sight glass. Below a certain residual reel height, the cleaning process would no longer function correctly and the belt must be replaced. In addition, a dialogue box in the machine visualisation system indicates that the belt must be replaced.
---	---

Application

Material residues are removed from the outside of the discharge nozzle or mixing tube of a dispensing unit by immersing in the unwinding cleaning belt. Additional info: Material residues at the outlet of dispensing systems prevent on the one hand the correct dispensing of material beads, especially in narrow gaps, and on the other hand a needle check will return wrong results in case of residues and incorrectly signal a needle misalignment. The needle check would fail, resulting in the replacement of the needle or mixing tube.

2.3.8 Switchable remote access



Illustration 24: Switchable remote access

1 Key switch with two positions:

active: can be accessed from an external network; Additional lamp (see picture "Additional lamp") lights up

inactive: Access only possible from internal network; Additional lamp does not light



Illustration 25: Additional lamp

2.3.9 Material preparation technology

2.3.9.1 PPS – Configuration and function

Pail preparation systems (PPS)

Application:

A pail preparation system (PPS) is a unit controlled by special control algorithms that works through all processes such as stockpiling material, preparing it and dispensing it so that the materials (e.g. resin or hardener) can be used for further processing on mixing and dispensing machines. The preparation cycle is optimised according to chemical, physical and process-technical requirements. Here the prevention of disturbing air bubbles during pail change, the material-sparing removal of the sometimes high-viscosity and generally potted materials, together with the nearly pulsation-free supply of downstream dispensing components are the top priorities.

Mode of operation:

From a control-technical perspective, the PPS comes in two variants: on the one hand as a slave that is controlled by the dispensing machine; on the other hand as the master that controls the preparation processes itself and compares its state with the docked dispensing machine via just a few signals. From a preparation-technical perspective, there is a variant with and without vacuum chamber: with the vacuum chamber version, the pail to be prepared is in the chamber; with the other version, only a follow-up plate with attached vacuum connection is placed on the pail. Otherwise, the PPS is a compact arrangement of a double-cylinder arrangement with flanged follow-up plate, an optional vacuum chamber, progressing cavity pump, vacuum pump and addition pneumatic and electronic components.

Via individual modes of operation and a menu-guided input interface, the preparation tasks are made accessible to the operator so that it is possible to create individual preparation profiles via special parameter sets and work can be done in the automatic cycle. The signal "Material OK" communicates the end of the preparation to the dispensing machine so that the dispensing operation - if all other prerequisites have been fulfilled - can be started right away.

With the start of the material automatic, a cylinder arrangement moves the follow-up plate onto the pail surface. Here, the air previously included is removed through evacuation. Only when the follow-up plate defined sits on the material surface does a "Material OK" signal indicate the end of the preparation.

In its fully-completed state, a master PPS can control its own dispensing components and perform as a complete mixing and dispensing machine, which naturally can communicate via its own interface system with additional internal or external systems - for example, system components such as axles, transport systems, and thus controls the dispensing tasks as part of a higher-level system.



NOTICE

Machine configuration

The following description of this module shows only one possible application configuration. The illustration may differ from your customer-specific machine or module. This means that instead of a ruby gel filter, a diaphragm dryer can also be installed, and there are different versions of flow rate valves (FRV), etc. The basic function remains the same!

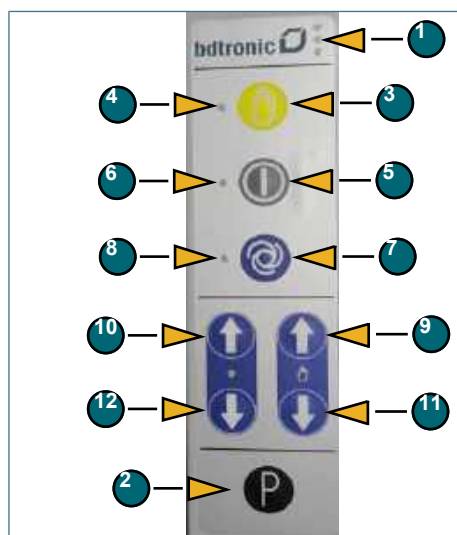


Illustration 26: Membrane keypad PPS

1	Status light	2	< Bleeding > button
3	< HAND > button	4	Indicator LED
5	< STANDBY > button	6	Indicator LED
7	< AUTOMATIC > button	8	Indicator LED
9	< Cylinder up > button	10	< Pail change > button
11	< Cylinder down > button	12	< Step request > button

Status light (1):

red LED lights up (top LED of the three-light combination): Error (e.g. "max" limit exceeded) is indicated, horn sounds

yellow LED lights up (middle LED of the three-light combination): Warning (e.g. "low" limit reached) is indicated

white LED lights up (bottom LED of the three-light combination): Note (e.g. login change, etc.)

< Bleeding > button (2):

Button for introducing compressed air under the follow-up plate: Old pail is thus pressed on the bottom.

< HAND > button (3):

Activate material operating mode HAND: all controlled consumers are switched off

Indicator LED (4):

LED lights up white as soon as HAND operating mode is active

< STANDBY > button (5):

Activate material operating mode STANDBY.

Indicator LED (6):

LED lights up white as soon as STANDBY operating mode is active: Follow-up plate is depressurized, consumers are switched automatically with active timer function.

< AUTOMATIC > button (7):

Start material preparation: Depending on the machine configuration, several subsequent processes such as lower ram press, draw vacuum, etc. are worked through until a "Material OK" signal indicates the end of the material preparation and there is sufficient supply pressure available for the dispensing components.

Indicator LED (8):

LED flashes white as long as material preparation procedure is under way.

LED lights up white as soon as material preparation procedure is complete.

< Cylinder up> button (9):

In the HAND operating mode, the ram press moves in the direction of the end position when this button is pressed.

< Pail change > button (10):

When this button is pressed, a pail change procedure is started and the ram press moves into the upper end position.

< Cylinder up> button (11):

In the HAND operating mode, the ram press moves in the direction of the lower end position when this button is pressed.

< Step request > button (12):

The function of this button is also indicated by a white LED: LED flashes means press button so that the next step can be started; LED lights up: Step was performed. Depending on the operator configuration, this button can be used to request one or several subsequent steps for an already-started material preparation.

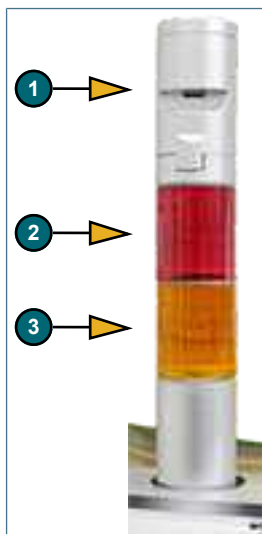
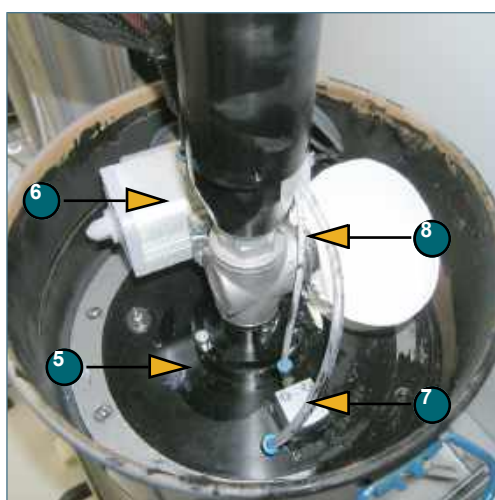
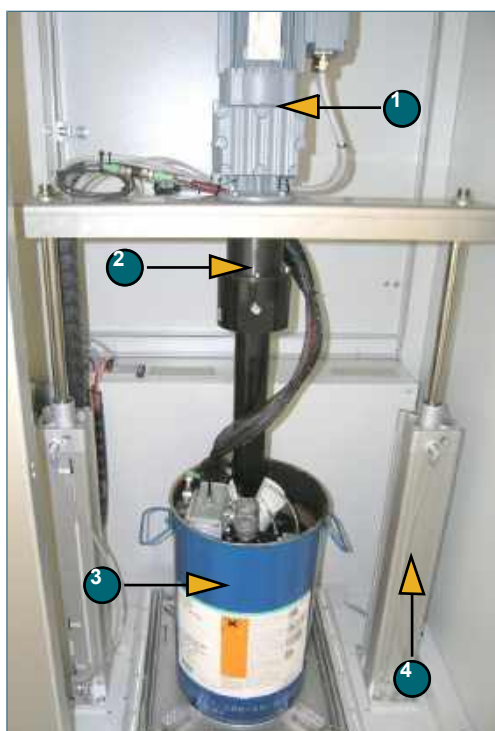


Illustration 27: Signal column

1	Acoustic signal generator (in the event of a fault)	2	Fault "MIN / MAX" limits (e.g. no material, pressure too high/ too low, speed)
---	---	---	--

3	Warning "LOW / HIGH" limits (e.g. material low, pressure too high/too low, speed)	4	
---	---	---	--



1	Pump drive	2	Worm pump
3	Pail	4	Lifting unit
5	Follow-up plate	6	Three-way ball cock (dispens- ing/venting)

7	Ventilation valve	8	Ventilation outlet
---	-------------------	---	--------------------

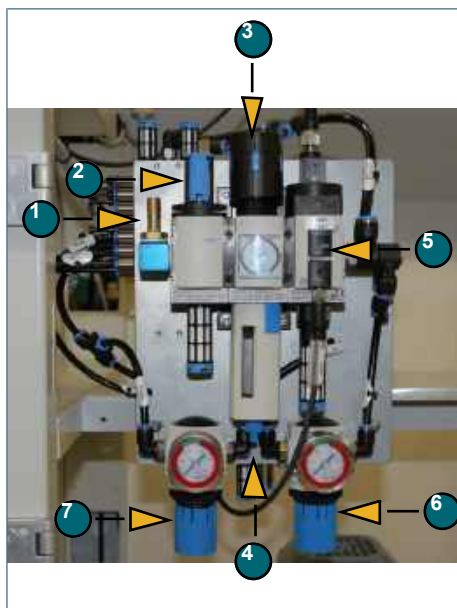


Illustration 28: Main distribution compressed air PPS

1	Supply connection compressed air	2	Manual start-up valve
3	Pressure regulator	4	Condensate drain
5	Electric start-up valve + pressure release	6	Pressure regulator ram press in automatic mode
7	Pressure regulator for old pail ventilation		

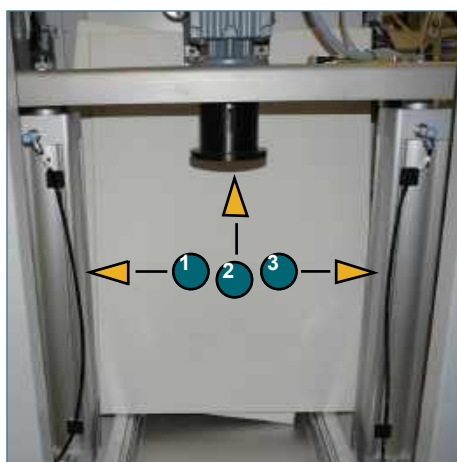


Illustration 29: Ram press

1	Pneumatic cylinder	2	Pump flange
---	--------------------	---	-------------

3	Pneumatic cylinder
---	--------------------

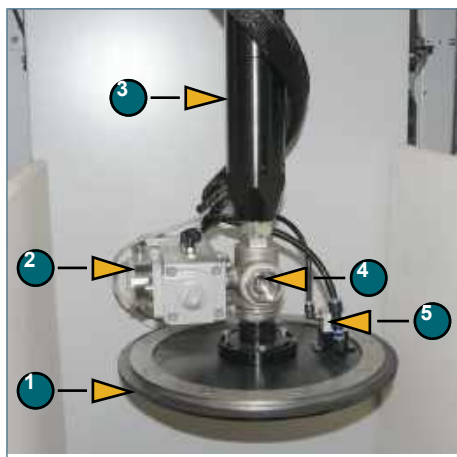


Illustration 30: Follow-up plate

1	Follow-up plate with scraper ring	2	Three-way ball cock
3	Progressing cavity pump	4	Ventilation outlet
5	Plate valve-upper part		

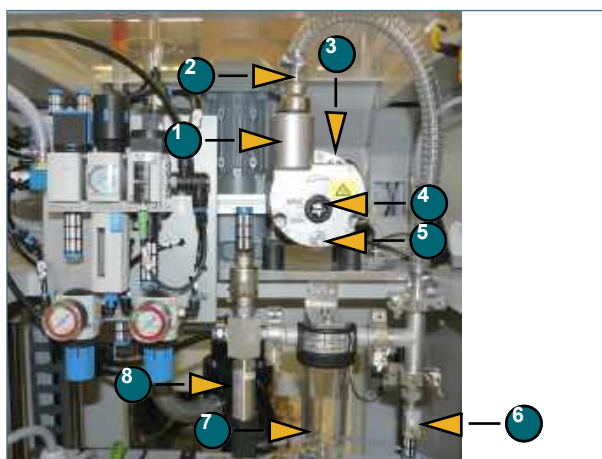


Illustration 31: Vacuum circuit PB0004

1	Exhaust air filter	2	Check valve
3	Fill opening	4	Oil inspection glass
5	Oil drain screw	6	Vacuum throttle
7	Condensate drain	8	Vacuum sensor



NOTICE

Selection of the vacuum pump

By default, a Type PB 0004 pump is installed. During configuration with vacuum chamber, a Type R5 pump is used.

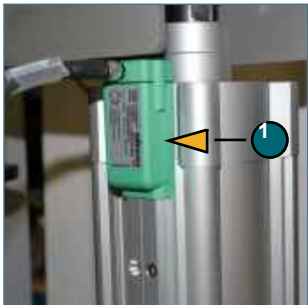


Illustration 32: Position measuring system

1	Measurement sensor		
---	--------------------	--	--

2.4 Function of static mixing principle

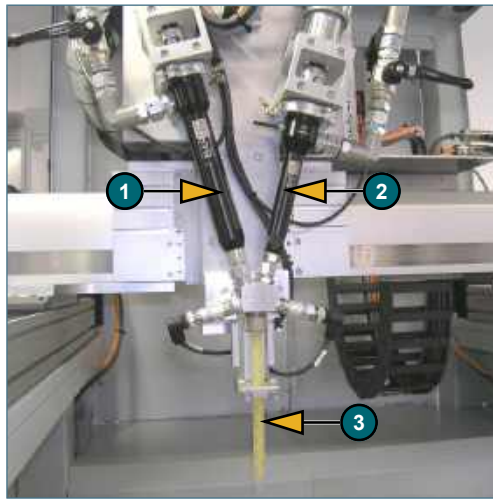


Illustration 33: Static mixing head with cavity pump

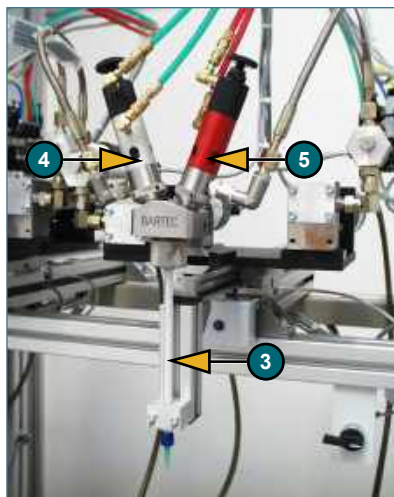


Illustration 34: Static mixing head with dispensing valves

Dispensing system in general

A static mixing and dispensing machine consists of:

- Material stock/preparation unit (e.g. pressurised container, MPS, PPS)
- Material conveying units (= dispensing pumps)
- Mixing unit (= mixing tube)
- An optional material outlet (e.g. mixing tube + optional outlet nozzle)

When the machine is switched on, the materials are prepared according to the specific application (e.g. vented, tempered, stirred). A control unit coordinates all processes. A "Material OK" signal indicates the preparation state. The following processes (e.g. filling, dispensing) are executed by dispensing programs per operator command. Optional moving systems (e.g. linear axes) move the material dispensing unit to the dispensing position. Various monitoring systems (e.g. potlife, pressure recording) ensure safe operation even in the event of a fault.

Filling process

Process description

After an operator command, the mixing tube (3) is supplied with material from the material stock via the resin pump (1) or resin valve (4) and hardener pump (2) or hardener valve (5). For the mixing head variant with valves, additional dispensing pumps, which are installed in front of the valves, supply these with material. In the mixing tube, the material streams (resin and hardener) are pressed separately into the spiral structure and are mixed mechanically with one another. The result is a nearly bubble-free, homogeneous mixture at the mixing tube outlet. The system reports "Chamber full".

Even now the potlife expires (about potlife, see under "Potlife").

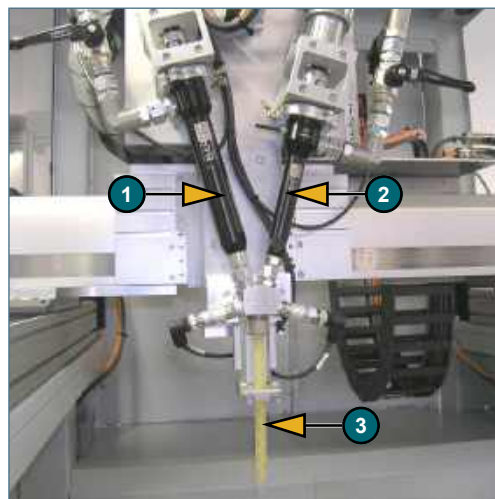


Illustration 35: Static mixing head with cavity pump

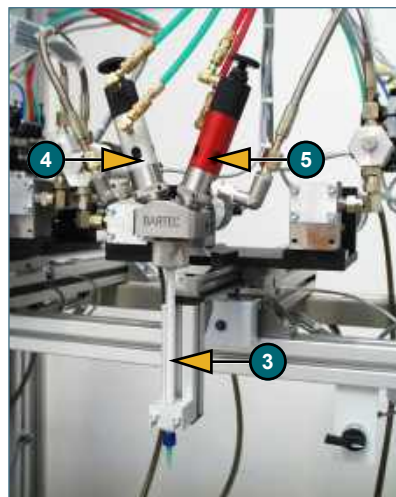


Illustration 36: Static mixing head with dispensing valves

Dispensing

Per operator command, the dispensing pumps start; the optional material valves are opened. The dispensing pumps push material through the optional open valves into the mixing tube input channels until the dispensing time has elapsed or the dispensing command is cancelled. The material

mixture is pressed out of the outlet nozzle. After the expiry of the dispensing time and the taking back of the dispensing command, the dispensing pumps and the optional valves close. The potlife starts again.

Potlife

Time monitoring (= potlife) starts as soon as the mixing tube is filled and after the dispensing shot has been dispensed. This timing relay has the task of preventing a hardening of the mixture in the mixing tube.

On expiry of the potlife, the system emits a tone and if axes are present, moves to a specified position in order to dispense so-called blindshots (= dispensing shots with extra parameters). For hand-guided systems, the operator himself initiates these blindshots. After processing of a number of potlife cycles, the system is set "Not ready". The potlife monitoring is stopped.

3 Operation

3.1 Danger notes

	WARNING Danger due to electric voltage of 400 V! Severe bodily injuries due to electrical shock. • Only have the machine commissioned by trained electricians. • Have damaged parts repaired by a trained electrician or have them replaced.
	DANGER Danger due to high electrical voltage! Severe to fatal bodily injury due to electric shock. • Only have the machine commissioned by trained electricians. • Have damaged parts repaired by a trained electrician or have them replaced.
	CAUTION Caution: automatic start-up! Injuries of all types. • Always maintain a safe distance. • Do not touch components or parts that are in motion. • Do not wear any clothing, loose hair, protective gloves or jewellery near the machine. • Only approach the danger point when the machine is shut down (the machine is disconnected from any supply lines, e.g. power, pneumatic, etc. or switched off).

**CAUTION****Compressed air!**

Optional pressure vessels and certain pneumatic machine parts (e.g. “Emergency purging”, “Emergency blowing”) may still be under pressure after a shutdown or a power failure. These must first be depressurised manually, e.g. by opening the pressure relief valve, to ensure an energy-free state of the machine.

Otherwise there is a risk of pressurised cleaning agents, reaction casting resins, etc., spraying out which may be harmful to health, depending on the application.

**CAUTION****Work on hydraulic machine parts!**

Bodily injuries of all types.



- All work on hydraulic machine parts may only be performed by trained specialists with special knowledge and experience in the area of hydraulics.
- The hydraulic equipment must be checked regularly and kept in proper condition.
- Before opening hydraulic lines and system sections, you must depressurize the system.
- According to the details from the hydraulic line manufacturer, you must change these at regular intervals even if no defects are detected.

3.2 Commissioning

3.2.1 Preparation for the operation of a system



NOTICE

Commissioning in general

The machine must be set up and connected properly and ready for operation.

The operating personnel must be trained according to the regulations.

Dispensing materials can be hazardous materials.

Therefore, read the safety data sheets for the materials used and adhere to the details and instructions they contain.

The safety data sheets for dispensing materials are in the appendix to the operating instructions.

For commissioning/working directly on components with external power supply (voltage, compressed air), special care is required: in case of doubt, disconnect the power supply first.

Under no circumstances may the target temperature of an optional temperature device exceed 80 °C.

When working with the machine where contact can be made with dispensing materials and detergents, it is essential to:



NOTICE

Wear eye protection!

Wear suitable eye protection for every dangerous situation, e.g. laser, infrared, etc.



NOTICE

Wear protective gloves!

Wear suitable protective gloves for every dangerous situation.

3.2.2 Connecting the power supply

- Insert the plug for the machine's power supply cable into a Schuko plug socket (230 V) or a CEE wall outlet (400 V/16 A or 400 V/32 A, see electrical schematic).
- Connect the compressed air supply hose of the machine with the branch point of the operational compressed air network.
- Connect any autonomous electrical devices (such as temperature devices, MPS masters, etc.) that have their own power supply cable to the operational voltage supply.

**NOTICE****Power supply cable**

The length of the machine power supply cable is fixed at 8 m to ensure adequate attenuation of a short-circuit current occurring in the event of a fault and must not be shortened. A mains transformer with a nominal apparent power of max. 2.5 MVA is assumed here. More powerful mains transformers require a re-evaluation of the short-circuit strength of the machine. In this case, it must be ensured that the short-circuit current specified on the machine rating plate cannot be exceeded due to the local mains infrastructure.

**NOTICE****Connection to the voltage supply system**

Only have the machine connected to the voltage supply by trained electricians.

**NOTICE****Connection to the compressed air system**

All work on system parts subject to compressed air may only be performed by trained personnel.

3.2.3 Info - Logging in to the machine

The „Login“ –feature is to get access to the information and functions of the plc and it's human machine interface. With this you can create users with appropriate permissions for machine configurations.

bdtronic-machine performs four access levels (Level 1 - 4) which can be individually assigned to the users.

No user login is active after machine start up despite there was an „Autologin“ – default set.

A level-4 login is authorized for creating further user logins.

There are default passwords for each individually level (also consult following table) which can be changed at any time by a level-4-login!

The rights are assigned as follows:

	Level 1	Level 2	Level 3	Level 4
Create and manage user logins				✓
Create and manage recipes		✓	✓	✓
Material preparation	✓	✓	✓	✓
Teach and move to positions			✓	✓
Use manual functions			✓	✓
Change parameters			✓	✓
Use all functions				✓
Default password	1	2	3	4



NOTICE

Assignment functions and operating modes!

Function execution often is linked to a special operating mode!.

3.3 Switching on the machine

Switching on the machine

The switch-on process serves to supply the machine with electrical and pneumatic energy so that the machine components are initialised and safety systems are activated.

Due to the widely differing designs of the machine types, there are also different switch-on procedures. In every case the machine reports an operation mode and a menu-guided operator interface via whose screens and tabs operating and service personnel can start additional flows.

Prerequisites:

- The machine must be set up, connected properly, and ready for operation.
- The machine has been checked for kinks, cracks in the material and power lines and hoses, and deemed OK.
- The operational extraction system is connected and is running at the specified performance.



NOTICE

Extraction system

Optional flow detection monitor the air flow quantity beginning at the sampling point of the extraction: as long as a defined flow is not guaranteed or an adjustable waiting time has not expired, the current supply of the components is not activated.



NOTICE

Extraction system

Failures of the extraction during ongoing production operation that are determined measurement-technically as reduced flow by flow detection cause the machine to shut down.

- The operating personnel has been trained according to regulations, especially about hazardous materials.

For work where operating personnel come into contact with dispensing materials and detergents, eye protection and protective gloves must be worn for fluid class 1; for special fluid class 2 materials, breathing protection designed for this must be worn.

The following *sequences* are worked through in tabular fashion, from top to bottom.

The following procedure is recommended:

	• Open hand switch-on valve of the compressed air inlet valve	> compressed air supply is switched on
	• Turn pressure regulator of the compressed air switch-on valve to operating pressure of the machine (normally 6 bar, see also chapter on "Technical data"): Compressed air-driven machine components are supplied	> Operating pressure is set
	• Release optional E-stop command device.	> E-STOP is released.
	• Turn main switch of the machine into the "On" position	> Power supply is supplied with voltage > Visualization starts with delay and makes itself known with start screen > Exception: B1000 reports with operating mode < PROCESS MONITORING>
	• Press optional < Machine on > button	> Control and load voltage is switched to the machine components: > Safety circuits are activated > < HAND > operating mode is active, as soon as safety is assured (access doors are closed, etc.)

	• Open access doors: SAFETY operating mode is active • Remove container from the scale for several seconds: Scale tares itself • Replace container • Close access doors: HAND operating mode is active	> Scale is ready for operation
	• Turn ball cocks of optional material container in material flow direction	> Ball cock is open
	• Press < Material > menu button • Press < Automatic material > symbol or < Automatic material > button on control unit of an MPS or PPS • With use of a PPS, press < Confirm step request > one time or several times after each step. This depends on the operating configuration; material preparation starts	> Material preparation starts
	• If handshake mode is desired: Check whether the “active” checkbox is checked in the group field of the handshake mask. Check the box if it is blank.	> The interface is active on the bdtronic side: Signals can be exchanged

3.4 Changing over languages

✓ The operator is logged in with authorisation level 4

1. Press the “bdtronic” symbol key
2. Press the wrench symbol key
 - ⇒ Tab “Settings” is opened

3. Press the "Language" symbol in the group field
 - ⇒ The selection field is displayed
4. Double click the desired language symbol key (e.g. "EN" or the English flag) for the English language version
 - ⇒ The menu is changed over to this language

or:

1. Click the desired language symbol key (e.g. "EN" or the English flag) once for the English language version
2. To confirm, press the symbol key with the green check mark
 - ⇒ The menu is changed over to this language
3. If the symbol key with the red cross is pressed, the selection is cancelled and the previous language version is retained

3.5 Starting material preparation

Start material preparation:

	NOTICE Flow of the material preparation cycle The material preparation cycle can be configured differently: in the simplest form, it starts immediately after pressing the automatic button (variant 1). Furthermore, it is possible to configure so that it starts after the first confirmation (variant 2) or so that each individual step is confirmed (variant 3). The following description covers all variants.
--	--

- ✓ Machine is already running in MANUAL operating mode
 - ✓ No warnings or error messages are displayed on the status bar or on the signal column
1. Press < *Automatic* > operating button (info: For two PPSes connected to one another, both buttons must be pressed!)
 - ⇒ < *Automatic* > operating button flashes
 - ⇒ < *Automatic* > operating button lights up
 - ⇒ "Material OK" display field lights up green
 - ⇒ Material preparation has ended (variant 1)
- or
1. Press < *Automatic* > operating button (info: For two PPSes connected to one another, both buttons must be pressed!)
 - ⇒ < *Automatic* > operating button flashes
 - ⇒ < Step request > button flashes (info: for 2 PPSes connected to one another, both flash!)

2. Press < *Step request* > button when it is blinking (info: For two PPSes connected to one another, both buttons must be pressed!)
 - ⇒ Preparation processes or first process starts
 - ⇒ LEDs of the < *Automatic* > and < *Confirm step request* > buttons light up as soon as preparation processes have ended
 - ⇒ "Material OK" display field lights up green
 - ⇒ Material preparation has ended (variant 2)
- or
1. Press < *Confirm step request* > button another time if it continues to flash (info: Machine requests confirmation of the individual steps one at a time)
 - ⇒ Next preparation step is performed
 - ⇒ LEDs of the < *Automatic* > and < *Confirm step request* > buttons light up as soon as preparation processes have ended
 - ⇒ "Material OK" display field lights up green
 2. Confirm < *Confirm step request* > buttons one after another until the LEDs of the < *Automatic* > and < *Confirm step request* > buttons light up
 - ⇒ Preparation processes are complete
- ⇒ "Material OK" display field lights up green (variant 3)

3.6 Operation modes

The machine recognises different operation modes:

Designation of the operation mode	Symbols on the visualisation
E-STOP	
MANUAL	
PROCESS MONITOR	
SETUP	
SECURE AUTOMATIC	
AUTOMATIC	

Table 1: Table of operation modes (without material operation modes)

3.6.1 Activating PROCESS MONITORING operation mode

Application of PROCESS MONITORING operation mode

With this operation mode, it is possible to test later automatic operation by manually performing certain processes such as drive program checking, pot life processing, etc. with the full safety of automatic operation: Doors are already locked. Also all dispensing, weighing and cleaning cycles can be processed manually

Activate PROCESS MONITORING mode:

1. Press the < *PROCESS MONITORING* > operating key

⇒ PROCESS MONITORING mode is active

or

1. Press the < *Operation modes* > symbol key on the status bar of the visualisation system

⇒ Three-symbol combination opens

2. Mark the < *PROCESS MONITORING* > symbol key

⇒ PROCESS MONITORING mode is active

or

1. switch on the B1000

⇒ PROCESS MONITORING mode is active

3.6.2 Activating MANUAL operation mode

Application

The MANUAL operation mode is optimised as a tool for the people setting up a system: many electrical and pneumatically-driven actuators (e.g. axles, cylinders, motors, etc.) are controlled by buttons of the visualisation. Here, the trained set-up technician must judge and be responsible for the flow and the logic of his own input sequences. This is also linked to an authorised login on the machine.

The access door(s) is (are) not normally locked, except in systems where it is intentional that a release button must be actuated to unlock (additional Info: the pressed release button only unlocks for a certain period of time, depending on the model. If operator intervention (i.e. In this case opening the door) does not follow within this time period, the door is automatically locked again!). In any case, the operator is protected from the system in terms of safety technology so that, if the monitored access door(s) is (are) opened, the load voltage is disconnected from the movement systems and processes: no axles or cylinders move, no process – such as plasma processing, hot riveting, etc. – is active or can be started.

Activating MANUAL operation mode:

1. Switch the machine on
2. Unlock the E-STOP command device

3. Ensure safety (close all access doors, etc.)

⇒ MANUAL mode is active

or

✓ AUTOMATIC mode is active

1. Press the < *Stop* > operating button

⇒ MANUAL mode is active

or

✓ PROCESS MONITORING mode is active

1. Press the < *Stop* > operating button

⇒ MANUAL mode is active

or

1. Press the < *Operation modes* > symbol key on the status bar of the visualisation system

⇒ Three-symbol combination opens

2. Mark the < *MANUAL* > button

⇒ MANUAL mode is active

3.6.3 Activating AUTOMATIC operation mode

Activate AUTOMATIC mode:

✓ MANUAL mode is active

1. Press the < *AUTOMATIC* > operating key

⇒ AUTOMATIC operation mode is active

or

✓ PROCESS MONITORING mode is active

1. Press the < *AUTOMATIC* > operating key

⇒ AUTOMATIC operation mode is active

or

1. Press the < *Operation modes* > symbol key on the status bar of the visualisation system

⇒ Three-symbol combination opens

2. Mark the < *AUTOMATIC* > symbol key

⇒ AUTOMATIC operation mode is active

or

✓ AUTOMATIC operation mode is active

1. Press the < *Automatic* > operating key

⇒ Blue LED flashes

- ⇒ Green and blue LEDs flash alternately
- ⇒ DEFSTOP operation mode is active
- ⇒ The machine process stops at the end of the current part of pallet process. The < *Automatic* > operating key can now be pressed to restart the normal operation mode or the < *Process monitoring* > key to switch the machine to PROCESS MONITORING mode. In the first case, the blue LED is lit and the green LED is out; in the second case, the green LED flashes and the blue LED is out.

3.6.4 Activating STANDBY operation mode

Application

The STANDBY operation mode is a pure material operation mode: it is only intended for material preparation systems such as the vessel rack, cartridge, MPS and PPS. This calls up a programmable timer whose program flow serves consumers for material preparation such as temperature devices, container stirrers, etc. depending on time or, in case of the PPS, as a prerequisite for special functions such as pressure relief of the dispensing circuit.

Activating STANDBY operation mode:

1. Switch the machine on
2. Unlock the E-STOP command device
3. Ensure safety (close all access doors, etc.)
 - ⇒ MANUAL operation mode is active
4. Press the < *STANDBY* > button

⇒ STANDBY operation mode is active

or

1. Press the < *Operation modes* > button on the status bar of the visualisation
 - ⇒ Three-symbol combination opens
2. Mark the < *STANDBY* > button

⇒ STANDBY operation mode is active

3.6.5 Activating SETUP operation mode

Application

This operation mode was only implemented for trained set-up and maintenance technicians and service personnel because, in contrast to MANUAL operation mode, the electrical and pneumatic actuators (e.g. axles, cylinders, etc.) can also be moved **with open doors**. Exception: Particular risky processes such as plasma handling, welding, hot riveting, etc. are inactive and cannot be started.

The personnel entrusted to work on the machine must be able to assess and take responsibility for the flow and logic of their input sequences. This is also linked to an authorised log-in on the machine.

Activating SETUP operation mode:

1. Switch the machine on
2. Unlock the E-STOP command device
3. Close access doors, ensure safety
4. Turn the key switch into the 'Setup' position

⇒ SETUP operation mode is active

or

1. Switch the machine on
2. Unlock the E-STOP command device
3. The key switch is in the 'Setup' position
4. Open the access door(s)
⇒ SAFETY operation mode is active
5. Press the confirmation button halfway

⇒ SETUP operation mode is active

or

✓ MANUAL or PROCESS MONITORING or AUTOMATIC operation mode is active

1. Turn the key switch into the 'Setup' position
⇒ SETUP operation mode is active
2. Open the access door(s)
⇒ SAFETY operation mode is active
3. Press the confirmation button halfway

⇒ SETUP operation mode is active

3.6.6 Activating SAFETY operation mode

Application

The machine switches into this operation mode when the operating personnel must be protected against risky processes such as axis movements, movements of all types, processes such as plasma handling, hot riveting, etc. This is always the case if machine accesses, such as doors, etc., must be opened because operator intervention is required. After restoring safety, the MANUAL, PROCESS MONITORING or AUTOMATIC operation mode is active.

A special form is SECURE AUTOMATIC operation mode which temporarily deactivates movement systems during the desired operator intervention, e.g. for loading purposes in ongoing AUTOMATIC operation until the operator has left the danger zone, e.g. via light curtain. After that, the normal production flow starts in AUTOMATIC operation mode.

Activating SAFETY operation mode

- ✓ The machine is switched on
- ✓ MANUAL operation mode is active
- 1. Open the access door(s)
 - ⇒ SAFETY operation mode is active
- or**
- ✓ PROCESS MONITORING or AUTOMATIC operation mode is active
- 1. Press the < Stop > operating button
 - ⇒ MANUAL operation mode is active
- 2. Open the access door(s)
 - ⇒ SAFETY operation mode is active

3.7 Commissioning activities

3.7.1 Referencing a multi-turn drive

After replacing a motor with absolute encoder, after a belt change in the drive train or after an axle change, the drive train in question must be referenced once. Particularly after a motor change where the motor ID also changes, the encoder loses its position (additional information: an absolute encoder stores the last position as an absolute distance).

For referencing, a reference mandrel is used which is clamped in a prism holder on the Z-axis. The mandrel is measured by the needle check assembly which thus determines a position that is forwarded to the drive section of the PLC for further use. In the system manager, the zero offset is corrected to the value from the absolute encoder.

The following procedure is recommended:

- ✓ MANUAL operation mode is active
- ✓ Login authorisation 'Level 4'
- 1. Open the "Axes" screen
- 2. Press key < Multiturn Reference >

⇒ An animated graphics application is opened

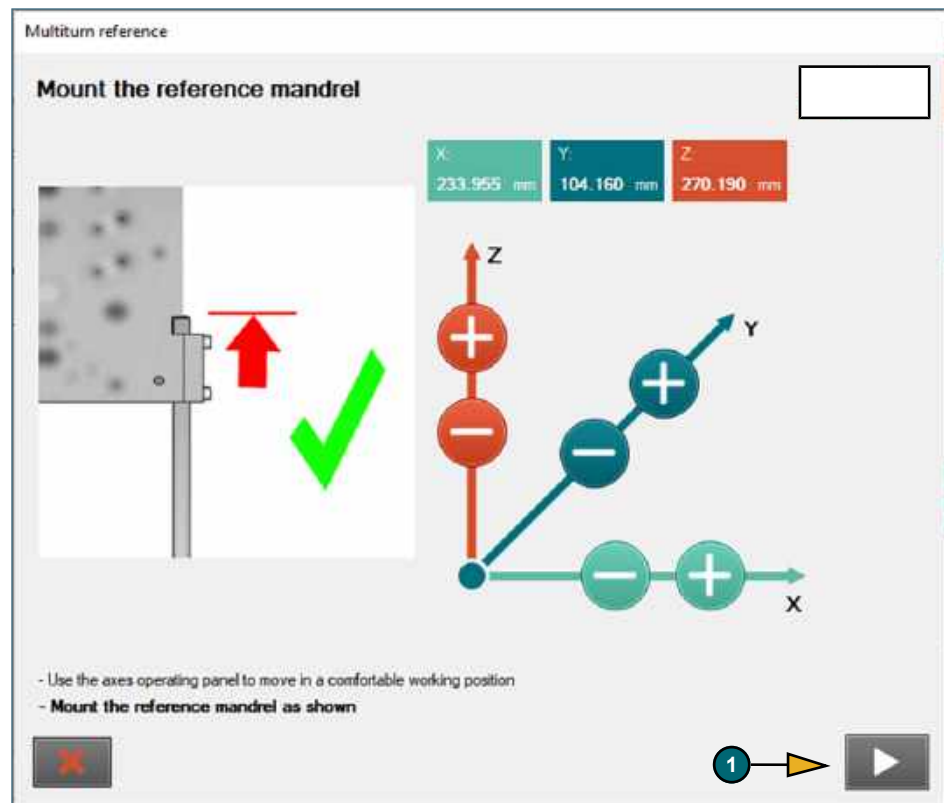


Illustration 37: Screen: Installing the reference mandrel, step 2/7

3. Move to a favourable handling position for the rest of the procedure using the manual movement keys "X+" and "-X", etc.
4. Remove the reference mandrel that is attached to the side of the Y-axis
5. Loosen the two screws of the prism holder
6. Push the reference mandrel into the prism holder on the side of the Z-axis as far as it will go and secure it with the two screws tightened to max. 3 Nm
7. Press the Continue key ("Screen: Installing the reference mandrel, step 2/ 7", item 1)

⇒ The next screen “Move to needle check position 4/7” is displayed

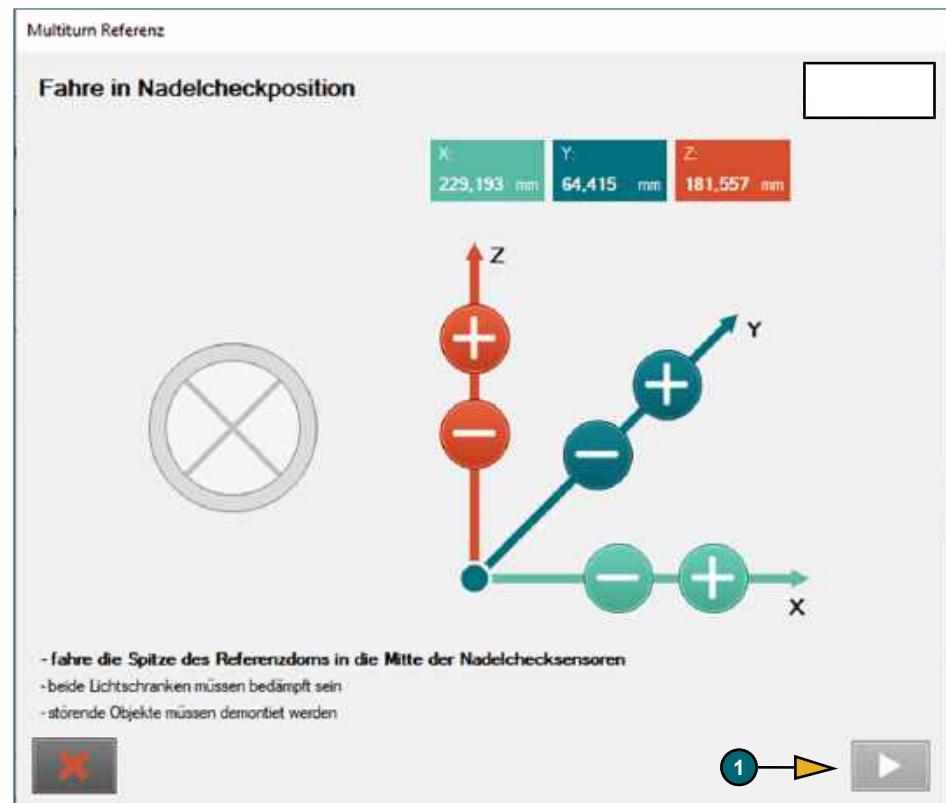


Illustration 38: Screen: Move to needle check position

8. Move into the needle check using the manual movement keys “X+” and “-X”, etc. so that both light barriers are interrupted at the same time (the two green lines intersect, see also “Screen: Measuring routine”, item 4 and item 5)

⇒ Screen: “Measuring routine” is displayed
9. Press the Continue key (“Screen: Move to needle check position”, item 1)

⇒ Screen: “Measuring routine” is displayed

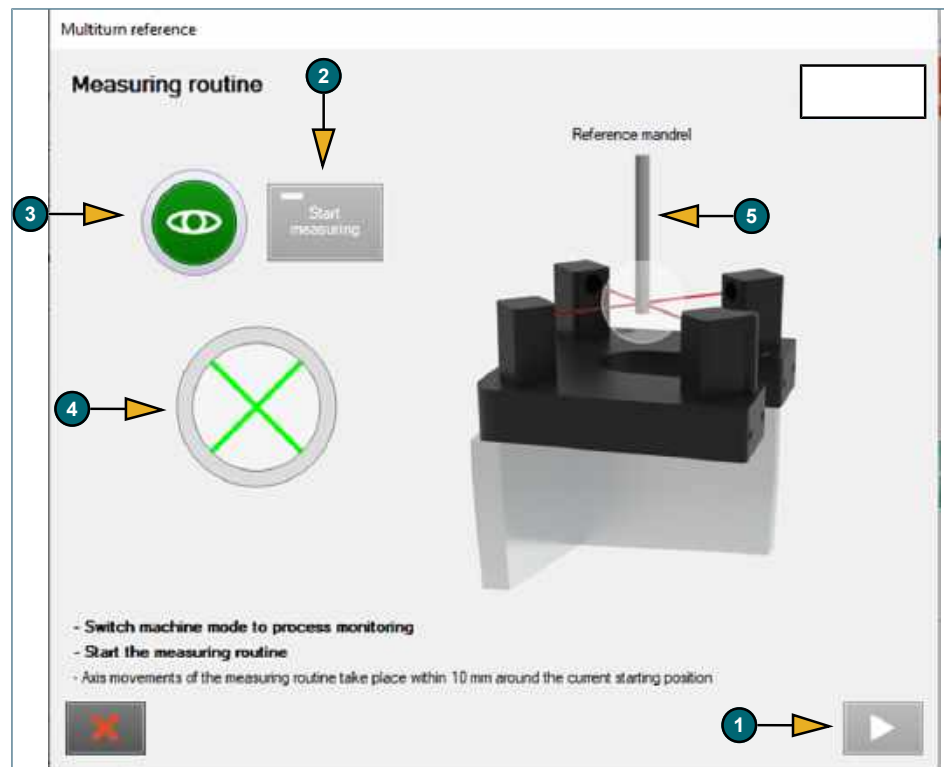


Illustration 39: Screen: Measuring routine 5/7

10. Switch to PROCESS MONITORING operation mode with key (“Screen: Measuring routine 5/7”, item 3)

⇒ Key < *Start measurement* > (Screen: “Measuring routine 5/7”, item 2) is enabled

11. Press key < *Start measurement* > (Mask: “Measuring routine 5/7”, item 2)

⇒ The measuring cycle starts. Attention: Follow the instructions

⇒ A new window “Teach in new position 6/7” is displayed

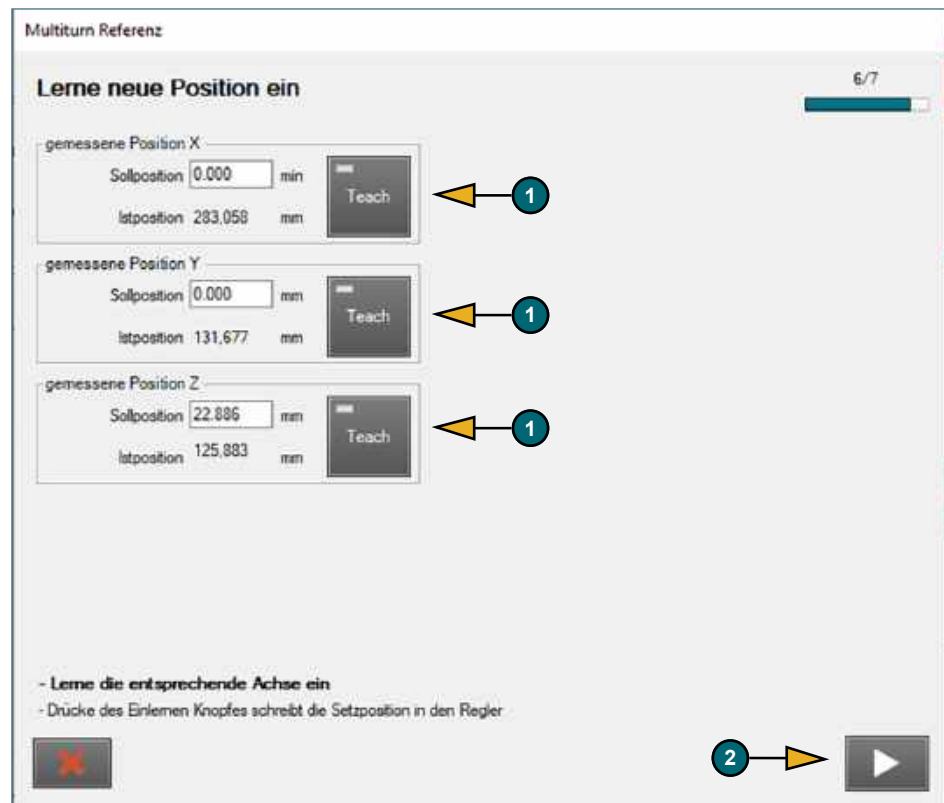


Illustration 40: Screen: Teach in new position 6/7

⇒ The measurement results are displayed

⇒ After a motor change, a flashing frame appears around the axis in question

12. Press key (“Screen: Teach in new position 6/7”, item 1) with flashing frame

⇒ The current position is overwritten

13. Press the Continue key (“Screen: Teach in new position 6/7”, item 2)

⇒ The window “Remove reference mandrel 7/7” is displayed

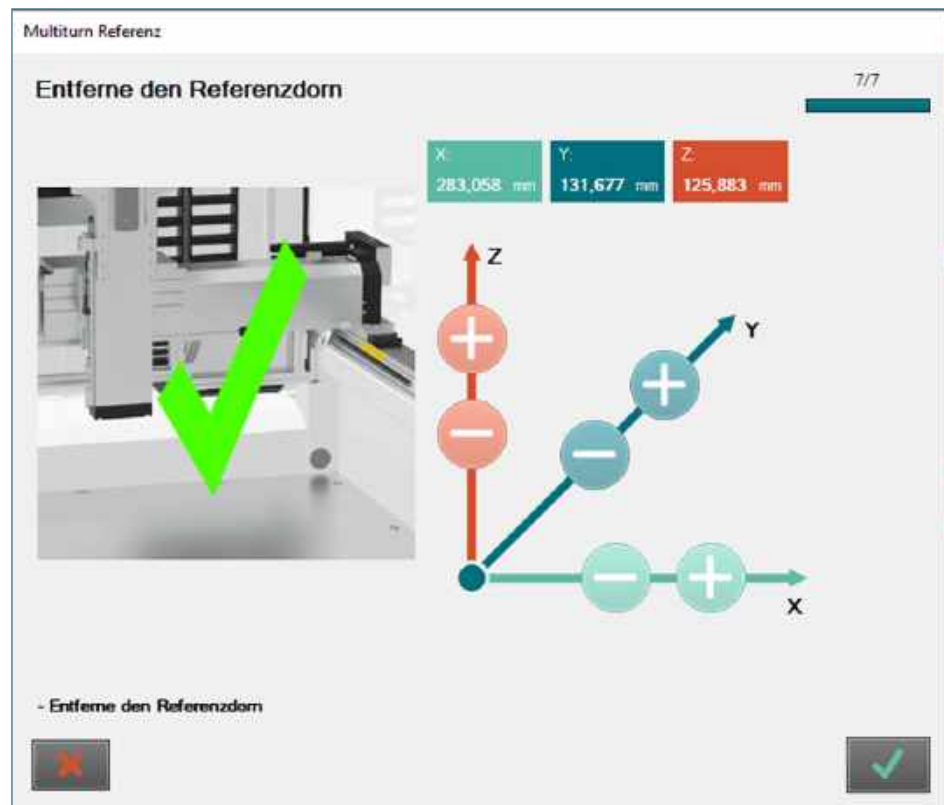


Illustration 41: Screen: Remove reference mandrel (7/7)

14. Loosen the two screws of the prism holder
15. Remove the reference mandrel
16. Tighten the two screws of the prism holder
17. Secure the reference mandrel in its holder on the side of the Y-axis

3.7.2 Automatic golden calibration

Golden calibration is the initial generation of a pump characteristic of the installed pump type within the defined working range and process environment of a dispensing pump.

Golden calibration is performed at the factory when the machine is commissioned or after a change of pump type. The working range of the pump is defined via the minimum and maximum dispensing rate during system design and is reflected in the machine configuration.

The measuring process proper is divided into several continuous intervals. 3 measurements are performed within each interval that must be within a tolerance of 5% of each other in order to generate a valid measured value. A weighing shot is dispensed into the cup on the scale at a specific dispensing rate, weighed and transferred to the system as a value. The average value is the transferred value. The green pump characteristic is formed from the values. Two tolerance bands (red for error, yellow for warning) are placed around this reference characteristic. This tolerance range defines the limits for subsequent recalibrations of the current reference characteristic.

Via the pump characteristic, the system software "knows" the dispensing rate within a defined range for each pump speed.

The golden calibration process is offered in 2 automation levels:

- **Fully automatic** with **integral scale** in the dispensing cell
- **Semi-automatic** by using an **external scale**.

The automatic procedure is described below:

Prerequisites:

- A precision scale must be installed at a defined weighing position in the dispensing system
- Furthermore the material supply to the dispensing system must have been completed in automatic mode and the "Material OK" signal must have been received
- A new mixing tube (for 2K static systems) or mixing chamber with agitator free of adhering material (2K dynamic), or clean dispensing nozzle (for 1K dispensing systems) must be installed

Sequence:

1. Log in with 'Level 4' authorisation
2. Activate the pump on the start page (set a green tick) if it is still inactive (red cross); material OK and system must be ready (= filled)
3. Activate PROCESS MONITORING mode
4. If necessary, call up the scale page and activate "Wipe off" if the material strand does not break off cleanly
5. Press the < *Golden calibration* > key
 - ⇒ Dialogue "Do you really want to perform a new calibration?"
6. Make the settings for "Interval", "Pause after dispensing", "Pause before reading out", "Min. Weight per shot"
7. Press the < *Start* > key
 - ⇒ The dialogue closes
 - ⇒ Data is stored in the controller and calibration is started
 - ⇒ A diagram with the green reference curve and the red and yellow tolerance bands is generated and displayed
 - ⇒ A green "Result" LED signals the end of the successful calibration process
 - ⇒ System is now ready for dispensing
 - ⇒ During each future filling, dispensing or blindshot process, the speed limit is monitored by the software

3.7.3 Automatic recalibration

Application

Recalibration must be carried out as soon as the weighing values are no longer within the tolerance range due to pump wear, replacement of pump components or of the complete pump. The same routine is followed as for

the "Golden Calibration" (= initial calibration). Here, too, a new pump characteristic is determined on the basis of the prevailing conditions. The warning and error limits of the golden calibration remain, but can be changed at any time.

A pump calibration is possible in 2 automation levels:

- **Fully automatic** with **integral scale** in the dispensing cell
- **Semi-automatic** by using an **external scale**.

The automatic procedure is described below:

Prerequisites:

- A precision scale must be installed at a defined weighing position in the dispensings system.
- Furthermore the material supply to the dispensing system must have been completed in automatic mode and the "Material OK" signal must have been received
- New mixing tube (for 2K static systems) or mixing chamber with agitator free of adhering material (2K dynamic), or clean dispensing nozzle (for 1K dispensing systems)

Sequence:

1. Log in with login authorisation 'Level 2' or higher
2. Activate the pump on the start page (set a green tick) if it is still inactive (red cross)
3. Press the < *Recalibration* > key
 - ⇒ Set intervals are processed: 3 measurements per interval, which may lie within the 5% deviation, otherwise NOK and abortion of the procedure with an error message
 - ⇒ The digram is refreshed
 - ⇒ The new characteristic is displayed with the warning and error limits
 - ⇒ A green "Result" LED signals the successful completion of the recalibration



NOTICE

Recalibration

After successful recalibration of the dispensing pump(s), all existing dispensing programs are automatically updated: no new parameterisation necessary!



NOTICE

Repeatability of the recalibration

If the pump wear is too severe, recalibration will not provide better values. Therefore, replace pump parts susceptible to wear, such as stator, rotor, seals, etc. in good time! Do not limit troubleshooting to the end of the dispensing line (outlet nozzle, mixing tube, mixing chamber), but also analyse the material feed for causes of errors.

3.7.4 Creating a new dispensing program

- ✓ The dispensing pumps must already have been calibrated!
- 1. Press key < New > of the dispensing program dialogue
- 2. Enter a name for the program (recommendation: The name should have the same designation as the dispensing rate, i.e. "1.5 g/s")
- 3. Press the OK key 2x
 - ⇒ The new name is displayed in the list
- 4. Highlight the name
- 5. Press key < Load >
 - ⇒ A new empty program is loaded
- 6. Switch to the header screen
- 7. In the "Fill" group field, describe the parameter "Filling volume factor".
Tip: With low-viscosity materials in combination with a dynamic mixer, enter a value ≥ 2 to fill the chamber air-free.
- 8. Only for dynamic systems: in the same group field, enter the value for "Del. mixer": Value range: 50% to 80%.
- 9. In the same field, enter "Decompression weight and number of decompression shots". Additional info: Decompression shots are only necessary if filling speed $\neq$ dispensing speed! It should take at least one second and be performed at dispensing speed!
- 10. Parametrise the settings in group field "Blindshot". Determine the potlife at the same time. Additional info: The dispensing pressure must not increase over several subsequent blindshots.
- 11. Enter the "Volume factor". Additional info: When using a dynamic mixer, enter at least "1.1" so that the chamber volume is reliably exchanged once.
- 12. Enter the number of blindshots and possible pre-warning time. Additional info: When using a dynamic mixer, the mixing chamber is cleaned after the entered number of blindshots has expired and the system is set to "System not active".
- 13. In the "Speed" group field, enter the dispensing speed (for dispensing, blindshots, manual dispensing functions, dispensing in automatic and preparation shots) and filling speed (for filling).
- 14. When using a dynamic mixer, enter a description in the "Cleaning" group field: Enter a time for purging and drying (= blowing)

15. With dynamic systems only: Switch to the “Mixer” screen. Enter speeds for purging and drying (default: 5000 rpm)
16. Switch to the “Settings” screen. Enter the values for “Max” and “Min”.
Additional info: The min value is set with the lower viscosity component, the max value with the higher viscosity component.

3.7.5 Teaching in the scale

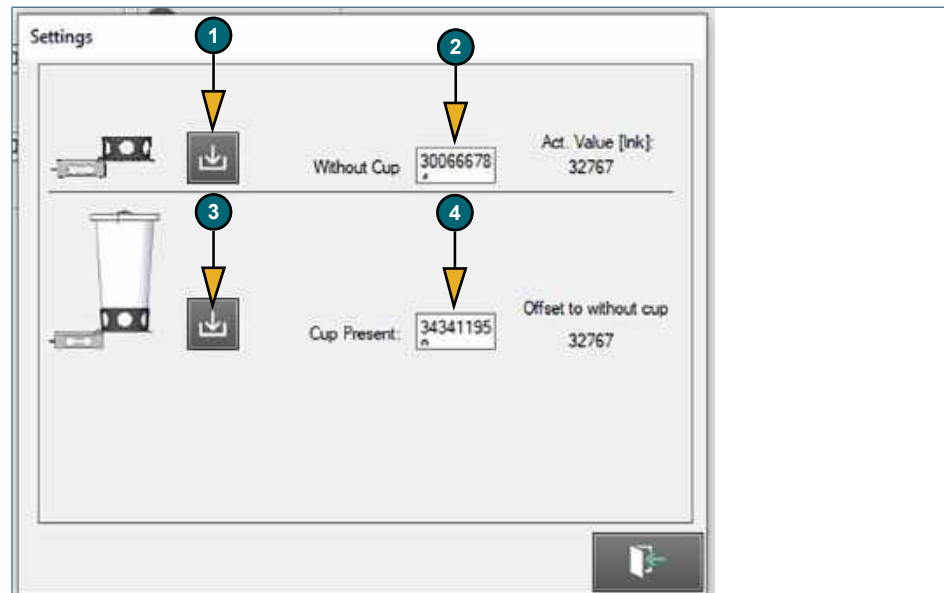


Illustration 42: Scale - Cup - Settings

- ✓ The weighing position has already been confirmed
 - ✓ An authorised user is logged in
1. Activate MANUAL mode
 2. Open access door of the dispensing cell
⇒ SAFETY mode is active
 3. Open the “Settings” dialogue
 4. Remove the weighing cup
 5. Press symbol key < Confirmation > (see also Figure: “Scale - Cup - Settings”, item 1)
⇒ The measured value is displayed in field (2)
⇒ The scale forwards the measured value without cup to the PLC
 6. Place an empty cup with optional wiper on the scale
 7. Press symbol key < Confirmation > (see also Figure: “Scale - Cup - Settings”, item 3)
⇒ The measured value is displayed in field (4)
⇒ The scale forwards the measured value with cup and optional auxiliary equipment to the PLC
 8. Close the dialogue with the key
 9. Press the < Save > menu key

- ⇒ All changes are saved
- 10. Close access door
 - ⇒ MANUAL operation mode is active

The next time the system is switched on, the message "Remove cup and tare scale" is displayed.

Then:

1. Activate MANUAL operation mode, if not already active
2. Open the access doors
3. Remove the cup from the scale for several seconds
4. Press key < Tare > on the weighing screen
 - ⇒ The scale tares without cup
5. Place a cup with optional wiper on the scale
 - ⇒ The PLC compares the value with the stored value
 - ⇒ The message "Cup present" is displayed: System detects the cup
6. Close access door
 - ⇒ MANUAL operation mode is active

3.7.6 Save values of weighing diagram in form

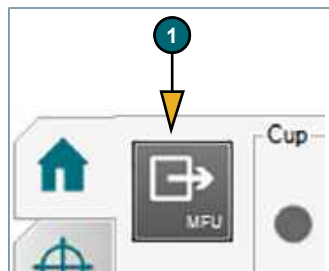


Illustration 43: Writing MFU file

Application

Storing the weighing values in a special form.

- ✓ A 'Level 3' authorised user is logged in
 - ✓ MANUAL operation mode is active
1. Press key < MFU > (see Figure "Writing MFU file", item 1)
 2. The Windows window "Select template file" is displayed
 3. On the hard disk C, find the path „C:\Seriennummer\visu\Log\Weighing_Dispatch_System_“
 4. Highlight form "FB 1.7.04-007_CMK_Pumpen" and open with key < Open >
 - ⇒ Excel file is opened
 5. Assign a new name
 6. Press the < Save > key

⇒ The dialogue closes and the file is saved

3.7.7 Carry out initial commissioning of pail

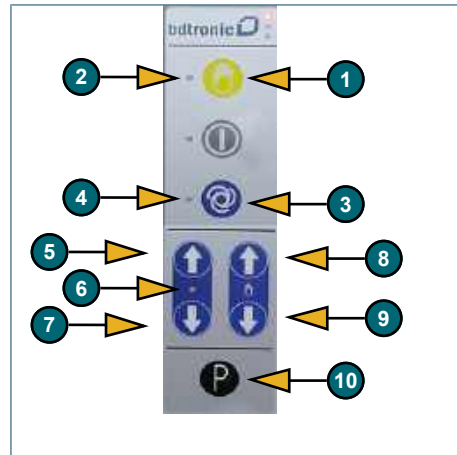


Illustration 44: Control keypad

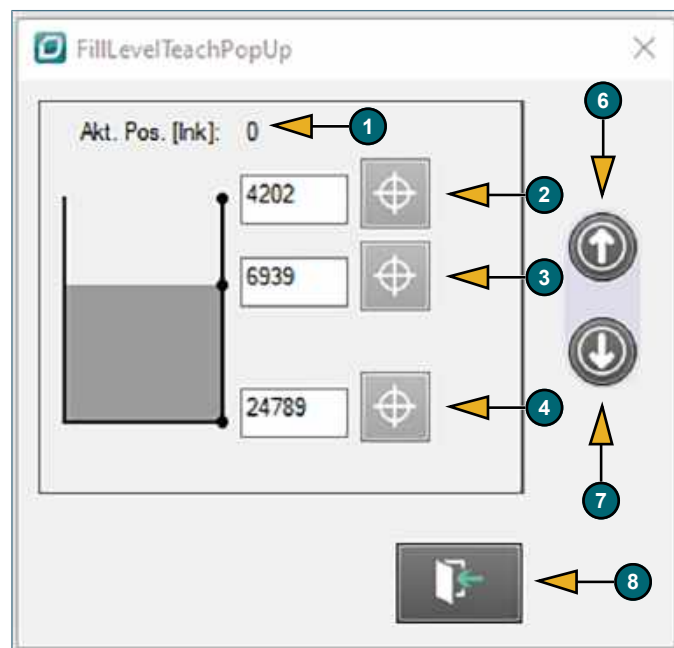


Illustration 45: Teach PPS fill level

- ✓ MANUAL material preparation mode is active
 - ✓ The follower plate valve is in the upper end position
 - ✓ The visualisation system of the PPS is open
1. Open the access doors
 2. Place an empty pail on the pail seat
 3. Close the access doors
 4. Press tab "Valves"
 5. In group field "Follower plate valve", press the < Open > key

- ⇒ The valve is opened
- 6. In group field "Inlet valve", press the < *Vacuum/atmosphere* > key
 - ⇒ The vacuum is prepared
- 7. Change to tab "Vacuum pump"
- 8. Start the vacuum pump with key < *Vacuum pump MANUAL* >
 - ⇒ The vacuum pump evacuates the line to the follower plate
- 9. With the "Arrow down" key (see Figure "Teach PPS fill level", item 7), move the follower plate onto the bottom of the pail
 - ⇒ The current position is displayed under "Current position" (see Figure "Teach PPS fill level", item 1)
- 10. Press the "Teach" key (see Figure "Teach PPS fill level", item 4)
 - ⇒ The current position is stored in the box to the left of the "Teach" key
 - ⇒ This position represents the empty level of the pail
- 11. Change to tab "Vacuum pump"
- 12. Stop the running vacuum pump with key < *Vacuum pump MANUAL* >
- 13. Change to tab "Valves"
- 14. Close the valve with key < *Close* > in the "Follower plate valve" group field
- 15. Press the < *OFF* > key of the "Inlet valve" group field
 - ⇒ The inlet valve is closed
- 16. Press operating key < *P* > (see Figure "Control keypad", item 10)
 - ⇒ Compressed air is pressed under the follower plate
- 17. Then with operating key < *P* > held depressed, press the "Arrow up" operating key (see Figure "Control keypad", item 8) at short intervals
 - ⇒ The follower plate moves upwards
- 18. Release the "Arrow up" operating key (see Figure "Control keypad", item 8) as soon as the follower plate is approx. 1 cm below the edge of the pail
- 19. Release operating key < *P* > (see Figure "Control keypad", item 10) as soon as the follower plate is just above the edge of the pail
- 20. Press the "Teach" key (see Figure "Teach PPS fill level", item 2)
 - ⇒ The edge of the pail has been taught
- 21. Prepare a new, full material pail: Remove the lid, place fleece on the pail surface, optionally place protective film on moisture cross-linking materials
- 22. Optionally, scan the component location
- 23. Optionally, scan the material data
- 24. Place the pail on the PPS seat
- 25. Close the door
- 26. Activate AUTOMATIC material mode
 - ⇒ Step prompt LED (see Figure "Control keypad" | item 6) flashes
- 27. Press the < *Arrow down* > key (see "Control keypad" figure | item 7)

- ⇒ An optional plate valve test is performed
 - ⇒ Valves are switched
 - ⇒ Vacuum pump is started
 - ⇒ Movement towards the pail surface is performed
28. As soon as the seating time has expired, activate MANUAL material mode
- ⇒ All consumers are disconnected or moved to the home position
29. Change to tab "Fill level"
30. Press the symbol key
- ⇒ Fill level teach-in pop-up opens
31. Press the middle teach-in button or enter the value in "Current position" in the middle input field
- ⇒ Material edge has been taught
 - ⇒ All three fill level marks have been taught
32. Press the < Save > key in the lower menu bar
- ⇒ The values are stored permanently
33. Activate AUTOMATIC material mode
34. Press the "Arrow up" key (see Figure "Control keypad", item 5) twice
- ⇒ The follower plate is moved to the upper position
 - ⇒ Step confirmation LED (see Figure "Control keypad", item 6) flashes
35. Press the < Arrow down > key (see "Control keypad" figure | item 7)
- ⇒ Optional plate valve test starts
 - ⇒ Normal piercing process is performed
 - ⇒ Signal "Mat OK" indicates readiness for dispensing

3.7.8 Calibration of the follower plate valve

The pneumatically controlled follower plate valve forms part of the follower plate. It is used to release the air in the space between the follower plate and the surface of the material in the pail when "piercing" a new pail. When the follower plate is raised from the empty old pail, compressed air is admitted to the material space so that the empty pail remains in position. The changing air space above the material is then emptied by means of a vacuum when the follower plate is lowered ("piercing"). The air flowing in through the open follower plate valve as the follower plate moves up then presses the residues pail into its seat so that the follower plate can move out.

During the calibration process described below, the switching behaviour of the follower plate valve is tested together with the connected mimic diagram (pneumatic line, start-up valve).

During commissioning of a PPS, values are determined here which are then used for each pail changing process. No pail is required for this, as no movement cylinder is activated during the calibration process.

Sequence of the calibration process:

- ✓ Log in with 'Level 4' authorisation
- 1. Open the "Valves" screen of the PPS or DPS visualisation system
- 2. Tick the "Active" check box in the "Plate valve check" group box
- 3. by pressing the symbol key
 - ⇒ The "Plate valve check" window opens
- 4. Enter the desired values in the "Hysteresis" input fields or accept the default values

3.7.8.1 Plate valve calibration – Start manually

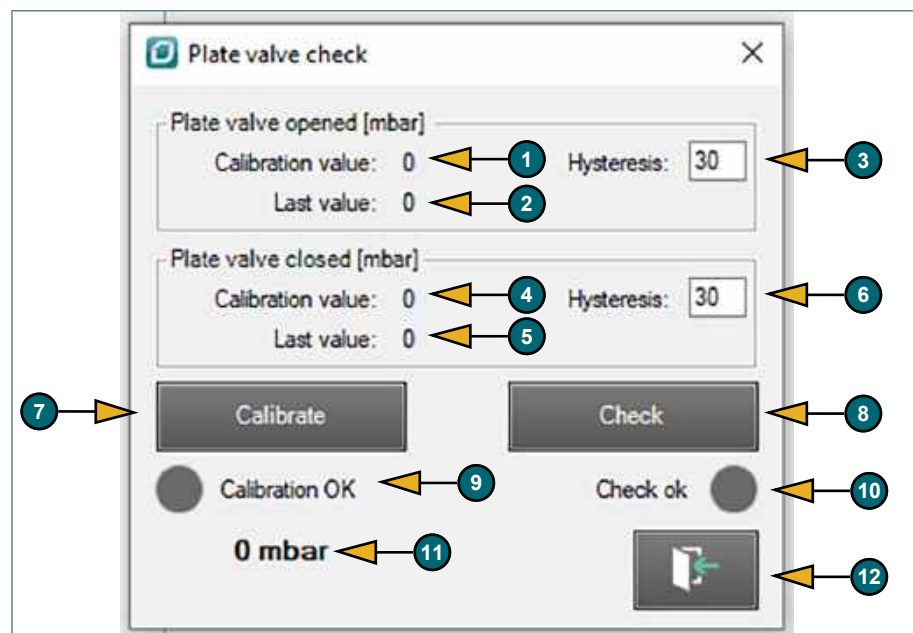


Illustration 46: Figure: Valves - Plate valve check

Application

As a pneumatic valve, the plate valve forms part of the follower plate. When the plate valve is open, either compressed air or the vacuum line is connected to the pail surface, depending on the position of the inlet valve. The correct function of the valve in connection with its further integration is therefore also important for the piercing process of every pail. A "calibration process" tests the configuration and stores values that are then used again in the subsequent test process.

The piercing process places the follower plate on the surface of the pail. During this process, air above the material in the pail is removed so that it does not interfere with the material supply to the dispensing pumps and thus jeopardise the quality of the dispensing result.

A plate valve check is carried out before each piercing. The piercing process only starts if the check is successful.

Calibration must be carried out when the pail preparation system is commissioned, after each change of the plate valve, the connected pneumatic hose or the vacuum pump.

The sequence of the calibration process is described below:

- ✓ The desired setpoint must be entered in group field "Automatic" in the "Current settings" screen of the vacuum pump tab
 - ✓ The "Active" check box in the "Valves" screen in group field "Plate check" is ticked
 - ✓ The hysteresis values are entered in the "Plate check" group field (see "Figure: Valves – Plate valve check", items 3 and 6)
 - ✓ The double cylinder of the follower plate movement is in the upper end position
 - ✓ MANUAL material preparation mode is active
1. Press the symbol key on the "Valves" screen in the "Plate check" group field
 - ⇒ The "Plate check" dialogue opens
 2. Check the entered hysteresis values (items 3 and 6) and correct, if necessary
 3. Press key < Calibration > (see "Figure: Valves – Plate valve check", item 7)
 - ⇒ The plate valve is closed
 - ⇒ While the vacuum pump is running, the system tries to fall below the setpoint value; this value is saved as the "calibration value" (see "Fig: Valves – Plate valve check", item 4)
 - ⇒ In the next step, the system tries to reach atmospheric pressure with the plate valve open and the vacuum pump still running. After a system-specific time, this value is also saved (see "Figure: Valves – Plate valve check", item 1)
 - ⇒ The vacuum pump then attempts to evacuate the system to the setpoint with the plate valve now closed
 - ⇒ Finally, with the inlet valve disconnected, i.e. neither compressed air nor vacuum is admitted, the system tries to reach atmospheric pressure again via the vent throttle. A venting timeout is polled for this. The process makes an error stop if the specified value is not reached within the timeout
 - ⇒ If the process is successful, calibration is completed
 - ⇒ The "Calibration OK" LED (see "Figure: Valves – Plate valve check", item 9) lights up green

3.7.8.2 Plate valve check – Start manually

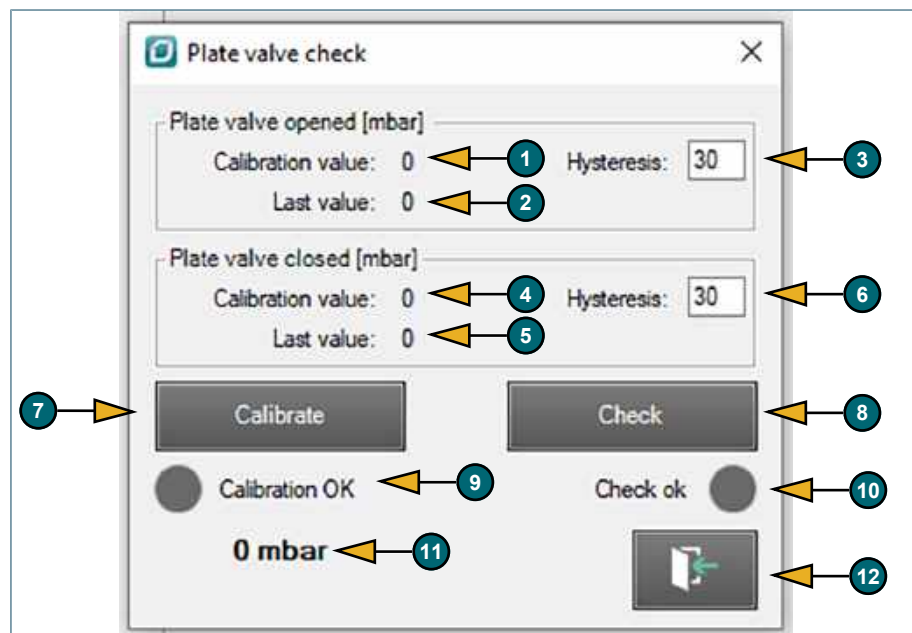


Illustration 47: Figure: Valves – Plate valve check

Application

The plate valve check presupposes a valid calibration of the plate valve. A plate valve check is carried out automatically before each piercing process. A successful check starts the piercing process immediately.

This check can also be started manually, as described below. This manual feature provides a troubleshooting option for the service technician. Except for a few details, manual starting with the < Check > key is similar to the automatic check sequence.

The manual starting process is described below:

- ✓ MANUAL material preparation mode is active
- ✓ The "Active" check box in the "Valves" screen in group field "Plate check" is ticked

1. Press key < Check > (see "Figure: Valves – Plate valve check", item 8)
 - ⇒ The plate valve is closed
 - ⇒ While the vacuum pump is running, the system tries to fall below the setpoint value; this value is saved (see "Fig: Valves – Plate valve check", item 5) and compared with the hysteresis settings
 - ⇒ If the previous step was successful, the system then tries to reach atmospheric pressure within a system-specific time with the plate valve open and the vacuum pump still running; the value is saved (see "Fig: Valves – Plate valve check", item 2) and compared with the hysteresis settings
 - ⇒ If the second check is also successful, the "Check OK" LED lights up green and the pail piercing process is started immediately

3.8 Manual functions .. Please read!

Manual functions

A frequently used mode is the manual operation of machines, abbreviated as MANUAL, in which the safety circuits remain active but the access doors are not locked. Depending on the machine configuration, certain functions – such as moving cylinder and axes, switching pneumatic ball valves, transport route elements such as lift positioning units, stoppers, conveyor belt segments – can be performed manually. A login in an authorisation level, usually level 3, is necessary for this! In advance, the operator of such functions must, of course, first be informed about the machine's mechanical condition, i.e. which actuators interact with which elements, which are mutually interlocked, which are not, what processes or individual steps present risks to the operator, etc. We recommend first checking the overall situation of that part of the machine that is to be manually influenced. Only then should action be taken by means of an intervention via the user interface or directly in the machine!

Some manual functions, such as the movement of axes (additional info: with reduced speed!) or the operation of pneumatic actuators can also be triggered in the SETUP mode. You access this by putting the key switch in the "Setup" position (only for B5xxx cells!). Due to the special situation, the machine access doors can be opened here with the axes in motion; the operator must also confirm that the activation switch is in the half-depressed position. When completely pressing or releasing the switch, the axes stop immediately!

As we are unable to predict all situations regarding the operation of our machines, the manual functions for our machines or machine components are taught in commissioning training by our service personnel to the extent that a basic understanding of the functions and processes is provided to participants – also with a view toward preventing hazardous situations.

Any damage caused by improper use of the machine or machine components in the modes MANUAL or SETUP, are the responsibility of the machine operator. He is obliged to document and keep current the level of knowledge of his operating personnel on a regular basis, if necessary immediately as well, through appropriate training measures, including by the manufacturer or distributor of the machine or machine component.

Only one operator per machine is allowed during the performance of manual functions!

In particular, the general safety instructions apply:



CAUTION

Caution during general handling of the machine's electronics!

- All work on the machine's electrical equipment may only be performed by trained electricians.
- This work may only be carried out when the machine is de-energised.
- All conductive parts and components are earthed.
- Regular checking and maintenance of the conductive connections must be performed by the specialist personnel.
- The switch cabinet must always be kept closed. This key may only be kept by authorised personnel.
- The current-conducting cables in the switch cabinet are marked up to the main switch.
- In case of the assembly or installation of machines, systems or tools from various manufacturers or suppliers, as well as after alteration by in-house or service personnel in which there were interventions into the electrical equipment, the operator must conduct a precise examination according to accident prevention regulation DGUV Regulation 3 in accordance with the applicable electro technical rules before commissioning.

Users themselves are responsible for the type, sequence and result of his actions and adhering to the safety provisions.

The manufacturer is not liable for damage due to improper or non-intended use of the machine.

3.8.1 Overview of the drive programs

Use of drive programs on bdtronic machines

Production systems process components via one or more sequential processes or processes running more or less in parallel, such as dispensing, plasma treatment, gripping, etc. in a more or less protected working area. Transport systems such as conveyors or infeed slides take on the tasks of component movement and axis portals of the tool handling system. The movement of the different paths for the tool, the activation of processes, etc. is realised by an NC program, called up by the controller, that processes from top to bottom. This was created by service technicians when running in the system, by means of customer requirements and consists of a list of NC command lines. The recognised rules for the syntax of NC programs are reflected here. A series of common drive programs are described below. The NC syntax in relevant factories of third parties is described in detail.

Subprograms are formulated like the main program, but they are called up via a subprogram name that can be freely selected and is the entry point for the code at the same time. It is terminated by the "M17" command.

3.8.1.1 Creating a drive program

Basically, we recommend the following steps:

1. Open an existing program in the "Drive program editor" screen
 2. The basic component for which the drive program is created is positioned on the fixture at the dispensing place
 3. The desired start point on the component is moved to using the manual axis travel functions
 4. Note these coordinates for subsequent entry in the start point register ("R100" for 'X', "R101" for 'Y' and "R102" for 'Z')
 5. Move to the paths to be dispensed (start and end of straight lines and arcs) with the manual axis travel functions and note the coordinates for this
 6. Read optional coordinates from a suitable CAD drawing format or generate them from a CAD file
 7. Enter an optional name of the program with a prefixed "%" character at the start of the program
 8. List NC sets with row number at the beginning of the lines or jump labels for starting sub-program call-ups, matching the desired program sequence with a separate line end (additional information: Measured coordinates or CAD coordinates for the start point and axis paths must now be inserted in the movement commands)
 9. Complete the main program with the "M30" command
 10. Save the program with the < *Save under* > symbol
 11. Open the folder structure with the < *New file* > symbol
 12. Double-click on the newly written program
- ⇒ The program name appears in the selection list of the drive program editor and can now be used for manually running through contours or for linking with a recipe



NOTICE

Additional information concerning drive program creation

bdtronic recommends copying already existing, executable drive programs that are present when the system is delivered in the C: _BDT\NC\customer folder, saving them under new names and then using them for further component-variant programs: Ready-made sub-programs are used for specific dispensing applications that can be transferred into an own application, if necessary, without having to reprogram everything.



NOTICE

Internal system drive programs

There are other drive programs present on the storage medium of the system under C:_BDT\NC\System for position travels for blindshot processing, weighing procedure, etc. These must never be changed by the user. They can be opened with a text editor for viewing purposes.

What follows is a small selection of frequently used NC commands:

3.8.1.2 Jump command "@121"

Com-mand	Example program fragment for Execute an "If not equal to" conditional jump		Comment
@121	Line no.	Command syntax	
	N100	@121 R499 K0 K9930	Compares the register content with "0": If not equal to "0", continue with line 'N9930', otherwise line 'N200'
	N200	M158	Resets all M commands
	...	...	...
	N9930	R450=R499	Write register R450 with contents of R499

3.8.1.3 Jump command "@124"

Com-mand	Example program fragment for Execute "If less than" conditional jump		Comment
@124	Line no.	Command syntax	
	N100	@124 R499 K2 K9930	Compares the register content with "2": If less than "2", continue with line 'N9930', otherwise line 'N200'
	N200	M158	Resets all M commands
	...	...	
	N9930	R450=R499	Write register R450 with contents of R499

3.8.1.4 Multiple branching "@111"

Com- mand syntax	Example program fragment for Multiple branching "@111"		Comment
@111	Line no.	Command syntax	
	N1296	@111 R499 K1 K2 K9930 K9931	Compares R499 with '1': If equal to '1', jump to 'N9930'; if equal to '2', jump to 'N9931'; if neither, then 'N1299'
	N1299	M158	Resets all M functions
	...	...	
	N9930	R450=R499	Save R499 to R450
	N9931	R451=R499	Save R499 on R451

3.8.1.5 Unconditional jump "@100"

Com- mand	Example program fragment for "Execute unconditional jump"		Comment
@100	Line no.	Command syntax	
	N1500	@100 K10000	Jump to line number 10000
	...	...	
	N10000	M30	Exit program

3.8.1.6 "M110" command

Com- mand	Example program fragment for M function* "Query M110 part present sensor" with handshake behaviour: M function queries PLC, PLC confirms if ful- filled.		Comment
M110	Line no.	Command syntax	
	N100	R595=0	Zero register
	N101	M110	Query PLC whether compon- ent detected
	N102	@714	NC processing stops when pro- cessing the M110 function. If con- firmed, continue with N103
	N103	@124 R595 K1 K590	Jump to line N110 if contents of R595 less than '1'
	...		
	N110	M30	Program end

*:



NOTICE

M commands

M commands form an interface between the NC program and the PLC application. The system programmer specifies the use of M commands and their parametrisation. Before using an M command, e.g. 'M110', 'M115', etc., contact bdtronic to obtain precise details for use.

3.8.1.7 Activating absolute dimension details

Com- mand	Example program fragment for "Toggle relative to absolute dimension de- tails"		Comment
	Line no.	Command syntax	
G90	N100	G90	Switches on the absolute dimension detail: All positions are calculated from the valid zero point; the relative dimension detail is switched off. Designates pos. that does not change with "0", e.g. 'Y0'.

3.8.1.8 Activating relative dimension details

Com- mand	Example program fragment for "Toggle absolute to relative dimension de- tails"		Comment
	Line no.	Command syntax	
G91	N100	G91	Switches on the relative dimension detail: All positions are calculated from the previous position; the absolute dimension detail is switched off. Designates pos. that does not change with "0", e.g. 'Y0'.

3.8.1.9 Moving axis in rapid traverse

Com-mand	Example program fragment for "Moving axis in rapid traverse"				Comment
G0	Line no.	Command syntax			
	N100	G90			Switches on the absolute dimension system
	N110	G0 X50 Y200 Z300			Moves X, Y, Z axes to position X50, Y200, Z300 at maximum speed
	...				

3.8.1.10 Travelling a straight line at feed speed

Com-mand	Example program fragment for "Travelling at feed speed"				Comment
G1	Line no.	Command syntax			
	N100	G91			Relative dimension details
	N110	G1 F2000 X100 Y200 Z100			Moves in a straight line at feed speed 'F': All involved axes move at the same time
	...				

3.8.1.11 Travel in a circle in clockwise direction (Mpt)

Com-mand	Example program fragment for "Travel in a circle in clockwise direction" (centre point programming)		Comment
G2	Line no.	Command syntax	
	N10	G91	
	N50	G2 F2200 X0 Y0 I0 J6.3	It is possible to move in segments and full circles. A full circle with a diameter of 6.3 mm is travelled at speed F2000 here. "X0", "Y0": Starting point is an end point; "I0": and "Jx.x": Start point relative to centre point ("I0": X amount; "Jx.x": Y amount)
	...		

3.8.1.12 Travel in a circle in anticlockwise direction (Mpt)

Com-mand	Example program fragment for "Travel in a circle in anticlockwise direction" (centre point programming)		Comment
G3	Line no.	Command syntax	
	N10	G91	
	N50	G3 F2200 X0 Y0 I0 J6.3	It is possible to move in segments and full circles. A full circle with a diameter of 6.3 mm is travelled at speed F2000 here. "X0", "Y0": Starting point is an end point; "I0": and "Jx.x": Start point relative to centre point ("I0": X amount; "Jx.x": Y amount)
	...		

3.8.1.13 Travel in a circle in clockwise direction (Radius)

Com-mand	Example program fragment for "Travel in a circle in clockwise direction" (radius programming)		Comment
G2	Line no.	Command syntax	
	N10	G91	
	N20	G1 X100 Y100 F2000	Travels to pos. (100,100) at speed F2000
	N50	G2 X200 B200	Travels a circle segment with radius 200 with end point at (300,100); speed is F2000. No full circles are possible.
	...		

3.8.1.14 Travel in a circle in anticlockwise direction (radius)

Com-mand	Example program fragment for "Travel in a circle in anticlockwise direction" (radius programming)		Comment
G3	Line no.	Command syntax	
	N10	G91	
	N20	G1 X100 Y100 F2000	Travel to pos. (X;Y) at speed F2000
	N50	G2 X200 B200	Travels a circle segment with end point at (300;100) and a radius of 200 mm. Speed remains F2000. No full circles are possible.

3.8.1.15 Dwell time

Com-mand	Example program fragment for "Dwell time"		Com-ment
G4	Line no.	Command syntax	
	...		
	N20	G4 F5	Waits at this position for 5 seconds
	...		

3.8.1.16 Exit drive program

Com-mand	Example program fragment for "Exit drive program"		Comment
M30	Line no.	Command syntax	
	...		
	N9930	M30	Exits the program and deletes all M functions without handshake

3.8.1.17 Example drive program structure

Line number or jump label	Command syntax	Comment
%VL 300		Optional name for program (info: Name is freely selectable)
N900	G90	Program line (=NC set) with NC word ("G90" in this case) and line number "N900"
...	...	Any other NC sets could appear here
N1310	L1000	e.g. call up an optional subprogram
...	...	Any other NC sets could appear here. They are processed after the return from the subprogram.
N1400	M3	Main program end
L1000		Code begin for "L1000" subprogram

Line number or jump label	Command syntax	Comment
...	...	Any amount of sets can appear here for the subprogram. These can also be subprogram call-ups.
M17		End of the "L1000" subprogram
...	...	Other subprograms can appear here that must be closed with an "M17" command.

**NOTICE****Subprogram call-ups**

The calling up of optional subprograms, e.g. "L1000", must always take place before the "M30" mark as this terminates the drive program.

**NOTICE****Register use in the drive program**

A specific part of the registers is occupied by the system or the PLC program. Normally the register range from R800 to R999 can be used by the user (exception: In multiple systems, the range from R800 to R899 is already occupied elsewhere!).

**NOTICE****NC programming**

Free curves cannot be programmed as in CNC programs.

3.8.1.18 Notes on how to change or create a drive program

- ✓ The drive program has already been opened in a text editor
 - ✓ MANUAL operation mode is active
 - ✓ Authorised user is logged on
1. Only use command syntax for NC programs
 2. Enclose comments in round brackets
 3. Begin each line with a line number (exception: Subprograms begin with the subprogram call-up name)
 4. Each subprogram ends with an "M17" command
 5. Each drive program ends with an "M30" command

6. Saving the drive program



NOTICE

Save symbol

The save symbol on the menu bar saves all changes except for drive program changes. To do this, use the save symbol or the "Save under" symbol of the drive program editor dialogue.

7.

3.8.1.19 Loading a drive program

- ✓ MANUAL operation mode is active
- ✓ Authorised user is logged on
- 1. Open the "Drive program editor" screen
- 2. Press the < *Load* > button
 - ⇒ The folder structure opens
- 3. Double-click the desired program
 - ⇒ The drive program opens in the system's own editor

3.8.1.20 Starting the drive program manually

- ✓ PROCESS MONITORING operation mode is active
- ✓ The drive program has already been loaded
- 1. Set the velocity regulator of the axes to the desired value
- 2. Start the drive program with the < *Play* > button for the entire sequence or with the < *Single step* > button for set-by-set processing



NOTICE

Sequence toggling

During processing, it is possible to toggle from the complete sequence to set mode and vice versa.

3.8.1.21 Inserting a drive program in the program selection list

- ✓ The drive program already exists in the C:_BDT\NC\customer path
- ✓ MANUAL or PROCESS MONITORING mode is active
- 1. Switch to "Axes" screen
- 2. Press the < *Load file* > symbol key (2nd icon from the left of the row of icons) of the "drive program" dialog
 - ⇒ The folder structure opens
- 3. Select the drive program
- 4. Press the < *Open* > button

- ⇒ The drive program name appears in the selection list
- 5. Press the < Save > symbol key
- ⇒ This drive program can already be linked in the Recipe Manager

3.8.1.22 Assigning drive program to a recipe

- ✓ MANUAL mode is active
- 1. Open the recipe screen
- 2. Mark an existing recipe or press the < New recipe > symbol key
- 3. Click on the "Drive program" tab
 - ⇒ The selection dialogue opens
- 4. Mark the drive program
- 5. Confirm with the green button
 - ⇒ The dialogue closes
- 6. Press the < Save > symbol key
- ⇒ The drive program has been assigned to the recipe

3.8.1.23 Renaming a drive program

- ✓ MANUAL mode is active
- ✓ Authorised user is logged on
- 1. Call up the "Axes" screen
- 2. In the "NCI Editor" tab, press the symbol key < Load file > (first icon from the left of the row of icons)
 - ⇒ Windows "Open" dialog appears
- 3. Mark the desired program
- 4. Press the < Open > dialog button
 - ⇒ The file is loaded in the Editor area
- 5. Press < Save as > button
 - ⇒ The folder structure opens
- 6. Enter the new name in the line
- 7. Press the save button
- ⇒ The file with this name appears in the selection list

3.8.1.24 Deleting a drive program

- ✓ MANUAL operation mode is active
- ✓ Authorised user is logged on
- 1. Mark drive program in selection list
- 2. Press the < Delete > button

⇒ The file name is removed from the selection list

3.8.2 Pail change with optional bar codes

Start pail change:

- ✓ Empty message appears on the status bar of the visualisation
 - ✓ Note: "Confirm lifter movement" appears on the status bar of the visualisation
 - ✓ LED of the < *Pail change* > button flashes
1. Press < *Pail change* > button
 - ⇒ Follower plate moves out of pail
 - ⇒ Compressed air is pressed under the follow-up plate
 - ⇒ Pail status is set to "not ready"
 - ⇒ Light of the < *Pail change* > button flashes
 2. Open door of the PPS
 - ⇒ SAFETY mode is active
 3. Remove old pail on carriage and dispose of properly
 4. Remove cover from new pail
 5. Place 2 special fleece on pail surface
 6. Place new pail on carriage



NOTICE

Pail change

Take care when installing a new pail that it is correctly positioned on the slider. It should be positioned level and in the middle of the slider. When tapping the pail, the operator must monitor the whole process and abort it by pressing the E-STOP button, if necessary, in the event of a fault.



NOTICE

Danger of confusing pails

Always make sure that new pail is placed in the intended place and provided with the intended material data.

Start optional material data recording:



NOTICE

Scan bar code flow

Depending on customer requirements, various data is recorded. This is reflected in a varying number of scanning processes! The subsequent description can thus be regarded only as an example.

1. Scan bar code label on follow-up plate unit with hand-held-scanner
 - ⇒ If scan successful: Bar code appears in part ID field
2. Scan bar code label on pail with hand-held-scanner
 - ⇒ If scan successful: Bar code appears in pail label field
3. Scan any additional labels with hand-held-scanner (info: This depends on the machine configuration!)
 - ⇒ If successful: Bar codes appear in the intended field
 - ⇒ If all bar codes have been read in successfully and check is OK: Result LED green is displayed
4. Close access doors: MANUAL mode is active

Start material preparation:



NOTICE

Flow of the material preparation cycle

The material preparation cycle can be configured differently: in the simplest form, it starts immediately after pressing the automatic button (variant 1). Furthermore, it is possible to configure so that it starts after the first confirmation (variant 2) or so that each individual step is confirmed (variant 3). The following description covers all variants.

1. Press < *Automatic* > operating button (info: For two PPSes connected to one another, both buttons must be pressed!)
 - ⇒ < *Automatic* > operating button flashes
 - ⇒ < *Automatic* > operating button lights up
 - ⇒ "Material OK" display field lights up green
 - ⇒ Material preparation has ended (variant 1)
- or
1. Press < *Automatic* > operating button (info: For two PPSes connected to one another, both buttons must be pressed!)
 - ⇒ < *Automatic* > operating button flashes
 - ⇒ < *Step request* > button flashes (info: for 2 PPSes connected to one another, both flash!)
2. Press < *Step request* > button when it is blinking (info: For two PPSes connected to one another, both buttons must be pressed!)
 - ⇒ Preparation processes or first process starts
 - ⇒ LEDs of the < *Automatic* > and < *Confirm step request* > buttons light up as soon as preparation processes have ended.
 - ⇒ Sequence for pail tapping is carried out.
- ⇒ "Material OK" display field lights up green

⇒ Material preparation has ended (variant 2)

or

1. Press < *Confirm step request* > button another time if it continues to flash (info: Machine requests confirmation of the individual steps one at a time)
 - ⇒ Next preparation step is performed
 - ⇒ LEDs of the < *Automatic* > and < *Confirm step request* > buttons light up as soon as preparation processes have ended
 - ⇒ "Material OK" display field lights up green
 2. Confirm < *Confirm step request* > buttons one after another until the LEDs of the < *Automatic* > and < *Confirm step request* > buttons light up
 - ⇒ Preparation processes are complete
- ⇒ "Material OK" display field lights up green (variant 3)

3.8.3 Relieving pail



NOTICE

Preventing sedimentation

This function is only used for the processing of materials that sediment. Thus you prevent the further sedimentation during longer production downtimes.

1. Press < *STANDBY* > button on the PPS control panel
 - ⇒ STANDBY operation mode is active
 - ⇒ Follow-up plate moves out of pail
- ⇒ Pail is depressurised
- or**
1. Press the < *Operation modes* > button on the status bar of the visualisation
 - ⇒ Three-symbol combination opens
 2. Press the < *STANDBY* > button
 - ⇒ STANDBY operation mode is active
 - ⇒ Follow-up plate moves out of pail
- ⇒ Pail is depressurised

3.8.4 Calibration of Mettler scales, Type 'WKC'

3.8.4.1 Internal calibration of the Mettler scales | Type WKC

The visualisation system function "Internal Calibration" is used when scales are put into operation for the first time and after a change of location. This means that every machine that is put into operation on the customer's premises, for example, must first go through this procedure.

- ✓ Scales of type 'WKC' are installed
- ✓ Serial interface via the BECKHOFF terminal is active: the serial number of the scales module appears on the visualisation system
- ✓ This function was enabled in the machine visualisation system during commissioning
- ✓ User is logged in with the login right 'Level 4'

1. Press quick key < *Internal calibration* >

⇒



NOTICE

Key status

If an internal calibration has already been carried out beforehand, the < *Internal calibration* > quick key lights up green, otherwise grey or red in the event of a fault.

- ⇒ The following message appears: "Calibration started"
- ⇒ If calibration is OK: Quick key < *Internal calibration* > lights up green
- ⇒ If the calibration has failed: Quick key < *Internal calibration* > lights up red

3.8.4.2 External calibration of the Mettler scales | Type WKC

The visualisation system function "External Calibration" is used if regular inspection equipment monitoring is carried out by the customer. The customer can carry out the calibration himself.

- ✓ Scales of type 'WKC' are installed
- ✓ Serial interface via the BECKHOFF terminal is active: the serial number of the scales module appears on the visualisation system
- ✓ This function was enabled in the machine visualisation system during commissioning
- ✓ Scales have already been tared using an empty cup: The scales show '0'
- ✓ A calibrated weight of at least 100 g is available
- ✓ User is logged in with the login right 'Level 4'

1. Enter the value of the calibrated weight, e.g. 100 g, in the “Calibration weight” input field
2. Press quick key < *External calibration* >
⇒



NOTICE

Status LEDs

If an external calibration has already been carried out beforehand, the < *External calibration* > quick key lights up green, otherwise grey or red in the event of a fault.

- ⇒ The following message appears: “Load weight”
- 3. Place the calibrated weight onto the scales
⇒ The following message appears: “Unload weight”
- 4. Remove the calibrated weight from the scales
 - ⇒ If calibration is OK: Quick key < *External calibration* > lights up green
 - ⇒ If the calibration has failed: Quick key < *External calibration* > lights up red

3.8.5 Weighing in

For two-component mixing and dispensing machines, the weighing-in serves to check the mixing ratio in weight percentages of the components A and B. From mixing ratio A: B emerge the specifications for setting the speeds of the dispensing pumps.

There are several ways to weigh in. The selection depends, among other things, on the mixing ratio itself (only for two-component systems). Furthermore, the dispensing power, the size of the mixing changer (two-component systems only) and the outlet nozzle, and finally also various physical and chemical material and environmental parameters play a role.

For single-component dispensing systems, you check the repetition accuracy of the dispensing shots. Naturally here the following calculation is omitted.



NOTICE

Weighing in

Always perform a weigh-in before the start of production in order to guarantee correct repetition results for the subsequent production operation and be able to locate faults that occur more quickly.

Example of the calculation of speed specifications:

Mixing ratio in weight percentages A to B = 100: 10;

Total dispensing quantity: 5 g in 2 seconds.

Calculation:

1. From the mixing ratio it follows that: 110 weight components for the mixture of A and B.
2. Dispensing output: $5 \text{ g} / 2 \text{ sec.} = 2.5 \text{ g/sec.}$ Mixture must be conveyed
3. Component A: $110 = 2.5 \text{ g/sec.}; 100 = A \text{ g/sec.}; A = 2.5 \text{ g/sec. times } 100 \text{ divided by } 110; B = 2.27 \text{ g/sec.}$
4. Component B: $110 = 2.5 \text{ g/sec.}; 10 = B \text{ g/sec. } B = 2.5 \text{ times } 10 \text{ divided by } 110; B = 0.227 \text{ g/sec.}$
5. Result: the speed of pump A must be so high that it can convey 2.27 g/sec. in the current design of the dispensing circuit; the speed of pump B must be accordingly high in order to be able to convey 0.227 g/sec. This is determined by the weighing-in.

3.8.5.1 Separate weighing via static mixing tube (automatic scale)

Principle

From the process, the first component is weighed in until the calculated weight is reached. Then the second component is weighed in. If their calculated weight is reached, the mixture weight is checked again for reproducibility. There are 3 slightly different methods of checking the weighing-in. This depends on the weight percentages and the viscosities of resin and hardener components.



NOTICE

Weighing process with weighing mask and weighing process only via scales

The quality check of the dispensing shots is ensured by the weigh-in described below. Depending on the construction, axis-driven dosing systems such as the B5xxx series and also the B2000 have their own weighing position at which the weighing-in is processed. Pure table systems cannot operate an additional weighing position due to the restricted amount of space and the restricted or non-existent movement possibilities. For the user, this means he needs to provide scales outside the dosing area, he must remove the dispensing shots into a container and place this on the scales himself in order to determine the weight. The following description refers solely to the weigh-in using a 3-axis transport system as is implemented in B5xxx or B2000 cells. Weighing in using separate scales is described in a separate chapter in the operating instructions.



NOTICE

Login authorisation and operation mode

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.

Sequence of weighing-in method 1 (equal parts of resin and hardener, e.g. 1:1 and viscosities nearly identical):

- ✓ Dispensing program was selected
 - ✓ PROCESS MONITORING operation mode is active
 - ✓ Material preparation reports OK
 - ✓ Dispensing time was set to component dispensing time, maximum 10 seconds
 - ✓ Weighing parameters were set
 - ✓ Empty container is on the scale
 - ✓ **Dispensing pressure and fill pressure levels for weighing-in process are set** (values must be determined depending on the material)
1. Open the pump tab for the component that according to the mixing ratio has the lower weight percentage
 2. Set the check box "active" with mouse pointer
 - ⇒ Pump is active
 3. Open the pump tab for the second component
 4. Reset the check box "active" with mouse pointer if set
 - ⇒ Pump is inactive
 5. Press the < *Fill* > button
 - ⇒ The dispensing unit moves into the filling position and fills the mixing tube with the selected component



NOTICE

Flood mixing tube

Assistance: With low-viscosity components (usually B), depending on the dispensing output and material properties, in addition to normal filling, it is necessary to flood the dispensing unit with additional dispensing processes so that the fill quantity is reached. To do this, start the dispensing processes one or more times with the < *Dispense with time* > button.

- ⇒ "Chamber full" is displayed on the mixing head symbol of the start page
6. Press the < *Dispense with time* > button several times
 - ⇒ Any trapped air can escape
 7. Press the < *Scale* > button
 - ⇒ Weighing screen opens
 8. Press the optional weighing tab for the selected component
 9. Press the < *Start weighing* > or < *Start weighing sequence 50* > button
 - ⇒ Dispensing unit moves into weighing position
 - ⇒ Number of preset weighing shots are dispensed



NOTICE

Type and source of the danger

Error indicator or weighing stop during weighing-in

Depending on the button that is pressed to start the weighing in (<Start weighing> or <Start weighing sequence 50> and the activity of optional special pressure limits that can be marked on the "Settings" screen), the process behaves differently if the weight or "Max." and "Min." dispensing pressure and special pressure limits is/are exceeded or underrun:

1 When triggered with the <Start weighing> button, the process stops as soon as a weighing limit is violated or one of the dispensing pressure limits "Max." or "Min." is exceeded or underrun; the special pressure limits are not considered!

2 When triggered with the <Start weighing sequence 50> button, the process only stops as soon as active special pressure limits are violated: any exceeding or underrunning of the weight limits is indicated with a result LED that is highlighted in red, the process continues to the processing of the 50 weighing shots.

10. If the weight is greater than the calculated weight: Reduce the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

11. If the weight is less than the calculated weight: Increase the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

12. Trigger weighing shots and change the speed until the calculated weight is displayed in stable fashion
13. Trigger the weighing sequence and observe whether the course of the weight remains stable
14. Set parameters for production operation of the weighed-in components
 - ⇒ **First component is weighed in**
15. Press the < Move to home position > button
 - ⇒ Dispensing unit moves into home position
 - ⇒ "Chamber empty" is displayed
 - ⇒ Potlife counter is stopped
16. Open the pump tab for the second component
17. Set the check box "active" with mouse pointer

- ⇒ Pump is active
- 18. Open the pump tab for the first component
- 19. Reset the check box "active" with mouse pointer
 - ⇒ Pump is inactive
- 20. Press the < *Fill* > button
 - ⇒ Dispensing unit moves into the filling position and fills the mixing tube with the second component
 - ⇒ "Chamber full" is displayed on the mixing head symbol of the start page



NOTICE

Flood mixing tube

Assistance: With low-viscosity components (usually B), depending on the dispensing output and material properties, in addition to normal filling, it is necessary to flood the dispensing unit with additional dispensing processes so that the fill quantity is reached. To do this, start the dispensing processes one or more times with the < *Dispense with time* > button.

- ⇒ Mixing tube is flooded with component
- 21. Press the < *Dispense with time* > button several times
 - ⇒ Any trapped air can escape
- 22. Press the < *Scale* > button
- 23. Open optional weighing tab for the second component
- 24. Press the < *Start weighing* > or < *Start weighing sequence 50* > button (see also note about error stop!)
 - ⇒ Dispensing unit moves into weighing position
 - ⇒ Number of preset weighing shots are dispensed
- 25. If the weight is greater than the calculated weight: Reduce the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

- 26. If the weight is less than the calculated weight: Increase the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

- 27. Trigger weighing shots and change the speed until the calculated weight is displayed in stable fashion

28. Trigger the weighing sequence and observe whether the course of the weight remains stable
29. Set parameters for production operation of the second weighed-in component
 - ⇒ **Second component is weighed in**
30. Press the *< Move to home position >* button
 - ⇒ Dispensing unit moves into home position
 - ⇒ "Chamber empty" is displayed
 - ⇒ Potlife counter is stopped
31. Open the pump tab for the first component
32. Set the check box "active" with mouse pointer
 - ⇒ Both pumps are now prepared for dispensing
33. Press the *< Fill >* button
 - ⇒ Mixing tube is filled with mixture
 - ⇒ "Chamber full" is displayed
34. Press the *< Scale >* button
35. Open optional weighing screen for mixture
36. Press the *< Start weighing >* or *< Start weighing sequence 50 >* button (see also note about error stop!)
 - ⇒ Dispensing unit moves into weighing position
 - ⇒ Number of preset weighing shots are dispensed
37. if weighing-in with the *< Start weighing >* button Increase the number of weighing shots (e.g. '25')
38. Press the *<Start weighing>* button and observe the course of the weight.
 - ⇒ If deviations are within the specified limits: **Mixture is weighed in**
39. If the mixture weight is not correct: simultaneously increase or decrease the speed of the two components by the same factor
40. Perform the weighing sequence and observe the course of the weight
41. Change the speed and trigger the weighing sequence until the calculated mixture weight is correct
42. Press the *< Move to home position >* button
 - ⇒ "Chamber empty" is displayed
 - ⇒ Potlife counter is stopped
43. Possibly adjust parameters for potlife, etc. to the subsequent production process

Flow weighing-in method 2 (equal parts of resin and hardener, e.g. 1:1 and viscosities radically different):

- ✓ Dispensing program was selected
- ✓ PROCESS MONITORING operation mode is active

- ✓ Material preparation reports OK
 - ✓ Dispensing time was set to component dispensing time, maximum 10 seconds
 - ✓ Weighing parameters were set
 - ✓ Empty container is on the scale
 - ✓ **Dispensing pressure and fill pressure levels for weighing-in process are set** (values must be determined depending on the material)
1. Open the pump tab for the component that according to the mixing ratio has the lower viscosity
 2. Set the check box "active" with mouse pointer
 - ⇒ Pump is active
 3. Open the pump tab for the second component
 4. Reset the check box "active" with mouse pointer if set
 - ⇒ Pump is inactive



NOTICE

Fill with weigh-in adapter

For operation with a duplex mixing head, a weigh-in adapter can be used. This is completed with an outlet nozzle. If the hole of the adapter is too narrow to allow the amount of material to flow through, it must be adapted accordingly. As a consequence, the mixing tube is omitted for the weighing-in process using the weighing-in adapter. Even in the description, the expression "mixing tube" must be replaced by "weighing-in adapter" - at least notionally.

5. Press the < *Fill* > button
 - ⇒ The dispensing unit moves into the filling position and fills the mixing tube with the selected component



NOTICE

Mixing tube drips

Assistance: Fill the mixing tube again at greatly increased speed so that any remaining air can escape. Reduce the speed again after another filling process.



NOTICE

Fill quantity specification too low

If the fill quantity of the component is too low: Unscrew the mixing tube and weigh in without the mixing tube!

- ⇒ "Chamber full" is displayed on the mixing head symbol of the start page

6. Press the < *Dispense with time* > button several times

- ⇒ Any trapped air can escape
- 7. Press the < *Scale* > button
 - ⇒ Weighing screen opens
- 8. Press the optional weighing tab for the selected component
- 9. Press the < *Start weighing* > or < *Start weighing sequence 50* > button (see also note about error stop!)
 - ⇒ Dispensing unit moves into weighing position
 - ⇒ Number of preset weighing shots are dispensed
- 10. If the weight is greater than the calculated weight: Reduce the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

- 11. If the weight is less than the calculated weight: Increase the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

- 12. Trigger weighing shots and change the speed until the calculated weight is displayed in stable fashion
- 13. Trigger the weighing sequence and observe whether the course of the weight remains stable
- 14. Set parameters for production operation of the weighed-in components
 - ⇒ **First component is weighed in**
- 15. Press the < *Move to home position* > button
 - ⇒ Dispensing unit moves into home position
 - ⇒ "Chamber empty" is displayed
 - ⇒ Potlife counter is stopped
- 16. Open the pump tab for the second component
- 17. Set the check box "active" with mouse pointer
 - ⇒ Pump is active
- 18. Open the pump tab for the first component
- 19. Reset the check box "active" with mouse pointer
 - ⇒ Pump is inactive
- 20. Press the < *Fill* > button
 - ⇒ Dispensing unit moves into the filling position and fills the mixing tube with the second component

- ⇒ "Chamber full" is displayed on the mixing head symbol of the start page
- ⇒ Mixing tube is flooded with component
- 21. Press the < *Dispense with time* > button several times
 - ⇒ Any trapped air can escape
- 22. Press the < *Scale* > button
- 23. Open optional weighing tab for the second component
- 24. Press the < *Start weighing* > or < *Start weighing sequence 50* > button (see also note about error stop!)
 - ⇒ Dispensing unit moves into weighing position
 - ⇒ Number of preset weighing shots are dispensed
- 25. If the weight is greater than the calculated weight: Reduce the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

- 26. If the weight is less than the calculated weight: Increase the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

- 27. Trigger weighing shots and change the speed until the calculated weight is displayed in stable fashion
- 28. Trigger the weighing sequence and observe whether the course of the weight remains stable
- 29. Set parameters for production operation of the second weighed-in component
 - ⇒ **Second component is weighed in**
- 30. Press the < *Move to home position* > button
 - ⇒ Dispensing unit moves into home position
 - ⇒ "Chamber empty" is displayed
 - ⇒ Potlife counter is stopped
- 31. Open the pump tab for the first component
- 32. Set the check box "active" with mouse pointer
 - ⇒ Both pumps are now prepared for dispensing
- 33. Screw on the new mixing tube (info: the new needle position must be confirmed in the needlecheck register)
- 34. Press the < *Fill* > button

- ⇒ Mixing tube is filled with mixture
- ⇒ "Chamber full" is displayed
- 35. Press the *< Scale >* button
- 36. Open optional weighing screen for mixture
- 37. Press the *< Start weighing >* or *< Start weighing sequence 50 >* button (see also note about error stop!)
 - ⇒ Dispensing unit moves into weighing position
 - ⇒ Number of preset weighing shots are dispensed
- 38. If weighing should be started with the *< Start weighing >* button Increase the number of weighing shots (e.g. '25')
- 39. Press the *<Start weighing>* button and observe the course of the weight.
 - ⇒ If deviations are within the specified limits: **Mixture is weighed in**
- 40. If the mixing weight is not correct: deactivate lower viscosity component and weigh higher viscosity component
- 41. Perform the weighing sequence and observe the course of the weight
- 42. Change the speed and trigger the weighing sequence until the calculated mixture weight is correct
- 43. Press the *< Move to home position >* button
 - ⇒ "Chamber empty" is displayed
 - ⇒ Potlife counter is stopped
- 44. Possibly adjust parameters for potlife, etc. to the subsequent production process

Flow of weighing-in method 3 (resin and hardener are very different, e.g. 100:27 and viscosities radically different, e.g. 50 mPas and 50,000 mPas):

- ✓ Dispensing program was selected
 - ✓ PROCESS MONITORING operation mode is active
 - ✓ Material preparation reports OK
 - ✓ Dispensing time was set to component dispensing time, maximum 10 seconds
 - ✓ Weighing parameters were set
 - ✓ Empty container is on the scale
 - ✓ **Dispensing pressure and fill pressure levels for weighing-in process are set** (values must be determined depending on the material)
1. Open the pump tab for the component that has the lower viscosity according to the mixing ratio
 2. Set the check box "active" with mouse pointer
 - ⇒ Pump is active
 3. Open the pump tab for the second component

4. Reset the check box "active" with mouse pointer if set
⇒ Pump is inactive



NOTICE

Fill with weigh-in adapter

For operation with a duplex mixing head, a weigh-in adapter can be used. This is completed with an outlet nozzle. If the hole of the adapter is too narrow to allow the amount of material to flow through, it must be adapted accordingly. As a consequence, the mixing tube is omitted for the weighing-in process using the weighing-in adapter. Even in the description, the expression "mixing tube" must be replaced by "weighing-in adapter" - at least notionally.

5. Press the < *Fill* > button
⇒ The dispensing unit moves into the filling position and fills the mixing tube with the selected component



NOTICE

Mixing tube drips

Assistance: Fill the mixing tube again at greatly increased speed so that any remaining air can escape. Reduce the speed again after another filling process.



NOTICE

Fill quantity specification too low

If the fill quantity of the component is too low: Unscrew the mixing tube and fill without mixing tube.

- ⇒ "Chamber full" is displayed on the mixing head symbol of the start page
6. Press the < *Dispense with time* > button several times
⇒ Any trapped air can escape
7. Press the < *Scale* > button
⇒ Weighing screen opens
8. Press the optional weighing tab for the selected component
9. Press the < *Start weighing* > or < *Start weighing sequence 50* > button (see also note about error stop!)
⇒ Dispensing unit moves into weighing position
⇒ Number of preset weighing shots are dispensed
10. If the weight is greater than the calculated weight: Reduce the pump speed

**NOTICE****Speed adjustment during weighing-in**

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

11. If the weight is less than the calculated weight: Increase the pump speed

**NOTICE****Speed adjustment during weighing-in**

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

12. Trigger weighing shots and change the speed until the calculated weight is displayed in stable fashion
13. Trigger the weighing sequence and observe whether the course of the weight remains stable
14. Set parameters for production operation of the weighed-in components
 - ⇒ **First component is weighed in**
15. Open the pump tab for the second component
16. Set the check box "active" with mouse pointer
 - ⇒ Pump is active
17. Press the *< Fill >* button
 - ⇒ Dispensing unit moves into the filling position and fills the mixing tube with both components
 - ⇒ "Chamber full" is displayed on the mixing head symbol of the start page
 - ⇒ Mixing tube is flooded with mixture
18. Press the *< Dispense with time >* button several times
 - ⇒ Any trapped air can escape
19. Press the *< Scale >* button
20. Open optional weighing tab for mixture
21. Press the *< Start weighing >* or *< Start weighing sequence 50 >* button (see also note about error stop!)
 - ⇒ Dispensing unit moves into weighing position
 - ⇒ Number of preset weighing shots are dispensed
22. If the weight is greater than the calculated weight: Reduce the pump speed

**NOTICE****Speed adjustment during weighing-in**

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

23. If the weight is less than the calculated weight: Increase the pump speed



NOTICE

Speed adjustment during weighing-in

Also during ongoing weighing operation, it is possible to change the speed online in the pump dialog

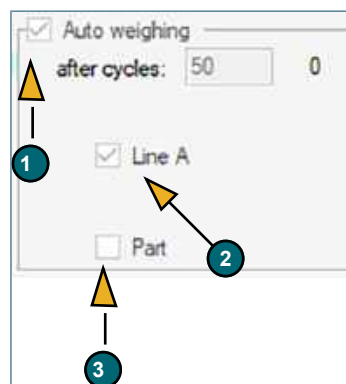
24. Trigger weighing shots and change the speed until the calculated weight is displayed in stable fashion
25. Trigger the weighing sequence and observe whether the course of the weight remains stable
26. Set parameters for production operation
- ⇒ **Mixture is weighed in**
27. Press the < Move to home position > button
- ⇒ Dispensing unit moves into home position
 - ⇒ "Chamber empty" is displayed
- ⇒ Potlife counter is stopped

3.8.6 Weight check according to the component simulation method

The weight check according to the component simulation method is an additional function on bdtronic dispensing systems for checking the shot weight and its repetition accuracy using the actual casting operation. It can be used in the Process Monitoring operation mode as a semi-automatic function for the machine setup engineer or as part of the production sequence in Automatic operation mode.

Using positions (Pos. 1), (Pos. 2) and (Pos. 3) of the "Automat. weighing" dialogue, the

marks (x) on positions (Pos. 1) and (Pos. 3) of the simulation weighing mode can be specified during the automatic sequence. Marks (x) at the positions (Pos. 1) and (Pos. 2) start the normal weighing sequence, without component simulation.

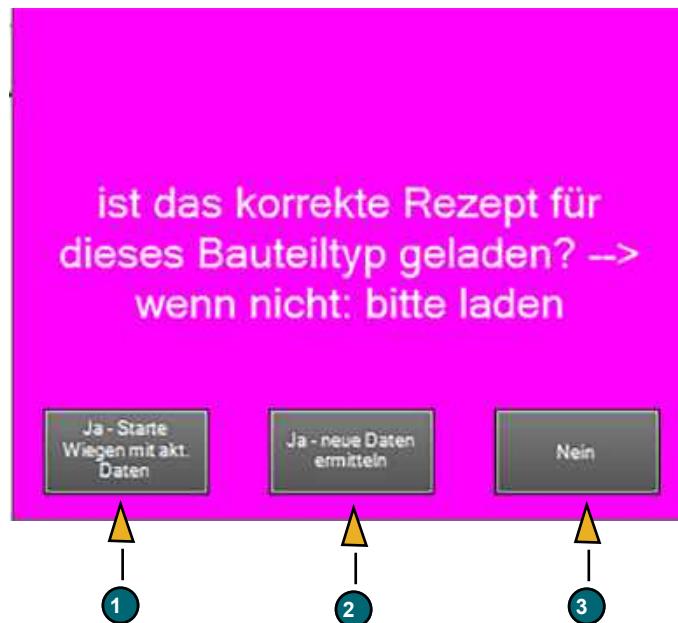


Weighing-in of the dispensing components before this type of weighing check must ensure that the correct shot weight of the individual components and the mixture (for 2K systems) are achieved.

A meaningful weight check is only obtained when the measuring process runs as if the specific underlying dispensing operation (e.g. dispensing a housing groove) were to be processed in order to take all influencing variables into account.

To this end, the processing of the recipe on which the dispensing task is based is simulated using measuring technology, and a series of dispensing shots is dispensed via the dispensing nozzle into a cup placed on a balancing scale. As the specific component is not run, but only the dispensing time(s) associated with the actual component, this is referred to as a "simulation". The actual dispensing time(s) was previously determined in what is referred to as a dry run on the component.

When starting a weighing sequence via one of the two start keys



Answering the prompt "Is the correct recipe loaded? If not, please load" with "No" means the correct recipe must be loaded as the very first action.

If it is already loaded, there is the choice between:

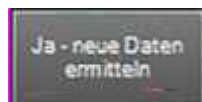
1. Simulation data already exists (see popup pos. 1) or
2. Simulation data has to be determined first (see popup pos. 2)

In situation (1.), the



key can be used to start the weighing sequence directly.

In situation (2.), the



key has to be used first to determine the required simulation data via a dry run on the component. Responding to the save prompt with "OK" stores the determined dispensing time(s) on the



tab under the consecutive shot number. In the next step, the



simulation weighing cycle is started via the key, as shown above under (1.).

In each case, the "Result" LED displays the result of the measurement



series in green as OK or in red as failed. In the event of an error, during the measurement series (e.g. pressure fault), a corresponding message is displayed in the Visu status bar. It is also stored as an entry under "Active Messages".



3.8.7 Manual functions, general

Switching off



End application

Designation of the symbol: < Switch off >



NOTICE

Login authorisation

Only an authorised user login can change configurations.

Application:

1. Press < Switch off > button

⇒ Application switches off

Activating MANUAL mode



Switch on Manual operation mode: there are two MANUAL operation modes:

- MANUAL for machine cell
- MANUAL for material preparation system such as MPS, PPS, vessel, cartridges

Designation of the symbol: < MANUAL >

Application:

1. Press the < MANUAL > button

⇒ MANUAL operation mode is active

or

1. Press the < Stop > operating button

⇒ MANUAL operation mode is active

Edit



Here the dialogue for overwriting the name of a selected element (e.g. program, recipe, action list, etc.) can be opened

Designation of the symbol: < Edit >



NOTICE

Login authorisation function execution

Only an authorised user login can execute functions.

Application:

1. Mark the element in the selection list
 - ⇒ The element is selected
2. Click the mouse pointer in the input line
 - ⇒ The virtual keyboard opens
3. Enter the name
4. Press the green < OK > symbol to confirm entry or the red < Cross symbol > to discard the entry
 - ⇒ The entry is taken over
5. Press the green < OK > symbol to confirm the name or the red < Cross symbol > to discard the name
 - ⇒ The new name appears in the selection list, the old name was removed

Copy



Starts the flow to copy data of the selected element (program, recipe, action list) to save it under a new name

Designation of the symbol: < Copy >



NOTICE

Login authorisation function execution

Only an authorised user login can execute functions.

Application:

1. Mark the element in the selection list
 - ⇒ The element is selected
2. Press the < *Copy* > button
 - ⇒ The input dialogue appears
3. Click the mouse pointer on the input field
 - ⇒ The virtual keyboard opens
4. Enter the name
5. Confirm the entry with the green < OK > button or discard with the red cross button
 - ⇒ The keyboard dialogue closes
6. Confirm the entry with the green < OK > button or discard with the red cross button

Load

Loads the selected file (e.g. programs, recipes, action lists, etc.) into system memory for further processing.

Designation of the symbol: < *Load* >

**NOTICE****Login authorisation function execution**

Only an authorised user login can execute functions.

Application:

1. Mark the element in the selection list
 - ⇒ The element is selected
2. Press the < *Load* > button
 - ⇒ Data of the element is loaded into system memory for further processing

Delete

Deletes the selected element (program, recipe, action list, etc.) from the selection list

Designation of the symbol: < *Delete* >

**NOTICE****Login authorisation function execution**

Only an authorised user login can execute functions.

Application:

1. Mark the element in the selection list
2. Press the < *Delete* > button
 - ⇒ The confirmation query appears
3. Delete the element with the < *OK* > button or discard the process with the < *Cancel* > button

⇒ The element is removed from the selection list

New

Calls up the dialogue to create a new element (e.g. program, recipe, action list, etc.)

Designation of the symbol: < *New* >

**NOTICE****Login authorisation function execution**

Only an authorised user login can execute functions.

Application:

1. Press the button
 - ⇒ The name dialogue opens
2. Click the mouse pointer in the input line
 - ⇒ The virtual keyboard opens
3. Enter the name
4. Press the green < *OK* > symbol to confirm the entry or the red < *Cross symbol* > to discard the entry
 - ⇒ The entry is taken over
5. Press the green < *OK* > symbol to confirm the name or the red < *Cross symbol* > to discard the name

⇒ The new element is displayed on the selection list

User login

Opens the login dialogue

Designation of the symbol: < *Call up the login screen* >

Application:

1. Press the < *Call up login screen* > button

⇒ The start screen of the "Login" submenu opens

E-STOP



Display of the E-STOP operation mode

Designation of the symbol: < *E-STOP* >

Application:

1. Press the < *E-STOP* > button

⇒ The E-STOP operation mode is active

Program assignment



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

1. Call up the recipe screen
2. Mark the desired recipe
3. Load the recipe into memory with the < *Load* > button
4. Click the mouse pointer on the program selection field
 - ⇒ The dialogue opens
5. Mark the desired program
6. Confirm the entry with the green button or discard with the red cancel button

⇒ The program was linked to the recipe



NOTICE

Save changes.

Only by pressing the save button are the changes taken over permanently into the system memory: when the system is restarted, the changes are available again (info: only an authorised user can start the saving process)

Create drive programs:



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

1. Call up "Drive program editor" screen
2. Mark drive program in selection list
3. Press the < Load > button
 - ⇒ Drive program is loaded into memory
4. Press < Save as > button
 - ⇒ The dialogue opens
5. Enter file name
6. Trigger storage process with < Save >
 - ⇒ New program is opened in editor
7. Call up "Axes" screen
8. Press the < New > button
 - ⇒ The dialogue opens
9. Mark new file
10. Make file known to the system with < Open > button
11. New file appears in selection

Manually moving driving axis X-



Starts movement of the (X-) axis

Designation of the symbol: < X- Hand >



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached
 - ⇒ Coordinates of the end position are displayed in the X-coordinate cell

Manually moving driving axis X+



Starts movement of the (X+) axis

Designation of the symbol: *< X+ Hand >*



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the X-coordinate cell

Manually moving driving axis Y-



Starts movement of the (Y-) axis

Designation of the symbol: *< Y- Hand >*



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the Y-coordinate cell

Manually moving driving axis Y+



Starts movement of the (Y+) axis

Designation of the symbol: *< Y+ Hand >*



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the Y-coordinate cell

Manually moving driving axis Z-



Starts movement of the (Z-) axis

Designation of the symbol: < Z- Hand >



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the Z-coordinate cell

Manually moving driving axis Z+



Starts movement of the (Z+) axis

Designation of the symbol: < Z+ Hand >



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the Z-coordinate cell

Create new recipe



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

1. Call up the recipe screen
2. Mark the recipe
3. Press the < Copy > button
 - ⇒ The virtual keyboard opens
 - ⇒ The input dialogue opens

4. Enter the name for the copy
5. Confirm the name with the green confirm button or discard with the red cancel button

⇒ The name is added to the selection list

Teach



Teach a new drive position

Designation of the symbol: < *Teach* >



NOTICE

Input and execution of functions and parameters.

Input and further processing of a parameter or function can only be performed with an authorised user login.

Application:

- ✓ The drive program editor appears
 - ✓ The process head unit is in the new position
1. Press the < *Teach* > button
- ⇒ The new position is saved in the drive program

Start reference run



Move to the preset reference point

Designation of the symbol: < *Reference run* >



NOTICE

Process monitoring and automatic operation modes!

This function or sequence can only be executed in process monitoring or automatic operation mode!

Application:

1. Press the < *Reference run* > button
 - ⇒ The axle system moves to the specified reference point
- ⇒ If the reference cycle can be ended, all other – even manual – axis moves are triggered.

Start button



The flow (e.g. drive program, action list, manual process) is started if all start conditions are fulfilled.

Designation of the symbol: < *Start* >



NOTICE

Login authorisation and operation mode

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.

Application:

- ✓ Possibly fulfil additional prerequisites

1. Press the < *Start* > button

⇒ The process (e.g. weighing cycle, etc.) or program execution (e.g. run through drive program) and processing of an action list is started

⇒ The process ends by itself or with the program or action list end

Stop procedure



Stops a running procedure such as drive program, action list, etc.

Designation of the symbol: < *Stop* >

Application:

1. Press the < *Stop* > button

⇒ The drive program is stopped

Add line



Add a new line in the editor

Designation of the symbol: < *New line* >



NOTICE

Login authorisation

Only an authorised user login can change configurations.

Application:

1. Mark the line in the selection list

⇒ The line is selected

2. Press the < *New line* > button

- Press the green **< OK >** symbol in order to confirm the entry or **< Cancel >** to discard the action.

⇒ A new line is present

Delete line



Delete a line from the editor

Designation of the symbol: **< Delete line >**



NOTICE

Login authorisation

Only an authorised user login can change configurations.

Application:

- Mark the line in the selection list
 - ⇒ The line is selected
- Press the **< Delete line >** button
- Press the green **< OK >** symbol in order to confirm the entry or the red **< Cancel >** symbol to discard the action.

⇒ The line is deleted

Safety



Indicator of SAFETY operation mode

Designation of the symbol: **< SAFETY >**

Application:

- Press the **< SAFETY >** button
 - ⇒ The "SAFETY" operation mode of the machine is active

Activating PROCESS MONITORING mode



Switch to PROCESS MONITORING mode (only for certain system cells!)

Designation of the symbol: **< PROCESS MONITORING >**



NOTICE

Process monitoring operation mode!

This function or routine can only be executed in process monitoring mode! An authorised user must be logged in for this!

Application:

1. Press the < *PROCESS MONITORING* > button
2. or press the < *PROCESS MONITORING* > operating button
 - ⇒ The pot life processing runs

⇒ The *PROCESS MONITORING* operation mode is active

Activating automatic mode

This symbol indicates the automatic operation mode: there are two automatic operation modes:

- Automatic for machine cell
- Automatic for material preparation system such as MPS, PPS, vessel, cartridges

Designation of the symbol: < *AUTOMATIC* >

**NOTICE****Process monitoring and automatic operation modes!**

This function or sequence can only be executed in process monitoring or automatic operation mode!

Application:

1. Press the < *AUTOMATIC* > button
2. or press the < *AUTOMATIC* > operating button
 - ⇒ The pot life processing runs

⇒ The "AUTOMATIC" operation mode is active

3.8.8 Manual functions | Dispensing system | Processes

3.8.8.1 Function key bar 1C | Static 2C

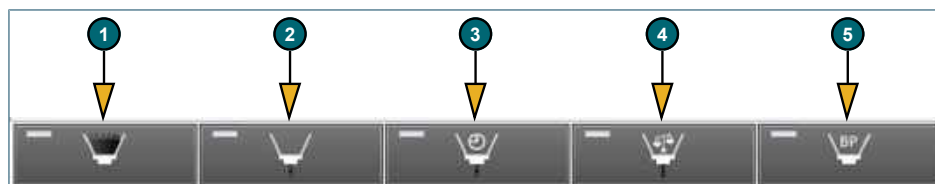


Illustration 48: Function key bar (here for single-component and static mixer)

1 < Filling > key	2 < Dispensing Event > key
3 < Dispensing with time > key	4 < Dispensing with weight > key
5 < Basic position > key	

Application

Manual triggering of the dispensing function.

3.8.8.1.1 Filling

< Filling > key (1):

This key triggers a filling process that floods the reactive volume of a dispensing system with material or material mixture so that this area is free of air. The parameters in the "Fill" group field of the "Head" screen also control the filling process. When the filling process has been completed successfully, the system signals "System ready". From now on, potlife monitoring is also running. This process is only carried out once and enables the operator to trigger dispensing processes, weighing shots, etc. via other function keys.

A prerequisite for successful filling is a prepared material system and, if operating an axis-controlled system, a previously confirmed position in which the filling process can take place. For systems without axes, an empty cup of sufficient size must be placed under the dispensing unit before starting the filling process! In case of doubt, this must also be able to accommodate blindshots due to potlife processing!

Pressing the key again or changing the operation mode within the runtime of the filling process interrupts it in such a way that the system status becomes undefined and potlife processing starts. After this has elapsed, the system moves to purge and is then set to "System not active".



NOTICE

Process monitoring operation mode!

This function or routine can only be executed in process monitoring mode! An authorised user must be logged in for this!



NOTICE

Automatic filling

Alternatively, the filling process can also be triggered automatically in AUTOMATIC mode. On tabletop machines, this is triggered via a handshake. In dispensing cells, it forms part of the automatic step chain and, depending on the customer's specifications, is processed directly after entering this operation mode or only when the first part to be sealed is detected.

3.8.8.1.2 Dispensing event

< Dispensing | Event > key (2):

When the key is pressed, dispensing starts and continues until the key is released. Dispensing then stops and the potlife counter counts down. This also happens without a previous filling process if the system status is dis-

played as "System undefined". In the regular process, filling is carried out first, then dispensing is started according to one of the three methods: event, time setting or weight setting.

**NOTICE****Operation mode during dispensing**

Possible operation modes: MANUAL, PROCESS MONITORING or SAFETY.

3.8.8.1.3 Dispensing with time**< Dispensing with time setting > key (3):**

The time setting is entered in the "Dispensing time" field of the "Head" screen.

After pressing the key, dispensing starts and stops automatically at the end of the "Dispensing time".

Pressing the key again or changing the operation mode during the current dispensing process ends the process immediately. Potlife processing starts.

The dispensing process starts even without previous filling, but with the status "System undefined"!

**NOTICE****Operation mode during dispensing**

Possible operation modes: MANUAL, PROCESS MONITORING or SAFETY.

3.8.8.1.4 Dispensing with weight setting**< Dispensing with weight setting > key (4):**

The weight setting is entered in the "Dispensing volume" field of the "Head" screen.

After pressing the key, dispensing starts and stops automatically at the end of the "Dispensing volume".

Pressing the key again or changing the operation mode during the current dispensing process ends the process immediately. Potlife processing starts.

The dispensing process starts even without previous filling, but with the status "System undefined"!

**NOTICE****Operation mode during dispensing**

Possible operation modes: MANUAL, PROCESS MONITORING or SAFETY.

3.8.8.1.5 Move to basic position

< Basic position > key (5):

Only single-component or static, two-component systems have this key. When the key is pressed, a system equipped with axes moves the dispensing system to purging position and stops potlife processing. The system status is displayed as "System not active".

Pressing the key again while the process is running interrupts it.

**NOTICE****Process monitoring operation mode!**

This function or routine can only be executed in process monitoring mode! An authorised user must be logged in for this!

3.8.9 Needle Check - Single-head System**General**

The needle check unit is an auxiliary module on bdtronic dispensing machines with which a dispensing needle or mixing tube can be checked for straight, vertical alignment and optionally also for length tolerance. In automatic mode, the needle check is incorporated seamlessly into the series of process steps to be executed. Incorrect alignments or lengths out of tolerance lead to an interruption in the production process.

This module is available as a 'master' version for single-head dispensing systems and as a 'master/slave' combination for a double-head dispensing system with manual or servo-controlled repositioning.

The set-up of these module variants differs correspondingly.

3.8.9.1 Set up 'Golden Needle'**General**

The procedures described below must be performed only once during commissioning of the system or when changing the dispensing system configuration in order to define the needle checking position set for the first time (= "Golden Needle").

Set-up procedure:

- ✓ Needle check module has been correctly installed and integrated electrically and pneumatically.
 - ✓ Software version of the PLC supports the "needle check" function
 - ✓ Dispensing needle or mixing tube has been installed
 - ✓ User with the highest login authorisation is logged in
 - ✓ PROCESS MONITORING mode is active
 - ✓ The needle diameter has been entered in the settings of the dispensing menu
1. Move the dispensing needle or mixing tube to the middle of the measuring circle of the 'tactile needle check' using the manual drive symbols of the "Axes" screen or using the membrane keys on the dispensing needle cell frame so that both light barriers are interrupted at the same time (additional information: The two intersecting lines in the "Deviation" symbolising the X and Y light barriers change to green).
 2. Press the < *New golden needle* > button; a search run is started during which the exact middle between the light barriers X and Y. The dispensing needle or mixing tube is then moved until light barriers X and Y are no longer interrupted.
- ⇒ The "Result" LED lights up green if the measurement of the 'Golden Needle' was performed successfully; the coordinates determined are stored in the "Positions" dialogue under "Golden Needle"
- ⇒ The "Error" LED lights up red if the measurement of the 'Golden Needle' was not performed successfully

3.8.9.2 Needle check procedure**General**

The 'tactile needle check' detects deviations in the alignment of the dispensing needle or mixing tube or in the needle length by reference to a defined valid needle position and tolerance specifications. A needle check can be performed both manually in PROCESS MONITORING mode and automatically in AUTOMATIC mode.

Manual check start procedure:

- ✓ "Current needle" position is defined
 - ✓ Dispensing needle or mixing tube is fixed to the dispensing unit
 - ✓ PROCESS MONITORING mode is active
 - ✓ The needle diameter has been entered in the settings of the dispensing menu
1. Press the < *Check* > operating key
 - ⇒ The needle or mixing tube is measured

- ⇒ Current values are compared with the values in the “Current needle” line
 - ⇒ The “Result” LED lights up green if the difference between “Current needle” position and “Last measurement” is within the specified tolerances (parameters “XY” and “Z”). The measuring point indicates the current position of the needle or mixing tube within the measuring ring (for “XY”) and within the bar (for “Z”).
 - ⇒ The “Error” LED lights up red if the difference between “Current needle” position and “Last measurement” is outside the specified tolerances (parameters “XY” and “Z”). In addition, the outer measurement circle lights up red in the event of an excessive “XY” deviation and the bar ends turn red in the event of an excessive “Z” deviation. The process is then aborted with an error message. The measuring point indicates the current position of the needle or mixing tube within the measuring ring (for “XY”) and within the bar (for “Z”).
- ⇒ Deviation values are updated in the line “Current deviation” and – if parameter “Add current deviation to NC offset” is marked – are added to the line “NC offset” and thus taken into consideration in the drive program.

Automatic check start procedure:

- ✓ “Current needle” position is defined
 - ✓ Dispensing needle or mixing tube is fixed to the dispensing unit
 - ✓ Parameter “Automatic active” of the needle check settings is marked
1. Press the < *Automatic ON* > operating key of the production process is not yet active
 - ⇒ The needle or mixing tube is measured
 - ⇒ Current values are compared with the values in the “Current needle” line
 - ⇒ The “Result” LED lights up green if the difference between “Current needle” position and “Last measurement” is within the specified tolerances (parameters “XY” and “Z”). The measuring point indicates the current position of the needle or mixing tube within the measuring ring (for “XY”) and within the bar (for “Z”).
 - ⇒ The “Error” LED lights up red if the difference between “Current needle” position and “Last measurement” is outside the specified tolerances (parameters “XY” and “Z”). In addition, the outer measurement circle lights up red in the event of an excessive “XY” deviation and the bar ends turn red in the event of an excessive “Z” deviation. The process is then aborted with an error message. The measuring point indicates the current position of the needle or mixing tube within the measuring ring (for “XY”) and within the bar (for “Z”).
- ⇒ Deviation values are updated in the line “Current deviation” and – if parameter “Add current deviation to NC offset” is marked – are added to the line “NC offset” and thus taken into consideration in the drive program.



NOTICE

Needle alignment

A needle that displays an excessive deviation after the needle check can possibly be bent back mechanically. Should this fail during a subsequent needle check, it is possible to define a new position with this – bent – needle by pressing the *< New needle >* button. A subsequent needle check should return a positive result.

3.8.9.3 Needle replacement procedure

General

The needle check identifies the deviation of the needle alignment or needle length from a defined position. If these values are within specified tolerances, the dispensing needle or mixing tube can continue to be used, otherwise it has to be cleaned, realigned or replaced with a new needle/mixing tube.

Needle replacement procedure:

- ✓ Production must have been stopped
 - ✓ PROCESS MONITORING mode is active
 - ✓ New, clean, straight needle is available
 - ✓ The needle diameter has been entered in the settings of the dispensing menu
1. Open the "Axes" screen
 2. Press the *< Move to set-up position >* symbol key
 - ⇒ Dispensing unit moves to set-up position
 3. Open doors
 - ⇒ SAFETY mode is active
 4. Unscrew old needle or mixing tube
 5. Clean mixing head attachment
 6. Screw on new needle or mixing tube (additional information: Align mixing tube roughly before it is screwed on)
 7. Close doors
 - ⇒ MANUAL mode is active
 8. Press the *< Process monitoring on >* operating button
 9. Press *< New needle >* button
 - ⇒ The system moves to the last "Current needle" position so that the two light barriers are interrupted
 - ⇒ Start measurement

- ⇒ The measurement results are written to line "Current needle" (additional information: If deviations in Z direction are detected, e.g. due to manufacturing tolerances of the mixing tube, this can be compensated with the parameter "Z offset needle search XY".)
- ⇒ Corrections for the NC are determined and updated in line "NC offset". Variables of the current drive program are updated.
- ⇒ Coordinates in line "Current deviation" are zeroed
- ⇒ The new needle position can be used for the subsequent needle checks.

3.8.10 Parameterisation of a blindshot bin/bucket on load cell

- ✓ Machine is in MANUAL mode.
- 1. Open lower machine doors.
- 2. In the machine visualisation, open the Blindshot Bin  submenu.
- 3. Press the Settings  symbol key
 - ⇒ The Teach Bin submenu appears.
- 4. Press the Teach "Empty Load Cell"  symbol key.
 - ⇒ Teaching of "Empty Load Cell" complete.
- 5. Position an empty blindshot bin/bucket on the load cell.
- 6. Press Teach symbol key for "Bin present".
 - ⇒ Teaching of "Blindshot bin" is complete.
- 7. Position a full blindshot bin/bucket on the load cell.
- 8. Press Teach symbol key for "Bin full".
 - ⇒ Teaching of "Blindshot bin full" is complete.
- ⇒ Teaching of load cell with blindshot bin is complete.

3.9 Dispensing machine symbols

Axis screen



Displays the jump into the axis screen.

Designation of the symbol: < *Call up axes screen* >

Application:

1. Press < *Call up axle screen* > button
- ⇒ Start screen of the "Axles" submenu opens

Current position



The current position of the process head

Designation of the symbol: < *Call up current position* >

Application:

1. Press < *Call up current position* > button

⇒ Start screen of the "Current position" submenu opens

Dispensing overview



Jump to dispensing overview

Designation of the symbol: < *Dispensing overview* >

1. Press < *Call up dispensing overview* > button

⇒ Jump to the topmost menu level "Dispensing overview"

A pump screen



Jump to A pump screen

Designation of the symbol: < *Call up pump A screen* >

Application:

1. Press < *Call up pump A screen* > button

⇒ Start screen of the "Pump A screen" submenu opens

B pump screen



Jump to B pump screen

Designation of the symbol: < *Call up pump B screen* >

Application:

1. Press < *Call up pump B screen* > button

⇒ Start screen of the "Pump B" submenu opens

Move to needle check screen



Jump to needle check screen

Designation of the symbol: < *Call up needle check screen* >

Needle check screen:

1. Press < *Call up needle check screen* > button

⇒ Start screen of the "Needle check" submenu opens

Cleaning station



Calls up the needle cleaning screen

Designation of the symbol: < *Call up cleaning screen* >

Application:

1. Press < *Call up cleaning screen* > button

⇒ Start screen of the "Cleaning" submenu opens

Pot life



Expiry of the pot life

Designation of the symbol: < *Pot life* >

✓ Mixing chamber must be filled

1. Press < *Blindshot* > button

⇒ The set blindshot time starts up

⇒ The blindshots entered are performed

⇒ Pot life is active

Balance screen



Jump to balance screen

Designation of the symbol: < *Call up balance* >

Application:

1. Press < *Call up balance* > button

⇒ Start screen of the "Balance" submenu opens

Weighing table page A



Listing of the weighing shots of the A-component page

Designation of the symbol: < *Weighing table page A* >

Application:

1. Press < *Weighing table page A* > button

⇒ Start screen of the "Weighing table page A" submenu opens

Weighing table page B

Listing of the weighing shots of the B-component page

Designation of the symbol: < *Weighing table page B* >

Application:

1. Press < *Weighing table page B* > button

⇒ Start screen of the "Weighing table page B" submenu opens

Recipe screen

Jump to recipe screen

Designation of the symbol: < *Call up recipe screen* >

Application:

1. Press < *Call up recipe screen* > button

⇒ Start screen of the "Recipe" submenu opens

Material screen

Call up of the material screen

Designation of the symbol: < *Call up material screen* >

Application:

1. Press < *Call up material screen* > button

⇒ Start screen of the "Material" submenu opens

3.10 PPS symbols

Material screen

Call up of the material screen

Designation of the symbol: < *Call up material screen* >

Application:

1. Press < *Call up material screen* > button

⇒ Start screen of the "Material" submenu opens

A material container screen

Jump to A material container screen

Designation of the symbol: < *Open A material container screen* >

Application:

1. Press the < *Open material container screen A* > symbol key

⇒ Start screen of the "Material container screen A" submenu opens

B material container screen

Jump to B material container screen

Designation of the symbol: < *Open B material container screen* >

Application:

1. Press the < *Open material container screen B* > symbol key

⇒ Start screen of the "Material container screen B" submenu opens

A vacuum pump screen

Jump to A vacuum pump screen

Designation of the symbol: < *Open vacuum pump A* >

Application:

1. Press the < *Open vacuum pump A* > symbol key

⇒ Start screen of the "Vacuum pump A" submenu opens

B vacuum pump screen

Jump to A vacuum pump screen

Designation of the symbol: < *Open vacuum pump B* >

Application:

1. Press the < *Open vacuum pump B* > symbol key

⇒ Start screen of the "Vacuum pump B" submenu opens

3.11 Menu structure

General

- The human/machine interface is implemented as a menu-guided operating system: an authorised user can access the running system's parameters and functions in real time.

Without logging in, no machine operator is logged in, unless login has been predetermined through the "Auto Log on" function!

- In addition, the message line communicates with the operator in order to display current states and messages. Here, errors are highlighted in red and warnings in yellow.
- Current sensor states are displayed in colour (green=active or grey=in-active) and analogue values in the corresponding font or as a symbol directly on the affected object on the screen.
- Functions and keys that are not currently available are displayed greyed out.
- Clicking on an input field opens a numerical keypad. Using its number keys, the value is input and taken over with the ✓ key or discarded with the X key.
- A navigation bar in the lower part of the screen and tab bars on the left inner part of each screen make it easier to open the available screens.

Symbol keys and dialogues of the human/machine interface:

	NOTICE Arrangement of the symbol keys and screen dialogues Symbol keys and screen dialogues are listed by topic and do not reflect the real arrangement system of the human/machine interface screens.
	NOTICE Login authorisation and operation mode The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.
	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.

3.11.1 Login

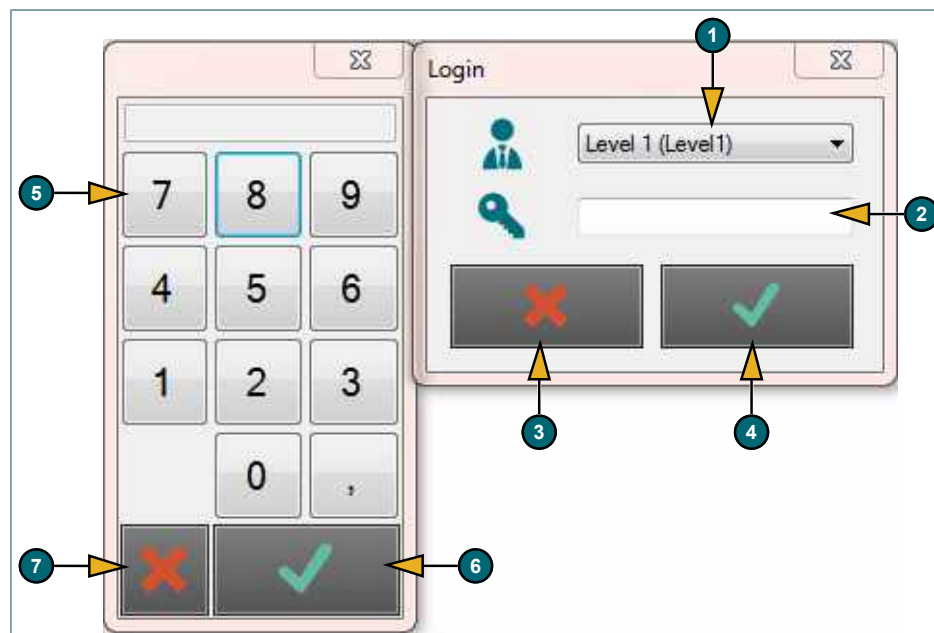


Illustration 49: Login dialogue

1	User selection list	2	Password input field
3	Cancel	4	Confirm log in
5	Input keyboard	6	Confirm input
7	Discard input		

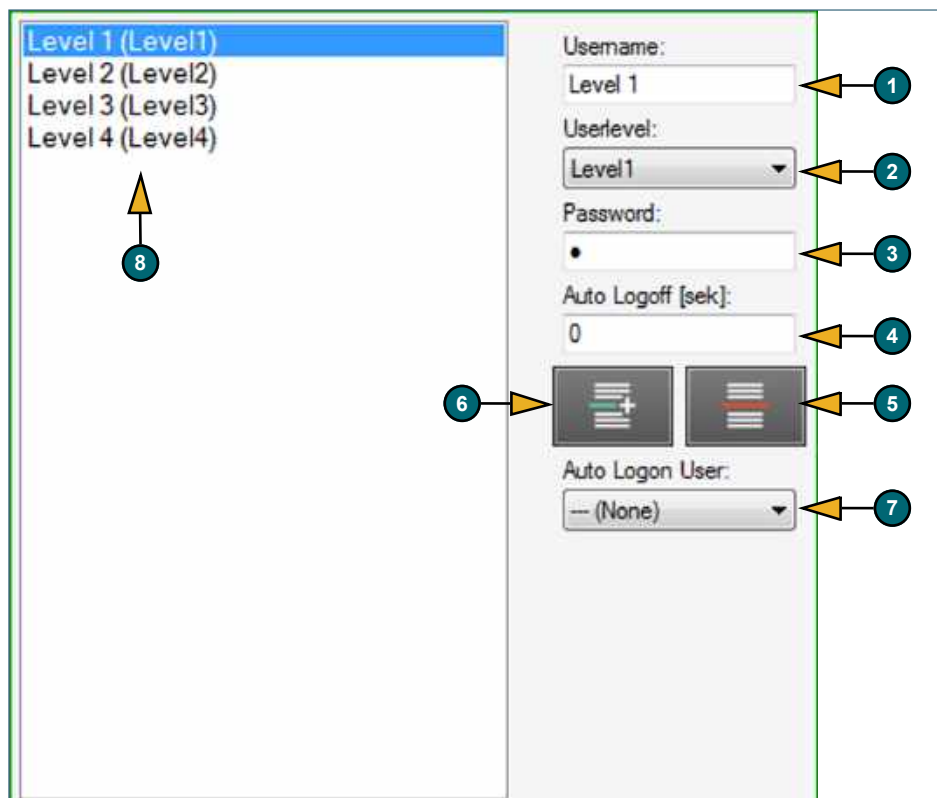
Task:

The login dialogue allows the operator of the machine to define various user profiles: up to four different authorisation levels can be assigned. This clearly defines which user may execute which function and influence which parameters. From the manufacturer side, the operating elements of the screens can be preconfigured in advance so that only when a particular authorisation level is reached do they become visible and executable by switching to a particular operation mode.

Log in as an authorised user:

1. Mark the user on the selection list (1)
 - ⇒ The user appears in the selection field
2. Click the password input field (2)
 - ⇒ The virtual keyboard appears (5)
3. Enter the numeric sequence of the password
 - ⇒ The numeric sequence appears on the display line
4. Press the green < OK > button (6) to confirm the input or press the red cross symbol (7) to discard it.
 - ⇒ The virtual keyboard closes
5. Press the green < OK > button of the log-in dialogue (4) to complete the input or press the red cross symbol (3) to discard it.

3.11.2 User



The screenshot shows a 'User' management dialog. On the left is a list box (8) containing 'Level 1 (Level1)', 'Level 2 (Level2)', 'Level 3 (Level3)', and 'Level 4 (Level4)'. On the right are several input fields and buttons. Callout 1 points to the 'Username:' text box containing 'Level 1'. Callout 2 points to the 'Userlevel:' dropdown menu showing 'Level1'. Callout 3 points to the 'Password:' text box with a masked character. Callout 4 points to the 'Auto Logoff [sek:]' text box containing '0'. Callout 5 points to the '< Delete User >' button. Callout 6 points to the '< Create User >' button. Callout 7 points to the 'Auto Logon User:' dropdown menu showing '-- (None)'. Callout 8 points to the list box on the left.

Illustration 50: User dialogue

1 "User" input field	2 "User level" selection field
3 "Password" input field	4 "Auto Log-Out" input field
5 < Delete User > button	6 < Create User > button
7 "Auto Logon User" selection field	8 Display list

Application

The user profiles for the machine are managed with this dialogue. The user can log into the machine using the "Log in" dialogue.

It is only possible to use all the options in this dialogue with the highest user level.

"User" input field (1):

Input a new user authorisation.

"User level" selection field (2):

Selection of an authorisation level.

"Password" input field (3):

Input of the user password.

"Auto Log-Out" input field (4):

Input of the time after which the logged-on user is logged off automatically.

< Delete User > button (5):

Deletes a user from list.

< Create User > button (6):

Creates a new user.

"Auto Logon User" selection field (7):

Preselection of the user who is to be logged on after the next system start.

Display list (8):

Display of the users already logged on.

The following tasks can be carried out via this dialogue:

Create new user:

- ✓ Only possible on authorisation 'Level 4' or 'Level 5'
- 1. Press the < Add user > button (pos. 6)
 - ⇒ The "New user" window opens
- 2. Click in the User input field (pos. 1)
 - ⇒ The virtual keyboard opens
- 3. Enter the sequence of letters for the name
- 4. Confirm the entry via the green < OK > button or discard with the red cross button
- 5. Click the arrow symbol on the "User level" list (pos. 2)
- 6. Mark the authorisation level
- 7. Click in the password input field (pos. 3)
 - ⇒ The virtual keyboard appears
- 8. Enter the numeric sequence for the password
- 9. Confirm the entry via the green < OK > button or discard with the red cross button
- 10. Optionally click on the "Log-Out" input field (pos. 4)
 - ⇒ The virtual keyboard appears
- 11. Enter the number of seconds for automatic log off
- 12. Confirm the entry via the green < OK > button
 - ⇒ The user is configured and appears on the selection list (pos. 8)

**NOTICE****Save changes.**

Only by pressing the save button are the changes taken over permanently into the system memory: when the system is restarted, the changes are available again (info: only an authorised user can start the saving process)

Delete user:

- ✓ Only possible on authorisation 'Level 4' or 'Level 5'
- 1. Mark the user on the selection list (pos. 8)
- 2. Press the < Delete user > button (pos. 5)
 - ⇒ The user profile is removed from the selection list

**NOTICE****Save changes.**

Only by pressing the save button are the changes taken over permanently into the system memory: when the system is restarted, the changes are available again (info: only an authorised user can start the saving process)

Set auto login user:

Info: After each system restart, the specified user is logged in first. This can also be the "No user" profile. Every user can then log in using the "Log in" dialogue.

Info: Only one user profile is active on a machine at one time.

- ✓ Only possible on authorisation level 'Level 5'.
- 1. Press the arrow key (pos. 7)
 - ⇒ The selection field opens
- 2. Mark the user profile
 - ⇒ The user profile is displayed on the selection field

**NOTICE****Save changes.**

Only by pressing the save button are the changes taken over permanently into the system memory: when the system is restarted, the changes are available again (info: only an authorised user can start the saving process)

3.11.3 Changing over languages

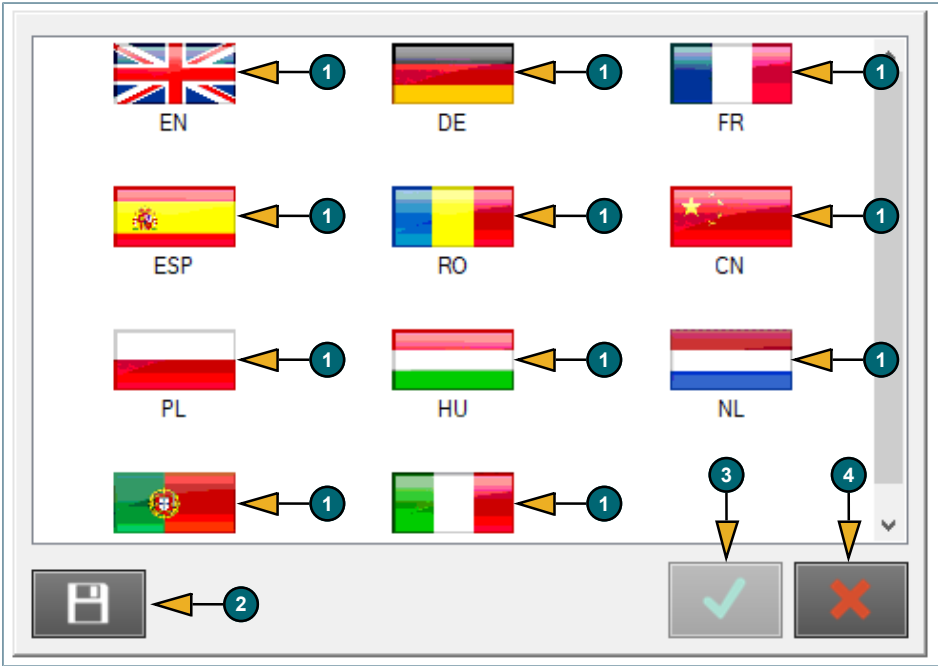


Illustration 51: Language selection

1	Symbol key	2	Symbol key
3	Symbol key	4	Symbol key

Application

Toggle the system menu language.

Symbol key (1):

Double clicking on the symbol changes to the language indicated by the flag or the international abbreviation (e.g. “EN” for English).



NOTICE

Login authorisation function execution

Only an authorised user login can execute functions.

Symbol key (2):

A new language file is created or an existing file is updated.



NOTICE

Login authorisation function execution

Only an authorised user login can execute functions.

Symbol key (3):

A single click on the language symbol key (1) to preselect. Then click the symbol key for confirmation (3): The language is switched over. This is an alternative to double-clicking symbol key (1).

**NOTICE****Login authorisation function execution**

Only an authorised user login can execute functions.

Symbol key (4):

Cancel the selection made by a single click on the language symbol key (1): No language change is requested, the current language is retained.

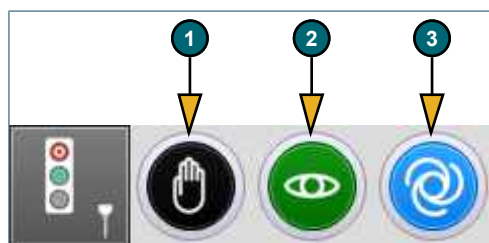
3.11.4 Graphic operation mode changeover - dispensing

Illustration 52: Operation mode changeover - dispensing

1	MANUAL operation mode	2	PROCESS MONITORING operation mode
3	AUTOMATIC operation mode		

Application

This symbol key allows you to switch back and forth between the individual operation modes.

**NOTICE****Input and execution of functions and parameters.**

Input and further processing of a parameter or function can only be performed with an authorised user login.

3.11.5 Graphic operation mode changeover - material preparation

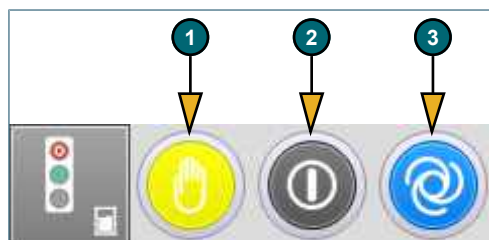


Illustration 53: Operation mode changeover - material preparation system

1	MANUAL operation mode	2	STANDBY operation mode (Note: Possibility of switching the consumers on and off using a timer application)
3	AUTOMATIC operation mode		

These operation modes are implemented for the material preparation systems such as MPS, PPS, CPS.

3.11.6 Dispensing

3.11.6.1 Starting screen

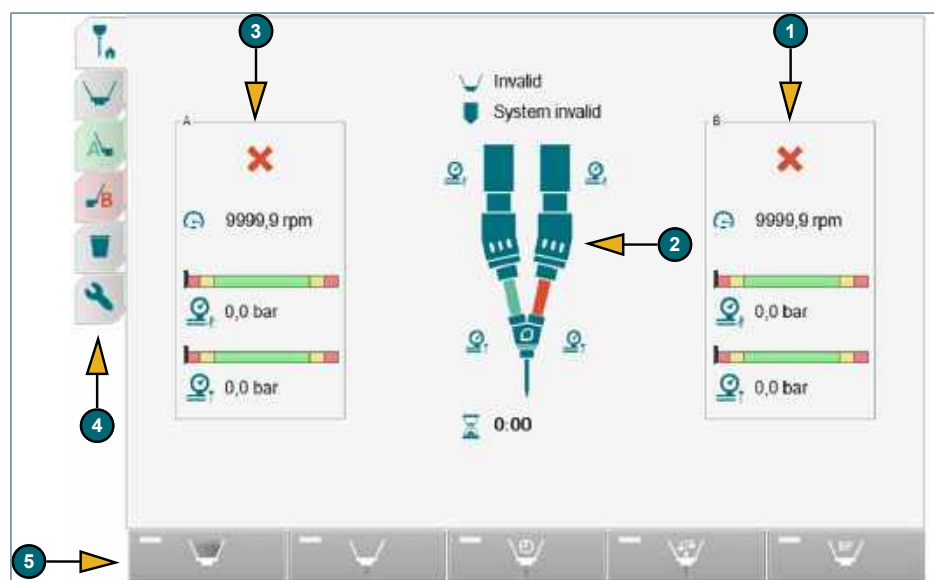


Illustration 54: Starting screen – Duplex with speed, pre-pressure, working pressure

1	Dispensing line A (active graphics field)	2	Static mixing head (active graphics field)
3	Dispensing line B (active graphics field)	4	Tab line
5	Function key bar		

Application

This is the starting screen of a dispensing application with static mixer. The settings pages for the assembly (mixers, dispensing pumps) can be called up via the active graphics fields (1 for pump A, 2 for mixers, 3 for pump B)

and via the tab lines (4). With the required login authorisation and operation mode, pressing the symbols in the function key bar starts the programmed process, such as filling, dispensing, etc. The tab line is also used to control special assemblies (such as blind shot bins, etc.) and for special settings (needle diameter, etc.) that are not linked as active graphics.

3.11.6.1.1 Starting screen - Activate dispensing pump A | with pre-pressure and dispensing pressure

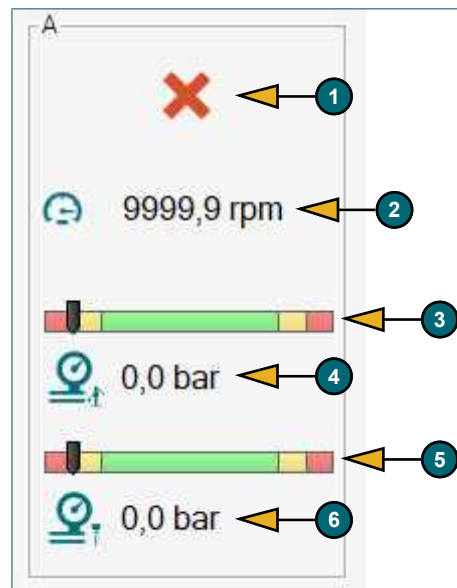


Illustration 55: Component A start dialog

1	Activation symbol key	2	Speed display field
3	Pre-pressure bar diagram, active symbol area	4	Pre-pressure display field
5	Dispensing pressure bar diagram, active symbol area	6	Dispensing pressure display field

Task:

Brief overview of the state of the components, switching of the components and jump to respective component tabs.

Activation symbol key (1):

Green tick mark: Assembly is active.

Red cross symbol: Assembly is not active.

Speed display field (2):

Display of the current pump speed.

Pre-pressure bar diagram, active symbol area (3):

The bar symbol has two functions:

1. Display of the current value of the monitored physical variable (black marker) within the limits set (coloured section: red = max. and min. limit area, error situation; yellow: Pre-warning area; green: normal working range between upper and pre-warning limit)
2. Jump to the associated component screen by clicking on the active symbol area

Pre-pressure display field (4):

Display of the current pre-pressure.

Dispensing pressure bar diagram, active symbol area (5):

The bar symbol has two functions:

1. Display of the current value of the monitored physical variable (black marker) within the limits set (coloured section: red = max. and min. limit area, error situation; yellow: Pre-warning area; green: normal working range between upper and pre-warning limit)
2. Jump to the associated component screen by clicking on the active symbol area

Dispensing pressure display field (6):

Display of the current dispensing pressure.

3.11.6.1.2 Start page - Static mixer - with system status - with potlife

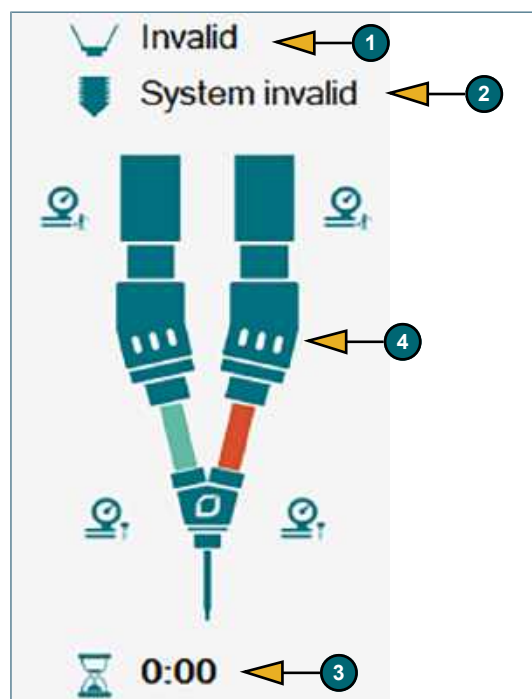


Illustration 56: Starting screen – Static mixer

1 Display field	2 Display field
3 Display field	4 Active graphics

Application

This is the static mixer. The settings page for this assembly can be called up via the active graphics.

Display field (1):

In this field the dispensing unit displays the task currently being performed. This can be a blind shot, weighing shot, timed dispensing shot, etc.

Display field (2):

This field displays the system status.

The various possible system statuses are listed below:

Designation	Status
System deselected	On double-head systems only: One dispensing system is active, the other is not active
System not active	Home position of the dispensing system. Dispensing system is not ready. Material in the mixing tube may have hardened.
Prepare system	Dispensing unit is filled with material
System prepared	Dispensing unit has been filled, decompression shot is still to be performed
System ready	Dispensing unit is ready for production
Prepare system A	Component A pump is currently dispensing or was the last pump to dispense, or if only A was dispensed in a two-component system
System A prepared	There is only component A in the dispensing unit
Prepare system B	Component B pump is currently dispensing or was the last pump to dispense, or if only B was dispensed in a two-component system
System B prepared	There is only component B in the dispensing unit
System error status	Manual intervention necessary after a dispensing problem

Display field (3):

Status of potlife processing is displayed:

Value '0:00': Potlife processing is not active. An additional acoustic signal sounds and the dispensing unit tries to dispense a blind shot at the corresponding taught position.



NOTICE

Process monitoring and automatic operation modes!

This function or sequence can only be executed in process monitoring or automatic operation mode!

Value <> '0:00': Potlife processing running; the remaining time until the end of the potlife is displayed.

Active graphics (4):

Jump to the "Head" screen.



NOTICE

Input and execution of functions and parameters.

Input and further processing of a parameter or function can only be performed with an authorised user login.

3.11.6.1.3 Starting screen - Activate dispensing pump B | with pre-pressure and dispensing pressure

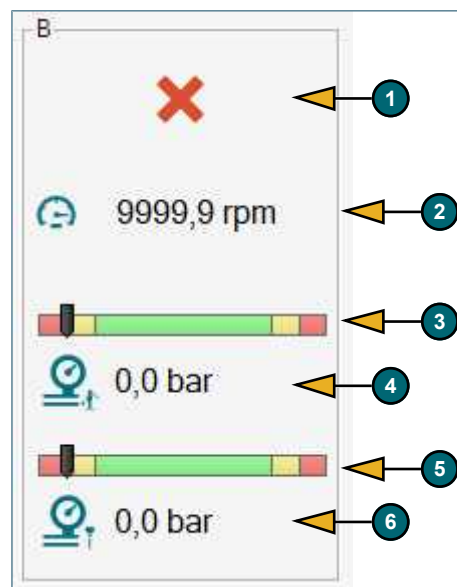


Illustration 57: Component B start dialog

1	Activation symbol key	2	Speed display field
3	Pre-pressure bar diagram, active symbol area	4	Pre-pressure display field
5	Dispensing pressure bar diagram, active symbol area	6	Dispensing pressure display field

Task:

Brief overview of the state of the components, switching of the components and jump to respective component tabs.

Activation symbol key (1):

Green tick mark: Assembly is active.

Red cross symbol: Assembly is not active.

Speed display field (2):

Display of the current speed of the dispensing pump.

Pre-pressure bar diagram, active symbol area (3):

The bar symbol has two functions:

1. Display of the current value of the monitored physical variable (black marker) within the limits set (coloured section: red = max. and min. limit area, error situation; yellow: Pre-warning area; green: normal working range between upper and pre-warning limit)
2. Jump to the associated component screen by clicking on the active symbol area

Pre-pressure display field (4):

Display of the current pre-pressure.

Dispensing pressure bar diagram, active symbol area (5):

The bar symbol has two functions:

1. Display of the current value of the monitored physical variable (black marker) within the limits set (coloured section: red = max. and min. limit area, error situation; yellow: Pre-warning area; green: normal working range between upper and pre-warning limit)
2. Jump to the associated component screen by clicking on the active symbol area

Dispensing pressure display field (6):

Display of the current dispensing pressure.

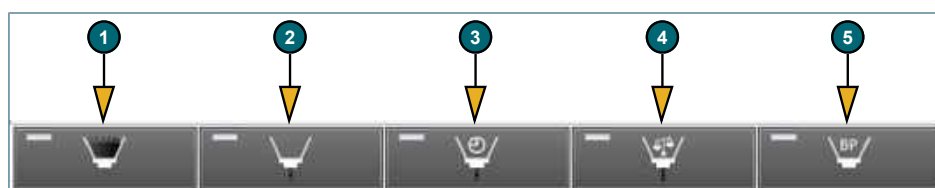
3.11.6.1.4 Function key bar 1C | Static 2C

Illustration 58: Function key bar (here for single-component and static mixer)

1 < Filling > key	2 < Dispensing Event > key
3 < Dispensing with time > key	4 < Dispensing with weight > key
5 < Basic position > key	

Application

Manual triggering of the dispensing function.

3.11.6.2 Head

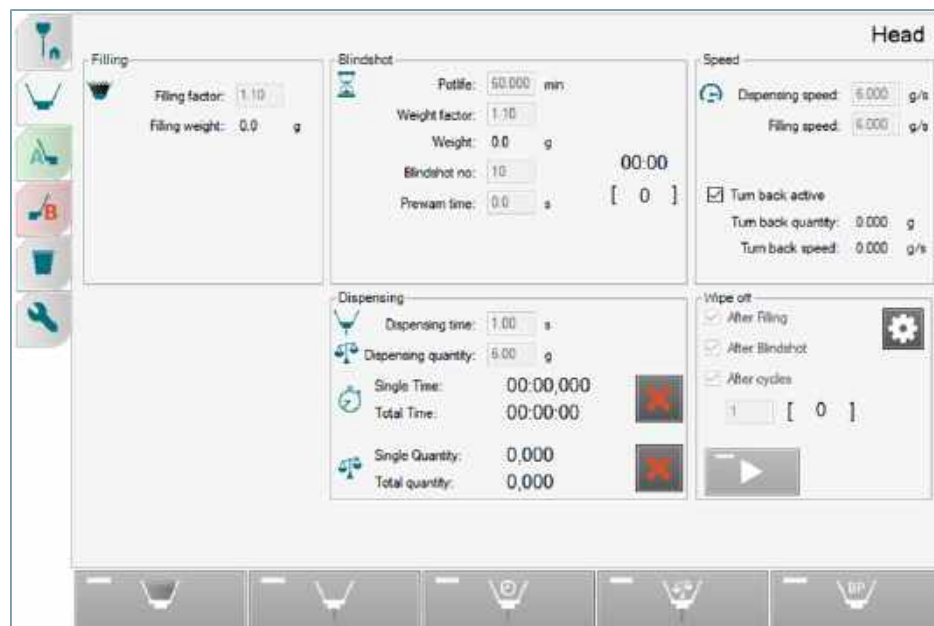


Illustration 59: Screen – Static head with turn back

3.11.6.2.1 Fill dialogue

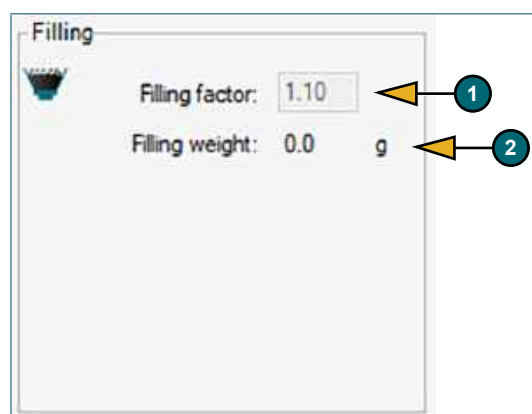


Illustration 60: Filling

1	"Filling volume factor" input field	2	"Filling weight" display field
---	-------------------------------------	---	--------------------------------

Application

The filling dialogue "operates" the important process for dispensing systems of flooding the mixing chamber (on dynamic mixers) or a mixing tube (on static mixers) or the outlet nozzle (on single-component systems) once until the mixing chamber, mixing tube or outlet nozzle is free of air. During subsequent dispensing/weighing/blind shots, only material mixture without air bubbles is displaced.

The system signals "System ready" when the filling process has been completed successfully. In AUTOMATIC operation mode, the filling process is performed automatically; in PROCESS MONITORING operation mode, only on request with the < Fill > symbol key. In all cases, potlife processing starts after the filling process!

Setting the parameters for the filling process is carried out once – when the system is commissioned – and is the responsibility of the service technician!



NOTICE

Input field

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.

"Filling volume factor" input field (1):

The "Filling volume factor" is an offset to be entered once by the service technician, who determines it on the basis of the material-specific properties and the dispensing environment. A value of '2.0' means that during filling, 2.0 times the reactive volume is dispensed. The prerequisite is a practically air bubble-free material. After expiry of the filling time, material or material mixture is available in the dispensing unit for dispensing processes.



NOTICE

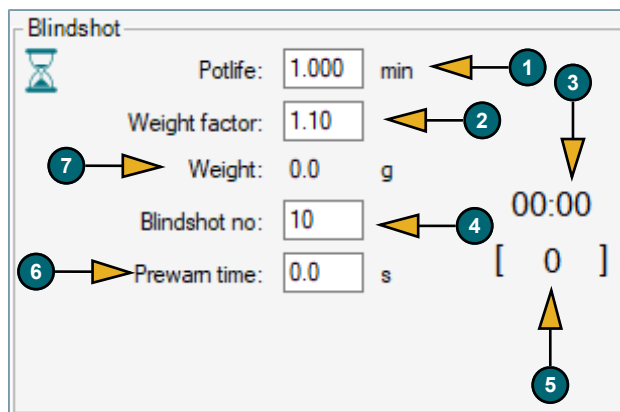
Input value(s)

This/these value(s) of the input field(s) can be permanently stored in a dispensing program. To do this, press the "Save" button (last button in the menu bar on the right) for a dispensing program that has already been loaded. The parameters are then available to the system unchanged even after a restart!

"Filling weight" display field (2):

Weight of the material volume to be conveyed calculated from the filling volume factor and the material densities.

3.11.6.2.2 Blind shot



The 'Blindshot' dialogue box contains the following fields and callouts:

- 1**: Points to the 'Potlife' input field (1.000 min).
- 2**: Points to the 'Weight factor' input field (1.10).
- 3**: Points to the 'Remaining potlife' display field (00:00).
- 4**: Points to the 'Blindshot no.' input field (10).
- 5**: Points to the 'Number of remaining blindshots' display field ([0]).
- 6**: Points to the 'Prewarn time' input field (0.0 s).
- 7**: Points to the 'Weight' display field (0.0 g).

Illustration 61: Blindshot dialogue

1 "Potlife" input field	2 "Volume factor" input field
3 "Remaining potlife" display field	4 "Number of blindshots" input field
5 "Number of remaining blindshots" display field	6 "Prewarn time" input field
7 "Weight" display field	

Application:

The behaviour during potlife processing of the dispensing machine is specified via the blindshot dialogue. The parameters are specified once during commissioning of the machine. The dynamised display fields indicate the state of the potlife processing.



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.

"Potlife" input field (1):

The potlife refers to the material or material mixture to be processed. The service technician takes this parameter from the material manufacturer's data sheet and adapts it accordingly. It specifies the time span within which a material system can be processed well; this means: after expiry of the potlife, the viscosity (fluidity) of the mixture has nearly doubled. There is a danger of hardening in the mixing system. That is why an automatic feature in the machine – "potlife processing" – ensures that a calculated proportion of the mixture volume that has exceeded a time limit is discarded by means of additional dispensing shots or "blindshots". The machine produces normally before and after the blindshots.

**NOTICE****Input value(s)**

This/these value(s) of the input field(s) can be permanently stored in a dispensing program. To do this, press the "Save" button (last button in the menu bar on the right) for a dispensing program that has already been loaded. The parameters are then available to the system unchanged even after a restart!

**NOTICE****System parameters**

This parameter is specified once when the system is run in.

“Volume factor” input field (2):

After expiry of the "potlife" the machine starts the set number of "blindshots". These are generally processed, i.e. dispensed, in the blindshot position. The quantity of material to be conveyed depends on the "reactive volume" and the specifics of a mixing tube (for static mixers). The "Volume factor" is an offset to be entered once by the service technician, who determines it on the basis of the material-specific properties and the dispensing environment. A value of '1.5' means that 1.5 times the reactive volume is dispensed during the blindshot, simply to be sure that the critical area of the material flow has been completely "exchanged".

**NOTICE****Volume factor**

Values below '1.1' cannot be entered! That is 110% of the "reactive volume".

**NOTICE****System parameters**

This parameter is specified once when the system is run in.

“Remaining potlife” display field (3):

This field displays the remaining time until potlife end.

“Number of blindshots” input field (4):

The machine dispenses a number of blindshots as soon as the "potlife" has elapsed. The purpose of the blindshot is to keep the dispensing system ready. The number is selected so as to avoid an unnecessary waste of material during longer stoppages in production. After the set number has expired, the system is emptied (for static mixers) / purged (for dynamic mixers).

**NOTICE****System parameters**

This parameter is specified once when the system is run in.

“Number of remaining blindshots” display field (5):

This field is updated during ongoing blindshot processing: The operator sees the number of shots still to be dispensed. If, in the meantime, another part is waiting to be dispensed, the counter is reset and starts again from the beginning. A running process is only interrupted by an error.

“Prewarn time” input field (6):

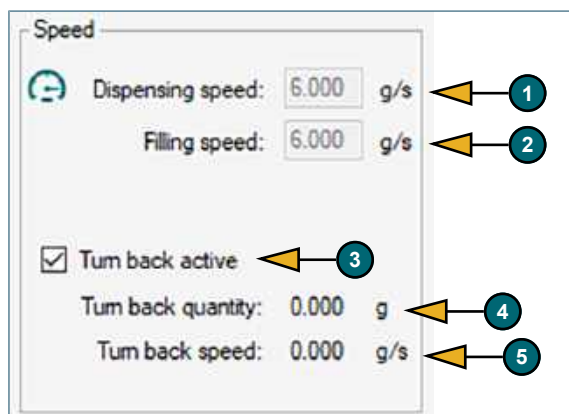
The prewarn time makes the machine operator aware of the upcoming pot-life processing: After expiry of the same, the machine starts with "blindshots". If the value is "0", the blindshot processing starts immediately after the expiry of the "potlife".

**NOTICE****Operation mode**

This function and the subsequent process can only be executed in PROCESS MONITORING or AUTOMATIC operation mode!

“Weight” display field (7):

Weight of the dispensed blindshot calculated from the volume factor and the material densities. This field is only updated when a blindshot is processed. With "system not ready", i.e. as long as the dispensing system has not yet been filled, '0.0 g' is displayed here.

3.11.6.2.3 Speed dialogue | with turn back


The screenshot shows a 'Speed' dialog box with the following fields and callouts:

- 1: 'Dispensing speed:' input field (value: 6.000 g/s)
- 2: 'Filling speed:' input field (value: 6.000 g/s)
- 3: 'Turn back active' check box (checked)
- 4: 'Turn back quantity:' display field (value: 0.000 g)
- 5: 'Turn back speed:' display field (value: 0.000 g/s)

Illustration 62: Speed with turn back

1 "Dispensing speed:" input field	2 "Filling speed:" input field
3 "Turn back active" check box	4 "Turn back volume" display field
5 "Turn back rate" display field	

Application

Partial parameterisation of the dispensing process.

"Dispensing speed:" input field (1):

Input of the dispensing speed during the dispensing process.

	NOTICE Input value(s) This/these value(s) of the input field(s) can be permanently stored in a dispensing program. To do this, press the "Save" button (last button in the menu bar on the right) for a dispensing program that has already been loaded. The parameters are then available to the system unchanged even after a restart!
	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.

"Filling speed:" input field (2):

Input of the dispensing speed during the filling process.

"Turn back active" check box (3):

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.

	NOTICE Login authorisation and operation mode The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.
---	--

**NOTICE****Turn back**

Turning back is performed also during manual dispensing.

"Turn back volume" display field (4):

Calculated weight of the dispensing compound conveyed during turning back.

"Turn back rate" display field (5):

Dispensing rate with the pump conveying the turn back volume.

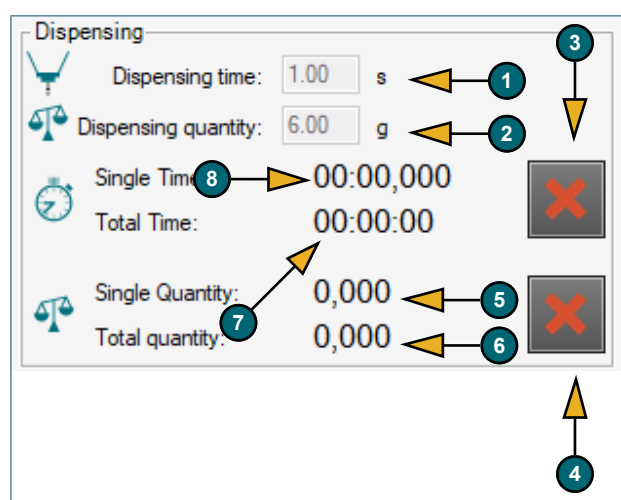
3.11.6.2.4 Dispensing dialog

Illustration 63: Dispensing dialog

1	"Dispensing time" input field	2	"Dispensing volume" input field
3	"Reset dispensing time" symbol key	4	"Reset dispensing volume" symbol key
5	"Single volume" display field	6	"Total volume" display field
7	"Individual time" display field	8	"Total time" display field

Application:

Setting the dispensing time and dispensing volume and resetting the counters for the manual dispensing time and dispensing volume.

"Dispensing time" input field (1):

With this field, the user determines the duration of the manual dispensing process: this time starts to run with the "Start dispensing" command, after expiry of this time, the dispensing process stops automatically.

**NOTICE****Input value(s)**

This/these value(s) of the input field(s) can be permanently stored in a dispensing program. To do this, press the "Save" button (last button in the menu bar on the right) for a dispensing program that has already been loaded. The parameters are then available to the system unchanged even after a restart!

"Dispensing volume" input field (2):

With this field, the user determines the dispensing volume for the manual dispensing process: After dispensing of the material volume, the process stops automatically.

"Reset dispensing time" symbol key (3):

Pressing this key resets the time display (see (7) or (8)) to zero.

**NOTICE****Login authorisation function execution**

Only an authorised user login can execute functions.

"Reset dispensing volume" symbol key (4):

Pressing this key resets the volume display (see (5) or (6)) to zero.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation function execution**

Only an authorised user login can execute functions.

"Single volume" display field (5):

The volume already dispensed during a dispensing process started manually with the  key is displayed here.

"Total volume" display field (6):

The volume already dispensed during all dispensing processes started manually with the  key – since the last reset – is displayed here.

"Individual time" display field (7):

The already elapsed dispensing time of a dispensing process started manually with key  or  is displayed here.

"Total time" display field (8):

The time of all dispensing processes started manually with key  or  – since the last reset – is displayed here.

3.11.6.2.5 Wiping | Dialogue

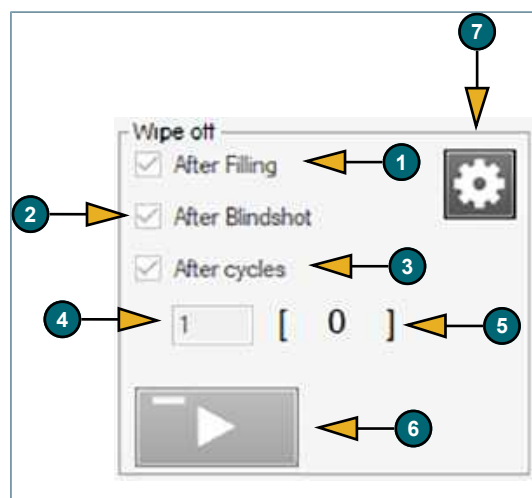


Illustration 64: Wiping dialogue

1 "After filling" checkbox	2 "After blind shot" checkbox
3 "After cycles" checkbox	4 Input field
5 Display field	6 Symbol key
7 Symbol key	

Task:

The wiping off of residual material on the mixing tube.

"After filling" checkbox (1):

Tick mark set: Wiping is performed after filling.

No tick mark: Wiping is not performed after filling.

"After blind shot" checkbox (2):

Tick mark set: Wiping is performed after the blind shot.

No tick mark: Wiping is not performed after the blind shot.

"After cycles" checkbox (3):

Tick mark set: Wiping is performed after the number of parts set in (3).

No tick mark: Wiping is not performed after the number of parts set in (3).

Input field (4):

Number of parts after which wiping is performed.

Display field (5):

Number of parts already processed. They are counted in ascending order.

Symbol key (6):

Once-only start of the wiping cycle.

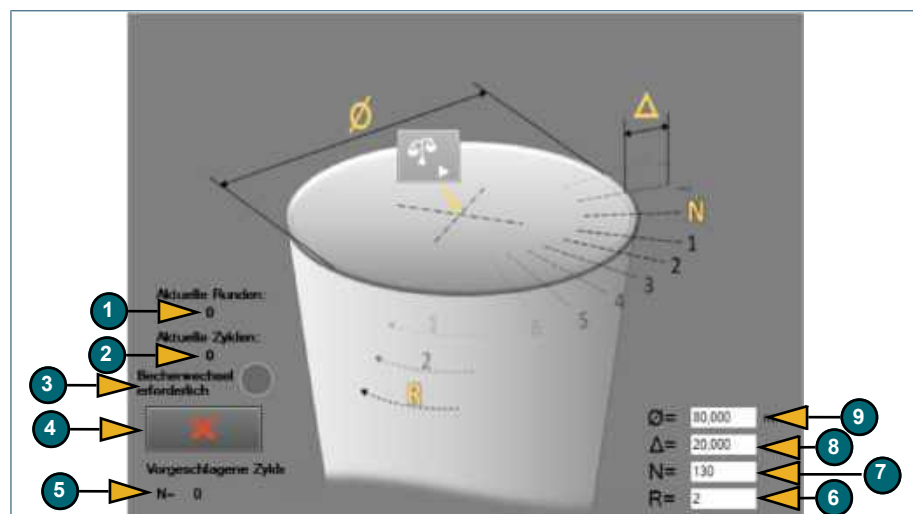
**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Symbol key (7):

Pressing the symbol key opens the window for configuration of cup wiping.

3.11.6.2.5.1 Cup edge wipe off setting dialogue



1	Display field	2	Display field
3	Indicator LED	4	Symbol key
5	Display field	6	"R" input field
7	"N" input field	8	Input field
9	"Diameter" input field		

Application:

Wiping off the cup edge is shown over the entire circumference of the cup edge. Wiping takes place over a given number of rounds and cycles. This number of wiping cycles is determined and set by the system on the basis of the entered parameters, for example: Diameter and number of rounds.

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Display field (1):

Displays the current number of rounds of the wipe-off process.

Display field (2):

Displays the current number of cycles.

Indicator LED (3):

Red LED: Wipe-off cycles reached. The cup must be replaced. A pop-up window appears. Insert a new cup and acknowledge the pop-up window.

Grey LED: No status feedback.

Symbol key (4):

By activating the symbol key, the cycles counter and the rounds counter are reset to "0".

Display field (5):

Display of the cycles calculated and suggested by the system.

"R" input field (6):

Input of the number of rounds for wiping off the dispensing needle on the cup edge.

"N" input field (7):

Input of the number of cycles for wiping off the dispensing needle on the cup edge.

Input field (8):

Input of the travel distance that the dispensing needle is to travel along the cup edge.

Input field (9):

Input of the diameter of the cup.

3.11.6.3 Pump A without valve



Illustration 65: Pump A

		2	"Pre-pressure" display field
3	"Dispensing pressure" display field		

Application

In this screen, the pressure limits for dispensing can be set and the dispensing pump can be switched on and off. Depending on the system configuration, optional filling, dispensing and pre-pressure monitoring is offered here.

The resin dispensing valve is actuated manually here.

"Pre-pressure" display field (2):

Display of the current value.

"Dispensing pressure" display field (3):

Display of the current value.

3.11.6.3.1 Pump A settings

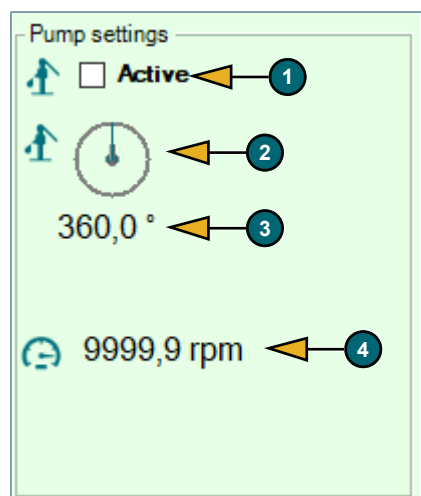


Illustration 66: Pump A settings

1 "Active" check box	2 Dynamic display
3 Display field	4 Display field

Application

Activate dispensing pump.

"Active" check box (1):

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.



NOTICE

Login authorisation and operation mode

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

Dynamic display (2):

Dynamic display of the rotor position.

Display field (3):

Display of the rotor angle from 0° to 360°.

Display field (4):

Display of the current speed.

3.11.6.3.2 Filling pressure A

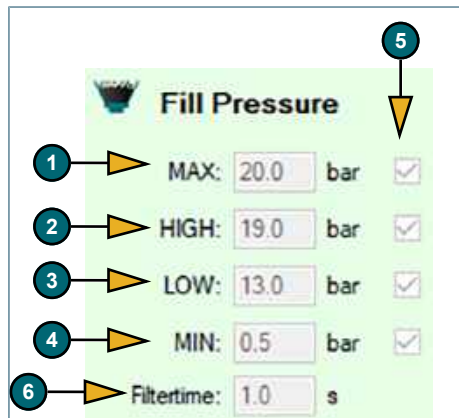


Illustration 67: Filling pressure A

1 "MAX" input field	2 "HIGH" input field
3 "LOW" input field	4 "MIN" input field
5 "Alarm" check box	6 "Filtration time" input field
7 Not installed	

Application

Define limit values of a physical variable (temperature, pressure, filling level, etc.) so that the operator is made aware of the situation by means of "alarms" and the running process is terminated.

"MAX" input field (1):

Input of the limit value to be monitored.

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.

**NOTICE****Alarm signalling on "MAX"**

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on undershooting the limit value.

If the check box is not set, the system maintains its operating state on reaching the value in the input field: no error display, no other signalling!

"HIGH" or "HOCH" input field (2):

Input of the limit value to be monitored.

**NOTICE****Alarm signalling**

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is undershot. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"LOW" or "NIEDRIG" input field (3):

Input of the limit value to be monitored.

**NOTICE****Alarm signalling**

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is exceeded. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"MIN" input field (4):

Input of the limit value to be monitored.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

**NOTICE****Alarm signalling on "MIN"**

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on exceeding the limit value.

If the check box is not set, the system maintains its operating state on reaching the limit value in the input field: no error display, no other signalling!

Alarm checkbox (5):

Tick mark set: Alarm is set: If an alarm occurs, it is displayed in yellow in the event of a warning in the status bar and an optional yellow signal lamp lights up as long as the alarm is present; conversely, an error is indicated with a red message in the status bar and an optional red signal lamp additionally lights up. In this case, the process also stops.

No tick mark: Errors and warnings are ignored.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

"Filter time" input field (6):

Time specification with which the querying control unit tolerates fluctuations from a value (e.g. measured value of a physical variable, limit value, etc.). A value is only taken over as valid after this elapses.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

Display field (7):

Display of the current value.

**NOTICE****Current value**

Value indicates the current programming of the physical value.

The filling pressure is measured at the outlet from the dispensing pump. It is generated during the filling process of the dispensing system and, depending on the material and the dispensing system, is reduced to the dispensing pressure level by suitable measures such as decompression shots, so that the first subsequent dispensing shots can be dispensed with high quality.

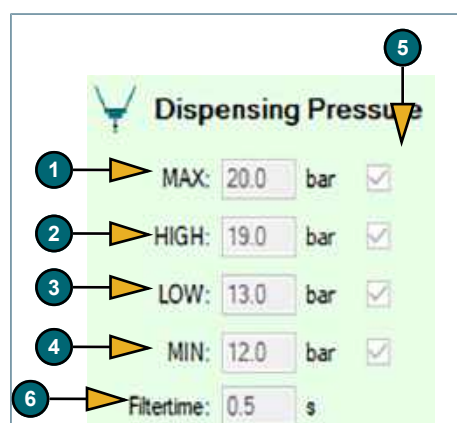
3.11.6.3.3 Dispensing pressure A

Illustration 68: Dispensing pressure A

1 "MAX" input field	2 "HIGH" input field
3 "LOW" input field	4 "MIN" input field
5 "Alarm" check box	6 "Filtration time" input field
7 Not installed	

Application

Define limit values of a physical variable (temperature, pressure, filling level, etc.) so that the operator is made aware of the situation by means of "alarms" and the running process is terminated.

"MAX" input field (1):

Input of the limit value to be monitored.

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
---	--

	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.
---	---

	NOTICE Alarm signalling on "MAX" With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on undershooting the limit value. If the check box is not set, the system maintains its operating state on reaching the value in the input field: no error display, no other signalling!
---	---

"HIGH" or "HOCH" input field (2):

Input of the limit value to be monitored.

	NOTICE Alarm signalling On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is undershot. In this case, an optional yellow/orange signal lamp lights up. The machine continues in normal operation.
---	---

"LOW" or "NIEDRIG" input field (3):

Input of the limit value to be monitored.

	NOTICE Alarm signalling On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is exceeded. In this case, an optional yellow/orange signal lamp lights up. The machine continues in normal operation.
---	--

"MIN" input field (4):

Input of the limit value to be monitored.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

**NOTICE****Alarm signalling on "MIN"**

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on exceeding the limit value.

If the check box is not set, the system maintains its operating state on reaching the limit value in the input field: no error display, no other signalling!

Alarm checkbox (5):

Tick mark set: Alarm is set: If an alarm occurs, it is displayed in yellow in the event of a warning in the status bar and an optional yellow signal lamp lights up as long as the alarm is present; conversely, an error is indicated with a red message in the status bar and an optional red signal lamp additionally lights up. In this case, the process also stops.

No tick mark: Errors and warnings are ignored.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

"Filter time" input field (6):

Time specification with which the querying control unit tolerates fluctuations from a value (e.g. measured value of a physical variable, limit value, etc.). A value is only taken over as valid after this elapses.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

Display field (7):

Display of the current value.

**NOTICE****Current value**

Value indicates the current programming of the physical value.

The dispensing pressure is measured at the outlet from the dispensing pump.

3.11.6.4 Pump B without valve

Illustration 69: Pump B

		2 "Filling pressure" display field
3 "Dispensing pressure" display field		

Application

In this screen, the pressure limits for dispensing can be set and the dispensing pump can be switched on and off. Depending on the system configuration, optional filling, dispensing and pre-pressure monitoring is offered here.

"Pre-pressure" display field (2):

Display of the current value.

"Dispensing pressure" display field (3):

Display of the current value.

3.11.6.4.1 Pump B settings

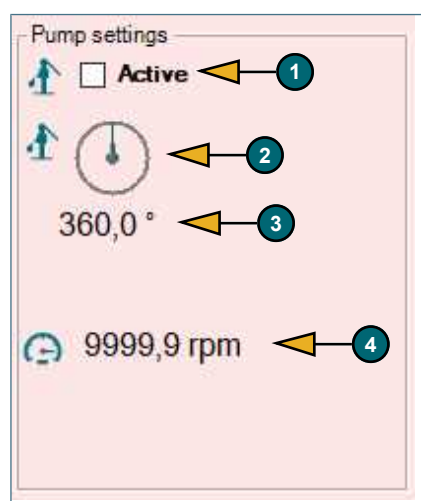


Illustration 70: Pump B settings

1 "Active" check box	2 Dynamic display
3 Display field	4 Display field

Application

Activate dispensing pump.

"Active" check box (1):

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.



NOTICE

Login authorisation and operation mode

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

Dynamic display (2):

Dynamic display of the rotor position.

Display field (3):

Display of the rotor angle from 0° to 360°.

Display field (4):

Display of the current speed.

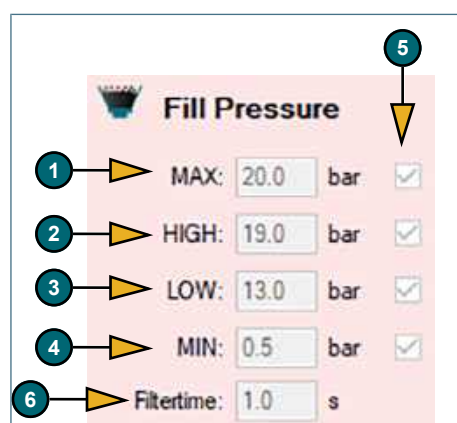
3.11.6.4.2 Filling pressure B

Illustration 71: Filling pressure B

1 "MAX" input field	2 "HIGH" input field
3 "LOW" input field	4 "MIN" input field
5 "Alarm" check box	6 "Filtration time" input field
7 Not installed	

Application

Define limit values of a physical variable (temperature, pressure, filling level, etc.) so that the operator is made aware of the situation by means of "alarms" and the running process is terminated.

"MAX" input field (1):

Input of the limit value to be monitored.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

**NOTICE****Alarm signalling on "MAX"**

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on undershooting the limit value.

If the check box is not set, the system maintains its operating state on reaching the value in the input field: no error display, no other signalling!

"HIGH" or "HOCH" input field (2):

Input of the limit value to be monitored.

**NOTICE****Alarm signalling**

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is undershot. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"LOW" or "NIEDRIG" input field (3):

Input of the limit value to be monitored.

**NOTICE****Alarm signalling**

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is exceeded. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"MIN" input field (4):

Input of the limit value to be monitored.

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
---	---

	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.
---	--

	NOTICE Alarm signalling on "MIN" With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on exceeding the limit value. If the check box is not set, the system maintains its operating state on reaching the limit value in the input field: no error display, no other signalling!
---	--

Alarm checkbox (5):

Tick mark set: Alarm is set: If an alarm occurs, it is displayed in yellow in the event of a warning in the status bar and an optional yellow signal lamp lights up as long as the alarm is present; conversely, an error is indicated with a red message in the status bar and an optional red signal lamp additionally lights up. In this case, the process also stops.

No tick mark: Errors and warnings are ignored.

	NOTICE Login authorisation and operation mode The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.
---	--

"Filter time" input field (6):

Time specification with which the querying control unit tolerates fluctuations from a value (e.g. measured value of a physical variable, limit value, etc.). A value is only taken over as valid after this elapses.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

Display field (7):

Display of the current value.

**NOTICE****Current value**

Value indicates the current programming of the physical value.

The filling pressure is measured at the outlet from the dispensing pump. It is generated during the filling process of the dispensing system and, depending on the material and the dispensing system, is reduced to the dispensing pressure level by suitable measures such as decompression shots, so that the first subsequent dispensing shots can be dispensed with high quality.

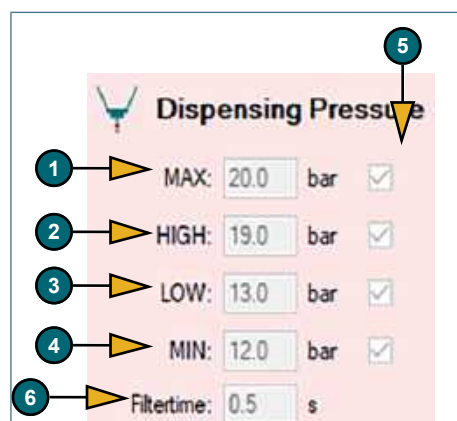
3.11.6.4.3 Dispensing pressure B

Illustration 72: Dispensing pressure B

1 "MAX" input field	2 "HIGH" input field
3 "LOW" input field	4 "MIN" input field
5 "Alarm" check box	6 "Filtration time" input field
7 Not installed	

Application

Define limit values of a physical variable (temperature, pressure, filling level, etc.) so that the operator is made aware of the situation by means of "alarms" and the running process is terminated.

"MAX" input field (1):

Input of the limit value to be monitored.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

**NOTICE****Alarm signalling on "MAX"**

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on undershooting the limit value.

If the check box is not set, the system maintains its operating state on reaching the value in the input field: no error display, no other signalling!

"HIGH" or "HOCH" input field (2):

Input of the limit value to be monitored.

**NOTICE****Alarm signalling**

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is undershot. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"LOW" or "NIEDRIG" input field (3):

Input of the limit value to be monitored.

	NOTICE Alarm signalling On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is exceeded. In this case, an optional yellow/orange signal lamp lights up. The machine continues in normal operation.
---	---

"MIN" input field (4):

Input of the limit value to be monitored.

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
---	---

	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.
--	--

	NOTICE Alarm signalling on "MIN" With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on exceeding the limit value. If the check box is not set, the system maintains its operating state on reaching the limit value in the input field: no error display, no other signalling!
---	--

Alarm checkbox (5):

Tick mark set: Alarm is set: If an alarm occurs, it is displayed in yellow in the event of a warning in the status bar and an optional yellow signal lamp lights up as long as the alarm is present; conversely, an error is indicated with a red message in the status bar and an optional red signal lamp additionally lights up. In this case, the process also stops.

No tick mark: Errors and warnings are ignored.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

"Filter time" input field (6):

Time specification with which the querying control unit tolerates fluctuations from a value (e.g. measured value of a physical variable, limit value, etc.). A value is only taken over as valid after this elapses.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

Display field (7):

Display of the current value.

**NOTICE****Current value**

Value indicates the current programming of the physical value.

The dispensing pressure is measured at the outlet from the dispensing pump.

3.11.6.5 Parameterisation of blind shot container on load cell

3.11.6.5.1 Blind shot container dialogue

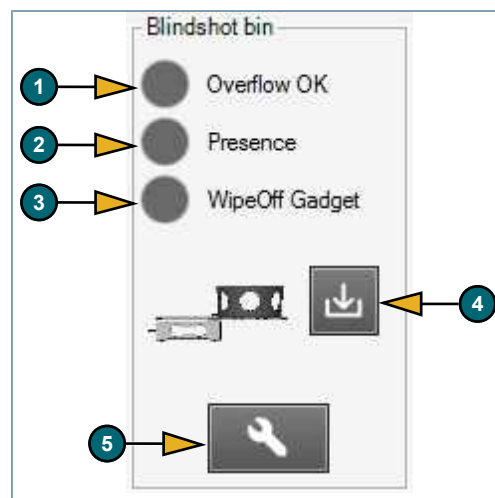


Illustration 73: Blind shot container dialogue

1 "Overflow OK" indicator LED	2 "Present" indicator LED
3 "Wiper" indicator LED	4 "Teach" symbol key
5 "Settings" symbol key	

Task:

This dialogue displays the states of the sensors for the blindshot container.



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

"Overflow OK" indicator LED (1):

green LED: Overflow is OK.

grey LED: No status signal present.

"Present" indicator LED (2):

green LED: Blindshot container is present.

grey LED: No blindshot container present.

"Wiper" indicator LED (3):

green LED: Wiper is in position.

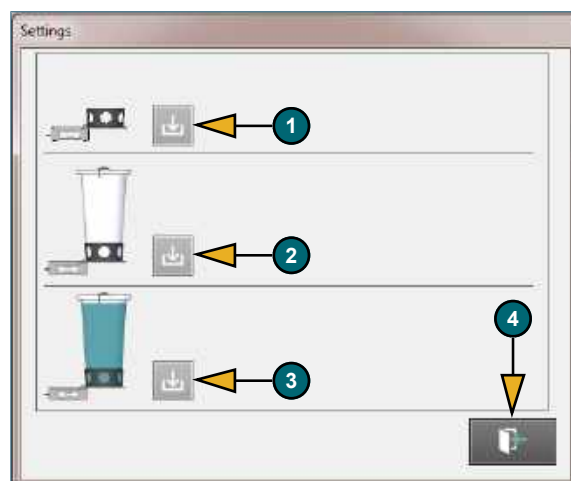
grey LED: Wiper not in position.

"Teach" symbol key (4):

Pressing the symbol key "teaches" the load cell.

"Settings" symbol key (5):

Pressing the symbol key opens the "Settings" dialogue window.

3.11.6.5.2 Blind shot container settings dialogue

1	"Teach load cell" symbol key	2	"Teach blindshot bin present" symbol key
3	"Teach blindshot bin full" symbol key	4	"Exit" symbol key

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

"Teach load cell" symbol key (1):

Pressing the symbol key opens the "Teach load cell" submenu.

"Teach blindshot bin present" symbol key (2):

Pressing the symbol key opens the "Teach blindshot bin present" submenu.

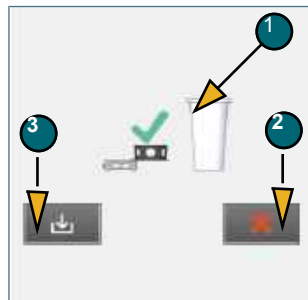
"Teach blindshot bin full" symbol key (3):

Pressing the symbol key opens the "Teach blindshot bin full" submenu.

"Exit" symbol key (4):

Pressing will close the "Settings" submenu.

Teach load cell



1	Animation field	2	"Close" symbol key
3	"Teach" symbol key		

Animation field (1):

Visual depiction of the teaching process for the load cell.

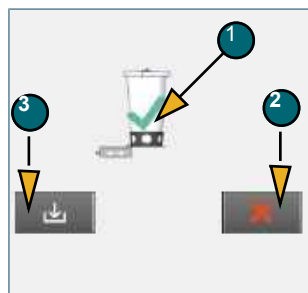
"Close" symbol key (2):

Pressing the symbol key closes the dialogue window.

"Teach" symbol key (3):

Teaching of the load cell is carried out by pressing the symbol key.

Blindshot bin present



1	Animation field	2	"Close" symbol key
3	"Teach" symbol key		

Animation field (1):

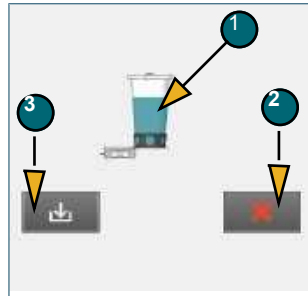
Visual depiction of the teaching process for the blindshot bin.

"Close" symbol key (2):

Pressing the symbol key closes the dialogue window.

"Teach" symbol key (3):

Teaching of the blindshot bin is carried out by pressing the symbol key.

Blindshot bin full

1	Animation field	2	"Close" symbol key
3	"Teach" symbol key		

Animation field (1):

Visual depiction of the teaching process for bin full.

"Close" symbol key (2):

Pressing the symbol key closes the dialogue window.

"Teach" symbol key (3):

Teaching of blindshot bin full is carried out by pressing the symbol key.

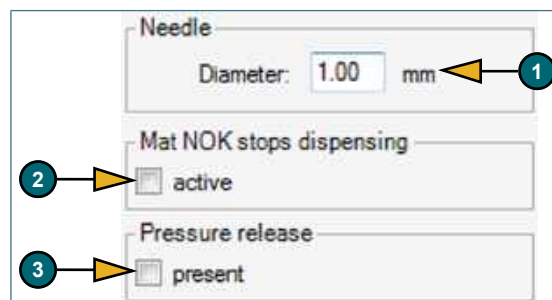
3.11.6.6 Behaviour after Mat NOK

Illustration 74: Behaviour after Mat NOK

1	Input field	2	Check box
3	Check box		

Task:

Material-dependent production control

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

Input field (1):

Input of the needle diameter in use.

Check box (2):

Tick mark set: Production stops dispensing after MAT NOK

No tick mark: no effect

Check box (3):

Tick mark set: Pressure is reduced.

No tick mark: no effect

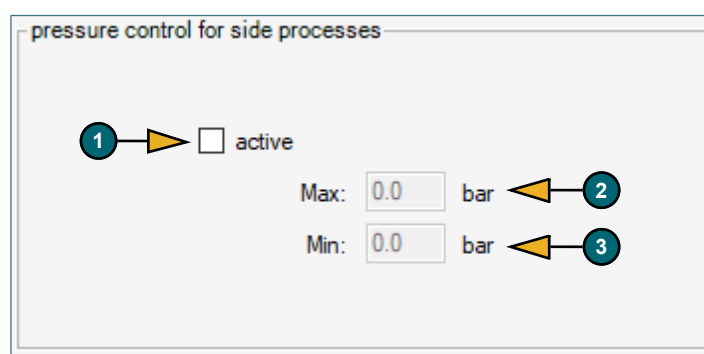
3.11.6.6.1 Pressure limits for ancillary processes

Illustration 75: Pressure limits for ancillary processes

1 "Active" check box	2 "Max." input field
3 "Min" input field	

Application

Parameterisation of the individual component weighing process.

"Active" check box (1):

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

"Max." input field (2):

Input of the maximum permitted pressure during the individual component weighing process. If exceeded: Process stops due to an error.

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.

"Min" input field (3):

Input of the minimum permitted pressure during the individual component weighing process. If exceeded: Process stops due to an error.

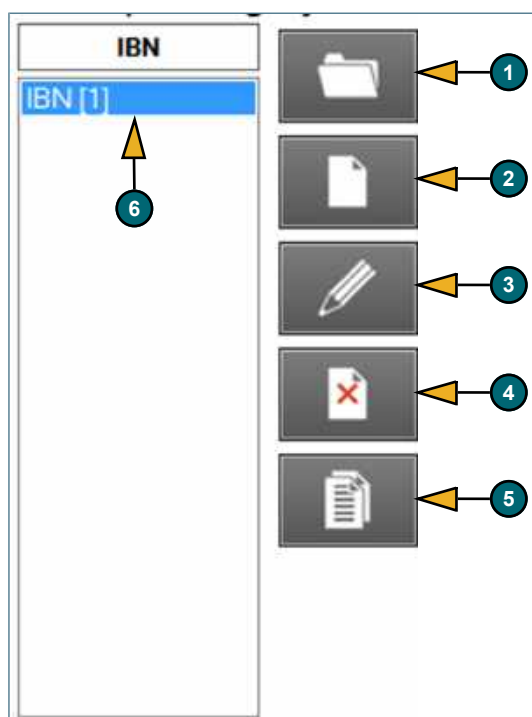
3.11.6.7 Dispensing programs

Illustration 76: Dispensing programs dialogue

1 "Load" symbol key	2 "New" symbol key
3 "Edit" symbol key	4 "Delete" symbol key

5	"Copy" symbol key	6	"Dispensing program" display field
---	-------------------	---	------------------------------------

Task:

Managing the dispensing program.

**NOTICE****Login authorisation**

Only an authorised user login can change configurations.

"Load" symbol key (1):

Loads the selected file.

"New" symbol key (2):

Opens new input dialog.

"Edit" symbol key (3):

Opens the dialog for editing.

"Delete" symbol key (4):

Deletes the selected element.

"Copy" symbol key (5):

Copies the selected element.

"Dispensing program" display field (6):

Shows the selected dispensing program.

**NOTICE****Detailed description**

A detailed description of the individual manual functions can be found under "Manual functions: General".

3.11.7 Weighing screen

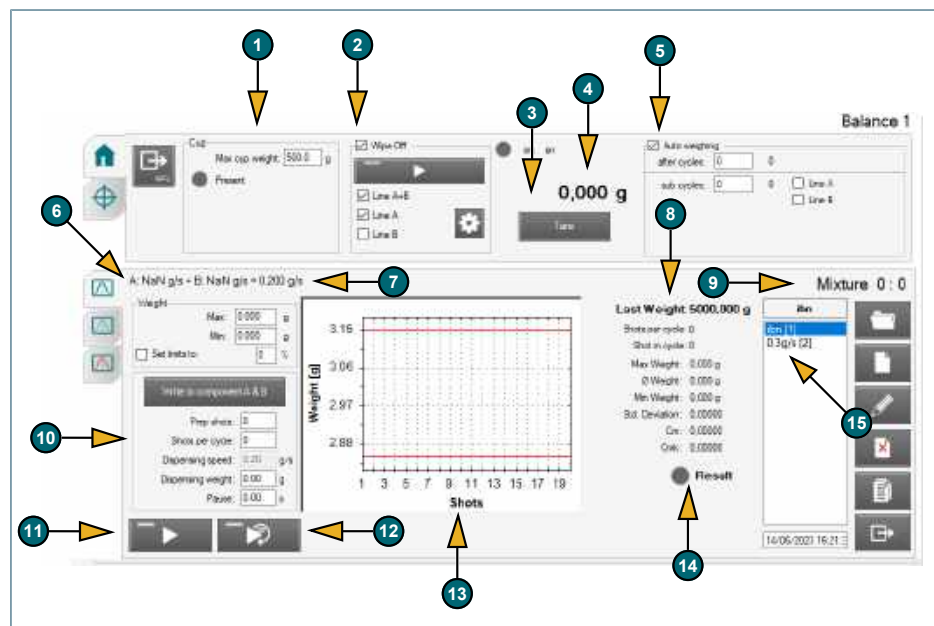


Illustration 77: Weighing screen with mixing tab

1 "Cup" group field	2 "Wiper" group field
3 < Tare > short-cut key	4 Display field
5 "Automatic weighing" group field	6 "Weight" group field
7 Display field	8 Display fields
9 Display field	10 "Weighing parameters" group field
11 < Weighing > start key	12 < Weighing sequence 50 > start key
13 Weighing diagram	14 "Result" indicator LED
15 Weighing program	

Application

This screen maps the quality assurance process for a bdtronic dispensing system both from the commissioning side and for the customer side. Important settings for recording and evaluating the weighing shots are performed via the communication interface to the precision scale or precision load cell. The results are presented both graphically in the form of a real-time diagram animation and statistically as deviations (standard deviation, Cm, Cmk) and as single variable evaluation (min. weight, max. weight), so that a comprehensive statement about the quality status of the dispensing system is possible.

During production, weighing tests of both the individual components and the mixture are automatically processed, so that complete control of the dispensing operation becomes part of the process task.

A red "Result" LED together with self-explanatory error texts in the status bar of the visualisation screen indicate in good time whether the permitted limit values have been exceeded and prompt the machine operator to take action with an error stop.

Of course, smooth operation always presupposes careful material preparation and regularly serviced wear components such as the dispensing pump!

“Cup” group field (1):

A description is given in a later sub-chapter “Cup group field”.

“Wiper” group field (2):

A description is given in a later sub-chapter “Wiper group field”.

< Tare > short-cut key (3):

Tares scale without cup and optional wiper ring due to the settings to be determined in the "Cup" group field beforehand.

Display field (4):

Display of the cup weight.

“Automatic weighing” group field (5):

A description is given in a later sub-chapter “Automatic weighing group field”.

“Weight” group field (6):

A description is given in a later sub-chapter “Weighing group field”.

Display field (7):

Displays settings for the dispensing speed.

Display fields (8):

Last weight: Last weighed weighing shot.

Shots per cycle: Display of the number of shots entered in the “Weighing parameters” group field.

Current shot: Incrementing of the number entered in “Shots per cycle”.

Max. weight: Highest weight determined in the series of measurements of the weighing sequence.

Ø weight: Average weight calculated from the series of measurements of the weighing sequence.

Min. weight: Lowest weight determined in the series of measurements of the weighing sequence.

Std. deviation: Standard deviation calculated from the series of measurements of the weighing sequence.

Cm: Scatter values determined from the series of measurements of the weighing sequence.

Cmk: Standard deviation determined from the series of measurements of the weighing sequence in relation to the weighing setpoint.

Display field (9):

Tab name with display of the mixing ratio (with mixture screen only!).

“Weighing parameters” group field (10):

A description is given in a later sub-chapter “Weighing parameters group field”.

< Weighing > start key (11):

Only if the system status "System ready" (for mixture weighing) or "A prepared" (for component A weighing) or "B prepared" (for component B weighing) is displayed and if automatic preparation is not active can the settings in the group fields "Weight" and "Weighing parameters" be processed. Otherwise the system automatically sets the system status. For this the corresponding dispensing and weighing programs must be loaded.

During mixture weighing, the pressure limits of the normal dispensing process are monitored, for individual components the pressure limits of the secondary processes.

< Weighing sequence 50 > start key (12):

Here, 50 weighing shots are subsequently dispensed without consideration of the weighing limits. This weighing sequence only starts if the status "System ready" (for mixture weighing) or "A prepared" (for component A weighing) or "B prepared" (for component B weighing) is displayed and the corresponding dispensing and weighing program is loaded.

During mixture weighing, the pressure limits of the normal dispensing process are monitored, for individual components the pressure limits of the secondary processes.

Weighing diagram (13):

Plot of the actual values of the weighing shots as a continuous line. The scaling of the axes ("shots" and "weight") is a function of the specifications "shots per cycle", "dispensing weight" and "max" and "min" limits.

"Result" indicator LED (14):

LED lights up green: The current weighing result is within the tolerance limits.

LED lights up red: Current weighing result exceeds the weighing limits. or error stop due to e.g. pressure fault.

Weighing program (15):

Collection of settings in the weighing masks. The name should be chosen so that it matches that of the corresponding dispensing program. For example, a dispensing program should be named according to the dispensing speed to be processed, i.e. "1.5 g/s", and this name should also be selected for the weighing program, i.e. "1.5 g/s". Before processing a dispensing task with a dispensing speed of 1.5 grams per second with automatic weighing check, the corresponding weighing program ("1.5/g") must therefore also be loaded!

3.11.7.1 Weighing cup with cup weight detection

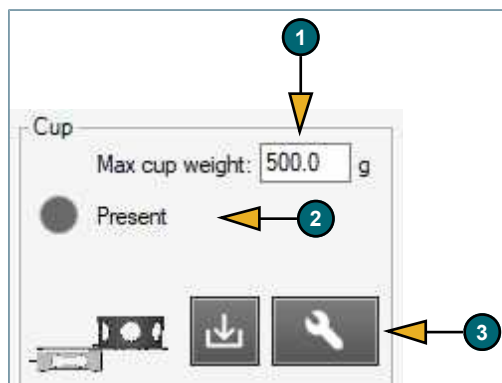


Illustration 78: Scale - Cup

1 "Max. cup weight" input field	2 "Present" indicator LED
3 Symbol key	

Application

Teach in cup detection function.

"Max. cup weight" input field (1):

Input of the permitted cup weight. If this weight is exceeded, an error stop occurs and the operator must change the cup.



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.

"Present" indicator LED (2):

LED lights up green: Cup is in position.

LED is grey: No cup present.

Symbol key (3):

Opens the settings dialogue for teaching in the scale statuses (scale with and without cup).



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.

3.11.7.2 Weighing cup – Wiping off

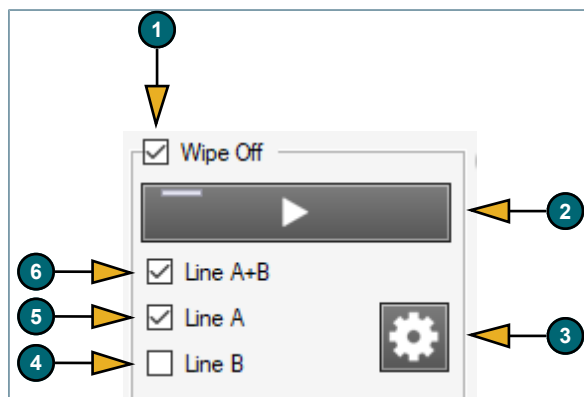


Illustration 79: Wiping off

1	"Wipe off" checkbox	2	< Start > short-cut key
3	< Settings > symbol key	4	"Line A" check box
5	"Line B" check box	6	"Line A+B" check box

Application

Configure and activate wiping process for weighing cup. The settings are also used for calibration.

"Wipe off" checkbox (1):

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.



NOTICE

Login authorisation and operation mode

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

< Start > short-cut key (2):

Start the wiping process manually.



NOTICE

Process monitoring operation mode!

This function or sequence can only be executed in MANUAL or PROCESS MONITORING operation mode! An authorised user must be logged in for this!

< Settings > symbol key (3):

A settings dialogue opens in which the process can be parameterised.

This is a task that is carried out once by the service technician running in the new system.

"Line A" check box (4):

Tick mark set: After the weighing shot, wiping is performed so that any build-up (drops) of material on the dispensing nozzle that would not be dispensed into the cup without wiping are also weighed. Otherwise, they would only be weighed during the next shot, which would be wrong. This setting is displayed as an additional green "Wiping active" LED on the "Calibration A" screen.

"Line B" check box (5):

Tick mark set: After the weighing shot, wiping is performed so that any build-up (drops) of material on the dispensing nozzle that would not be dispensed into the cup without wiping are also weighed. Otherwise, they would only be weighed during the next shot, which would be wrong. This setting is displayed as an additional green "Wiping active" LED on the "Calibration B" screen.

"Line A+B" check box (6):

Tick mark set: After the weighing shot, wiping is performed so that any build-up (drops) of material on the dispensing nozzle that would not be dispensed into the cup without wiping are also weighed. Otherwise, they would only be weighed during the next shot, which would be wrong.

3.11.7.3 Automatic weighing with part simulation

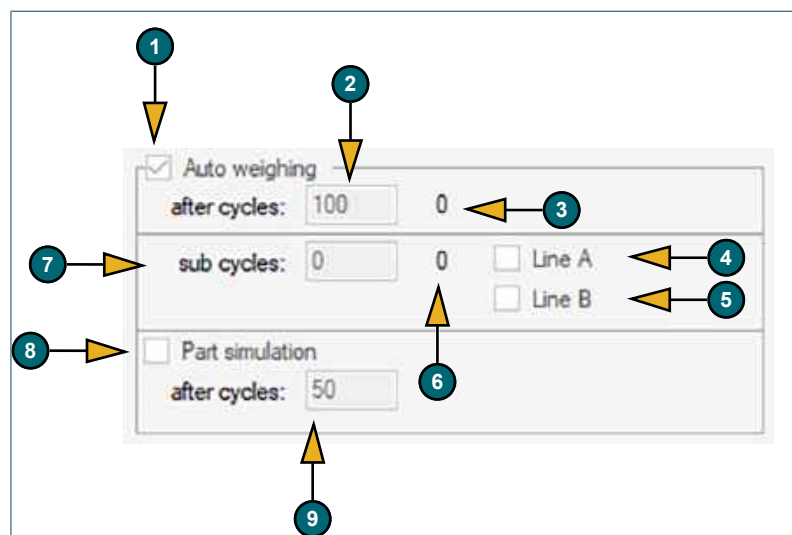


Illustration 80: Automatic weighing with part weighing simulation

1 "Automatic weighing" check box	2 "After cycles" input field
3 Display field	4 "Line A" check box
5 "Line B" check box	6 Display field
7 "Under cycles" input field	8 "Part simulation" check box
9 "After cycles" input field	

Application

Activate and configure automatic weighing check.

"Automatic weighing" check box (1):

Tick mark set:

If only check box (1) is ticked: On 2K systems, the mixture weight is checked each time the machine goes to AUTOMATIC mode, on 1K systems only the actual single line. Depending on the ticks in check boxes 4, 5 and 8, the sequence changes as follows:

No tick mark:

No automatic weight check is performed in AUTOMATIC mode. Only in PROCESS MONITORING mode can the weight check be started via the corresponding key on the weighing screen. The "individual settings" of the machine settings still apply there!

"After cycles" input field (2):

After processing the specified number of cycles (number of parts or pallets), the automatic weighing check for the mixture weight is started if the "Automatic weighing" check box is ticked.

Case example 1: Dynamic system

On a dynamic mixer (case example: Purging is not set in automatic processing; "System ready" is displayed):

- On entering AUTOMATIC mode, needle cleaning and a needle check are performed if the system has been configured accordingly
- An algorithm determines the system status
- As soon as "System ready" is displayed, the system is ready for dispensing
- The dispensing unit moves over the scale and performs the parameterised number of preparation shots
- When the preparation shots have been processed and the pressure values were OK, the weighing parameters are processed
- If the "Result" LED turns green at the end of the weighing cycle, the weighing check is OK
- The next pending process is processed

Case example 2: Static system

On a static mixer (case example: "System ready" is displayed, both densities are the same, mixing ratio is 1 to 1):

- On entering AUTOMATIC mode, needle cleaning and a needle check are performed if the system has been configured accordingly
- An algorithm determines the system status
- As soon as "System ready" is displayed, the system is ready for dispensing
- The dispensing unit moves over the scale and performs the parameterised number of preparation shots

- When the preparation shots have been processed and the pressure values were OK, the weighing parameters are processed
- If the "Result" LED turns green at the end of the weighing cycle, the weighing check is OK
- The next pending process is processed

Display field (3):

The number of cycles to be processed is counted down. When zero is reached, the weighing check is started by moving the dispensing unit to the blind shot position.

"Line A" check box (4):

Tick mark set:

- On entering AUTOMATIC mode, needle cleaning and a needle check are performed if the system has been configured accordingly
- An algorithm determines the system status
- Depending on the check marks in the "Individual settings" group field, the line for component A is prepared. For example, a dynamic system first moves to the purging position if configured in this way.
- When status "A prepared" has been reached, the weighing cycle starts with the settings of the weighing parameters
- A green "Result" LED signals the successful completion of the weighing sequence, a red LED an error stop
- If the single component was OK, a mixture check is performed (on 2K systems only!)

No tick mark:

No automatic weight check is performed in AUTOMATIC mode. Only in PROCESS MONITORING mode can the weight check be started via the corresponding key on the weighing screen. The "individual settings" of the machine settings still apply there!

"Line B" check box (5):

Tick mark set:

- On entering AUTOMATIC mode, needle cleaning and a needle check are performed if the system has been configured accordingly
- An algorithm determines the system status
- Depending on the check marks in the "Individual settings" group field, the line for component B is prepared. For example, a dynamic system first moves to the purging position if configured in this way.
- When status "B prepared" has been reached, the weighing cycle starts with the settings of the weighing parameters
- A green "Result" LED signals the successful completion of the weighing sequence, a red LED an error stop
- If the single component was OK, a mixture check is performed (on 2K systems only!)

No tick mark:

No automatic weight check is performed in AUTOMATIC mode. Only in PROCESS MONITORING mode can the weight check be started via the corresponding key on the weighing screen. The "individual settings" of the machine settings still apply there!

Display field (6):

The number of cycles to be processed is entered in (2) and is counted downwards. When zero is reached in (2), the counter in (6) is incremented by 1. When the input value in (7) is reached, a weighing check of the lines selected in (4) and/or (5) is started by moving the dispensing unit to the blindshot position. This is followed by a mixture check (only for 2K systems!).

"Under cycles" input field (7):

Input of the factor with which the parameter "After cycles" is multiplied in order to start the strand check(s) when the resulting number is reached. A "0" as input prevents the selected lines from being checked. A number greater than "0" for unmarked lines (see (4) and (5)) also prevents the line check.

"Part simulation" check box (8):

Tick mark set:

In addition to the optional marks in 4 and/or 5, an NC run is performed 1:1 in the weighing position. The X, Y, Z axes are thereby coupled to virtual axes, i.e. they are not moved.

No tick mark:

Depending on the marks in 4 and 5, the weight checks are performed.

"After cycles" input field (9):

If check box 1 is checked, optional line checks are started first (if 4 and 5 are checked) after processing the specified number of cycles (number of parts or number of pallets), then the mixture check, then the NC run in the weighing position.

3.11.7.4 Scale - Weighing limits

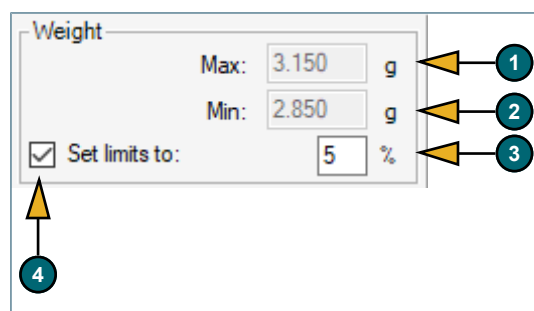


Illustration 81: Weight limits

1 "Max." input field	2 "Min" input field
3 "Set limits to:" input field	4 "Set limits to:" check box

Application

Setting weighing limits.

“Max” input field (1):

Manual input or display of the calculated, maximum permitted weight (see also (3)).

**NOTICE****Manual input**

The input is only possible when check box (4) is **not** ticked!

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

“Min” input field (2):

Manual input or display of the calculated, minimum permitted weight (see also (3)).

**NOTICE****Manual input**

The input is only possible when check box (4) is **not** ticked!

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

“Set limits to:” input field (3):

Input of the permitted tolerances for the dispensing weight.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

“Set limits to:” check box (4):

Tick mark set:

On the basis of the tolerance entered in (3), the permitted 'Min:' and 'Max:' weight limits are automatically calculated and displayed in the fields (1) and (2).

No tick mark:

No calculation of the weight limits.



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.

3.11.7.5 Scale - Actual values - Evaluations - Result

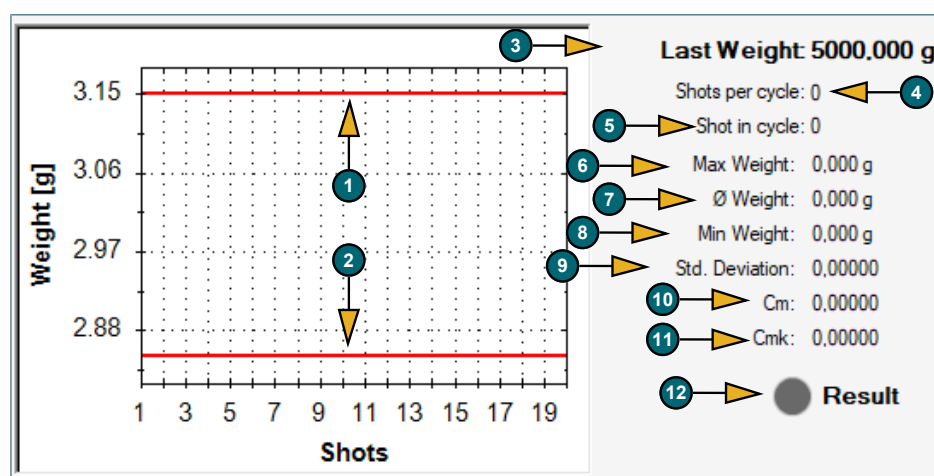


Illustration 82: Scale - Actual values - Evaluations - Result

1 Weight upper limit	2 Weight lower limit
3 "Last weight" display field	4 "Shots per cycle" display field
5 "Current shot" display field	6 "Max. weight" display field
7 "Ø weight" display field	8 "Min. weight" display field
9 "Standard deviation" display field	10 "Cm" display field
11 "Cmk" display field	12 "Result" indicator LED

Application

Weighing information, evaluations.

Weight upper limit (1):

Marker for the upper weight limit entered for parameter "Max". Exceeding this limit causes a weighing stop with error.

Weight lower limit (2):

Marker for the lower weight limit entered for parameter "Min". Exceeding this limit causes a weighing stop with error.

“Last weight” display field (3):

Display of the weight of the weighing shot before the current weighing shot.

“Shots per cycle” display field(4):

Display of the number of weighing shots of the active weighing sequence.

“Current shot” display field (5):

Weight of the current weighing shot.

"Max. weight" display field (6):

Display of the highest weight measured within the current weighing sequence.

“Ø weight” display field (7):

Display of the average weight within the current weighing sequence.

"Min. weight" display field (8):

Display of the lowest weight measured within the current weighing sequence.

"Standard deviation" display field (9):

Display of the calculated standard deviation of previous weighing shots of the current weighing sequence.

“Cm” display field (10):

Display of the calculated “Cm” value of previous weighing shots of the current weighing sequence.

“Cmk” display field (11):

Display of the calculated “Cmk value of previous weighing shots of the current weighing sequence.

"Result" indicator LED (12):

LED lights up green: Weighing shot lies between the entered “Min” and “Max” limit values.

LED lights up red: Weighing shot lies outside the entered “Min” and “Max” limit values. The consequence is an error stop.

3.11.7.6 Scales parameters

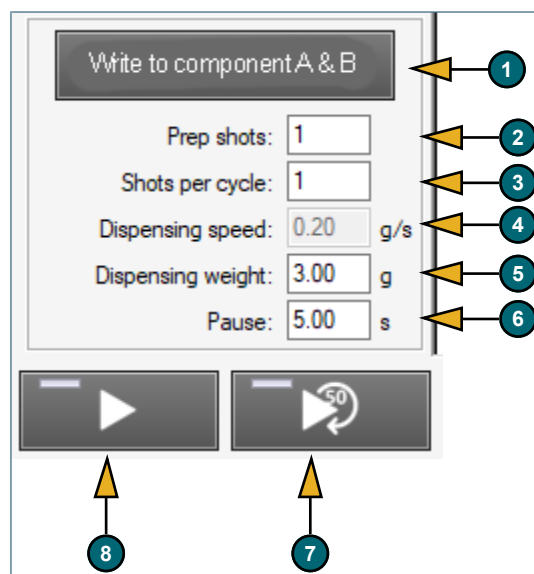


Illustration 83: Weighing parameter

1 < Write to component A & B > key	2 "Preparatory shots" input field
3 "Shots per cycle" input field	4 "Dispensing speed" display field
5 "Dispensing weight" input field	6 "Pause" input field
7 Symbol key	8 Symbol key

Application

Parameterise weighing sequence and manual weighing start.



NOTICE

Settings in 'Level 4'

Changes to subsequent parameters or the triggering of a function etc. can only be carried out by a user who is logged in with at least 'Level 4' authorisation!

< Write to component A & B > (1):

Weights and dispensing speeds for the individual components are calculated and entered there ("Dispensing weight" and "Dispensing speed"). In addition, the settings are copied so that you only have to enter them once for a weighing method.

"Preparatory shots" input field (2):

Input of the number of preparation shots (additional info: a preparation shot is dispensed with the parameter specifications of a weighing shot, but is not evaluated. This serves for the material flow. Several can be entered.)

"Shots per cycle" input field (3):

Input of the number of subsequent weighing shots.

"Dispensing speed" display field (4):

Display of the dispensing speed stored in the loaded dispensing program.

"Dispensing weight" input field (5):

Input of the weight to be checked.

"Pause" input field (6):

Input of a time after which the weight value is accepted (additional info: scale settling time).

Symbol key (7):

Processing of 50 subsequent weighing shots, without observing the "Min" and "Max" weight limits, i.e. with optional preparation shot. During weighing, the "Max" and "Min" pressure limits of the active dialogue "Pressure limits for secondary processes" are observed.

Symbol key (8):

Running of the parameterised weighing cycle, i.e. with optional preparation shot.

When weighing individual components, the "Max" and "Min" pressure limits of the active dialogue "Pressure limits for secondary processes" are observed, when weighing mixtures, the dispensing pressure limits of dispensing pumps A and B are observed.

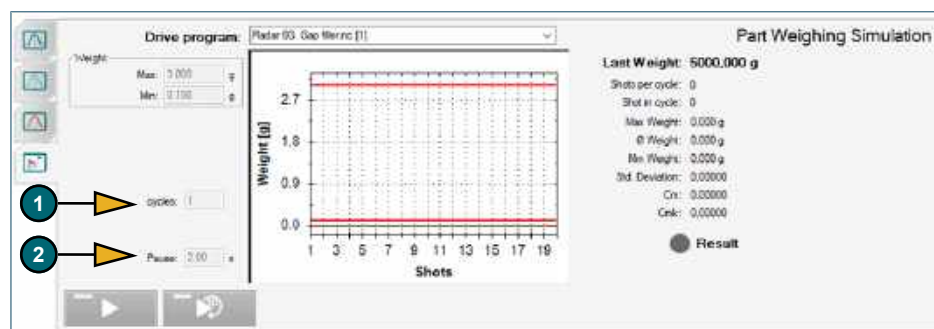
3.11.7.7 Scale - Part weighing simulation

Illustration 84: Scale - Part weighing simulation

1 "No of cycles" input field	2 "Pause" input field
------------------------------	-----------------------

Application

Parameterise part weighing simulation.

3.11.7.8 Scale - Weighing programs

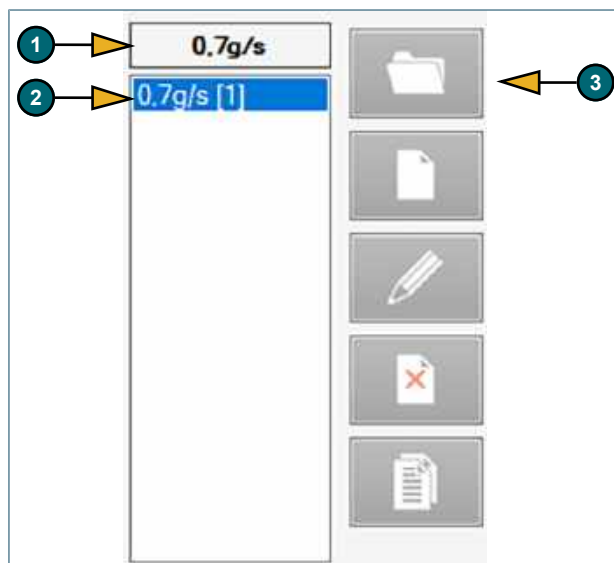


Illustration 85: Scale - Weighing programs

1	Display field	2	Selection list
3	< Load > short-cut key		

Application

Weighing programs save the settings in the weighing screens.

Display field (1):

Display of the currently selected weighing program.

Selection list (2):

Highlight program in the selection list.

< Load > short-cut key (3):

Load selected program.



NOTICE

Login authorisation function execution

Only an authorised user login can execute functions.

3.11.8 Needle check

3.11.8.1 Master positions

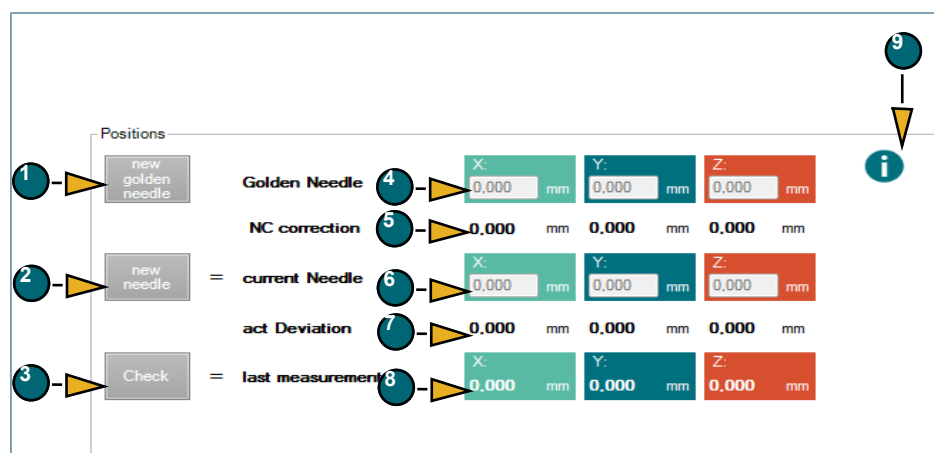


Illustration 86: Positions (Master)

1 < New golden needle > quick key	2 < New needle > quick key
3 < Check > quick key	4 "Golden Needle" input and display fields
5 "NC offset" display fields	6 "Current needle" input and display fields
7 "Current deviation" display fields	8 "Last measurement" display fields
9 < Info > symbol key	10 <Reset> quick key

Application

Central screen for triggering the functions of the needle check (information: This is the master version of the screen).

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.

< New golden needle > quick key (1):

Confirm the position reached as the "New golden needle" position.

< New needle > quick key (2):

Pressing this key moves the needle to the "Current needle" position for measurement of the needle. The results are entered as new coordinates in the line "Current needle". In addition, offset values are determined that are displayed in line "NC offset" and that are included in the corresponding variables of the drive program. 0/0/0 is displayed in line "Current deviations".

< Check > quick key (3):

Pressing the key starts a measuring cycle that compares the current measured values with the values in "Current needle" position. The measuring point is set in the measuring ring or measuring bar: Changing of the bar ends to red and "Error" LED means Z deviation outside the tolerances, changing of the outer measuring ring to red and "Error" LED means XY deviation outside the tolerances. Lighting up of the "Result" LED green means positive test result.

"Golden Needle" input and display fields (4):

Possibility for manual input of coordinates or for displaying the reached coordinates of the "New golden needle" position.

"NC offset" display fields (5):

Display of the calculated offset values for the current valid needle check position. These coordinates are included in the drive program variables.

"Current needle" input and display fields (6):

Possibility for manual input of coordinates or for displaying the reached coordinates of the "Current needle" position.

"Current deviation" display fields (7):

Display of the deviation/difference between the "Current needle" position and the "Last measurement" values: The Status LED "Error" lights up red if the deviation is outside the tolerances in parameter "XY" or the "Result" LED lights up green if the deviation is within the tolerances.

"Last measurement" display fields (8):

Display of the deviation values determined by the current measurement.

< Info > symbol key (9):

Pressing the keys displays a graphic image of the different needle positions for a better understanding.

<Reset> quick key (10):

Press the key to set the "New needle" position equal to the "New golden needle" position (additional information: This function is useful if a new product is to be set up with the same needle! The "New needle" position is then overwritten by the "New golden needle" position; NC offset values are deleted.)

3.11.8.2 Deviation dialogue

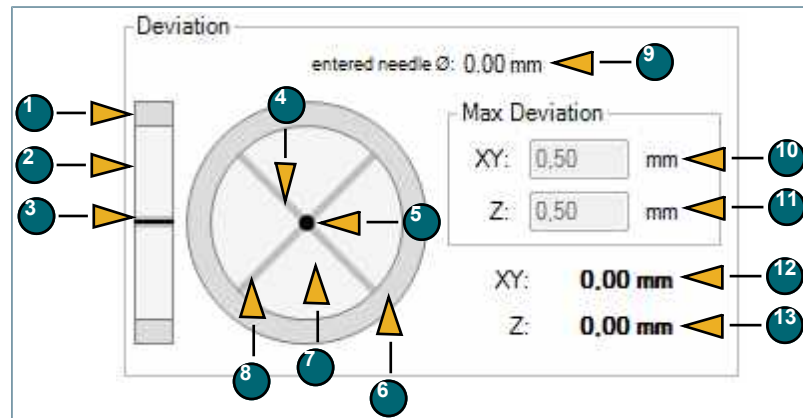


Illustration 87: Deviation dialogue (master/slave)

1	Needle length NOK range	2	Needle length OK range
3	Current needle Z position	4	Light barrier "X"
5	Current needle XY position	6	Needle XY deviation NOK range
7	Needle XY deviation OK range	8	Light barrier "Y"
9	"Needle diameter" display field	10	Input field "XY"
11	"Z" input field	12	"XY" display field
13	"Z" display field		

Application

Parametrisation of the tactile needle check.

Needle length NOK range (1):

Display of the inadmissible Z deviation of the measured needle from the confirmed needle position: A red bar indicates an error.

Needle length OK range (2):

Display of the permissible Z deviation of the measured needle from the confirmed needle position: If the measuring bar is within the light grey range, the Z deviation is OK.

Current needle Z position (3):

The black measuring bar moves downwards starting from the middle for a negative deviation, or in the opposite direction for a positive deviation. If it remains in the middle, no deviation was determined.

Light barrier in X (4):

Measurement of the needle in the X-direction.

Light barrier green: Needle is recorded.

Light barrier grey: Needle not recorded

Current needle XY position (5):

The black measuring point wanders – depending on the size and direction of the deviation – into one of the 4 segments of the measuring ring or remains in the coordinate origin if no deviation was determined.

Needle XY deviation NOK range (6):

Display of the inadmissible XY deviation of the measured needle from the confirmed needle position: A red ring indicates an error.

Needle XY deviation OK range (7):

Display of the permissible XY deviation of the measured needle from the confirmed needle position: If the measuring bar is within the light grey range, the XY deviation is OK.

Light barrier in Y (8):

Measurement of the needle in the Y-direction.

Light barrier green: Needle is recorded.

Light barrier grey: Needle not recorded

"Needle diameter" display field (9):

Display of the current needle diameter.

"XY" input field (10):

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.

Input of the permitted deviation of the needle in X or Y direction only for the needle check, not for a repositioning sequence! The system preset ± 0.1 mm applies here. Exception: if a value of below 0.2 mm is pre-filled in this input field, half of that value is taken and included in the repositioning calculation.

"Z" input field (11):

Input of the permitted deviation in Z direction only for the needle check, not for a repositioning sequence! The system preset ± 0.1 mm applies here. Exception: if a value of below 0.2 mm is pre-filled in this input field, half of that value is taken and included in the repositioning calculation.

"XY" display field (12) and "Z" display field (13):

Display of the current deviation from the confirmed needle position.

3.11.8.3 Settings

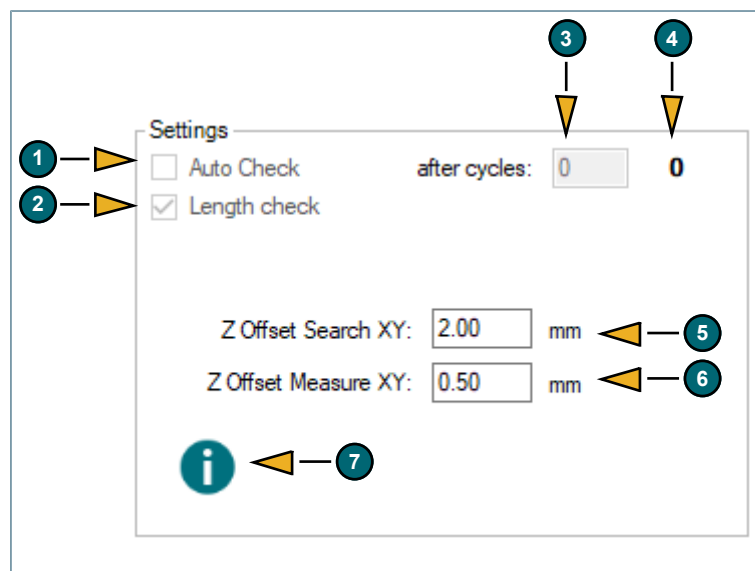


Illustration 88: Settings

1 "Auto Check" checkbox	2 "Length check" checkbox
3 "After cycles" input field	4 Counter status display
5 "Z offset needle search XY" input field	6 "Z offset measurement XY" input field
7 Info	

Application:

Configuration of the needle check for automatic mode.



NOTICE

Login authorisation and operation mode

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.

"Auto Check" checkbox (1):

Tick mark set: before start of production, a needle check with/without length check is run, depending on pos. 2.

No tick mark: No needle check.

"Length check" checkbox (2):

Tick mark set: The needle length is also checked during a needle check.

No tick mark: No length check is carried out.

"After cycles" input field (3):

Input of the number of components after whose processing a needle check is started.

Counter status display (4):

Display of the components processed since the last reset.

"Z offset needle search XY" input field (5):

This value indicates how far it is necessary to travel in Z direction in order to interrupt the light barriers XY during the needle search.

"Z offset measurement XY" input field (6):

This value is set by the set-up technician according to the needle geometry.

Info (7):

Graphics show important steps in the needle check.

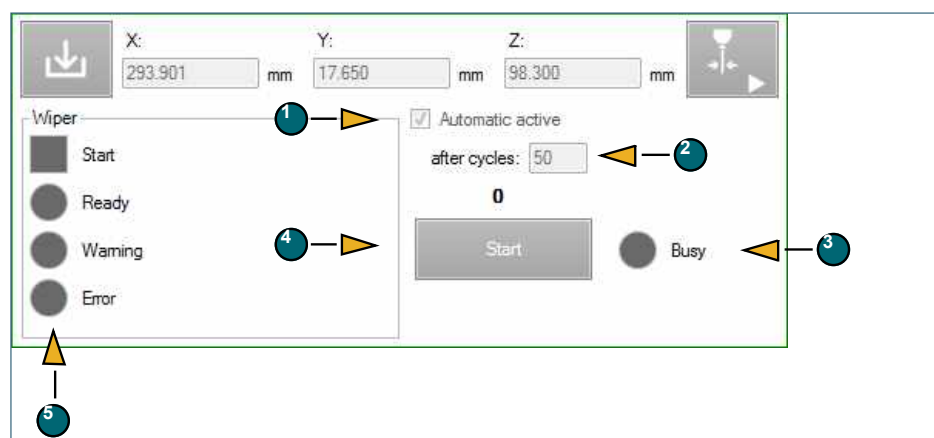
3.11.9 Cleaning station with belt

Illustration 89: Belt cleaner

1	"Automatic active" check box	2	"After cycles" input field
3	"Busy" indicator LED	4	< Start > quick key
5	Indicator LEDs		

Application

Configure belt cleaning station.

"Automatic active" check box (1):

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

When AUTOMATIC operation mode is activated, the cleaning process is stopped after the set number of parts to be cleaned (see also under "After cycles" input field): Axis system moves dispensing needle into cleaning position, belt surrounds needle geometry, soiling is wiped off as dispensing needle moves out of cleaning position again, soiled part of the cleaning belt is wound on.

"After cycles" input field (2):

Input of the number of parts after whose processing the cleaning process starts.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

"Busy" indicator LED (3):

LED indicates the status of the process or the detected process step:

Green LED: Process or process step running: The same process or process step cannot be started again at the moment.

Grey LED: Process or process step stopped or was not started.

< Start > quick key (4):**NOTICE****Quick key**

The function is performed immediately on pressing the button or symbol key if both login authorisation and operation mode have been preset.

Cleaning process starts as soon as the button is pressed: Axis system moves dispensing needle into cleaning position, belt surrounds needle geometry, soiling is wiped off as dispensing needle moves out of cleaning position again, soiled part of the cleaning belt is wound on. This manual function is based on the PROCESS MONITORING operation mode.

Indicator LEDs (5):

"Start" LED:

LED green: Device has been activated.

LED grey: Device is inactive.

"Ready":

LED green: Device is ready for cleaning.

LED grey: Device is not ready.

"Warning":

Red LED: Prewarning of end of cleaning belt.

LED grey: Device has no maintenance request.

"Error":

Red LED: End of cleaning belt reached or sequence error: Device stops, operator intervention required.

LED grey: No error detected or no maintenance request.

3.11.10 Basic settings

3.11.10.1 Calibration A with scale

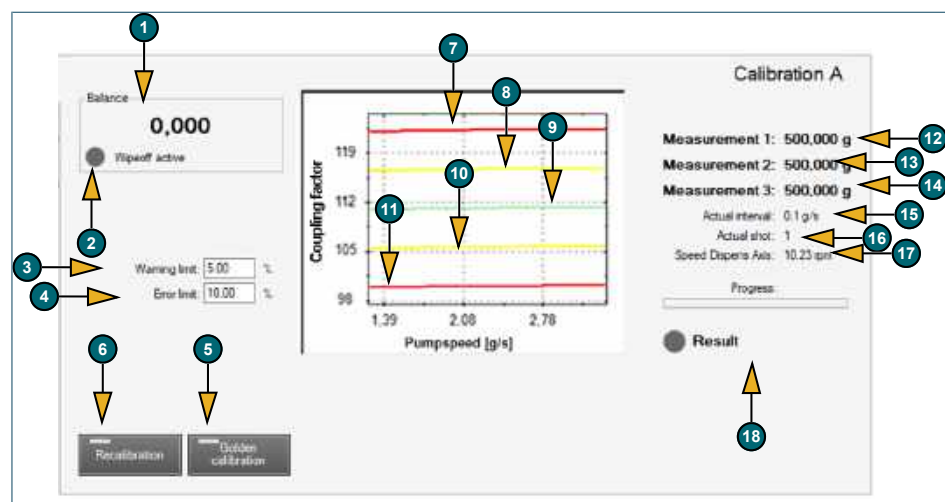


Illustration 90: Calibration A

1 Display field	2 "Wiping active" indicator LED
3 "Warning limit" input field	4 "Error limit" input field
5 < Golden calibration > short-cut key	6 < Recalibration > short-cut key

7	Max. error limit	8	Max. warning limit
9	Reference curve	10	Min. warning limit
11	Min. error limit	12	"Measurement 1" display field
13	"Measurement 2" display field	14	"Measurement 3" display field
15	"Current interval" display field	16	"Current shot" display field
17	"Current speed" display field	18	"Result" indicator LED

Application

Parameterise and perform calibration.

Display field (1):

Current weighing result.

"Wiping active" indicator LED (2):

LED lights up green: Wiping is active for this component.

LED is grey: Wiping is not active.

"Warning limit" input field (3):

Warning limit in % relative to the reference curve (green). If exceeded: Error message.

"Error limit" input field (4):

Error limit in % relative to the reference curve (green). If exceeded: Error stop.

< Golden calibration > short-cut key (5):

Opens a pop-up window for input of the data for the golden calibration (see under "**New golden calibration**"). Additional info: The pump reference curve is recorded for the first time during the golden calibration. It serves as the reference for all speed calculations for the dispensing pump.

< Recalibration > short-cut key (6):

PROCESS MONITORING operation mode must be active to start the recalibration. Recalibration is started when the speed values can no longer be reached within the limits due to wear or in the event of replacement of the pump or changing of important pump components such as stator or rotor.

No recalibration without a first golden calibration!

The procedure is as follows:

1. Activate PROCESS MONITORING operation mode
2. Press the < *Recalibration* > key
 - ⇒ Each interval is measured three times
 - ⇒ The measurement result is OK if it is within the 5% limit
 - ⇒ The mean value is the reference value that is plotted as the curve point

⇒ A green "Result" LED indicates the existence of a new, valid reference curve. The warning and error limits are retained.

Max. error limit (7):

Limit line for the weighing values: An error stop is triggered in the event of the line being exceeded.

Max. warning limit (8):

Limit line for the weighing values: A warning message is triggered in the event of the line being exceeded.

Reference curve (9):

The pump characteristic generated by interval measurements with different speeds. Recording starts after input of the calibration specifications.

Min. warning limit (10):

Limit line for the weighing values: A warning message is triggered in the event of the values dropping below the line.

Min. error limit (11):

Limit line for the weighing values: An error stop is triggered in the event of the values dropping below the line.

"Measurement 1" display field (12):

1. Measured value for the current interval

"Measurement 2" display field (13):

2. Measured value for the current interval

"Measurement 3" display field (14):

3. Measured value for the current interval

"Current interval" display field (15):

Dispensing speed input for this interval.

"Current shot" display field (16):

Counter for the number of the dispensing shot (0 to 3).

'Actual speed' display field (17):

Speed at which the pump is driven.

"Result" indicator LED (18):

LED lights up green: Weighing value within the limits.

LED is grey: Weighing value outside the limits: Error!

3.11.10.1.1 Machine settings – New golden calibration

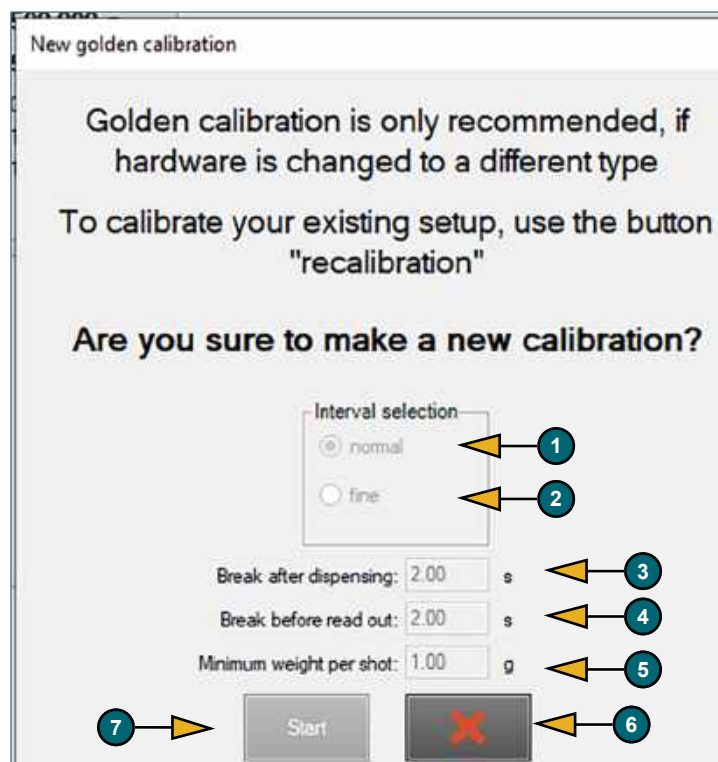


Illustration 91: New golden calibration

1 "Normal" selection field	2 "Fine" selection field
3 "Pause after dispensing" input field	4 "Pause before reading out" input field
5 "Min. weight per shot" input field	6 "Cancel" key
7 Start key	

Application

Presettings for golden calibration.

"Normal" selection field (1):

When selected: Curve recording is split into 4 areas.

"Fine" selection field (2):

When selected: Curve recording is split into 8 areas.

"Pause after dispensing" input field (3):

Input of a time difference between the dispensing stop and the time from which no more material is added.

"Pause before reading out" input field (4):

Without active wiping, this is the time after the expiry of the parameter "Pause after dispensing" until a stable weighing value is obtained.

Without active wiping, this is the time difference between the end of wiping and the time at which a stable weighing value is obtained.

"Min. weight per shot" input field (5):

In case of doubt, determine this weight input by trials. Background: For small quantities, a higher weight may be more favourable than the 0.1 g default setting.

Cancel key (6):

Changes to settings are discarded.

< Start > key (7):

Settings are permanently stored in system variables of the controller core. The golden calibration is started. After two preparation shots, 3 measurements are taken per interval, with at least one full rotation resulting in a weight value. Three consecutive weighing values within an interval must lie within the 5% tolerance limit to one another. The mean of these three values is then the reference value. The green reference curve is updated after each interval, so that at the end a complete green line has been created. This contains the correction values for the speed calculation of the pump when dispensing in the specified dispensing rate range.

A green "Result" LED signals the successful calibration process!

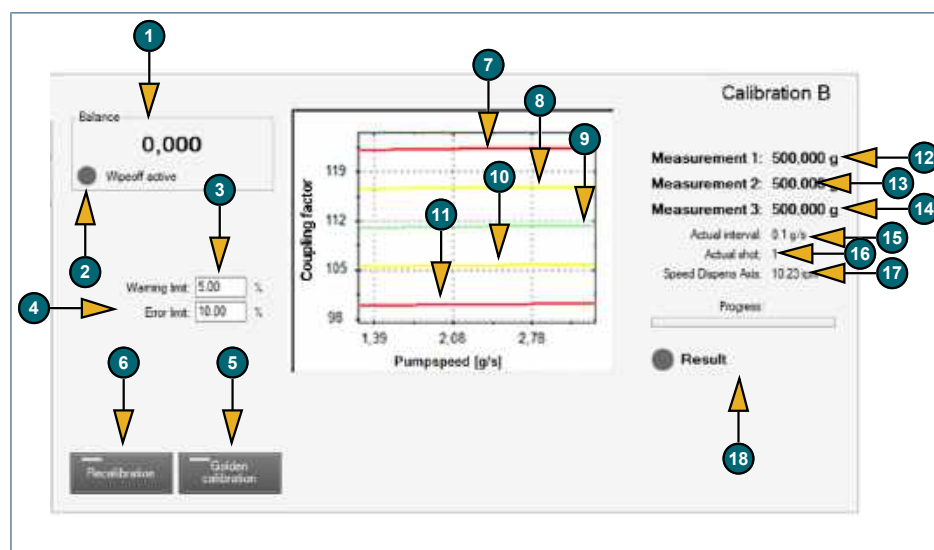
3.11.10.2 Calibration B with scale

Illustration 92: Calibration B

1 Display field	2 "Wiping active" indicator LED
3 "Warning limit" input field	4 "Error limit" input field
5 < Golden calibration > short-cut key	6 < Recalibration > short-cut key
7 Max. error limit	8 Max. warning limit
9 Reference curve	10 Min. warning limit
11 Min. error limit	12 "Measurement 1" display field
13 "Measurement 2" display field	14 "Measurement 3" display field
15 "Current interval" display field	16 "Current shot" display field

17	'Current speed' display field	18	"Result" indicator LED
----	-------------------------------	----	------------------------

Application

Parameterise and perform calibration.

Display field (1):

Current weighing result.

"Wiping active" indicator LED (2):

LED lights up green: Wiping is active for this component.

LED is grey: Wiping is not active.

"Warning limit" input field (3):

Warning limit in % relative to the reference curve (green). If exceeded: Error message.

"Error limit" input field (4):

Error limit in % relative to the reference curve (green). If exceeded: Error stop.

< Golden calibration > short-cut key (5):

Opens a pop-up window for input of the data for the golden calibration (see under "**New golden calibration**"). Additional info: The pump reference curve is recorded for the first time during the golden calibration. It serves as the reference for all speed calculations for the dispensing pump.

< Recalibration > short-cut key (6):

PROCESS MONITORING operation mode must be active to start the recalibration. Recalibration is started after replacement or changing of important components of the dispensing pump, such as stator or rotor.

Max. error limit (7):

Limit line for the weighing values: An error stop is triggered in the event of the line being exceeded.

Max. warning limit (8):

Limit line for the weighing values: A warning message is triggered in the event of the line being exceeded.

Reference curve (9):

The pump characteristic generated by interval measurements with different speeds. Recording starts after input of the calibration specifications.

Min. warning limit (10):

Limit line for the weighing values: A warning message is triggered in the event of the values dropping below the line.

Min. error limit (11):

Limit line for the weighing values: An error stop is triggered in the event of the values dropping below the line.

"Measurement 1" display field (12):

1. Measured value for the current interval

"Measurement 2" display field (13):

2. Measured value for the current interval

"Measurement 3" display field (14):

3. Measured value for the current interval

"Current interval" display field (15):

Dispensing speed input for this interval.

"Current shot" display field (16):

Counter for the number of the dispensing shot (0 to 3).

'Actual speed' display field (17):

Speed at which the pump is driven.

"Result" indicator LED (18):

LED lights up green: Weighing value within the limits.

LED is grey: Weighing value outside the limits: Error!

3.11.10.2.1 Machine settings – New golden calibration

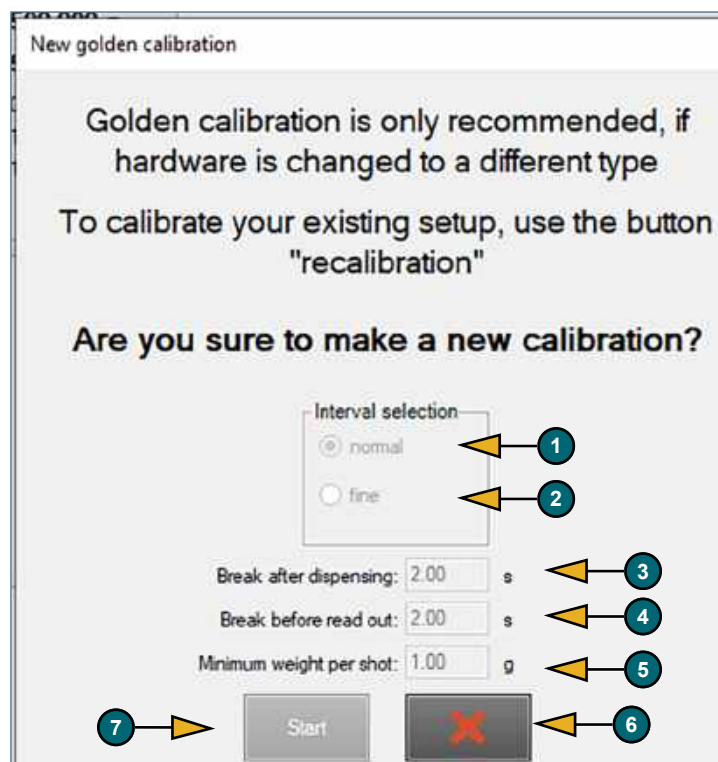


Illustration 93: New golden calibration

1	"Normal" selection field	2	"Fine" selection field
3	"Pause after dispensing" input field	4	"Pause before reading out" input field
5	"Min. weight per shot" input field	6	"Cancel" key
7	Start key		

Application

Presettings for golden calibration.

"Normal" selection field (1):

When selected: Curve recording is split into 4 areas.

"Fine" selection field (2):

When selected: Curve recording is split into 8 areas.

"Pause after dispensing" input field (3):

Input of a time difference between the dispensing stop and the time from which no more material is added.

"Pause before reading out" input field (4):

Without active wiping, this is the time after the expiry of the parameter "Pause after dispensing" until a stable weighing value is obtained.

Without active wiping, this is the time difference between the end of wiping and the time at which a stable weighing value is obtained.

"Min. weight per shot" input field (5):

In case of doubt, determine this weight input by trials. Background: For small quantities, a higher weight may be more favourable than the 0.1 g default setting.

Cancel key (6):

Changes to settings are discarded.

< Start > key (7):

Settings are permanently stored in system variables of the controller core. The golden calibration is started. After two preparation shots, 3 measurements are taken per interval, with at least one full rotation resulting in a weight value. Three consecutive weighing values within an interval must lie within the 5% tolerance limit to one another. The mean of these three values is then the reference value. The green reference curve is updated after each interval, so that at the end a complete green line has been created. This contains the correction values for the speed calculation of the pump when dispensing in the specified dispensing rate range.

A green "Result" LED signals the successful calibration process!

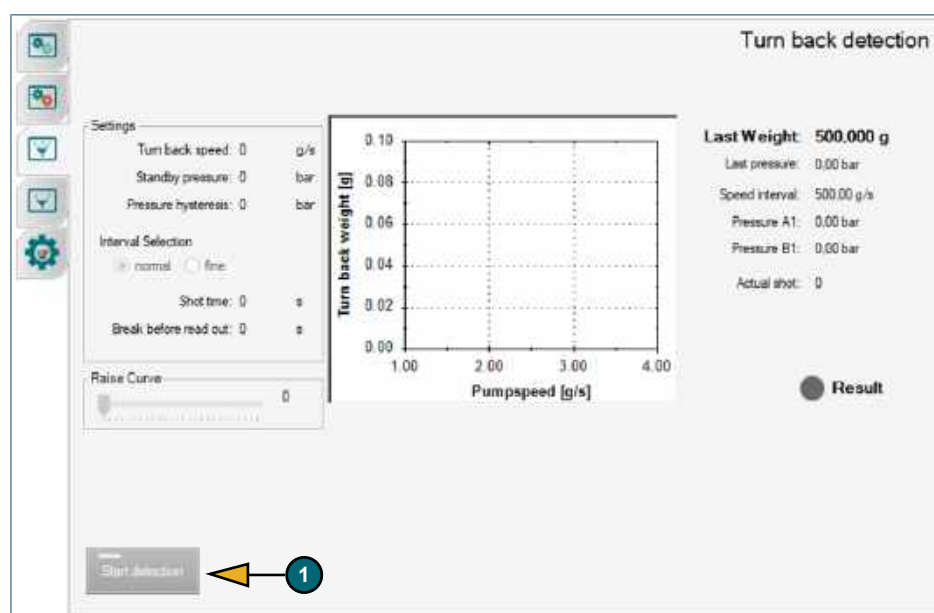
3.11.10.3 Basic settings – Turn-back determination

Illustration 94: Basic settings – Turn– back determination

1 "Turn-back determination" key

Application

The turn-back behaviour of the dispensing line is determined. This is particularly useful in conjunction with speed-dependent dispensing, because it significantly shortens the dispensing process, especially with complex traversing geometries.

Background: In the case of a dispensing shot into a static mixing tube, the pressure builds up slowly at the start of dispensing and then slowly decreases after the end of dispensing. In order to accelerate the pressure reduction, the pump is rotated backwards. For this, the "Turn-back determination" function determines the turn-back weights of the individual dispensing rates in the mixture.

As with calibration, the maximum required dispensing rate is divided into intervals and these are processed one after the other.

"Turn-back determination" key (1):

Routine for determining the turn-back values is started.



NOTICE

Login authorisation function execution

Only an authorised user login can execute functions.

3.11.10.3.1 Settings

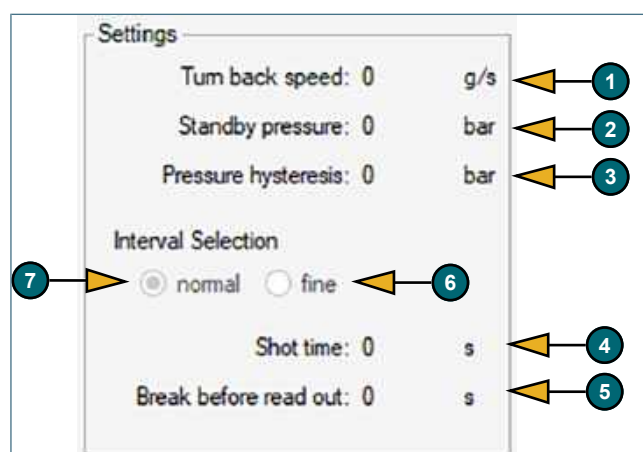


Illustration 95: Turn-back determination – Settings

1 "Turn back rate" display field	2 "Rest pressure" display field
3 "Hysteresis pressure" display field	4 "Shot time" display field
5 "Pause before reading out" display field	6 "Fine" selection field
7 "Normal" selection field	

"Turn back rate" display field (1):

Display of the current dispensing rate during the turn back process.

"Rest pressure" display field (2):

Display of the residual pressure in the dispensing line that is present in the mixing tube with a certain delay after the dispensing process has stopped.

"Hysteresis pressure" display field (3):

Permitted tolerances around the rest pressure.

"Shot time" display field (4):

Display of the duration of the dispensing rate with which the turn back process is performed.

"Pause before reading out" display field (5):

This waiting time starts after the end of the turn back process. Only then is the determined weight value saved.

"Fine" selection field (6):

A measurement series is a sequence of 8 measurements.

**NOTICE****Login authorisation**

Only an authorised user login can change configurations.

"Normal" selection field (7):

A measurement series is a sequence of 4 measurements.

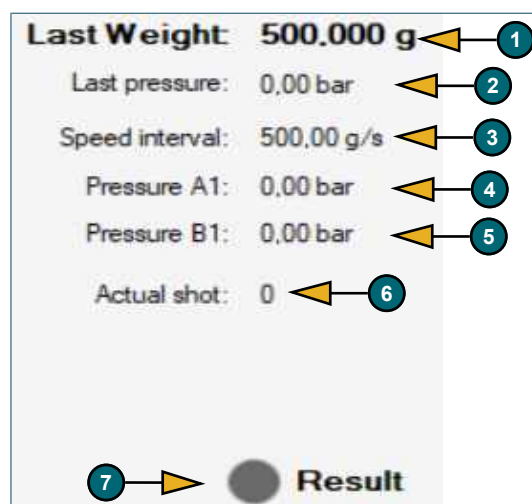
3.11.10.3.2 Evaluation

Illustration 96: Turn-back determination – Evaluation

1 "Last" display field	2 "Last pressure" display field
3 "Speed interval" display field	4 "Pressure A1" display field
5 "Pressure B1" display field	6 "Current shot" display field
7 "Result" indicator LED	

Application

Display of the process.

"Last" display field (1):

Display of the last turn-back volume determined.

"Last pressure" display field (2):

Display of the target pressure reached during the previous measurement.

"Speed interval" display field (3):

Dispensing rate set when determining the current value.

"Pressure A" display field (4):

Display of the system pressure in dispensing line A.

"Pressure B" display field (5):

Display of the system pressure in dispensing line B.

"Current shot" display field (6):

Counter reading.

"Result" indicator LED (7):

LED lights up green: Turn-back determination completed, values within the tolerance band.

LED lights up red: Turn-back determination failed, must be repeated.

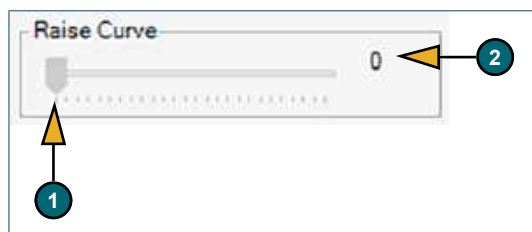
3.11.10.3.3 Raise curve

Illustration 97: Turn-back determination – Raise curve

1	Settings bar	2	Display field
---	--------------	---	---------------

Application

For critical applications (low dispensing rate, low-viscosity materials), the turn back volume can be increased by up to 25% to prevent dripping at rest pressures close to 0.1 bar.

Settings bar (1):

Values from 0 to 25% can be set in 1% steps. The turn back volume is increased by this percentage.



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.

Display field (2):

Percentages from 0 to 25 are displayed.

3.11.10.4 Machine settings

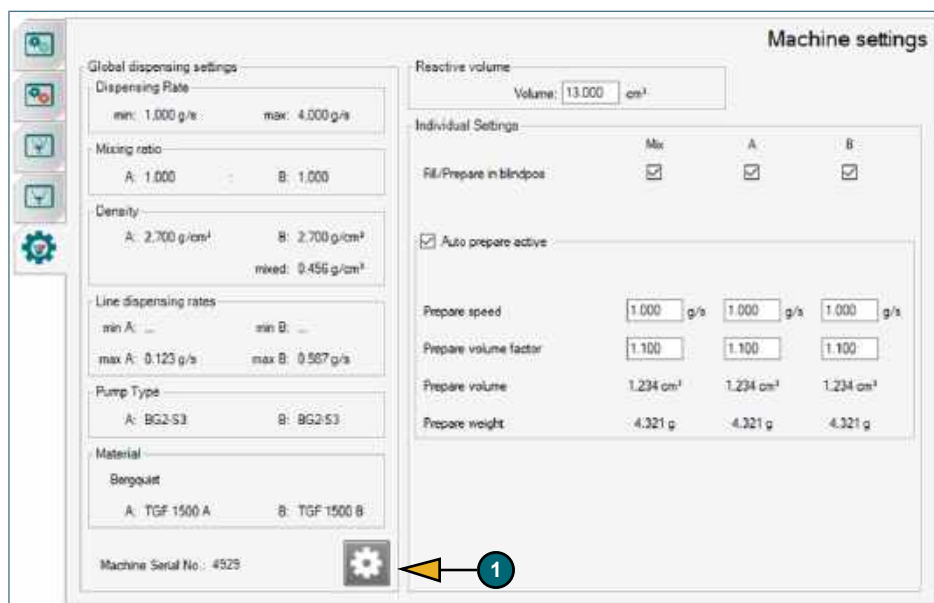


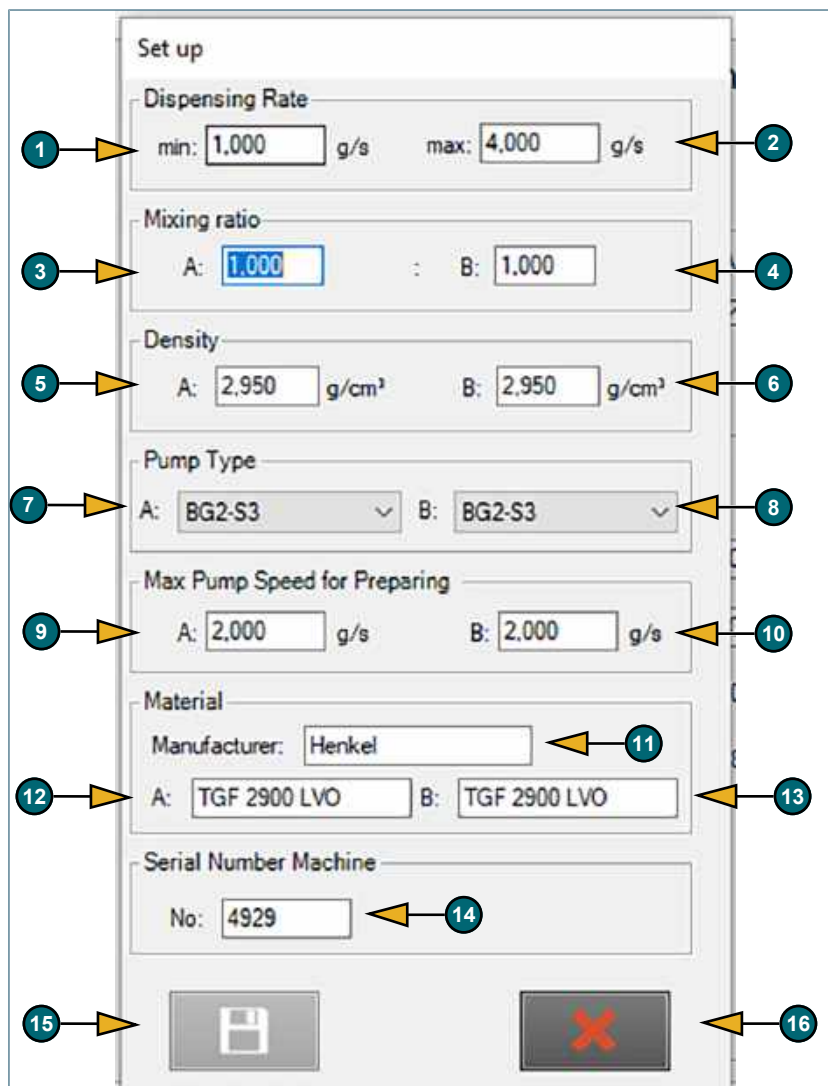
Illustration 98: Machine settings screen

1 < Setup > key

Application

The settings in this screen prepare the dispensing system for the dispensing task. The machine commissioning engineer first describes the fields and marks and the setup dialogue that can be called up from here with the < Setup > symbol key (see "Machine settings screen", 1). All other dispensing settings are entered in the other dispensing screens, such as the "Head" screen, the "Pump" screens, etc.

3.11.10.4.1 Machine settings - Setup dialogue



Set up

Dispensing Rate

min: 1.000 g/s max: 4.000 g/s

Mixing ratio

A: 1.000 : B: 1.000

Density

A: 2.950 g/cm³ B: 2.950 g/cm³

Pump Type

A: BG2-S3 B: BG2-S3

Max Pump Speed for Preparing

A: 2.000 g/s B: 2.000 g/s

Material

Manufacturer: Henkel

A: TGF 2900 LVO B: TGF 2900 LVO

Serial Number Machine

No: 4929

< Save > symbol key

Cancel button (red X)

Illustration 99: Machine settings - Setup

1 "Min. dispensing rate" input field	2 "Max. dispensing rate" input field
3 "Mixing ratio A" input field	4 "Mixing ratio B" input field
5 "Density A" input field	6 "Density B" input field
7 "Pump Type A" selection field	8 "Pump Type B" selection field
9 "Max. pump speed A for preparing" input field	10 "Max. pump speed B for preparing" input field
11 "Manufacturer" input field	12 Material designation input field for component A
13 Material designation input field for component B	14 Input field for machine serial number "No"
15 < Save > symbol key	

Application

The service technician carrying out the initial commissioning starts to parameterise the system via this page.

"Min. dispensing rate" input field (1):

Input of the smallest mixture rate according to the design expertise of bdtronic Dosing Technology, that the system should be able to dispense.



NOTICE

Settings only at Level 5!

Changes to these parameters can only be made by a user logged in with 'Level 5' authorisation!

"Max. dispensing rate" input field (2):

Input of the largest mixture rate according to the design expertise of bdtronic Dosing Technology, that the system should be able to dispense.



NOTICE

Settings only at Level 5!

Changes to these parameters can only be made by a user logged in with 'Level 5' authorisation!

"Mixing ratio A" input field (3):

Input of the A component of the mixing ratio (for two-component material systems only) that can be taken from the material technical data sheet.

"Mixing ratio" input field (4):

Input of the B component of the mixing ratio (for two-component material systems only) that can be taken from the material technical data sheet.

"Density A" input field (5):

Input of the material density of the A component that can be taken from the material technical data sheet.

"Density B" input field (6):

Input of the material density of the B component that can be taken from the material technical data sheet.

Selection field (7):

Selection of the pump type for the A component that can deliver the required dispensing rate.

**NOTICE****Settings only at Level 5!**

Changes to these parameters can only be made by a user logged in with 'Level 5' authorisation!

Selection field (8):

Selection of the pump type for the B component that can deliver the required dispensing rate.

**NOTICE****Settings only at Level 5!**

Changes to these parameters can only be made by a user logged in with 'Level 5' authorisation!

"Max. pump speed A for preparing" input field (9):

Input of the maximum dispensing rate that the selected pump is allowed to deliver during preparation. This value results from the maximum mixture dispensing specification (see 1) and the mixing ratio (see 9).

**NOTICE****Dispensing times under 1.0 second**

Dispensing times under 1.0 s can only be realised to a limited extent, on the one hand because of the switching times of the dispensing assemblies such as valves etc., and on the other hand because of the flow capability of some reaction casting resins! Due to the associated poorer repeat accuracy of the dispensing shots, dispensing times greater than or equal to 1.0 s should always be used!

**NOTICE****Settings only at Level 5!**

Changes to these parameters can only be made by a user logged in with 'Level 5' authorisation!

"Max. pump speed B for preparing" input field (10):

Input of the maximum dispensing rate that the selected pump is allowed to deliver during preparation. This value results from the maximum mixture dispensing specification and the mixing ratio.

**NOTICE****Dispensing times under 1.0 second**

Dispensing times under 1.0 s can only be realised to a limited extent, on the one hand because of the switching times of the dispensing assemblies such as valves etc., and on the other hand because of the flow capability of some reaction casting resins! Due to the associated poorer repeat accuracy of the dispensing shots, dispensing times greater than or equal to 1.0 s should always be used!

**NOTICE****Settings only at Level 5!**

Changes to these parameters can only be made by a user logged in with 'Level 5' authorisation!

"Manufacturer" input field (11):

Input of the name of the material manufacturer (see manufacturer's technical data sheet).

Material designation input field for component A (12):

Input of the material designation for component A (see manufacturer's technical data sheet).

Material designation input field for component B (13):

Input of the material designation for component B (see manufacturer's technical data sheet).

Input field for machine serial number "No" (14):

Input field of the machine serial number (see also type plate!).

< Save > symbol key (15):

Input changes are only stored in the system variables after a positive response to the query in "Load Configuration", closing the prompt "Please restart Twincat" with the < OK > key and a subsequent system restart.

How do I restart?

- Close the visualisation system window with the "Close" button in the window.
- Right-click with the mouse on the TwinCat icon
- Press "Restart" in the pop-up menu
- Open the visualisation system via the symbol in the taskbar

**NOTICE****Settings only at Level 5!**

Changes to these parameters can only be made by a user logged in with 'Level 5' authorisation!

Cancel key (16):

Input changes are discarded.

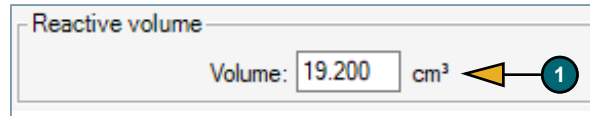
3.11.10.4.2 Reactive volume

Illustration 100: Reactive volume

1	"Reactive volume" input field
---	-------------------------------

Application

The "reactive volume" is the actual process space that must be flooded during the one-time filling with material or purged with cleaning agent or prepared when preparing the system, depending on the setting in the "Individual settings" group field:

- For a dynamic head, this would be the difference in volume between an empty mixing chamber and the agitator volume plus the volume of the outlet nozzle.
- For a static mixing head this would be the mixing volume of the static mixing tube plus an optional outlet nozzle at the end of the mixing tube.
- For single-component dosing units, only the volume of the outlet nozzle counts.

**NOTICE****Parameter assignment**

The value for this parameter is specified by a bdtronic service technician during commissioning.

3.11.10.4.3 Individual settings

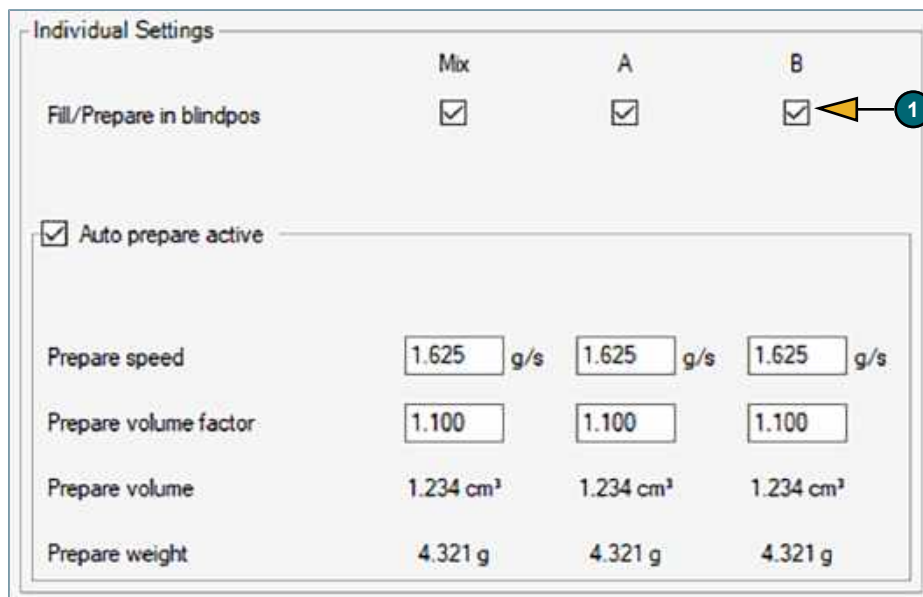


Illustration 101: Individual settings, static

1 "Filling/preparation in blind shot position" check box

Application

Parameterisation of the preparation activities for the dispensing system. This means that the software behind this settings screen knows the system history and the operator does not have to switch the dispensing pumps himself; instead, they are activated and deactivated automatically as required.



NOTICE

Settings in 'Level 4'

Changes to subsequent parameters or the triggering of a function etc. can only be carried out by a user who is logged in with at least 'Level 4' authorisation!

"Filling/preparation in blind shot position" check box (1):

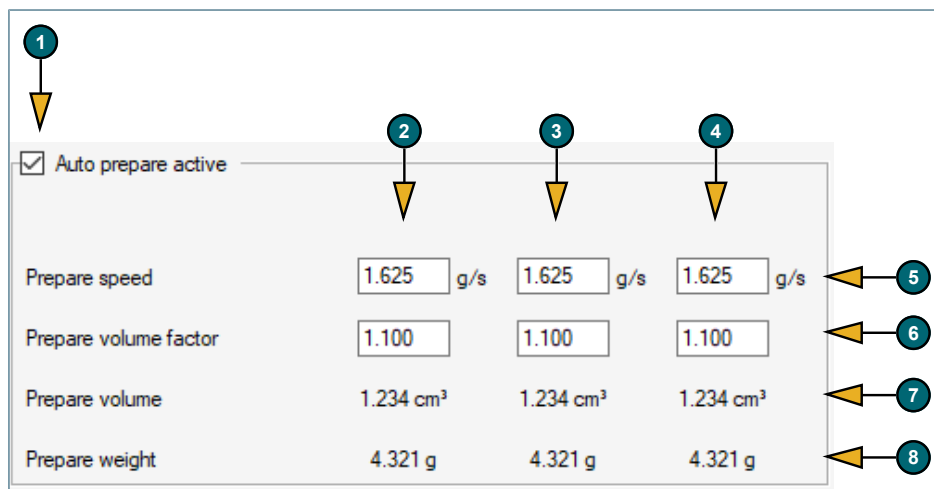
Tick mark set:

The one-time filling of the dispensing unit and running of the preparation cycles for the mixture or the individual components A or B are carried out in blind shot position: The dispensing unit is positioned there before starting.

No tick mark:

The one-time filling of the dispensing unit and running of the preparation cycles for the mixture or the individual components A or B are carried out in purging position: The dispensing unit is positioned there before starting.

3.11.10.4.4 Automatic preparation active



	2	3	4
☒ Auto prepare active			
Prepare speed	1.625 g/s	1.625 g/s	1.625 g/s
Prepare volume factor	1.100	1.100	1.100
Prepare volume	1.234 cm ³	1.234 cm ³	1.234 cm ³
Prepare weight	4.321 g	4.321 g	4.321 g

Illustration 102: Automatic preparation active

1 "Automatic preparation active" check box		
5 "Preparation speed" input field	6 "Preparation volume factor" input field	
7 "Preparation volume" display fields	8 "Preparation weight" display fields	

Application

Further parameterisation of the preparation activities for the dispensing system.



NOTICE

Settings in 'Level 4'

Changes to subsequent parameters or the triggering of a function etc. can only be carried out by a user who is logged in with at least 'Level 4' authorisation!

"Automatic preparation active" check box (1):

Tick mark set:

Automatic preparation is activated: Together with the settings in the following fields, the dispensing system is prepared for dispensing and weighing tasks.

Example of a preparation process:

The dispensing unit moves to purging position to prepare the weighing of single component B. There only the B component is dispensed with the specification e.g. '1 g/s' (this is displayed as "Prepare system status B") until the mixing tube is completely prepared with component B (this is displayed as "System B prepared"). Now the weighing process starts.



NOTICE

General information on automatic preparation

The system status is always checked when entering AUTOMATIC operation mode. If this status is not "System ready" or "System error state / System undefined", the "System ready" status is automatically established by a filling process or before each single component weighing by preparation with the target component.

No tick mark:

It may be necessary to switch on the pumps and perform the preparation manually, step-by-step.

"Preparation speed" input field (5):

Input of the dispensing rate with which the component is prepared. The input fields in the line from left to right are for: Mixture – Component A, Component B. A "3.000" in the right-hand field, for example, means that the dispensing unit should be flooded with 3.0 g/s of component B.

"Preparation volume factor" input field (6):

This is the "over-purge factor". The input fields in the line from left to right are for: Mixture – Component A, Component B. A "1.500" in the right-hand field, for example, means that the dispensing unit should be prepared with 1.5 times the quantity of the reactive volume. If the reactive volume is e.g. 7.000 cm³, then this means 10.500 cm³. This ensures that, for example, in the case of a large mixing chamber, the entire mixing chamber volume and the dispensing nozzle have been flooded with B.

"Preparation volume" display fields (7):

This is the calculated preparation volume. It is calculated from reactive volume x "Preparation volume factor". The display fields in the line from left to right are for: Mixture – Component A, Component B.

"Preparation weight" display fields (8):

This is the weight calculated from "Preparation volume" and the density/densities. The display fields in the line from left to right are for: Mixture – Component A, Component B.

3.11.11 Material preparation PPS

3.11.11.1 Start page 1 x 2K

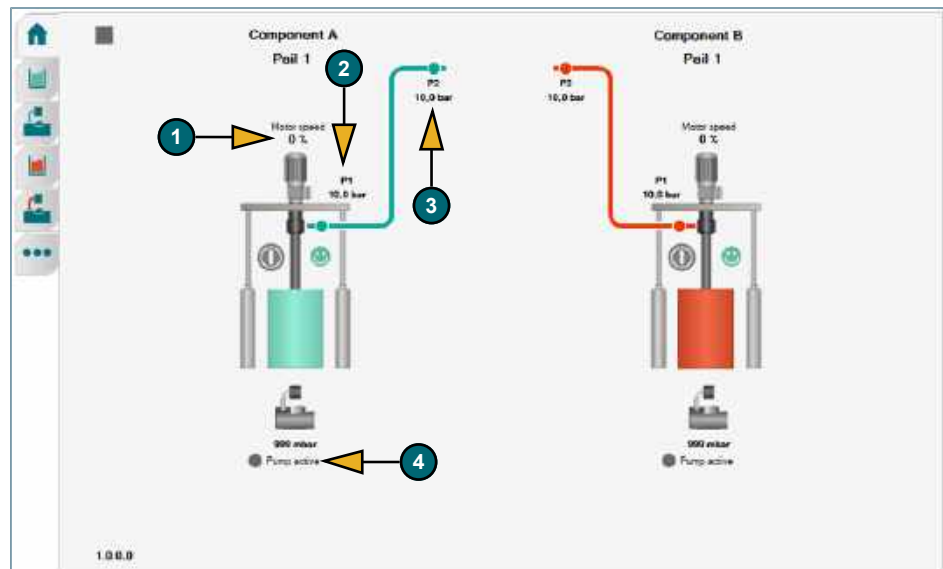


Illustration 103: PPS - Start page

1 to 4	Display fields
--------	----------------

Application

Start page for the machine operator. The current actual values or states at the consumers or sensors are displayed.

3.11.11.2 Component A - Pressure



Illustration 104: Component A - Pressure

1	"Pressure P1" display field	2	"Pressure P2" display field
---	-----------------------------	---	-----------------------------

Application

Setting pressure limits for the two pressure sensors P1 and P2.



NOTICE

Screens for the components

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.

"Pressure P1" display field (1):

Current pressure at the output of the conveyor pump of the PPS.

"Pressure P2" display field (2):

Current pressure before the dispensing pump to be fed.

3.11.11.2.1 Pump outlet port P1

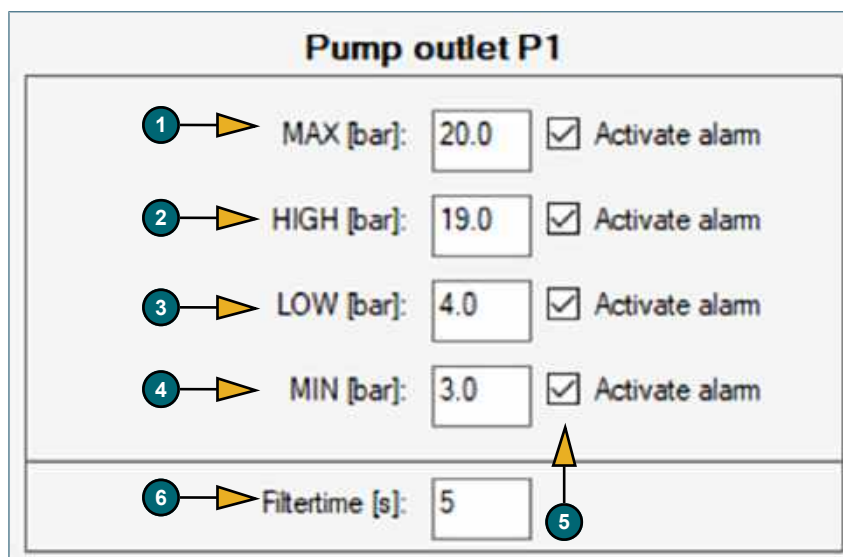


Illustration 105: P1

1 "MAX" input field	2 "HIGH" input field
3 "LOW" input field	4 "MIN" input field
5 "Activate alarm" check box	6 "Filtration time" input field

Application:

Setting the pressure limits for the conveyor pump output in controlled operation.

Application

Limit values of a physical variable (temperature, pressure, fill level, etc.), so that – by means of the so-called alarms – the operator is aware of the situation and the ongoing process can be terminated. This prevents unnecessary waiting times and an increase in the production of bad parts.

"MAX" input field (1):

Input of the limit value to be monitored.



NOTICE

Input field

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.



NOTICE

Alarm signalling on "MAX"

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on undershooting the limit value.

If the check box is not set, the system maintains its operating state on reaching the value in the input field: no error display, no other signalling!

"HIGH" or "HOCH" input field (2):

Input of the limit value to be monitored.



NOTICE

Alarm signalling

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is undershot. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"LOW" or "NIEDRIG" input field (3):

Input of the limit value to be monitored.

**NOTICE****Alarm signalling**

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is exceeded. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"MIN" input field (4):

Input of the limit value to be monitored.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

**NOTICE****Alarm signalling on "MIN"**

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on exceeding the limit value.

If the check box is not set, the system maintains its operating state on reaching the limit value in the input field: no error display, no other signalling!

**NOTICE****Current value**

Value indicates the current programming of the physical value.

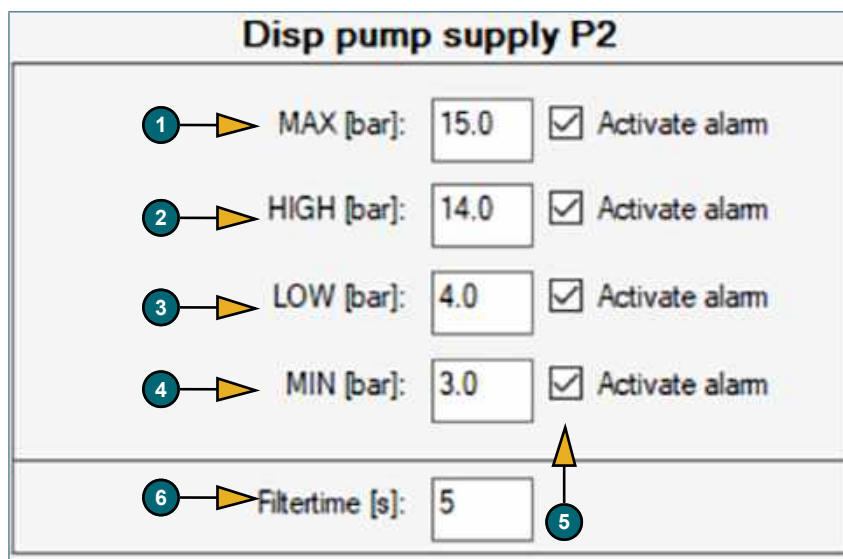
Alarm checkbox (5):

Tick mark set: Alarm is set: If an alarm occurs, it is displayed in yellow in the event of a warning in the status bar and an optional yellow signal lamp lights up as long as the alarm is present; conversely, an error is indicated with a red message in the status bar and an optional red signal lamp additionally lights up. In this case, the process also stops.

No tick mark: Errors and warnings are ignored.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

3.11.11.2.2 Dispensing pump supply P2


Disp pump supply P2

1	MAX [bar]:	15.0	☒ Activate alarm
2	HIGH [bar]:	14.0	☒ Activate alarm
3	LOW [bar]:	4.0	☒ Activate alarm
4	MIN [bar]:	3.0	☒ Activate alarm
6	Filtertime [s]:	5	

5 (points to the 'Activate alarm' checkbox for the MIN setting)

Illustration 106: P2

1 "MAX" input field	2 "HIGH" input field
3 "LOW" input field	4 "MIN" input field
5 "Activate alarm" check box	6 "Filtration time" input field

Application:

Setting the pressure limits for supplying the dispensing pump in controlled operation.

Application

Limit values of a physical variable (temperature, pressure, fill level, etc.), so that – by means of the so-called alarms – the operator is aware of the situation and the ongoing process can be terminated. This prevents unnecessary waiting times and an increase in the production of bad parts.

"MAX" input field (1):

Input of the limit value to be monitored.



NOTICE

Input field

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.



NOTICE

Alarm signalling on "MAX"

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on undershooting the limit value.

If the check box is not set, the system maintains its operating state on reaching the value in the input field: no error display, no other signalling!

"HIGH" or "HOCH" input field (2):

Input of the limit value to be monitored.



NOTICE

Alarm signalling

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is undershot. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"LOW" or "NIEDRIG" input field (3):

Input of the limit value to be monitored.

**NOTICE****Alarm signalling**

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is exceeded. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

"MIN" input field (4):

Input of the limit value to be monitored.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

**NOTICE****Alarm signalling on "MIN"**

With the combination of an input field and set check box, the PLC interrupts operation on reaching the limit value in the input field and waits for an operator intervention. In this process, an optional signalling unit (signal lamp and/or horn) is activated and a red alarm message that first has to be acknowledged is displayed. Normal operation can only be resumed on exceeding the limit value.

If the check box is not set, the system maintains its operating state on reaching the limit value in the input field: no error display, no other signalling!

**NOTICE****Current value**

Value indicates the current programming of the physical value.

Alarm checkbox (5):

Tick mark set: Alarm is set: If an alarm occurs, it is displayed in yellow in the event of a warning in the status bar and an optional yellow signal lamp lights up as long as the alarm is present; conversely, an error is indicated with a red message in the status bar and an optional red signal lamp additionally lights up. In this case, the process also stops.

No tick mark: Errors and warnings are ignored.



NOTICE

Login authorisation and operation mode

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

3.11.11.3 Component A - Drive

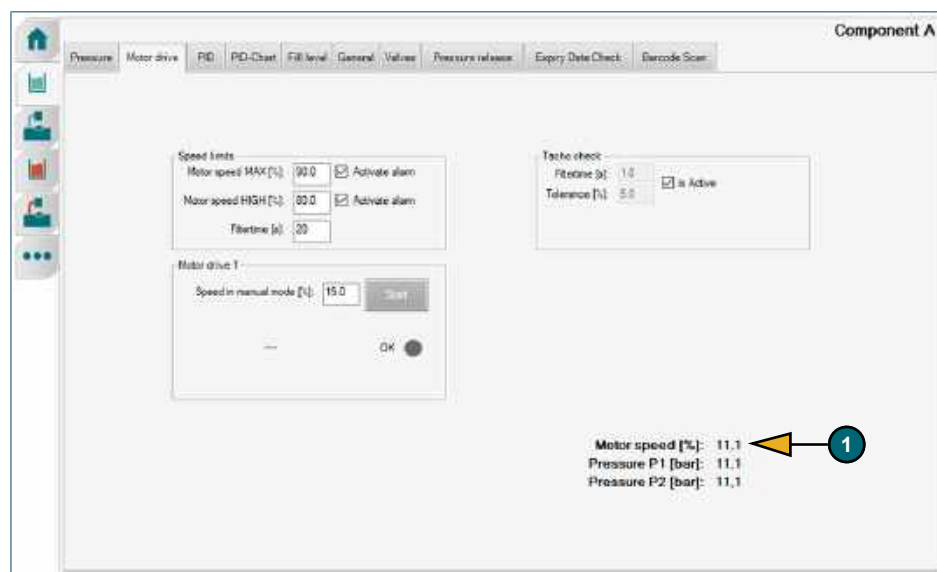


Illustration 107: Component A - Drive

1 Display field

Application

Partial parametrisation and manual functions.



NOTICE

Screens for the components

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.

Display field (1):

Display of the current barrel pump speed.

**NOTICE****Group field information**

Further detailed descriptions of the group fields follow.

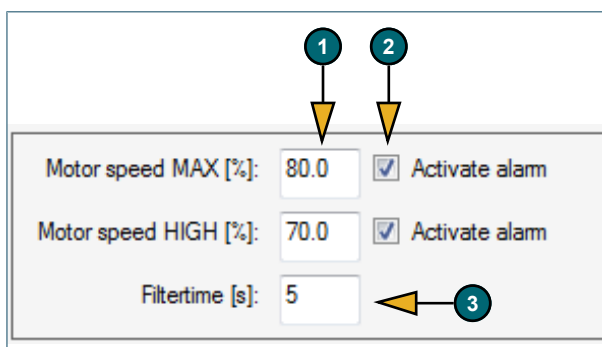
3.11.11.3.1 Component A - drive - speed limits


Illustration 108: Speed setting

1	Speed limits input fields	2	Alarm check box
3	Filter time input field		

Application:

Setting the speed limits for the monitoring of the pump in controlled operation.

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

Speed limits input field (1):

"MAX.":

Input of a value that stops the preparation process if exceeded and sets the "Material OK" signal inactive: an error message is displayed.

**NOTICE****Alarm signalling at "MIN" and "MAX"**

If the check box is set, the machine interrupts operation when the limit value is reached and waits for operator intervention. An optional message device (signal light and/or horn) is activated.

In addition, an alarm message is displayed that must first be acknowledged.

Normal operation can only be resumed on undershooting a maximum limit or exceeding a minimum limit.

If the check box is not set, the machine maintains its operating state on reaching the limit value: no error display, no signalling.

Speed limits input field (1):

"HIGH":

Input of a speed value that, when reached, triggers a warning message.

"HIGH" or "HOCH" input field (2):

Input of the limit value to be monitored.

**NOTICE****Alarm signalling**

On reaching the value, a warning message is output if the alarm box is checked. This remains present until the limit value is undershot. In this case, an optional yellow/orange signal lamp lights up.

The machine continues in normal operation.

Filter time input field (3):

Through the description of this field, a time screen is specified within which the PLC fluctuations from a set value are tolerated.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

3.11.11.3.2 Drive - motor drive

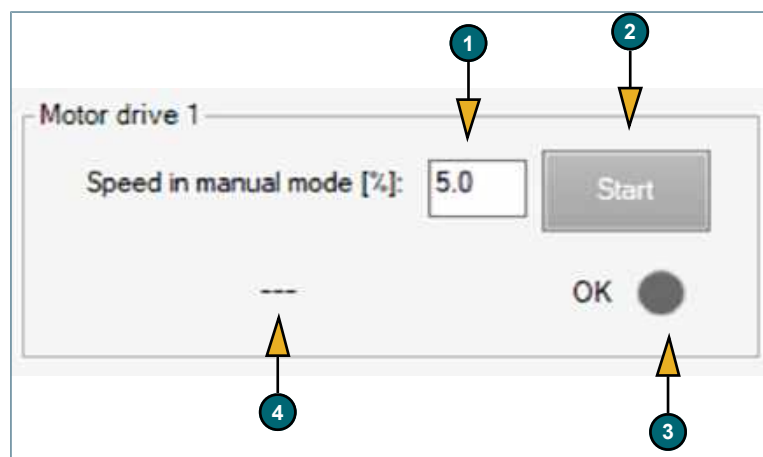


Illustration 109: Motor drive 1

1 "Speed manual" input field	2 < Start > short-cut key
3 "OK" indicator LED	4 Display field

Application

Switching the pump drive on or off manually.

"Speed manual" input field (1):

Input of the motor speed when starting the pump manually.

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
---	---

	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.
---	--

< Start > short-cut key (2):

The pump drive is started with the speed input or running drive is stopped.

	NOTICE Login authorisation function execution Only an authorised user login can execute functions.
---	---



NOTICE

Hand button

Do not use this key to convey material. It is used to **briefly** check the direction of motor rotation after a motor change. Additional info: Certain types of pumps, such as the eccentric screw pump, are not suitable for operation without material, as otherwise the rotor-stator system may be damaged.

"OK" display field (3):

LED lights up green: Drive is ready.

LED is grey: Drive is not ready.

Display field (4):

Status display of the pump drive for diagnostic purposes (see also operating instructions of the drive manufacturer, drive type "ATV 320").

3.11.11.3.3 Drive - tacho monitoring

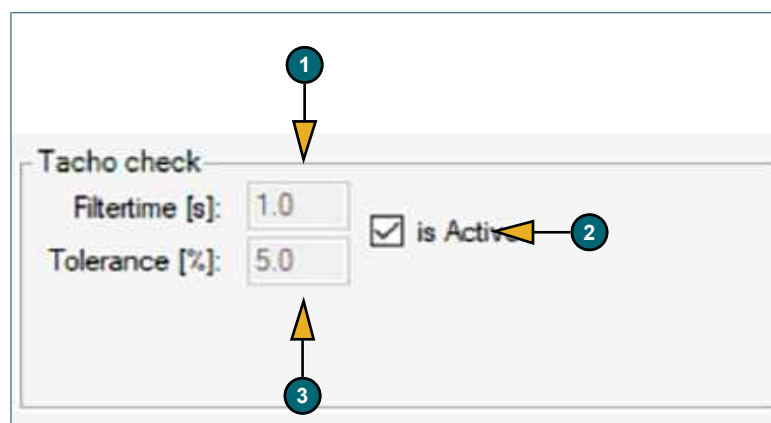


Illustration 110: Tacho monitoring

1 "Filtration time" input field	2 "Active" check box
3 "Tolerance" input field	

Application

Activating tacho monitoring of the drive regulator.

"Filtration time" input field (1):

"Filter time" input field (6):

Time specification with which the querying control unit tolerates fluctuations from a value (e.g. measured value of a physical variable, limit value, etc.). A value is only taken over as valid after this elapses.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

“Active” check box (2):

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

"Tolerance" input field (3):

Input of the permitted deviation from the speed setpoint.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

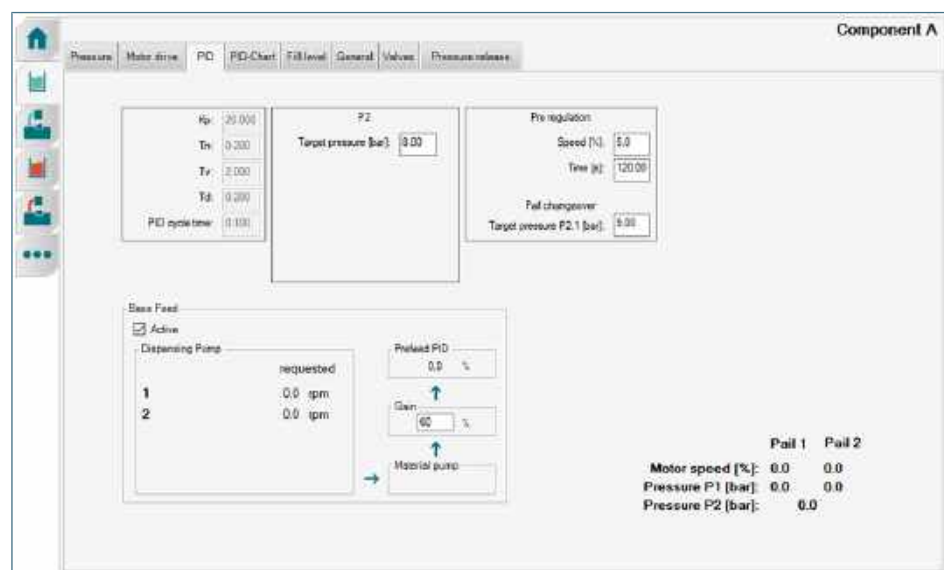
3.11.11.4 Component A - PID controller

Illustration 111: Component A - PID controller

Application

Parametrising PID controller behaviour.



NOTICE

Screens for the components

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.

3.11.11.4.1 PID dialog

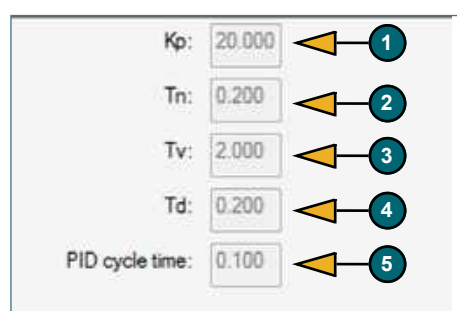


Illustration 112: PID dialog

1	Proportional factor input field	2	Reset time input field
3	Hold-back time input field	4	Differential percentage input field
5	Control setting time input field		

Application:

Parameterisation of a PID control circuit.

3.11.11.4.2 PID regulator - target pressure

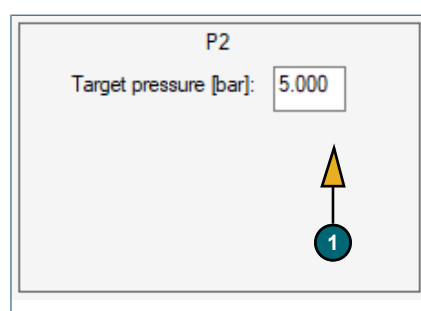


Illustration 113: PID regulator - target pressure

1	"Target pressure" input field
---	-------------------------------

Application

Input of the pressure to be reached.

“Target pressure” input field (1):

Input of the pressure to which the PID regulator regulates before the input (pressure sensor P2) of the dispensing pump.

The service technician running in the machine will determine this on the basis of the customer's specifications, material specifics and dispensing application.

3.11.11.4.3 Basic supply

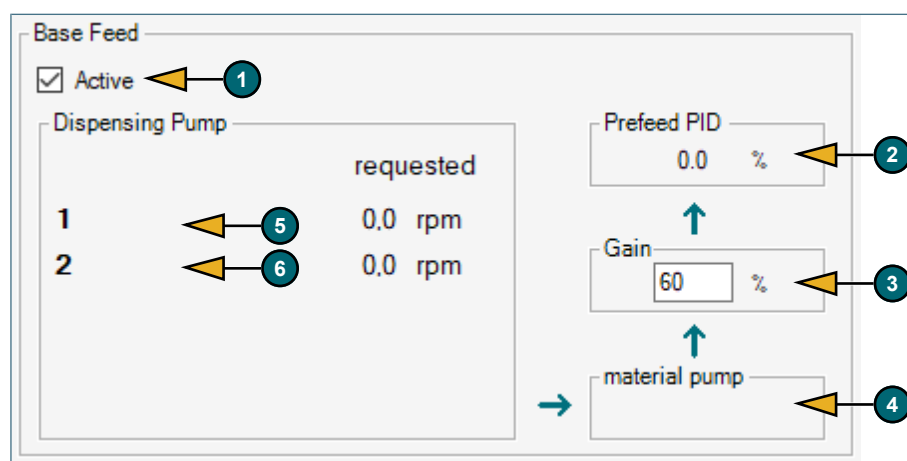


Illustration 114: Basic supply

1 "Active" check box	2 "PID pilot feed" display field
3 "Gain" input field	4 "Material pump" display field
5 "Dispensing pump" display field	6 "Dispensing pump" display field

Application

Parametrising the basic supply.

It intervenes as soon as the "MatOK" signal is given and a dispensing process is started. In parallel to the PID regulator, an additional volume of material is introduced into the dispensing line in order to minimise the control deviation and to prevent excessive overshooting of the control loop.

"Active" check box (1):

Tick mark set: Basic supply process intervenes in the regulation.

No tick mark: PID regulation runs without basic supply. For certain applications, such as the use of the Minidis pump module, a basic supply is not necessary due to the minimal quantities of dispensing material required. This is because the expected control deviations can be easily compensated for via the PID control algorithm.

"PID pilot feed" display field (2):

Display of the calculated delivery (motor speed in %) of the material pump for the basic supply.

“Gain” input field (3):

This factor determines the percentage of the theoretically dispensed volume (required volume) that is directly conveyed by the material pump as a volume amount. Ideally, it is set in such a way that control deviations are kept as small as possible, but a PID amount remains to be controlled.

**NOTICE****Gain**

This value is determined once by the commissioning engineer of the system on the basis of material and dispensing set-up requirements and must not exceed 95%.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

**NOTICE****Value assignment of the field**

A maximum value of '95' can be entered in this input field!

“Material pump” display field (4):

Display of the preselected type of material pump (e.g. “BG7-S3”).

**NOTICE****Value - info assignment**

The information in this screen section is missing, as it is only entered at runtime of the system!

“Dispensing pump” display field (5):

Display of the preselected type of dispensing pump (e.g. “BG1-S3”). This is the requesting pump that, in the case of dispensing, takes material from the supply line and conveys it directly in the direction of the mixing device or component.

**NOTICE****Value - info assignment**

The information in this screen section is missing, as it is only entered at runtime of the system!

“Dispensing pump” display field (6):

Display of the preselected type of dispensing pump (e.g. “BG1-S3”).

This is only displayed for tandem systems. It is the second requesting pump that, in the case of dispensing, takes material from the supply line and conveys it directly in the direction of the mixing device or component.

**NOTICE****Value - info assignment**

The information in this screen section is missing, as it is only entered at runtime of the system!

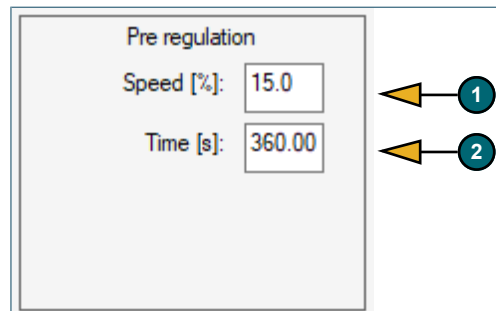
3.11.11.4.4 Pilot control


Illustration 115: Pilot control

1 "Speed" input field	2 "Time" input field
-----------------------	----------------------

Application

Parameter for the pilot control process.

“Speed” input field (1):

When entering automatic material preparation, pilot control starts when the actual pressure is lower than the set pressure. The material pump starts with the entered percentage (here 10%) of the maximum speed.

**NOTICE****Input and execution of functions and parameters.**

Input and further processing of a parameter or function can only be performed with an authorised user login.

"Time" input field (2):

When entering automatic material preparation, pilot control starts when the actual pressure is lower than the set pressure. The pump runs at the entered speed (see 1) for this length of time to reach the set pressure. If it does not reach the set pressure, an error stop is triggered. When the set pressure is reached, the "MatOK" signal gives the enable for dispensing. The PID regulator now operates continuously to balance the pressure in the event of pressure fluctuations due to material removal during the dispensing process.

3.11.11.5 Component A - fill level



Illustration 116: Component A - fill level

Application

Calibration of the pail and fill level detection.



NOTICE

Screens for the components

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.

3.11.11.5.1 Fill level

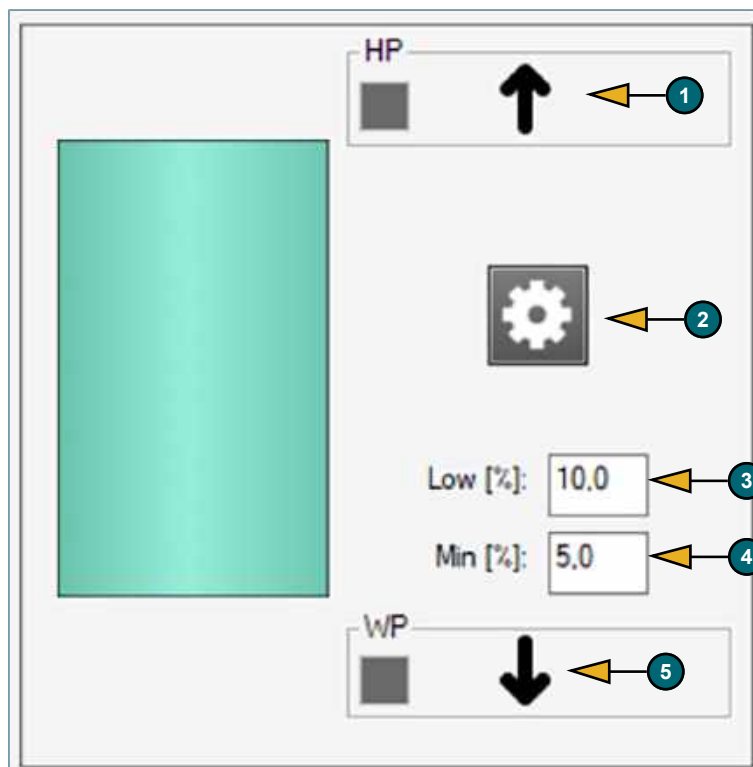


Illustration 117: Fill level

1 "HP" indicator LED	2 Configuration field
3 "Low" input field	4 "Min" input field
5 "WP" indicator LED	

Application

Parametrising fill level limits.

"HP" indicator LED (1):

LED lights up green: Ram press is controlled upwards.

LED is grey: No signal default.

Configuration field (2):

For teaching the container (barrel height, filling height before piercing, empty height).



NOTICE

Login authorisation

Only an authorised user login can change configurations.

"Low" input field (3):

Input of the early warning limit from which an impending barrel change is reported. In the case of a machine configuration with buffer storage, this limit triggers the start of filling.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

"Min" input field (4):

Input of the warning limit after which a barrel change must be carried out, otherwise the machine stops.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

"WP" indicator LED (5):

LED lights up green: Ram press is controlled for downward travel.

LED is grey: No signal input.

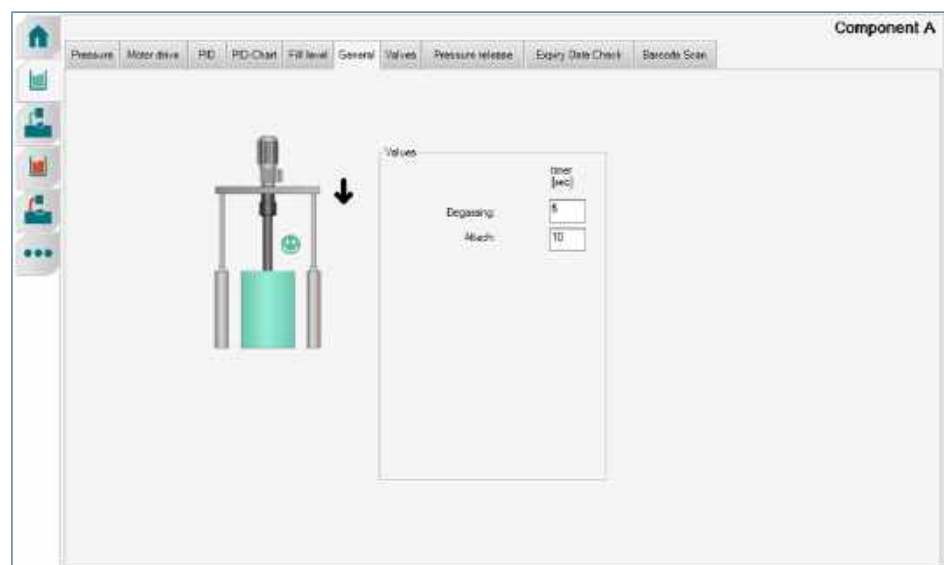
3.11.11.6 Component A - general

Illustration 118: Component A - general

Application

Parametrising procedure for new container piercing, status displays.

**NOTICE****Screens for the components**

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.

**NOTICE****Group field information**

Further detailed descriptions of the group fields follow.

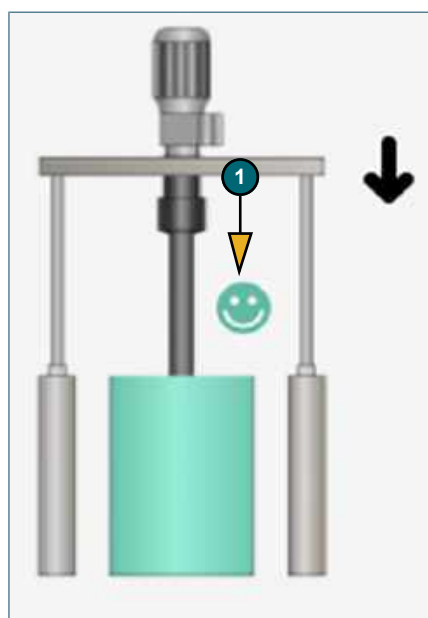
3.11.11.6.1 General - container

Illustration 119: Container - graphic

1	Symbol display field
---	----------------------

Application

Status display.

Symbol display field (1):

Symbol colour green: Ready, i.e. container is prepared and suitable for removal.

Symbol colour red: Undefined, i.e. no dispensing material can be taken from this barrel.

3.11.11.6.2 Values

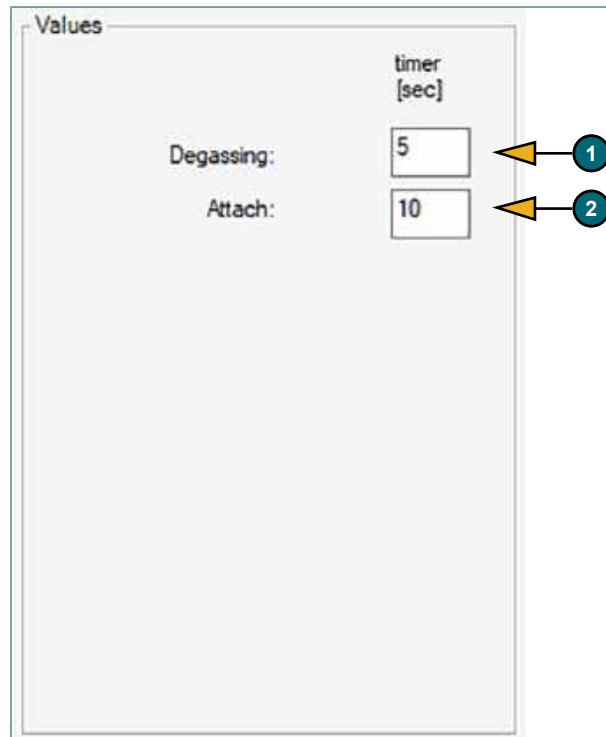


Illustration 120: General - values

1 "Timer Degas" input field	2 "Timer Mount" input field
-------------------------------	-------------------------------

Application

Parametrising the movement of the follow-up plate into the container.

These values are determined **once** by the service technician when the machine is run in. They can be overwritten at any time with the 'Level 4' authorisation. This changes the respective piercing process times!

"Timer | Degas" input field (1):

The follow-up plate touches the fleece previously placed on the material surface. The vacuum pump is running. After the input vacuum setpoint has been reached, evacuation takes place for this period of time, i.e. air is extracted between the follow-up plate and the container surface via valves in the follow-up plate. The position of the follow-up plate is checked.

"Timer | Mount" input field (2):

The follow-up plate remains on the container surface for this length of time.

3.11.11.7 Component A - Valves - Pail

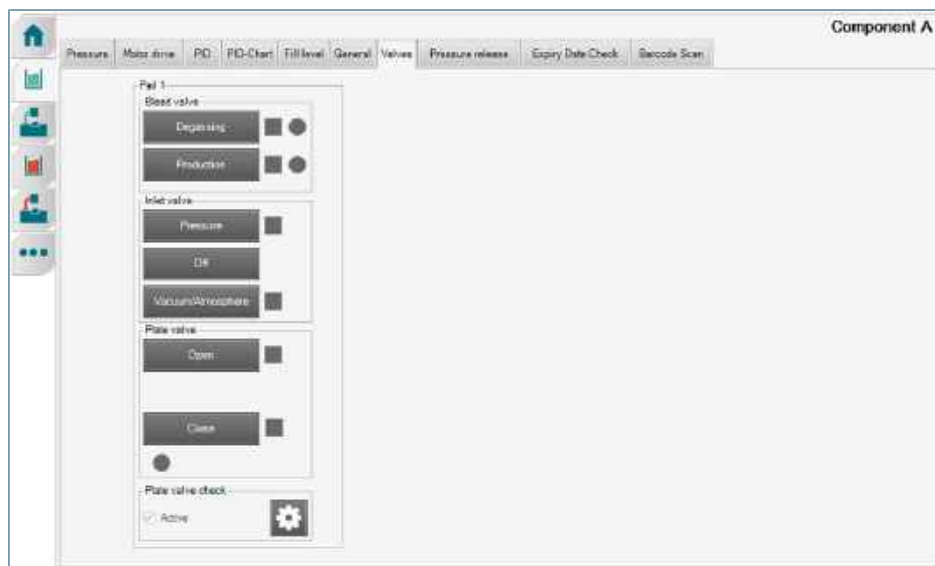


Illustration 121: Comp. A - Valves

Application

Manually operate pneumatic actuators (valves, cylinders), calibrate and check plate valve.



NOTICE

Screens for the components

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.

3.11.11.7.1 Valves - Pail

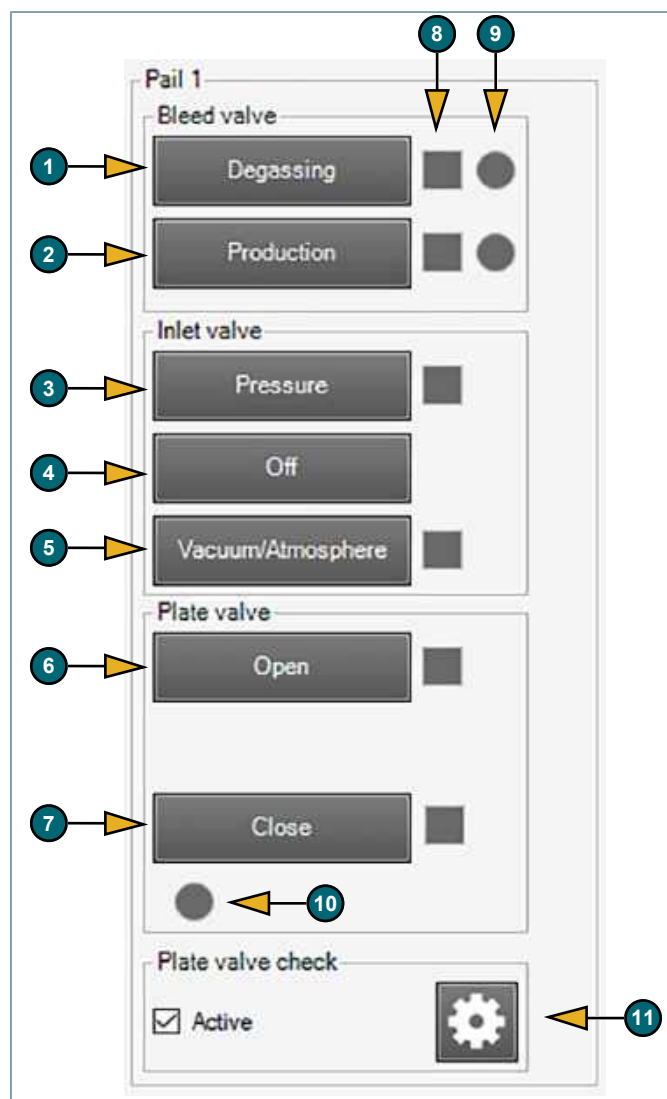


Illustration 122: Comp. A - Valves

1 < Degas > key	2 < Production > key
3 < Pressure > key	4 < Off > key
5 < Vacuum/Atmosphere > key	6 < Open > key
7 < Close > key	8 Indicator LED
9 Indicator LED	10 Indicator LED
11 Symbol key	

Application

Manually operating the valves.



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

< Degas > key (1):

The ball valve switches from the dispensing to the venting position: The material/air mixture can be conveyed into an optional collection container. As a result, the conveyor pump only draws in clean material when the follow-up plate is resting on the container surface.

< Production > key (2):

The ball valve switches from the venting to the dispensing position: Material is conveyed to the dispensing pump.

< Pressure > key (3):

A pneumatic valve switches compressed air to the plate valve or shuts it off. The compressed air reaches the container surface via the downstream plate valve. This function is used when the follow-up plate is moved out.

< Off > key (4):

The pneumatic valve switches to the centre position and locks.

< Vacuum/Atmosphere > key (5):

"Vacuum": A pneumatic valve switches the access to the suction line of the evacuation pump. The container surface is drawn off via the downstream plate valve. This function is used when piercing the new container.

"Atmosphere": The same pneumatic valve switches atmospheric pressure.

< Open > key (6):

The follow-up plate valve is opened and switches the pressure status (pressure, atmosphere or vacuum) that is present from the inlet valve, to the container surface.

< Close > key (7):

The follow-up plate valve is closed and shuts off the medium (pressure, atmosphere or vacuum) that is present from the inlet valve.

Indicator LED (8):

LED lights up green: Valve is activated.

LED is grey: Valve is not activated.

Indicator LED (9):

LED lights up green: Valve is activated.

LED is grey: Valve is not activated.

Indicator LED (10):

LED lights up green: Valve is in end position.

LED is grey: Valve is not in end position.

Symbol key (11):

"Plate check dialogue" is opened.

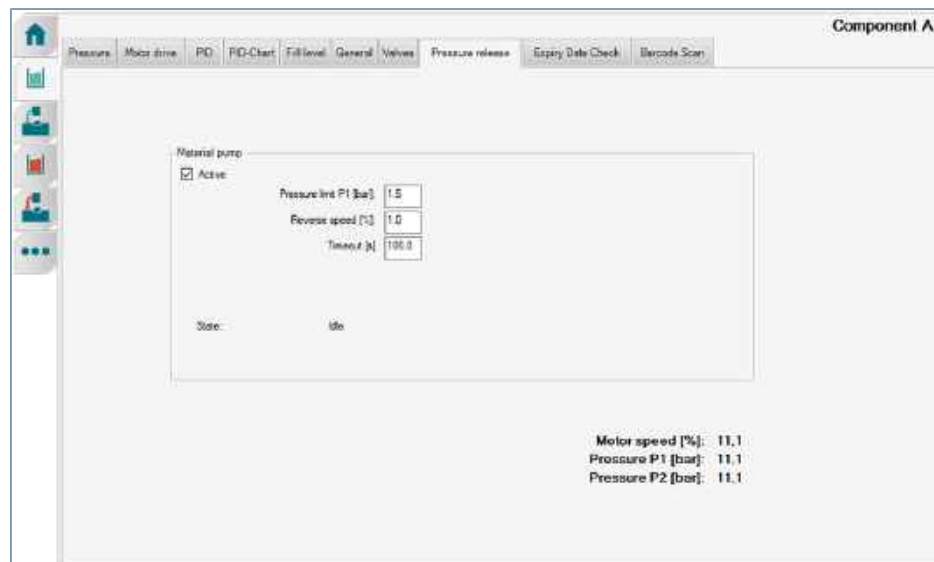
3.11.11.8 Component A - Pressure relief via material pump

Illustration 123: Pressure relief - material pump

Application

Setting the pressure limit for the pressure relief function.

**NOTICE****Screens for the components**

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.

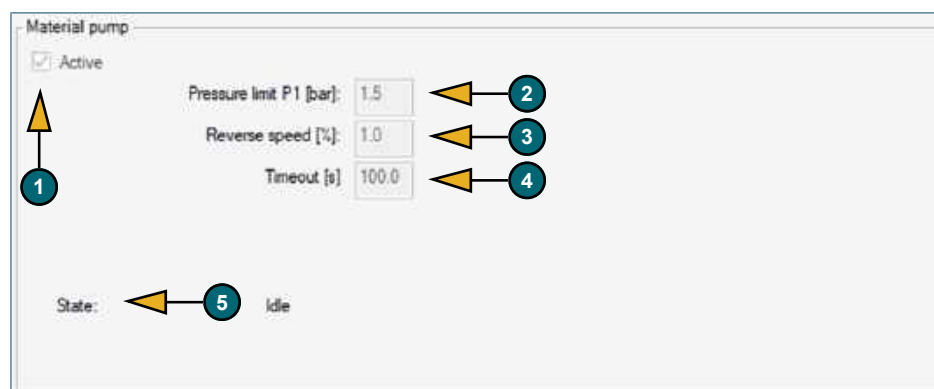
3.11.11.8.1 Pressure relief | via material pump

Illustration 124: Material pump

1 "Active" check box	2 "Pressure limit P1" input field
3 "Reverse speed" input field	4 "Timeout" input field

5	"Status" display field		
---	------------------------	--	--

Application

Parameterisation of the pressure relief via the material pump of the PPS.

"Active" check box (1):

Tick mark set: Pressure relief is possible.

No tick mark: No pressure relief possible.

"Pressure limit P1" input field (2):

Input of the pressure that must be reached during pressure relief.

"Reverse speed" input field (3):

The material pump runs in reverse at this speed until the pressure limit (see 2) is reached within a set time (see also 3).

"Time limit" input field (4):

The set pressure must be reached within this time, otherwise cancellation due to error.

"Status" display field (5):

Display of the statuses signalled by the drive amplifier.

3.11.11.9 Component A evacuation - vacuum pump

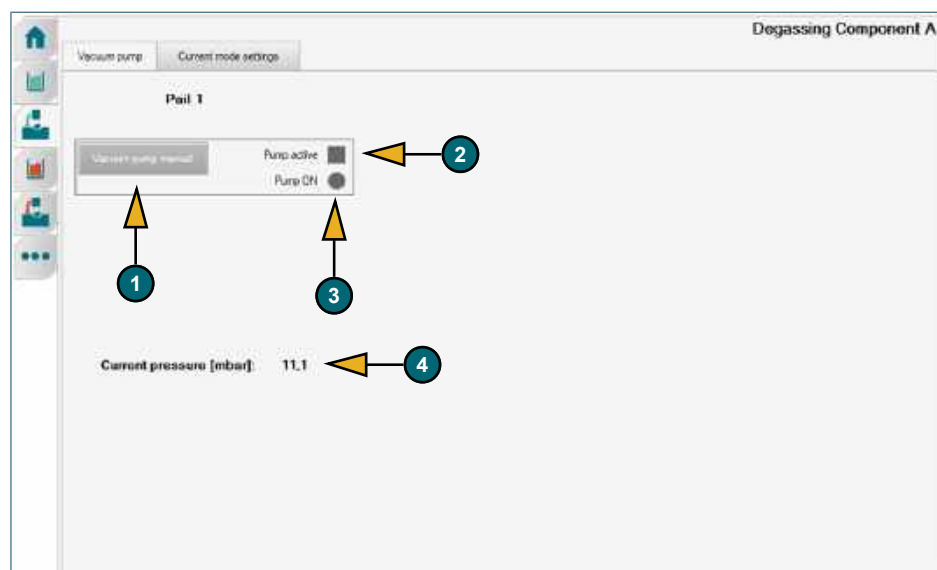


Illustration 125: Component A evacuation - vacuum pump

1	< Vacuum pump manual > button	2	"Pump active" display field
3	"Pump on" display field	4	Display field

Application

Display of a value for a configuration as a single solution.

**NOTICE****Screens for the components**

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.

< Vacuum pump manual > key (1):

Switches the vacuum pump on or off manually.

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

"Pump active" display field (2):

LED lights up green: Pump is activated.

LED is grey: Pump is not activated.

"Pump on" display field (3):

LED lights up green: Pump has been switched on.

LED is grey: Pump is off.

Display field (4):

Actual pressure display from pail 1.

3.11.11.10 Component A evacuation - Current setting Affects

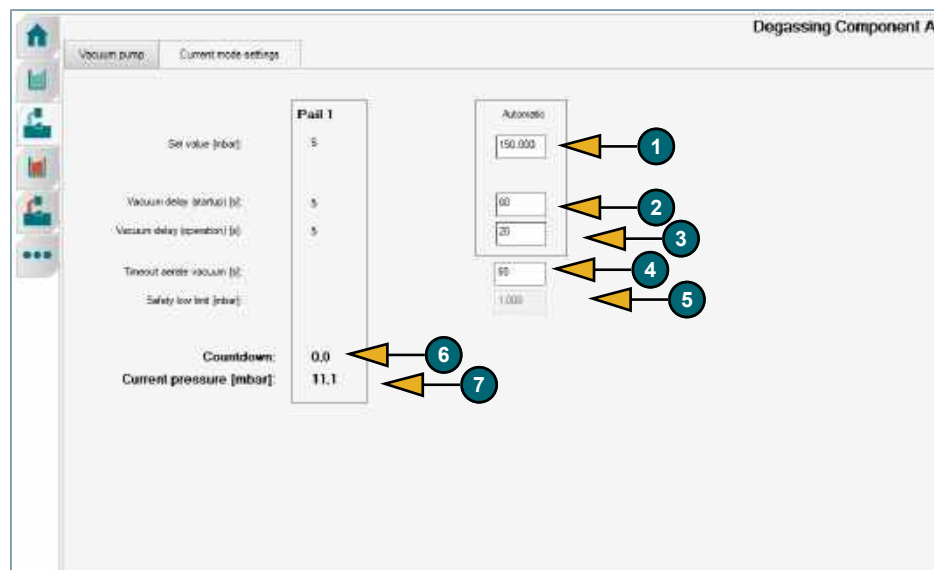


Illustration 126: Component A evacuation - Current setting Affects

1	"Automatic set value" input field	2	"Automatic vacuum delay (start-up)" input field
3	"Automatic vacuum delay (operation)" input field	4	"Time-out ventilation valve" input field
5	"Safety min limit" input field	6	"Countdown" display field
7	"Actual pressure" display field		

Application:

Parameterisation of the evacuation process for the material preparation phase.



NOTICE

Screens for the components

The screens for material components A and B are identical in content. For reasons of reducing the amount of reading, only the image for the A-component is shown.



NOTICE

Input field

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

"Automatic set value" input field (1):

Input of the set point for the vacuum regulation in the AUTOMATIC operating mode: The vacuum circuit regulates to this set point.

"Automatic vacuum delay (start-up)" input field (2):

Input of a time within which the permissible fluctuation of the vacuum value on start-up of the evacuation flow in AUTOMATIC operation does not cause an error message or cancellation.

"Automatic vacuum delay (operation)" input field (3):

Input of a time within which the permissible fluctuation of the vacuum value during ongoing evacuation process during AUTOMATIC operation does not cause an error message or cancellation.

"Ventilation time timeout" input field (4):

Input of a time within which the atmospheric pressure must be reached; otherwise cancellation due to error.

"Safety lower limit" input field (5):

Input of a lower limit value for the vacuum regulation: an dropping below this limit causes a cancellation due to error.

"Countdown" display field (6):

Display of the remaining ventilation time.

"Actual pressure" display field (7):

Display of the current pressure values in the evacuation circuit supplied by the vacuum sensor.

3.11.11.11 Configuration

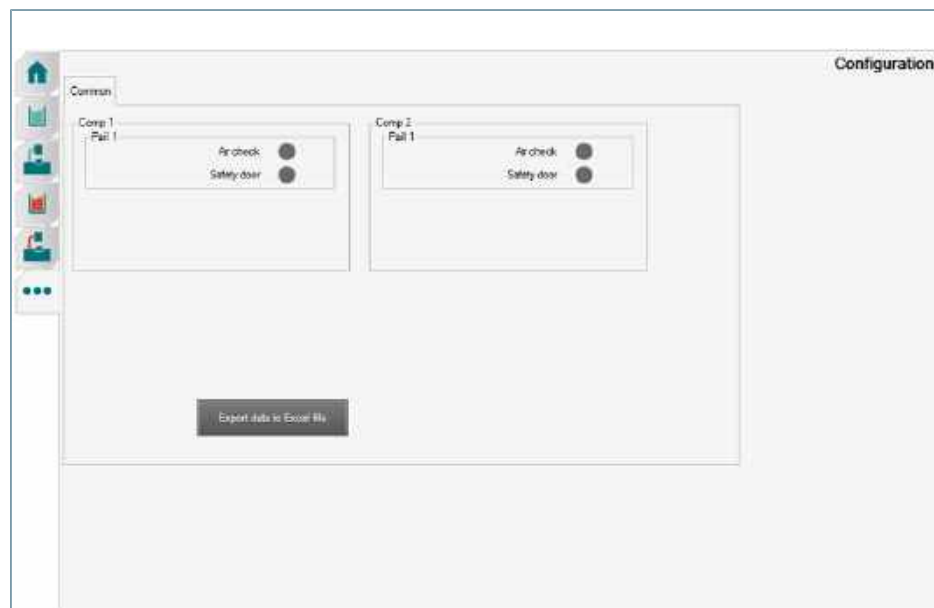


Illustration 127: Configuration

Application

Display of general machine states.

3.11.11.11.1 General - Component 1



Illustration 128: Configuration - Pail 1

1 "Compressed air monitoring" indicator LED	2 "Safety door" indicator LED
---	-------------------------------

Application

Status indicators.

"Compressed air monitoring" indicator LED (1):

LED lights up green: Pail 1 is under compressed air.

LED is grey: Compressed air is missing or too low.

"Safety door" indicator LED (2):

LED lights up green: Safety circuit for pail 1 is OK; access door is closed.

LED is grey: Safety circuit interrupted, door is open.

3.11.11.11.2 General - Component 2

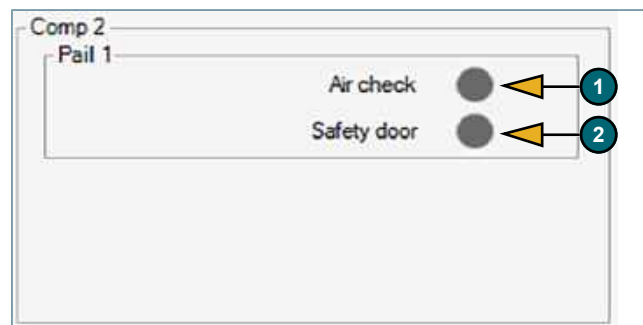


Illustration 129: Configuration - Pail 2

1 "Compressed air monitoring" indicator LED	2 "Safety door" indicator LED
---	-------------------------------

Application

Status indicators.

"Compressed air monitoring" indicator LED (1):

LED lights up green: Pail 1 is under compressed air.

LED is grey: Compressed air is missing or too low.

"Safety door" indicator LED (2):

LED lights up green: Safety circuit for pail 1 is OK; access door is closed.

LED is grey: Safety circuit interrupted, door is open.

3.11.11.12 "Barcode scan" dialogue

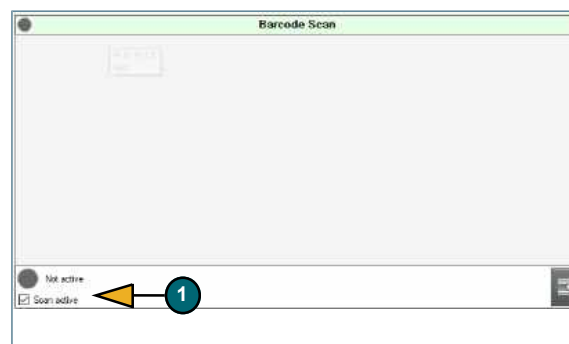


NOTICE

Input and execution of functions and parameters.

Input and further processing of a parameter or function can only be performed with an authorised user login.

3.11.11.12.1 Activating the scan procedure



1 Check box		
-------------	--	--

Check box (1):

Tick mark set: Barcode scanning is active.

No tick mark: Barcode scanning is inactive.

3.11.11.12.2 LED status indicator

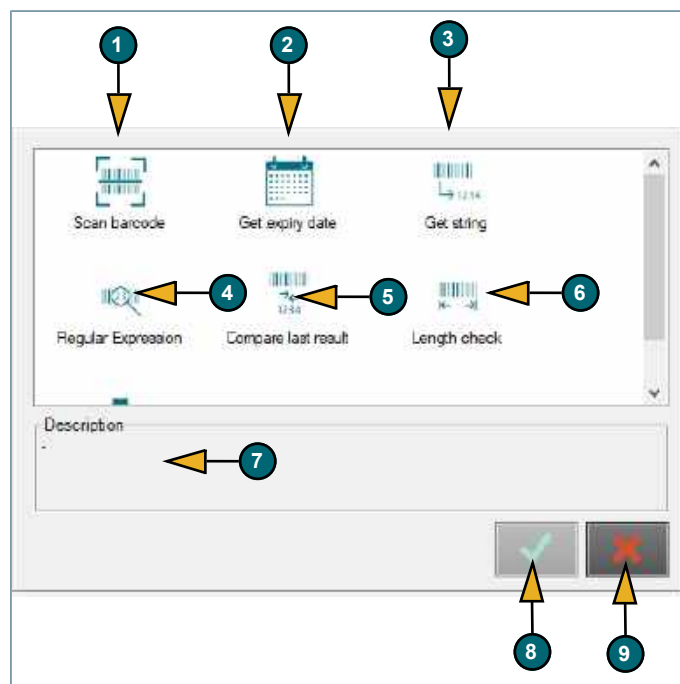
LED	Description
grey	Scan procedure is deactivated.
grey yellow	Flashing LED Scan procedure is ready The operator can start scanning.
yellow	The barcode is scanned NOTICE! With longer barcodes, this may take several seconds.
green	The scan procedure is finished and is OK.
red	The scan procedure is faulty.

3.11.11.12.3 Creating a scan procedure

1	Symbol key		
---	------------	--	--

Symbol key (1):

Pressing the symbol key opens the "Search folder" submenu.



1 "Scan barcode" symbol selection	2 "Read out expiry date" symbol selection
3 "Read out text" symbol selection	4 "Regular expression" symbol selection
5 "Compare last result" symbol selection	6 "Length check" symbol selection
7 Display field	8 Symbol key
9 Symbol key	

"Scan barcode" symbol selection (1):

The operator must scan a barcode.

"Read out expiry date" symbol selection (2):

The expiry date is read out of the barcode scanned last.

This is performed using configurable search criteria.

"Read out text" symbol selection (3):

A specific part of the last scanned barcode is read out.

This is performed using configurable search criteria.

"Regular expression" symbol selection (4):

A specific part of the last scanned barcode is read out.

This is performed using a configurable search pattern.

“Compare last result” symbol selection (5):

The last result is compared with a text.

“Length check” symbol selection (6):

The length of the last result is checked.

Display field (7):

Description of the respective search criterion.

Symbol key (8):

By pressing the symbol key, the selected scan module is added to the scan process list.

Symbol key (9):

Pressing the symbol key closes the submenu for the scan selection.

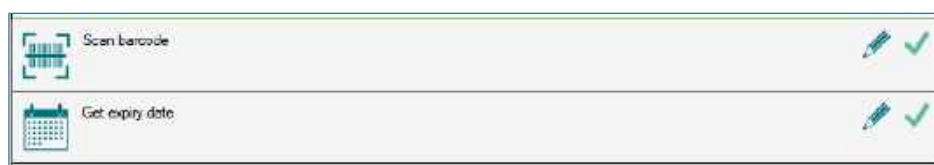
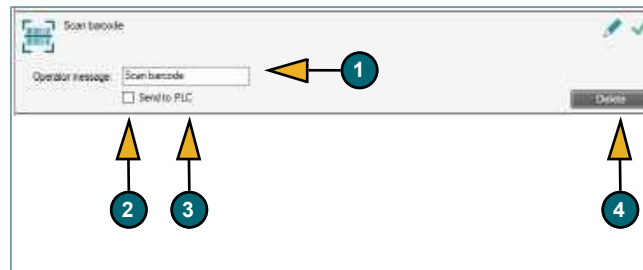
3.11.11.12.4 Edit scan modules

Illustration 130: Example

Symbol	Description
	Edit module. Clicking the Settings menu expands and collapses it.
	The module is activated and is run through during the scan procedure. Clicking on the module deactivates it.
	The module is deactivated and is ignored during the scan procedure. Clicking on the module activates it.

3.11.11.12.5 Barcode scanning dialogue



1	"Operator message" input field	2	Menu selection
3	Check box	4	Symbol key

"Operator message" input field (1):

The entered text is displayed next to the symbol and is also output as an information message for the operator as soon as the module is run through.

Menu selection (2):

If the barcode is to be sent to the PLC, the corresponding barcode type must be selected. **NOTICE! Each type may only be selected once per component.**

Check box (3):

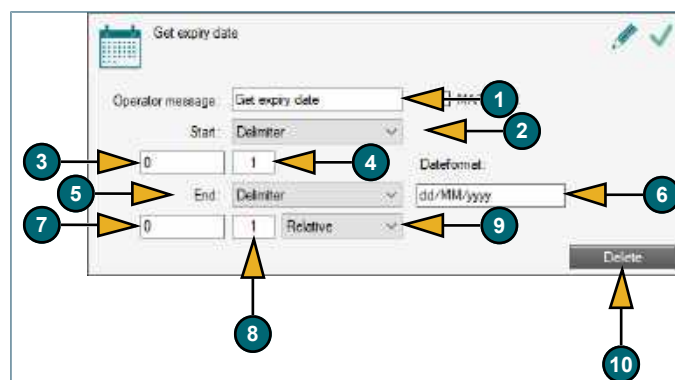
Tick mark set: The scanned barcode is sent to the PLC to forward it via a handshake/MES.

No tick mark: The barcode is not sent to the PLC.

Symbol key (4):

The scan module is removed from the list by pressing the symbol key.

3.11.11.12.6 Read out expiry date dialogue



1	"Operator message" input field	2	Menu selection field
3	"Start" input field	4	"Position point" input field
5	Menu selection field	6	"Date format" input field

7	"End" input field	8	"Position point" input field
9	Menu selection field	10	Symbol key

"Operator message" input field (1):

The entered text is displayed next to the symbol and is also output as an information message for the operator as soon as the module is run through.

Menu selection field (2):

Selection of the corresponding separator for processing the date input.

"Start" input field (3):

Input at the beginning of the expiry date.

'Position point' input field (4):

Input at which the position of the expiry date begins in the barcode.

Menu selection field (5):

Selection of the corresponding separator for processing the date input.

"Date format" input field (6):

Input of the date format.

Example: dd/MM/yyyy = 24/01/2020

dd = day, two digits

MM = month, two digits

yyyy = year, four digits

"End" input field (7):

Input for the end of the expiry date.

'Position point' input field (8):

Input at which the position of the expiry date ends in the barcode.

Menu selection field (9):

Selection of the corresponding procedure for processing the barcode.

Symbol key (10):

The scan module is removed from the list by pressing the symbol key.

Example 1: Separator – separator (absolute)

Operator message: Ablaufdatum auslesen	The expiry date begins after the second "@" of the barcode.
Start: Delimiter	The expiry date ends before the third "@" of the barcode.
@ 2	
End: Delimiter	
@ 3 Absolute	
Barcode	@123@24/12/2020@56789@
Start	Second "@"
End	Third "@"
Date format	dd/MM/yyyy
Result	Day (dd) = 24 Month (MM) = 12 Year (yyyy) = 2020

Example 2: Separator – separator (relative)

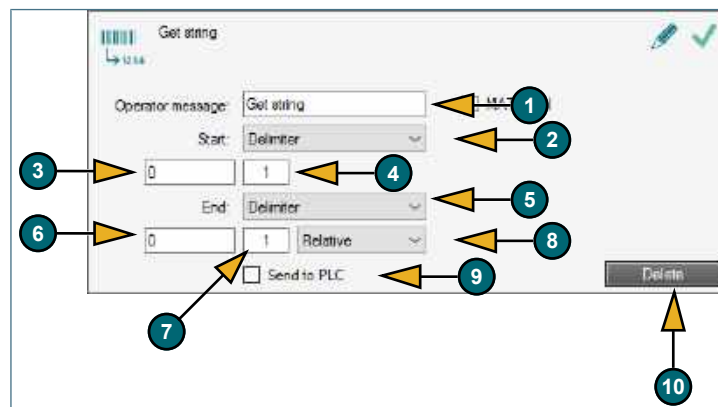
Operator message: Ablaufdatum auslesen	The expiry date begins after the second "@" of the barcode.
Start: Position	The expiry date ends before the first "@" after the start position.
6	
End: Position	
10 Relative	
Barcode	@123@12242020@56789@
Start	Second "@"
End	First "@" after the start position = 12242020@56789@
Date format	MMddyyyy
Result	Day (dd) = 24 Month (MM) = 12 Year (yyyy) = 2020

Example 3: Position – number

Operator message: Get expiry date	The expiry data begins at the 6th position of the barcode.
Start: Position	
6	
End: Count	The expiry date has a length of 10 digits.
10	

Barcode	@123@12-24-2020@56789@
Start	6th place
End	10th place from the start position
Date format	MM-dd-yyyy
Result	Day (dd) = 24 Month (MM) = 12 Year (yyyy) = 2020

3.11.11.12.7 Read out the text dialogue



1 "Operator message" input field	2 Menu selection field
3 "Start" input field	4 "Position point" input field
5 Menu selection field	6 "End" input field
7 "Position point" input field	8 Menu selection field
9 Check box	10 Symbol key

"Operator message" input field (1):

The entered text is displayed next to the symbol and is also output as an information message for the operator as soon as the module is run through.

Menu selection field (2):

Selection of the corresponding separator for processing the text input.

"Start" input field (3):

Input at the beginning of the text.

'Position point' input field (4):

Input indicating the position at which the text that begins the barcode.

Menu selection field (5):

Selection of the corresponding separator for processing the text input.

“End” input field (6):

Input for the end of the text.

'Position point' input field (7):

Input that indicates the position where the text ends in the barcode.

Menu selection field (8):

Selection of the corresponding procedure for processing the barcode.

Check box (9):

Tick mark set: The scanned barcode is sent to the PLC to forward it via a handshake/MES.

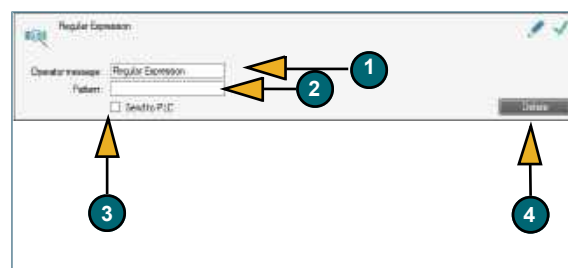
No tick mark: The barcode is not sent to the PLC.

Symbol key (10):

The scan module is removed from the list by pressing the symbol key.

**NOTICE****Example of reading out text:**

Reading out of the text operates according to the same principle as reading out an expiry date. See the examples below this dialogue. **Note!** The information on the format is lost when the text is read out.

3.11.11.12.8 Regular expression dialogue

1	“Operator message” input field	2	“Pattern” input field
3	Check box	4	Symbol key

“Operator message” input field (1):

The entered text is displayed next to the symbol and is also output as an information message for the operator as soon as the module is run through.

“Pattern” input field (2):

Input of a certain pattern (text position) from the barcode.

Check box (3):

Tick mark set: The scanned barcode is sent to the PLC to forward it via a handshake/MES.

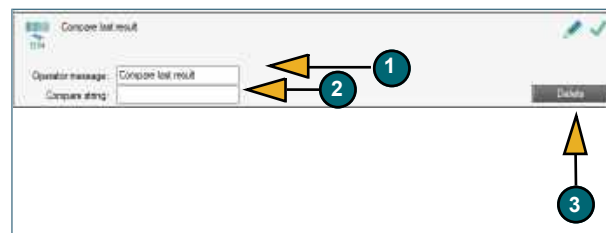
No tick mark: The barcode is not sent to the PLC.

Symbol key (4):

The scan module is removed from the list by pressing the symbol key.

Example:

Pattern:	12(.*)23	The result of the search pattern is the text that lies between "12" and "23".
Barcode	@7@12PartA23@24102020@	
Pattern	12(.*)23	
Result	Part A	

3.11.11.12.9 Comparison of last result dialogue

1	"Operator message" input field	2	"Comparative text" input field
3	Symbol key		

"Operator message" input field (1):

The entered text is displayed next to the symbol and is also output as an information message for the operator as soon as the module is run through.

"Comparative text" input field (2):

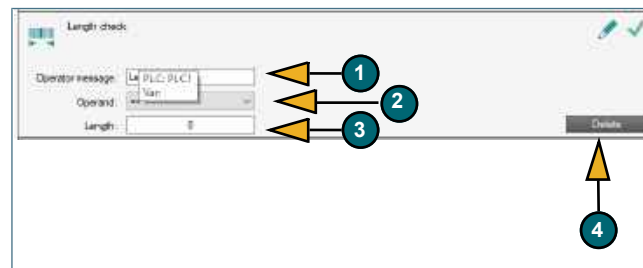
Input of the comparative text.

Symbol key (3):

The scan module is removed from the list by pressing the symbol key.

Example:

Barcode "OK"	Barcode "NOK"
Barcode: COMP 1	Barcode: COMP 2
Comparative text: COMP 1	Comparative text: COMP 1
Result: OK	Result: NOK

3.11.11.12.10 Length check dialogue

1 "Operator message" input field	2 Menu selection
3 "Length" input field	4 Symbol key

"Operator message" input field (1):

The entered text is displayed next to the symbol and is also output as an information message for the operator as soon as the module is run through.

Menu selection (2):

Selection of the operands.

"Length" input field (3):

Input of the length for checking the operands.

Symbol key (4):

The scan module is removed from the list by pressing the symbol key.

Example:

Barcode "OK"	Barcode "NOK"
Barcode: 0123456789	Barcode: 0123456789
Operand: "=="	Operand: ">"
Longer: 10	Longer: 10
Result: OK	Result: NOK

3.11.12 Transport

3.11.12.1 Rotary table

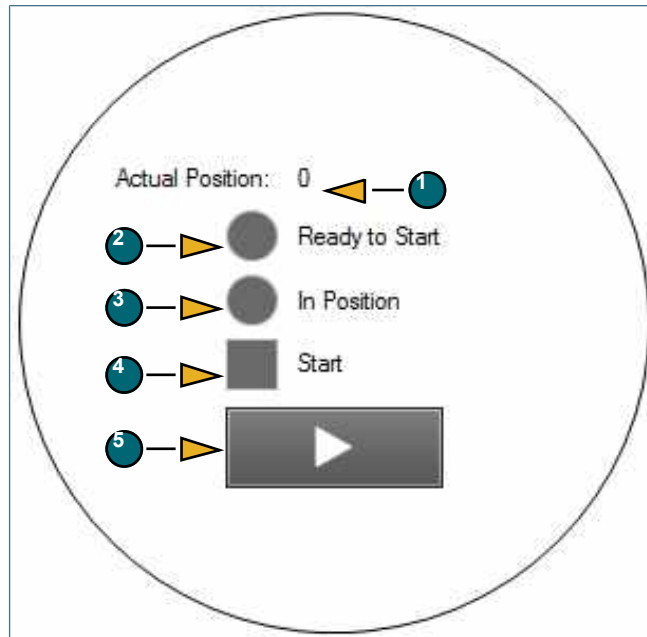


Illustration 131: Rotary table

1 "Act. Position"	2 "Ready to start" indicator LED
3 "In Position" indicator LED	4 "Start" indicator LED
5 <Start> quick key	

Application

Activate rotary table manually.

"Act. Position" (1):

Display of the position number on the rotary table.

"Ready to start" indicator LED (2):

Green LED: Voltage supplied to rotary table.

Grey LED: No power supply.

"In Position" indicator LED (3):

Green LED: Rotary table has reached position.

"Start" indicator LED (4):

Green LED: Rotary table turning.

Grey LED: No rotary movement.

<Start> quick key:

Rotary movement is triggered manually with this key.

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

**NOTICE****Stop by operating intervention**

Rotary table stops immediately if the light curtain is broken during its rotary movement. To continue moving, a new safety position (such as the weighing position, the start or blind shot position) must first be moved to manually via the “Axes” screen. The interrupted movement process of the rotary plate can then be completed using the <Start> quick key.

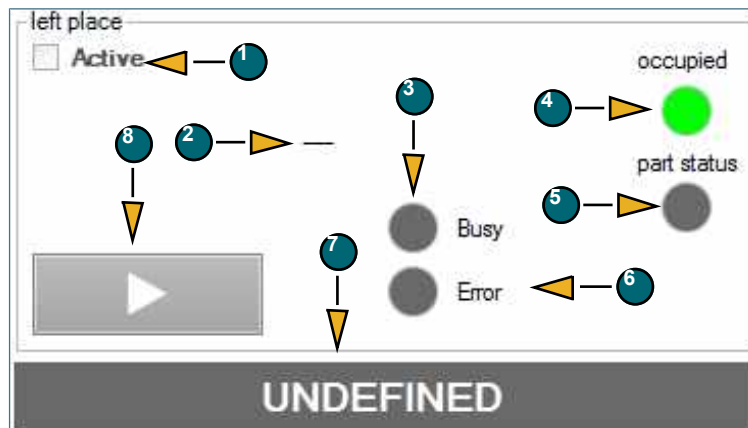
3.11.12.2 Workplace, left

Illustration 132: Workplace, left

1	“Active” checkbox	2	Text display field
3	“Busy” indicator LED	4	“Occupied” indicator LED
5	“Part status” indicator LED	6	“Error” indicator LED
7	Display field	8	<Start> quick key

Application

Activate workplace, scan part manually.

“Active” checkbox (1):

Tick mark set: Workplace can be used.

No tick mark: Workplace is locked.

Text display field (2):

Display of decoded scan information.

“Busy” indicator LED (3):

Green LED: Scan process running.

Grey LED: No scan process started.

“Occupied” indicator LED (4):

Green LED: Part has been detected.

Grey LED: No part detected.

“Part status” indicator LED (5):

Green LED: No error detected, part properly dispensed.

Red LED: Error state, part processing failed.

Yellow LED: Part being processed.

“Error” indicator LED (6):

Red LED: Scan error.

Grey LED: Scan error-free.

Display field (7):

Information to operator:

“Undefined”: Background colour grey: Part place is not occupied.

“OK”: Background colour green: Process was in order.

“NOK”: Background colour red: Process failed.

“Bypass”: Background colour yellow.

<Start> quick key:

Trigger scan process manually.

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

3.11.13 Axis dialogues

Coordinate input

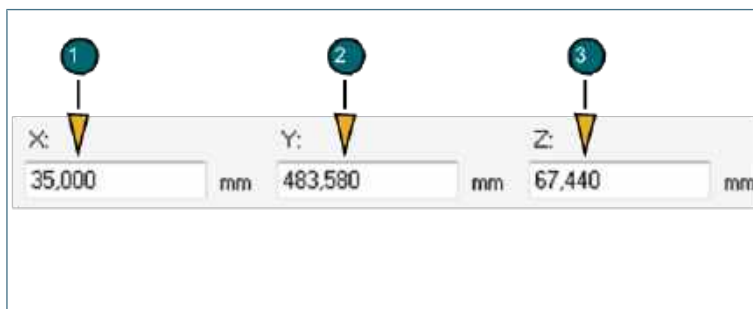


Illustration 133: Coordinate line

1	X-coordinates	2	Y-coordinates
3	Z-coordinates		

Task

Input possibility for coordinates and display of manual and automatically moved-to coordinates of the axis system.



NOTICE

Login authorisation and operation mode

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.



NOTICE

Input field

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

X-coordinate (1):

Input or display of the coordinates of the X axis moved to

X-coordinate (2):

Input or display of the coordinates of the Y axis moved to

X-coordinate (3):

Input or display of the coordinates of the Z axis moved to

Inching function

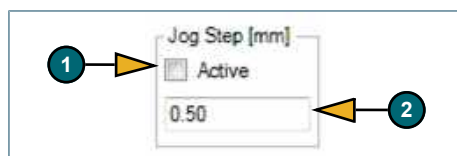


Illustration 134: Inching function

1	Inching function active	2	Step size
---	-------------------------	---	-----------

Task

Settings for inching function



NOTICE

Login authorisation and operation mode

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.

Inching function active (1):

Set tick mark:

Inching function is active: Axes move with limited speed; only the set step size is moved per button-press.

No tick mark:

Normal axis operation

Inching speed



Illustration 135: Inching speed

1	Speed regulator	2	Speed indicator
---	-----------------	---	-----------------

Task

Set speed for the axis moving in inching function.



NOTICE

Login authorisation and operation mode

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.

Speed regulator (1):

Regulator to the left:

Speed is reduced.

Regulator to the right:

Speed increases up to the limit.

Speed indicator (2):

Display of the speed in percent of maximum speed.

Override

Illustration 136: Override

1	Speed indicator	2	Speed controller
---	-----------------	---	------------------

Task

With this dialog, you can limit the maximum speed of the drive system.

**NOTICE****Login authorisation and operation mode**

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.

Speed indicator (1):

Numeric value displays the current setting as a % of the end value.

Speed regulator (2):

By dragging the controller symbol to the right, the speed is increased; dragging it to the left reduces the speed.

Function keys and other operating elements:

Start reference run



Move to the preset reference point

Designation of the symbol: < *Reference run* >



NOTICE

Process monitoring and automatic operation modes!

This function or sequence can only be executed in process monitoring or automatic operation mode!

Application:

1. Press the < *Reference run* > button

⇒ The axle system moves to the specified reference point

⇒ If the reference cycle can be ended, all other – even manual – axis moves are triggered.

Manually moving driving axis X-



Starts movement of the (X-) axis

Designation of the symbol: < *X- Hand* >



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the X-coordinate cell

Manually moving driving axis X+



Starts movement of the (X+) axis

Designation of the symbol: < *X+ Hand* >

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the X-coordinate cell

Manually moving driving axis Y-

Starts movement of the (Y-) axis

Designation of the symbol: < Y- Hand >

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the Y-coordinate cell

Manually moving driving axis Y+

Starts movement of the (Y+) axis

Designation of the symbol: < Y+ Hand >

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the Y-coordinate cell

Manually moving driving axis Z-

Starts movement of the (Z-) axis

Designation of the symbol: < Z- Hand >



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the Z-coordinate cell

Manually moving driving axis Z+



Starts movement of the (Z+) axis

Designation of the symbol: < Z+ Hand >



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

Application:

1. Press the button until the desired position or drive distance is reached

⇒ Coordinates of the end position are displayed in the Z-coordinate cell

Start button



The flow (e.g. drive program, action list, manual process) is started if all start conditions are fulfilled.

Designation of the symbol: < Start >



NOTICE

Login authorisation and operation mode

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.

Application:

- ✓ Possibly fulfil additional prerequisites

1. Press the < Start > button

⇒ The process (e.g. weighing cycle, etc.) or program execution (e.g. run through drive program) and processing of an action list is started

⇒ The process ends by itself or with the program or action list end

3.11.13.1 Management of servo axis positions

Servo-driven, 3-coordinate axis systems can reach each point within their allowed movement space X/Y/Z with a specified speed and acceleration component. Using predefined positions, i.e. points with fixed X/Y/Z coordinates that you determine when setting up the machine, the same coordinate triple (= point) of the movement area will always be addressed. For example, a “blind shot position” contains the coordinates for the point over a beaker or tube where the quantity of specified blind shots are dosed in at the end of the pot time. Each task can have multiple positions. These can also be subsequently adjusted.

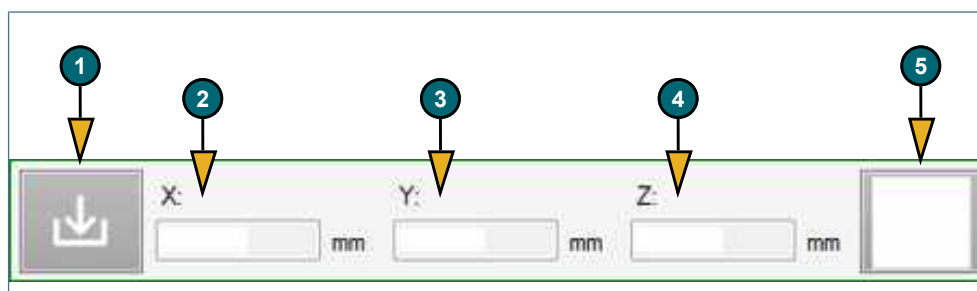


Illustration 137: Neutral position dialogue

1 < Confirm position > quick key	2 Input combo box “X”
3 Input combo box “Y”	4 Input combo box “Z”
5 < Move to position > quick key	

Application

Definition of the positions for axle movements and manual triggering of a positioning movement.



NOTICE

Position description

The “local” assignment of a position is done either by means of a name in a frame around the position line or through a specially-created combination symbol (e.g. the scales icon for weighing position with a small triangle-to-the-right icon) for the triggering of the position travel.

< Confirm position > quick key (1):

Key pressed: The “X” and “Y”, “Z” coordinates are confirmed as a position by the next system start.

	NOTICE Save changes. Only by pressing the save button are the changes taken over permanently into the system memory: when the system is restarted, the changes are available again (info: only an authorised user can start the saving process)
---	--

“X” input combo field (2):

Input or display of the desired X coordinates is done by means of the manual axis move button or via automated system-measured coordinates.

	NOTICE Input field This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1"). The input limits are specified by the software.
---	---

	NOTICE Login authorisation parameter change! Only an authorised logged in user can change parameters.
--	--

Input combo box “Y” (3):

Input or display of the desired Y coordinates is done by means of the manual axis move button or via automated system-measured coordinates.

Input combo box “Z” (4):

Input or display of the desired Z coordinates is done by means of the manual axis move button.

< Move to position > quick key (5):

Key pressed: Move the axes to the position for the stored coordinates “X” “Y” “Z”.

	NOTICE Symbols Depending on the position (e.g., setup, weighing position, etc.), a special dual-symbol combination of is used. The triangle shown below and to the right of the icon on the right always indicates the position functionality, i.e. it is the same for all combinations.
---	---



< *Blind shot position* > quick key

Moving to the blind shot position

	< <i>Purge position</i> > quick key	Moving to the purge position
	< <i>Weighing position</i> > quick key	Moving to the weighing position
	< <i>Service position</i> > quick key	Moving to the service position
	< <i>Automatic position</i> > quick key	Moving to the automatic start position

3.11.13.2 Manual operation of the axes

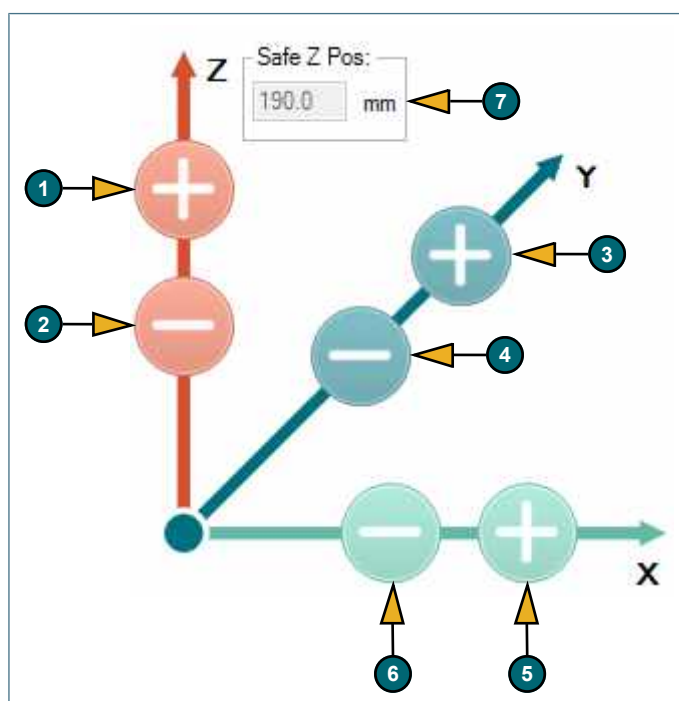


Illustration 138: Manual driving axes

1 < Z+ > symbol key	2 < Z- > symbol key
3 < Y+ > symbol key	4 < Y- > symbol key
5 < X+ > symbol key	6 < X- > symbol key
7 "Safe Z" input field	

Application

Move axes manually.

< Z+ > symbol key (1):

The cell axis moves in this direction at the preset speed as long as the key is pressed, or for the preset distance and at the preset jog speed if jog mode is active.

**NOTICE****Hand operation mode!**

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

< Z- > symbol key (2):

The cell axis moves in this direction at the preset speed as long as the key is pressed, or for the preset distance and at the preset jog speed if jog mode is active.

< Y+ > symbol key (3):

The cell axis moves in this direction at the preset speed as long as the key is pressed, or for the preset distance and at the preset jog speed if jog mode is active.

< Y- > symbol key (4):

The cell axis moves in this direction at the preset speed as long as the key is pressed, or for the preset distance and at the preset jog speed if jog mode is active.

< X+ > symbol key (5):

The cell axis moves in this direction at the preset speed as long as the key is pressed, or for the preset distance and at the preset jog speed if jog mode is active.

< X- > symbol key (6):

The cell axis moves in this direction at the preset speed as long as the key is pressed, or for the preset distance and at the preset jog speed if jog mode is active.

"Safe Z" input field (5):

Input of a height that is selected such that crash between the tool on the Z axis and another object in the system is not possible in any process step.

3.11.13.3 Drive program editor

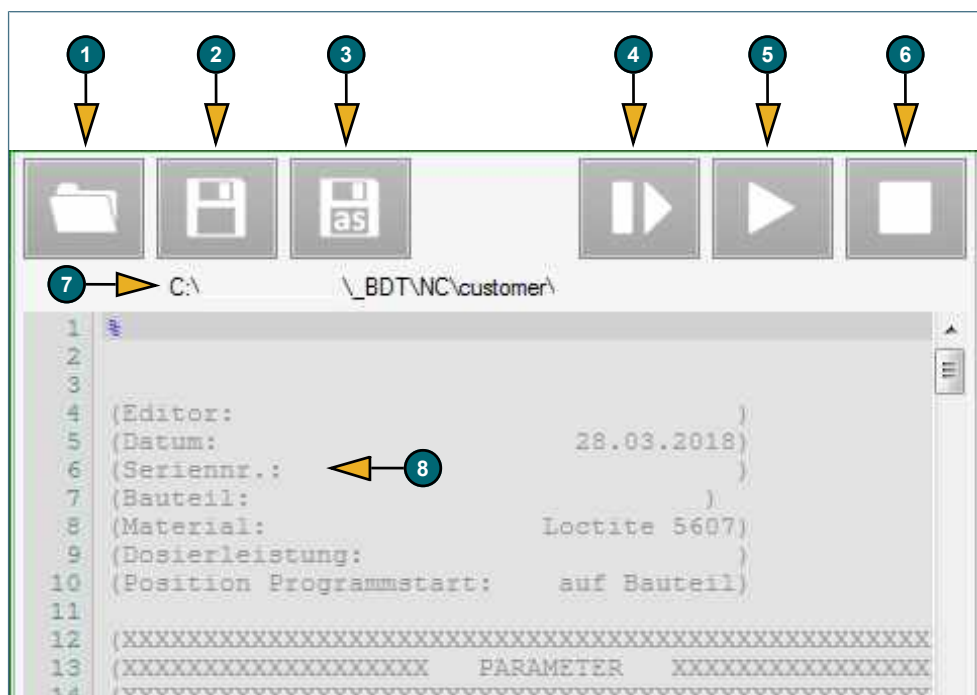


Illustration 139: Drive program editor

1 < (Windows) "Open" dialog > quick key	2 < Save > quick key
3 < (Windows) "Save as" dialog > quick key	4 < Step execution > quick key
5 < Start drive program > quick key	6 < Stop drive program > quick key
7 Path display field (Windows)	8 Drive program editor work area

Application

Manual editing **of previously existing drive programs** and sequence testing of them in the PROCESS MONITORING mode.



NOTICE

Drive program creation

Only previously saved drive programs can be edited with this editor. These can be copied into in the folder C:\Serial number of the system_BDT or, for example, via a USB stick. A "new program" is generated from an existing one using the < Save as > symbol key. This includes, of course, the contents of the source program. The program saved as a copy can then be edited applying the rules and commands of the NC programming (see also under "**An overview of drive programs**").

< (Windows) “Open” dialog > quick key (1):

Key pressed: The “Open” dialog of the Windows interface displays the files saved in the displayed path. Using the < *Open* > dialog button you can load a previously selected NC program into the editor area. The < *Cancel* > dialog button dialog pops out of the dialog without having to load a program.



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

< Save > quick key (2):

Key pressed: Saves the file version currently loaded in the Editor area in the displayed path (see pos. 7) on the hard disk of the machine.

< (Windows) “Save as” dialog > quick key (3):

Key pressed: Opens the “Save as” dialog of the Windows interface. A new file name can be entered in the “File name” field and stored with the dialog button < *Save* > on the hard drive, or saved as the file under the same name on the hard disk – but only after a system query. In any case, the file name extension is specified as “.nc”.

The save operation is terminated with the dialog button < *Cancel* >.

< Step execution > quick key (4):

Key pressed: The program loaded in the Editor can be executed step-by-step, command-line-by-command line; with each keystroke a new command line is summoned up. Only after completion of the current execution operation is the next keystroke accepted!



NOTICE

Process monitoring operation mode!

This function or routine can only be executed in process monitoring mode! An authorised user must be logged in for this!

< Start drive program > quick key (5):

Button is pressed: The drive program loaded in the Editor is started; all command lines are executed in the programmed sequence until the end of the program is reached. Thus, with metering pumps switched on a complete drive program sequence is executed, but without the component transport. By first switching off the metering pump(s) in MANUAL mode, you can simulate a so-called “dry run” of the program in order to assess the essential movement pattern of the program.



NOTICE

Process monitoring operation mode!

This function or routine can only be executed in process monitoring mode! An authorised user must be logged in for this!

< Stop drive program > quick key (6):

Key pressed: a running drive program is stopped. The PROCESS MONITORING mode remains active and thus the potlife processing will continue to be executed. In case of a potlife alarm, is the default number of blind shots in the position taught for this purpose is dosed.

Path display field (Windows) (7):

Shows the path on the hard disk where the file to be edited is stored.

Drive program editor work area (8):

This is an 'onboard' editor for loading and actively editing previously existing NC drive programs here. All commands of the NC programming can be used.



NOTICE

Login authorisation and operation mode

The execution of these and subsequent function(s) and flow(s) depends on the login authorisation and operation mode set; the symbol appears greyed out if functionality cannot be executed.

3.11.13.4 Drive program management

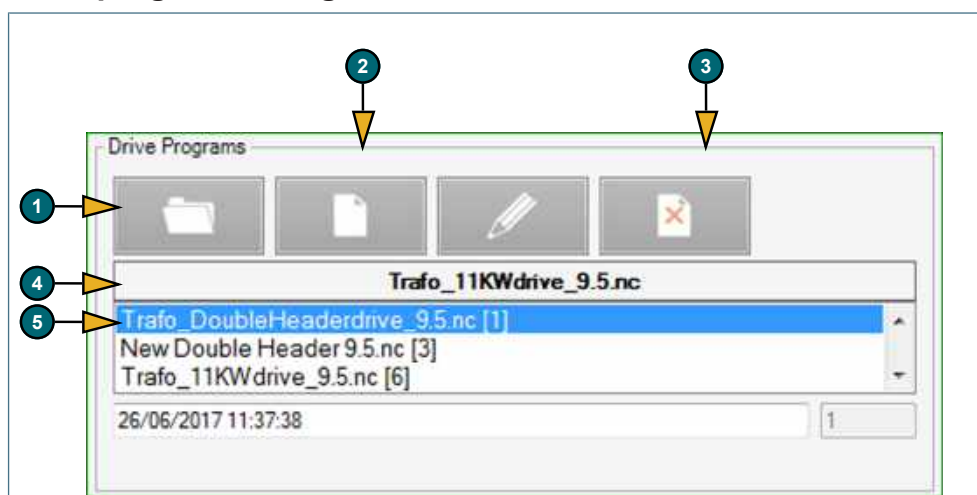


Illustration 140: Managing drive programs

1 < Load program > quick key	2 < Open file dialog > quick key
3 < Delete file > quick key	4 File display field
5 File selection list	

Application

Files can be loaded from the hard disk into the editor area through this dialog. These can be found in the selection dialog "Drive program" in the recipe screen, where they can be linked to a recipe.

< Load program > quick key (1):

Key pressed: Selected program (see pos. 5) is loaded to the display line (pos. 4).



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

< File open dialog > quick key (2):

Key pressed: The Windows "Open dialog" shows, via the Explorer view, shows the _BDT/NC/Customer. The NC programs are stored there.

< Delete file > quick key (3):

Key pressed: The marked line (see pos. 5) is removed from the selection.

File display field (4):

Display the file downloaded with the "Load program" command (see pos. 1).

File selection list (5):

Display of the files downloaded via the Windows Open dialog.

3.11.14 Recipe dialogue

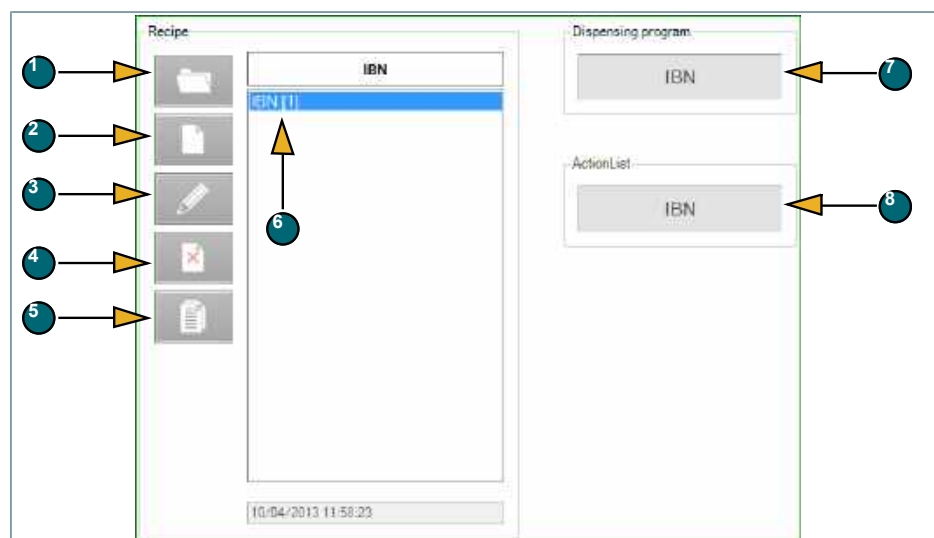


Illustration 141: Recipe dialogue

1	Load	2	Create
3	Change name	4	Delete
5	Copy	6	Select
7	Select another program	8	Select another program

Task:

Management of part recipes for the production process. A part recipe includes all information (e.g. drive programs, dispensing programs, sensor settings, etc.) that are significant for the production process for a particular part variant.



NOTICE

Dialogue display

The appearance of this dialogue depends largely on the application: additional fields may be displayed (e.g. camera check, etc.) that allow the linking of additional data records or additional information to the recipe.



NOTICE

Login authorisation parameter change!

Only an authorised logged in user can change parameters.



NOTICE

Hand operation mode!

This and the following function(s) and flow(s) can only be executed in manual operation mode! An authorised user must be logged in for this!

**NOTICE****Save changes.**

Only by pressing the save button are the changes taken over permanently into the system memory: when the system is restarted, the changes are available again (info: only an authorised user can start the saving process)

Load (1):

1. Mark the entry on the list
 2. Press the *< Load >* symbol key
 - ⇒ The recipe or program, etc. is loaded into PLC memory
- ⇒ The entry is displayed on the status bar of the menu

Create (2):

1. Press the *< New >* symbol key
 - ⇒ The virtual keyboard opens
 2. Enter the name for the recipe, program, etc.
 3. Confirm the input with the green *< OK >* key or discard with the red cross key
 - ⇒ The "New" window appears if confirmed with the *< OK >* key
 4. Confirm the name with the green *< OK >* key or discard with the red cross key
 - ⇒ The name is displayed in the recipe or program list, etc.
- ⇒ The recipe or program, etc. is available for further processing

Change (3):

1. Mark the entry in the recipe or program list, etc.
2. Press the *< Change >* symbol key
 - ⇒ The virtual keyboard appears
3. Enter the name
4. Confirm the name with the green *< OK >* key or discard with the red cross key
 - ⇒ The "New" window appears if confirmed with the *< OK >* key
5. Confirm the name with the green *< OK >* key or discard with the red cross key
 - ⇒ The name is displayed in the recipe or program list, etc.
 - ⇒ The recipe or program, etc. is available for further processing

Delete (4):

1. Mark the entry in the recipe or program list, etc.
2. Press the *< Delete >* symbol key

- ⇒ The following message appears: "Really delete recipe or program, etc.?"
- 3. Delete the recipe or program, etc., with the < OK > key
- 4. Discard the process with the < Cancel > key
- ⇒ The entry is removed from the recipe or program list, etc.

Copy (5):

1. Mark the recipe or program, etc. on the list
 2. Press the < Copy > symbol key
 - ⇒ The virtual keyboard opens
 3. Enter the name for the copied recipe, program, etc.
 4. Confirm the input with the green OK symbol or discard it with the red cross symbol
 - ⇒ The "Copy recipe" window opens
 5. Press the green OK key to start the copy procedure or press the red cross to discard it
- ⇒ The new recipe or program, etc. is transferred to the list and displayed
 - ⇒ The recipe or program, etc. is available for further processing

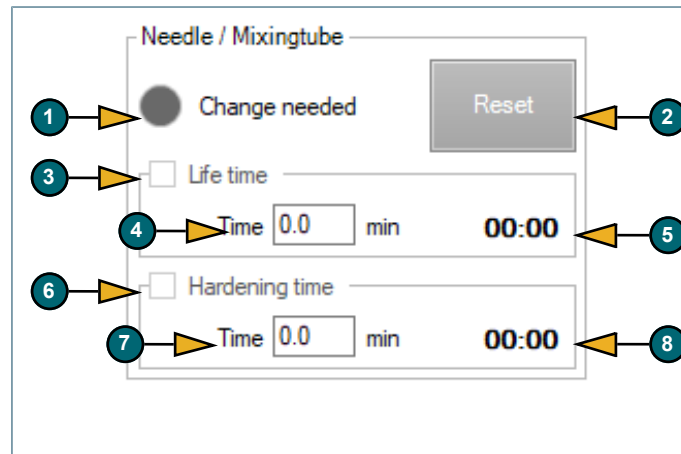
Select (6)

1. Mark the recipe or program name, etc. on the list
- ⇒ The entry can be edited with other functions (e.g. 'Change', 'Copy', etc.)

Select another program (7) or (8):

1. Click the field
 - ⇒ The dialogue opens
 2. Mark the entry
 3. Adopt with the green OK key or discard with the red cross key
- ⇒ The dialogue closes
 - ⇒ The entry is linked if confirmed with the OK key

3.11.14.1 Mixing tube change indicator



Task:

Setting the mixing tube change indicator.



NOTICE

Input and execution of functions and parameters.

Input and further processing of a parameter or function can only be performed with an authorised user login.

Display field (1):

Indicates when a mixing tube change is necessary.

Button (2):

Resets the run times after the mixing tube change.

Check box (3):

Tick mark set: Prompt for mixing tube changing (referred to the total service life of the mixing tube) is switched to **Active**.

No tick mark: Prompt for mixing tube changing (referred to the total service life of the mixing tube) is switched to **Inactive**.

Input field (4):

Input of the time after which a prompt for mixing tube changing is to be displayed.

Display field (5):

Displays the countdown of the current time until the mixing tube change.

Check box (6):

Tick mark set: Prompt for mixing tube changing (referred to the time since the last dispensing event or blind shot) is switched to **Active**.

No tick mark: Prompt for mixing tube changing (referred to the time since the last dispensing event or blind shot) is switched to **Inactive**.

Input field (7):

Input of the time after which a prompt for mixing tube changing is to be displayed.

Display field (8):

Displays the countdown of the current time until the mixing tube change.

3.11.14.2 Part counter

Illustration 142: Part counter

1 "Bad" display field	2 "Good" display field
3 "Total" display field	4 <Reset> quick key
5 "Bad" display field	6 "Good" display field
7 "Total" display field	8 <Reset> quick key

Application

Display of the production results.

"Bad" display field (1):

Display of the bad parts from ongoing production.

"Good" display field (2):

Display of the good parts from ongoing production.

"Total" display field (3):

Display all processed parts.

<Reset> quick key (4):

	NOTICE Quick key The function is performed immediately on pressing the button or symbol key if both login authorisation and operation mode have been preset.
---	---

This resets the entries of the shift counter.

“Bad” display field (5):

Display of all the bad parts since the last reset.

“Good” display field (6):

Display of all the good parts since the last reset.

"Total" display field (7):

Display of all the processed parts since the last reset.

<Reset> quick key (8):

	NOTICE Quick key The function is performed immediately on pressing the button or symbol key if both login authorisation and operation mode have been preset.
--	---

This resets the entries of the overall counter.

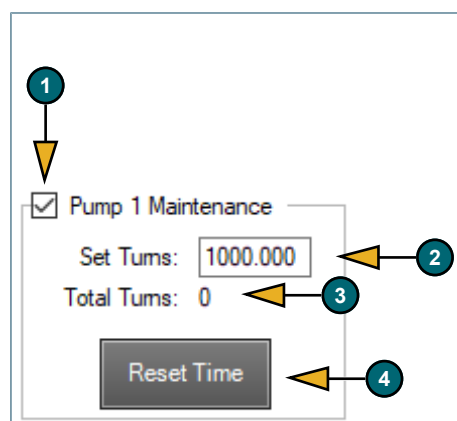
3.11.14.3 Maintenance dialogue - Pump change

Illustration 143: Pumpe 1 Wartung -en

1	Check box	2	Input field
3	Display field	4	Taste < Reset Time >

Application

Monitoring of pump activity.

Check box (1):

Tick mark set: Function, module (e.g. sensor system, etc.) is active and will also be processed in semi- or fully-automated sequences.

No tick mark: The module function will not be further processed.

**NOTICE****Login authorisation and operation mode**

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

Input field (2):

Setting the pump rotation up to the alarm signal.

**NOTICE****Input field**

This and subsequent input field(s) can be filled in with both a floating point number and just numerals (e.g. "1.0" or "1").

The input limits are specified by the software.

**NOTICE****Login authorisation parameter change!**

Only an authorised logged in user can change parameters.

Display field (3):

Display of pump rotations already completed.

Taste < Reset Time > (4):

Resetting the pump speed to the initial value after pump replacement.

**NOTICE****Login authorisation function execution**

Only an authorised user login can execute functions.

3.11.15 Active messages

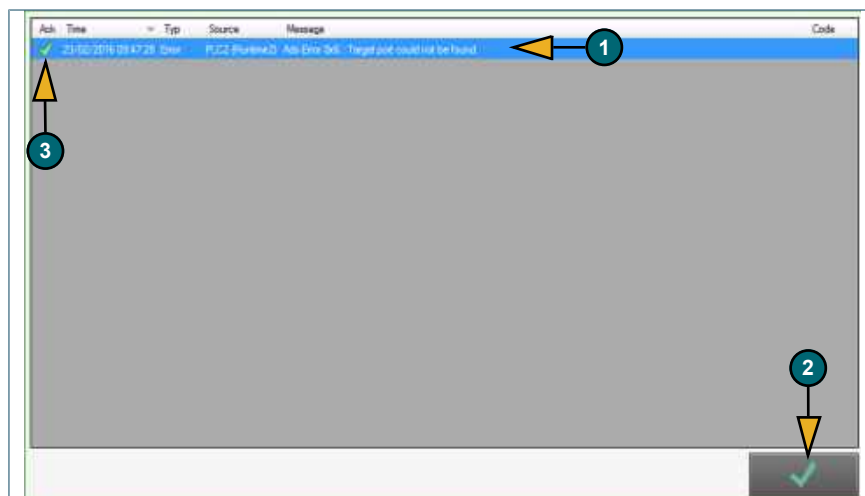


Illustration 144: Active messages

1	Message line	2	Confirmation key
3	Confirmation field		

Application

Listing and confirmation of the current messages with different background colours.

Message line (1):

Red background: Alarm message: The effect on the machine depends on the program.

Yellow background: Warning message: The effect on the machine depends on the program.

White background: Other information: No effect on the machine.

Confirmation key (2):

Pressing the key confirms all listed messages. Alarms and warnings that are still active are displayed again after confirmation.

Confirmation field (3):

Ticked field: This message is still active.

Delete tick mark: This message is removed from the list. If the condition that brought about this message is still active, the message appears again.

3.11.16 Log file settings

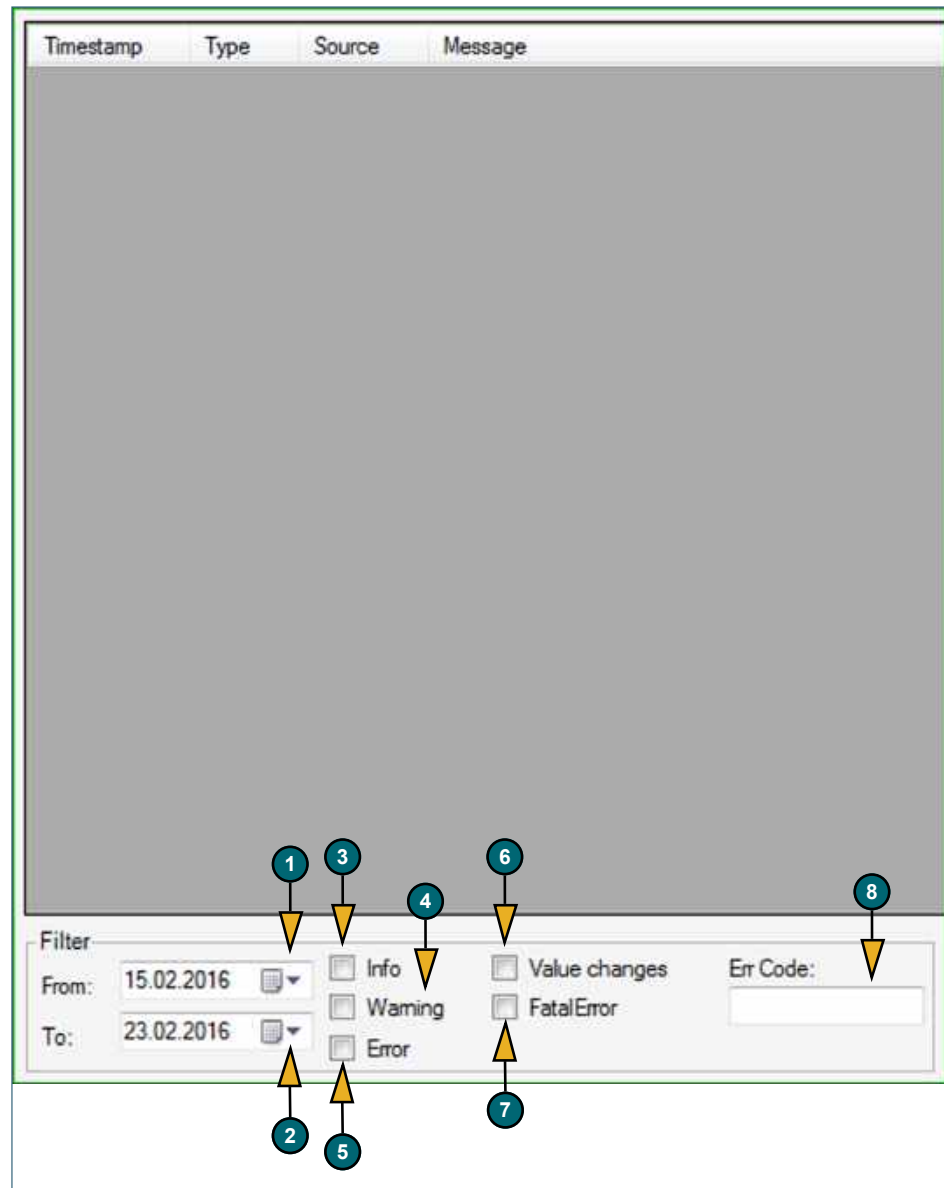


Illustration 145: Log file

1	Time filter, beginning	2	Time filter, end
3– 7	Checkboxes	8	"Err. code" input field

Application

Sets the filter conditions for the log file display list.

Time filter, beginning (1):

Select a start date for recording the log information.



NOTICE

Login authorisation and operation mode

The execution of the function of this and subsequent check box elements depends on the login authorisation and operating mode set; symbol appears greyed out if functionality cannot be executed.

Time filter, end (2):

Select a date limit for recording the log information.

Checkboxes (3– 7):

Tick mark set: Incoming information of this category is listed.

No tick mark: Incoming information of this category is not listed.

"Err. code" input field (8):

The error code of a message is found at the end of its line (e.g. 57-0/433). When an error code is entered (here: "57-0/433"), the growing message list is presorted with this entry.



NOTICE

Error codes

It is also possible to enter wildcards here, e.g. *433.

3.12 Production flow

- ✓ MANUAL operation mode is active
- ✓ Material preparation is ok
- 1. Activating AUTOMATIC operation mode
 - ⇒ AUTOMATIC operation mode is active
 - ⇒ Filling process starts in the filling position
 - ⇒ Dispensing system is ready
 - ⇒ Needle check is being carried out
 - ⇒ Weighing sequence is carried out
- 2. Insert components into component nests
 - ⇒ QR codes are scanned
 - ⇒ If QR codes OK, start button blinks green
- 3. Press the start button
 - ⇒ Component status is displayed
 - ⇒ Turntable rotates when all other processes are complete

- ⇒ Recipe is being carried out
- 4. Reload the turntable
- 5. Exit the light barrier
 - ⇒ QR codes are scanned
 - ⇒ If QR codes OK, start button blinks green; otherwise start button lights up permanently green and prevents further action
- 6. Press the start button
 - ⇒ Component status is displayed
 - ⇒ Turntable rotates
- 7. Remove components and sort them depending on the component status
- 8. Continue as already described

3.13 Shutting down the dispensing machine

Shutting down a machine

Machines are shut down repeatedly for production downtimes, maintenance or service purposes. For this, on the one hand the machine's energy store (electrical energy, compressed air, material pressure) must be emptied/switched off partially automatically, partially manually; on the other hand, particular prerequisites for a proper start-up must be created: depending on the machine configuration, this can be cleaning processes that must be performed while the machine is shut down.

Prerequisite:

production process has been stopped: no component processing and no optional product transport

The following *sequences* are worked through in tabular fashion, from top to bottom.

1. Machine is in < HAND > or < PROCESS MONITORING > operating mode	Perform homing (see also chapter "Homing").	> "Chamber empty" or "System not ready" is displayed on status bar > Potlife monitoring was deactivated
	Press < Material > menu button: Material screen opens	> Automatic material preparation stops > "Material OK" display field appears greyed out
	Press < <i>Machine off</i> > button	> Control and load voltage are disconnected > Compressed air supply is disconnected

Exception: for dispensing systems with dynamic mixing systems, for the supply of the < Emergency purge > and < Emergency blow > buttons, some compressed must remain in the system

1. Machine must be shut down via the system 2. Material preparation must be in HAND operating mode or in STANDBY for PPS	• Turn main switch into "Off" position or press < Switch off > button of the "Info" screen: all electrically-controlled and supplied components are disconnected from the voltage (info: depending on the system, it may be necessary to let a wait time of several minutes elapse, in which the internal flows of the PLC such as UPS shut down and are processed until the machine can be switched back on via the main switch Only if all LEDs of the controller core have gone out is the OFF state reached! Only this person is authorised to switch back on!) • Disconnect voltage supply line	> Machine is out of operation > System can be switched on again
--	--	---

After a system-dependent wait time, all electrical energy stores are discharged; exception: Buffer batteries in the controller core and optional external uninterruptible power supplies (UPSes). If necessary, these UPSes must be switched off separately and disconnected from the machine.

	• Disconnect the voltage supply line: Components are disconnected from the voltage (Information: Depending on the system, it may be necessary to wait several minutes during which the internal processes of the PLC such as UPS shut down, etc. are completed before the machine can be switched back on via the main switch! The OFF state is reached only when all LEDs of the controller core have gone out! Only then may the machine be switched on again!)	> The machine is out of operation > The machine can only be switched on again when the external power supply has been switched on via the "Machine ON" button or via the corresponding handshake signal
	• Close hand switch-on valve of the primary compressed air maintenance unit • Disconnect compressed air supply line: Machine is deenergized but for the named exceptions and any mechanically-stored energies via springs, etc.!	> Machine system is depressurized



CAUTION

Remaining energy on machine

Immediately after the shut down of the machine, particular components such as plasma heads, hot rivet heads, ovens, can have a certain residual energy, heat, voltage! Adhere to a safety wait time before maintenance and service work and protect yourself by wearing appropriate protective clothing.

4 Maintenance and service

This chapter is intended to inform machine personnel (operators, set-up and service technicians, etc.) about the necessary work on the modules of the machine. The auxiliary materials that require intervention due to wear, consumption or soiling. Here, it is essential to observe the notes and warnings for the safety of the person working on the machine or its modules!

Reasons for the upcoming maintenance may be:

- Shot weight not correct on weighing in
- Dispensing quantity fluctuations in the part
- Dripping after dispensing stop
- Insufficient hardening of the volume dispensed within the time
- Recorded supply pressure of a module drops impermissibly
- Recorded working pressure of a module increases impermissibly
- Air bubbles in the material circuit
- Working temperature of a module increases constantly
- Impermissible smear formation of the material dispensed in the part
- Escaping of dispensing material outside of the dispensing nozzle
- Unusual noises within a mixing module
- System fault messages

Replacement and wear parts

Material and energy-carrying components such as dispensing valves, mixing heads, dispensing pumps, cylinders, axes, transport belts, etc. are subject to application-dependent wear; particular components such as seals, gears, stirrers, shafts, rotors, stators, etc.

These assemblies must be serviced regularly, starting with initial commissioning, so that wear parts can be identified and preventative measures (e.g. cleaning, lubrication, replacement, etc.) can be taken in good time.

Only original replacement parts and recommended operating materials may be used!

**NOTICE****Replacement parts for bdtronic machines**

As defined by bdtronic, replacement parts are components and assemblies that perform their intended task due to the service life specified by the manufacturer and guaranteed limit values, including in combination with other components and assemblies of the machine, and that can only fail by exceeding one of their limit values or due to a higher-level fault (overload, incorrect load connections, etc.) and consequently must be replaced. These are normally movement systems (axes, pneumatic actuators, etc.), drive components (rectifiers, etc.), transport components (horn, stopper, cross-transport conveyors, etc.), control components (input/output assemblies), standard sensor system (light sensors, proximity switches, etc.). We also recommend keeping assemblies relevant for production in stock for quickly restarting production after the failure of an assembly.

Note about "Liability for defects"

The use of the terms "wear" and "wear parts" or "material conveying" parts and assemblies is described in the bdtronic Terms & Conditions.

The Terms & Conditions are stored on the homepage as a PDF document and can be downloaded.

Irrespective of these, identifiable assemblies or components that are subject to verifiable wear even when used as intended are excluded from the warranty.

**NOTICE****Concerning the term: "Material conveying"**

bdtronic technologies link the term "material conveying" with components and assemblies that come into direct contact with process-relevant materials such as resins, hardeners, cleaning agents for dispensing, impregnation and material preparation systems or compressed air and fixing components for hot riveting machines, or plasma gas for plasma heads, or directly involved for maintaining the process such as the plasma generation and forwarding components.

**NOTICE****Replacement part, wearing part lists**

Upon request, bdtronic GmbH will be pleased to generate a machine-related replacement and wearing part list that goes beyond the standard delivered replacement part documentation. The necessary effort for this is compensated under the normal terms.

**NOTICE****Packaging materials**

Particular modules such as dispensing pumps, hot rivet stamps etc. are provided as replacement and spare parts in special transport containers. We recommend continuing to use them as secure transport for return transport/repair. If possible, please only send these modules to bdtronic in these containers. Furthermore, these containers can be obtained from bdtronic.

The flows described by the manufacturer for dismounting, taking apart, maintenance, assembly and recommissioning must be adhered to.

bdtronic GmbH recommends keeping modules subject to wear available in stock to increase machine availability!

Cleaning**CAUTION****Flammable cleansing agents!**

Do not use any flammable cleansing agents such as acetone!

The operator/owner is liable for the consequences of improper cleaning!

If there are any questions or something is unclear with respect to cleaning, please contact bdtronic GmbH!

When working where there may be contact with dispensing materials and detergents, it is essential to:

**NOTICE****Wear eye protection!**

Wear suitable eye protection for every dangerous situation, e.g. laser, infrared etc.

	NOTICE Wear protective gloves! Wear suitable protective gloves for every dangerous situation.
Clean additional devices only according to the manufacturer's instructions (see chapter 10)!	
	NOTICE Particular cleansers! Info: Cleansing agent is to be used depending on the materials: Cleansing agent residues that may remain in the module, depending on the model, must be chemically compatible with the material that will be filled in order to counteract impermissible changes to the dispensing properties in further operation in good time!
	CAUTION Clean filter mats regularly. Filter mats for the air circulation in the switch cabinet must be cleaned or replaced regularly. Especially at high ambient temperatures, otherwise cooling is not effective.
	CAUTION Lubricants! Use only suitable lubricants! The operator/owner is liable for the consequences of any improper lubrication!
If there are any questions or something is unclear with respect to lubrication, please contact bdtronic GmbH! Do not put lubricants into contact with dispensing mass since violent chemical reactions are possible! When working where there may be contact with dispensing materials and detergents, it is essential to:	
	NOTICE Wear eye protection! Wear suitable eye protection for every dangerous situation, e.g. laser, infrared etc.

	NOTICE Wear protective gloves! Wear suitable protective gloves for every dangerous situation.
---	--

Lubricate additional devices according to manufacturer's instructions (see Chapter 10).

	NOTICE Only use recommended lubricants. Warning: Use only lubricants recommended by the manufacturer for re-filling, otherwise there is a risk of damaging the module in ongoing operation.
---	---

Important notes about disassembly

The following pages describe the maintenance work for the most important parts of the machine.

	CAUTION Climbing into a machine: Before climbing in, the main switch must be turned to the "OFF" position
---	---

	CAUTION During disassembly, pay special attention to the following points: • Do not use force • Only take apart and assemble individual modules on a soft, level, clean surface, e.g. wooden plate • Use a suitable tool or the included special tool • Only disassemble compressed air connections and material lines when they have been marked first (ensures correct reassembly) • Only trained personnel may perform work on electrical, pneumatic and hydraulic equipment • For removal of a motorised consumer (e.g. mixer motor), the control voltage must be switched off ("Machine off" key) or the main switch must be turned to the "OFF" position • General safety instructions and other generally valid rules for occupational safety, accident prevention and environmental protection must be adhered to
---	---



CAUTION

After disassembly, note the following points:

- Clean and dry individual parts with suitable means without applying force.
- Inspect all individual parts for damage and replace with new parts if necessary. Use only original bdtronic spare parts or spare parts recommended by bdtronic.
- Grease all threads with non-resinous grease.

Full maintenance or repair replacement

For precision modules such as dispensing pumps that are subject to particular wear in the application (e.g. leaks in various places, no maintenance success even after several changes of the same part), bdtronic recommends performing full maintenance on-site or a repair replacement of the entire module. In this case, it is reworked completely, i.e. all damaged parts are reworked or replaced.

After reassembly, the module can be used as a new spare part.

If the module reveals irreparable damage, bdtronic GmbH will immediately make an appropriate offer for a replacement.



CAUTION

Determine the necessity of connecting an extraction system!

Extraction connections must be provided at various points on the machine for cell, material preparation, hardening lines, etc.! This is mandatory for processing **Fluid 1 materials**. For special applications in the Ex sector, it is essential to observe the minimum extraction requirements! Operation of these machines without sufficient extraction is prohibited!



NOTICE

Maintenance instructions "on your own authority"

The following maintenance instructions have been formulated carefully with bdtronic service personnel and original modules so that a picture-by-picture procedure for the maintenance work is recommended in each case. Technical optimisations on the modules may, of course, lead to new procedures; these must first be "straightened out" by revising the present maintenance instructions. The bdtronic editorial team relies on feedback from the company's customers to correct mistakes and errors which have crept in and to deliver the most up-to-date version for the machine!

The maintenance requirements of supply parts are described in detail in the accompanying documents from our suppliers and must be performed at the recommended maintenance intervals and as described in the procedural in-

structions. In addition, the maintenance instructions from the system builder for special components such as the "door safety switch" and "light barrier" must be followed.

	CAUTION Maintenance work on the laser Before maintenance work on systems in which class 2 lasers or higher are used, the danger zone of the direct and indirect visual area must be sealed off effectively in order to protect the maintenance personnel from the effects of laser beams. The operator is obligated to provide training and instruction.
	CAUTION Maintenance of heat-generating devices When using heat-generating devices and components (e.g. hardening kilns, pre-warming kilns, etc.), maintenance and repair work may only be carried out if a temperature of less than 40 °C has been reached after switching off the devices or components. In any case, the wearing of sufficient protective clothing is recommended.

4.1 BDT folder backup

4.1.1 Backup group field

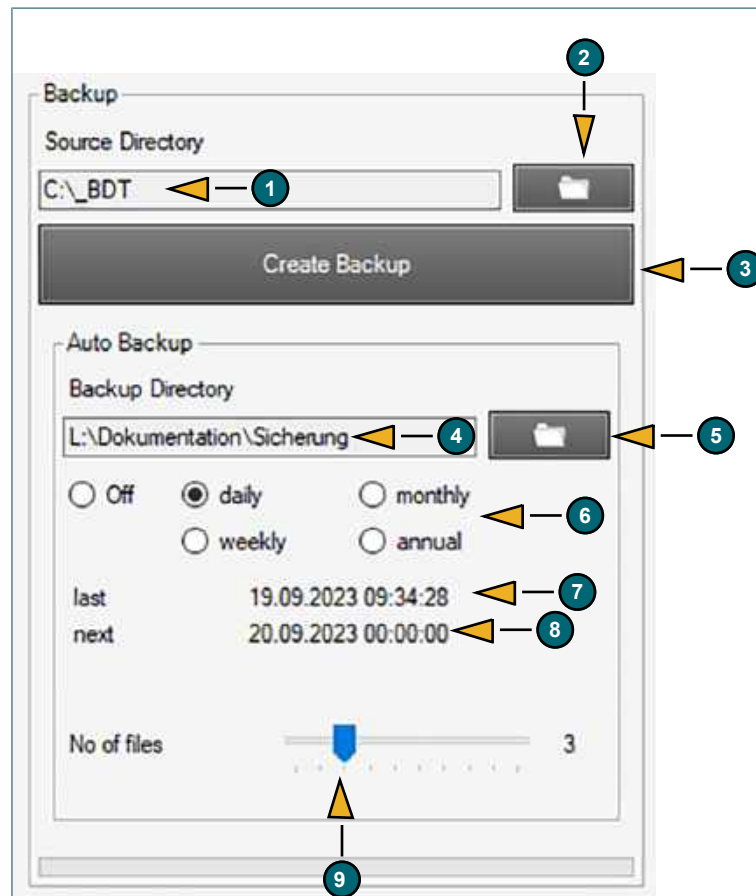


Illustration 146: BDT backup with target directory

1	Display path	2	WINDOWS symbol key
3	< Create backup > key	4	Display path
5	WINDOWS symbol key	6	Selections
7	Display field	8	Display field
9	Potentiometer		

Application

Manual or automatic backup of the BDT folder located on the machine IPC.

Display path (1), (4):

Display of the folder created in the WINDOWS directory structure.

WINDOWS symbol key (2), (5):

Open the WINDOWS directory structure.

< Create backup > key (3):

Depending on the data connection, a choice of save methods or just a "Save as" prompt appears.

**NOTICE****Input and execution of functions and parameters.**

Input and further processing of a parameter or function can only be performed with an authorised user login.

Selections (6):

"OFF": No automatic data backup possible.

"Daily": The next data backup is started on the next day at the time entered in display field (8)

"Weekly": The next data backup is started the next week at the time entered in display field (8)

"Monthly": The next data backup is started the next month at the time entered in display field (8)

"Annually": The next data backup is started on a date that is one year later that the previous backup and at the time entered in display field (8)

**NOTICE****Input and execution of functions and parameters.**

Input and further processing of a parameter or function can only be performed with an authorised user login.

Display field (7):

Date and time of the previous backup.

Display field (8):

Date and time of the next backup.

Setting controller (9):

Maximum number of ZIP folders in the data backup directory. The old data are deleted before a further backup is started.

**NOTICE****Input and execution of functions and parameters.**

Input and further processing of a parameter or function can only be performed with an authorised user login.

4.1.2 Manual backup of the “_BDT” folder on drive N of the bdtronic LAN

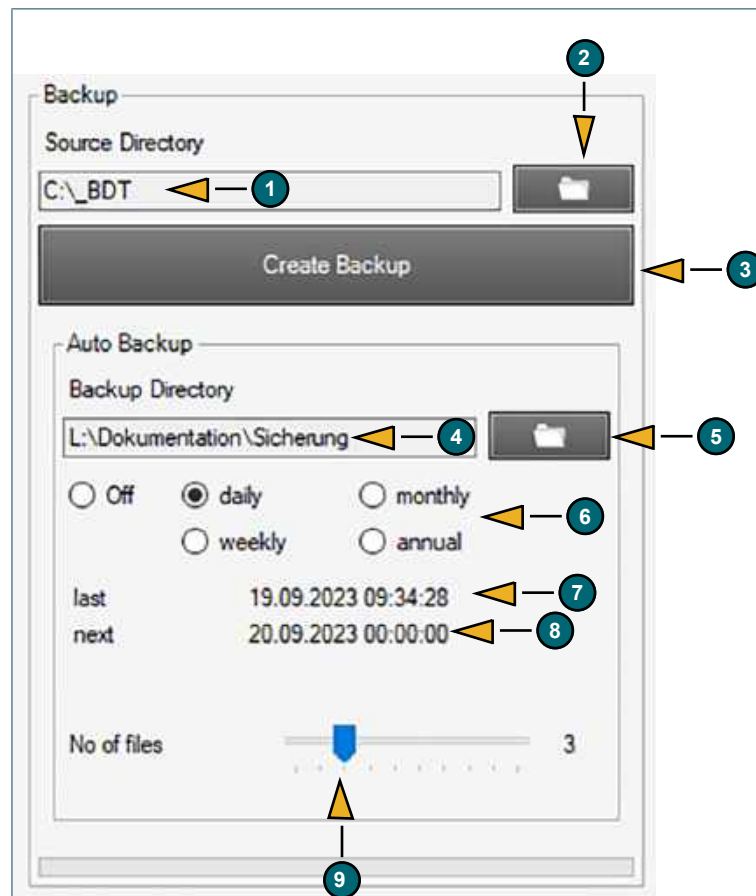


Illustration 147: Figure: BDT backup

- ✓ The correct designation of the source directory (see under “Figure: BDT backup”, item 1) has been entered

- ✓ **Data connection** to the bdtronic network (= LAN) is active

1. Press key < *Create backup* >

⇒ A selection dialogue field appears:

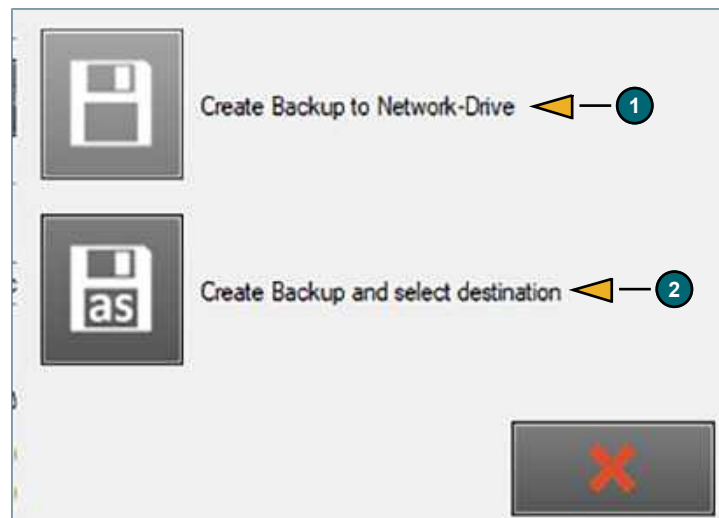


Illustration 148: Figure: Save buttons

2. Press key < *Create Backup to Network Drive* > (see “Figure: Save buttons”, item 1)

⇒ An input field for the serial number appears:



Illustration 149: Figure: Enter the serial number



NOTICE

Serial number prompt

After the first successful input of a serial number, this field no longer appears during the next backup!

3. Enter the serial number in field “Serial number” (see “Figure: Save buttons”, item 1)
4. Conclude the input with the confirmation key (see “Figure: Enter the serial number”, item 2)
 - ⇒ The backup starts with a green progress bar
 - ⇒ During the first backup, a folder with the machine serial number is created and a ZIP file is stored in it

4.1.3 Manual backup of the “_BDT” folder on a random medium

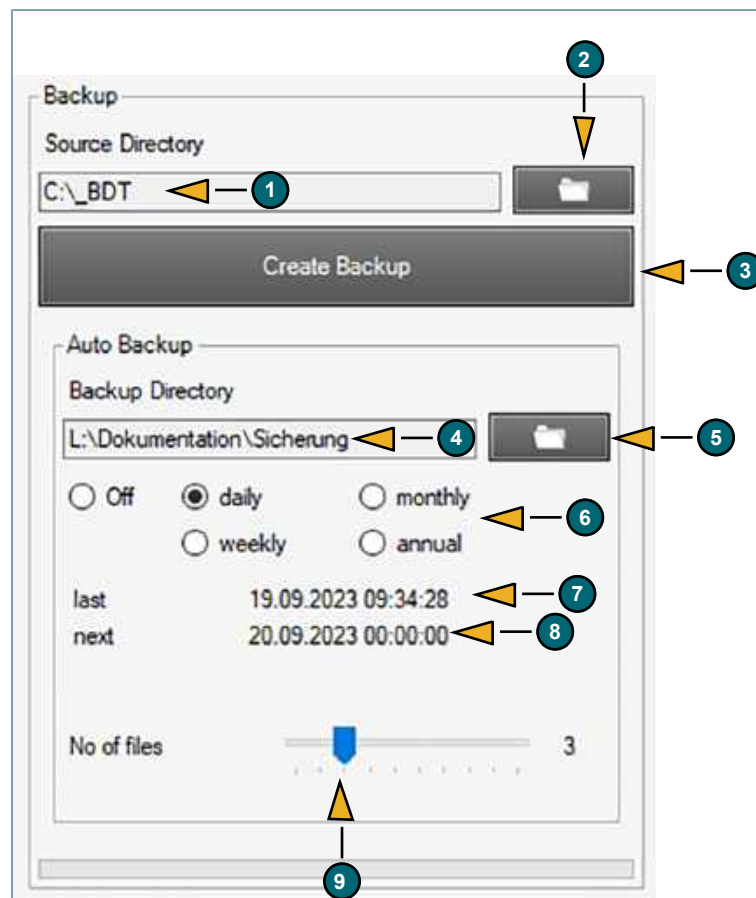


Illustration 150: Figure: BDT backup

- ✓ The correct designation of the source directory (see under “Figure: BDT backup”, item 1) has been entered
- ✓ A medium (USB stick, hard disk) has already been installed
- ✓ **Data connection** to the bdtronic network (= LAN) is **not active**

1. Press key < *Create backup* >
 - ⇒ The Windows “Save as” dialogue appears
2. Change to the storage location on the desired medium
3. Press the < *Save* > key
 - ⇒ The ZIP file is stored

4.1.4 Automatic backup of the “_BDT” folder

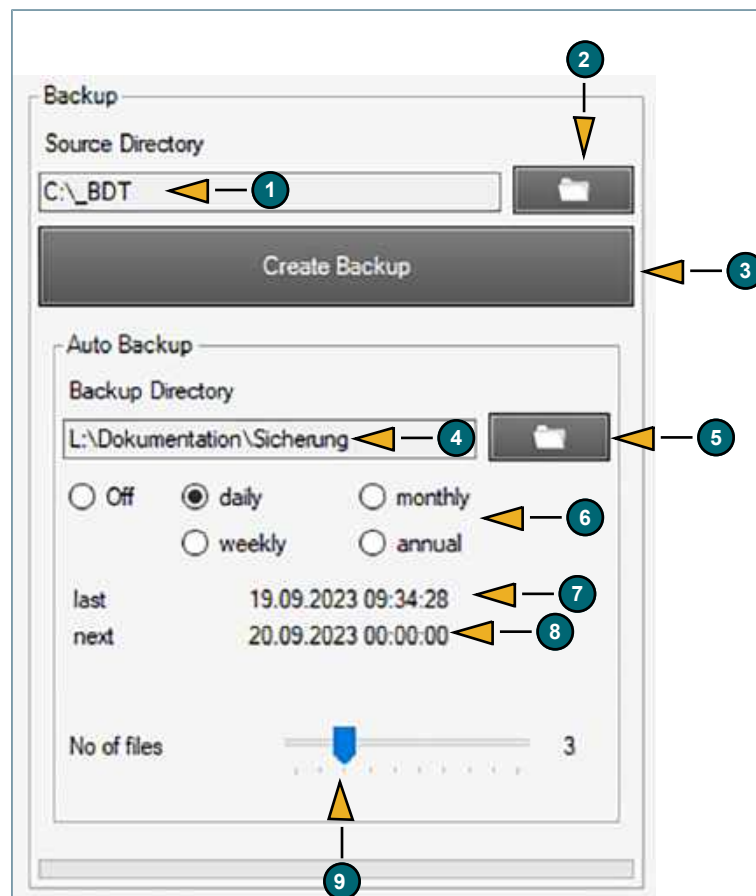


Illustration 151: Figure: BDT backup

- ✓ Source directory (see under “Figure: BDT backup”, item 1) has been correctly entered
- ✓ A valid data backup directory (see under “Figure: BDT backup”, item 4) has been correctly set up



NOTICE

Define the data backup directory

Pressing the symbol key (see “Figure: BDT backup”, item 5) allows the storage path to be changed!

1. Check the desired selection field
 - ⇒ Press the “Save” key in the menu bar
 - ⇒ All changes are permanently saved
 - ⇒ The next backup is automatically started at the time entered in line “Next ...” (see “Figure: BDT backup”, item 8)



NOTICE

Status of the machine computer at the start of the backup

If the machine computer is switched off or the visualisation system is closed at the start of the next backup, the backup is performed later!

4.2 Venting of material circuits | Dispensing units

Crucial to high-quality dispensing is a continuous flow of the dispensing material(s) into a part and its/their curing afterwards. This can be the dispensing of a bead into a housing groove, the filling of a cavity, application of adhesive strips, etc. The applied materials or material mixtures must be flowable and free of air bubbles. In the case of flowability, temperature-regulating additional assemblies and tubings are applied to the dispensing assemblies. The absence of air can be achieved by methods such as evacuation, accompanying stirring or mechanical removal via suitable sectional opening of the material strands from the place of storage to the material removal for the purpose of natural degassing. The documentation enclosed with the systems, such as the line plan - optionally the vacuum plan - provides a functional insight into the fluid mechanics employed and can be used to determine suitable separation points for venting the material sections.

Especially for material preparation, bdtronic has developed automated solutions such as an MPS for container-independent materials, a PPS for pail-bound materials or cartridge removal systems. Suitable algorithms are already applied at the supply point to reduce the amount of air in the dispensing material.

Once during commissioning or sometimes after pail refilling or changeover work, the extensive absence of air must be established manually with additional effort such as separating material line sections, venting pumps, valves, mixing heads, etc. before the actual dispensing process.

When venting, the natural degassing tendency of the materials should always be supported in such a way that material-bearing outlets of the affected assemblies are always positioned diagonally upwards for material removal. The operating conditions to be satisfied with regard to system temperature and pressure specifications are also further criteria for successful, process-stable venting and degassing.



NOTICE

Further sources of information

In addition to the explanations about the venting methods in this and the following chapters, further information can be found in individual maintenance manuals for these assemblies.

4.2.1 Venting a material line

Material line

A material line takes over the pressurised material flow and passes it on to the other end. The entire cross-section of the line is thereby pressurised. Temperature and line resistance, together with the upstream pressure, influence the viscosity of the material and thus also the flow rate. Compressible air between the material components impedes both the flow and the material distribution in such a way that no continuous strand leaves the line at the end, despite a sufficient inflow. From a dispensing point of view, more or less material is then dispensed or dispensed beads are interrupted as a result. The air in the material therefore has to be removed.

Venting – how do we do it?



Illustration 152: Venting the material line

- ✓ The line section up to the start of the line must already have been vented
- 1. Disconnect the discharge end of the line at a suitable point
- 2. Hold the end of the line over an empty cup
- 3. Start the material flow via the feed pressure or via a dispensing command
- 4. Switch off the feed pressure or cancel the dispensing command as soon as a continuous flow of material is dispensed air-free into the cup
- 5. Connect the line
 - ⇒ The section from the end of the line is free of air

4.2.2 Venting a progressive cavity pump



Illustration 153: Progressive cavity pump for dispensing applications

bdtronic offers a wide range of progressive cavity pumps for both dispensing and conveying in material preparation systems (PPS, MPS, DPS) with different dispensing or conveying capacities with different material characteristics. They are used as classic single-component dispensing pumps or as a component of static mixing heads or feed pumps for valve-equipped, static and dynamic mixing heads.

In material preparation systems, they take over the task of receiving and forwarding the dispensing materials from, for example, a pail (PPS) or pressure vessels (MPS).

The quality and reproducibility of an automated dispensing task is determined by the exact measurement of the required dispensing or delivery quantity and the pressure-stable forwarding to the dispensing or further processing point.

Compressible air pockets in the material hinder delivery of the exact dispensing volume. They therefore have to be removed in a venting process.

Procedure for a dispensing pump:

1. Position an empty cup under the venting screw (see also figure "Progressive cavity pump")
2. Loosen the venting screw using an open-jaw wrench
3. Start the material flow with a dispensing command
⇒ Material/air mixture escapes
4. Cancel the dispensing command as soon as only dispensing material is dispensed into the cup
5. Tighten the venting screw

Procedure after replacing a PPS conveyor pump:

Illustration 154: Venting a PPS pump

- ✓ The container must be in position
- 1. Open the access doors
- 2. Loosen the venting screw
- 3. Position an empty cup under the venting screw
- 4. Close the access doors
- 5. Press the < *Automatic* > button
- 6. Press the seating button
 - ⇒ The follower plate moves onto the pail surface
 - ⇒ Seating time starts
 - ⇒ Pilot control starts
 - ⇒ Observe the venting bore through the window
 - ⇒ Material/air mixture escapes
 - ⇒ If the pilot control time is not sufficient and air still escapes, change the operation mode (Automatic – Manual – Automatic) using the buttons < *Manual* > and < *Automatic* >
 - ⇒ A further pilot control time starts
- 7. If, at the end of the pilot control time or a further pilot control time only material is flowing, open the doors, remove the cup, screw in the venting screw, close the doors and press the < *Automatic* > button
 - ⇒ Automatic mode starts

Procedure after replacing an MPS conveyor pump (e.g. stator replacement)

- 1. If the container fill level is too low, top up by drawing in material using the suction lance
- 2. On tab “Drive”, press the < *Start* > button
 - ⇒ The pump starts in manual mode

3. When material starts running back into the container via the return line, press the *< Stop >* button on tab “Drive”
⇒ The return line is completely filled
4. Activate automatic material
⇒ The preparation cycle starts: Optional temperature control starts, stirring starts, recirculation starts, evacuation starts: Air is drawn out

4.2.3 Venting a Duplex-Plus mixing head



Illustration 155: Venting a Duplex-Plus

With simple mixing ratios, two-component dispensing materials are brought together via a static mixing head and mixed together in a disposable mixing tube. A solution is offered here by the bdtronic Duplex Plus mixing head, whose material streams (resin and hardener) are fed in directly via dispensing pumps screwed into the mixing head. The outlet of the mixing head is extended by the mixing tube.

Compressible air pockets in the material hinder delivery of the exact dispensing volume. They therefore have to be removed in a venting process. With this configuration, a two-step venting process has to be carried out, as first the two dispensing pumps and then the mixing head proper are vented.

Procedure for a dispensing pump:

Illustration 156: Vent the progressive cavity pump

1. Position an empty cup under the venting screw (see also figure “Venting a progressive cavity pump”)
2. Loosen the venting screw using an open-jaw wrench
3. Start the material flow with a dispensing command
 - ⇒ Material/air mixture escapes
4. Cancel the dispensing command as soon as only dispensing material is dispensed into the cup
5. Tighten the venting screw
6. Repeat the venting process for the second pump
 - ⇒ Both pumps are air-free
7. Position an empty cup under one of the venting screws of the mixing head (see also figure “Venting a Duplex-Plus”)
8. Loosen the venting screw using an open-jaw wrench
9. Start the material flow with a dispensing command
 - ⇒ Material/air mixture escapes
10. Cancel the dispensing command as soon as only dispensing material is dispensed into the cup
11. Tighten the venting screw
12. Position an empty cup under the second venting screw (see also figure “Venting a Duplex-Plus”)
13. Loosen the venting screw using an open-jaw wrench
14. Start the material flow with a dispensing command
 - ⇒ Material/air mixture escapes
15. Cancel the dispensing command as soon as only dispensing material is dispensed into the cup
16. Tighten the venting screw
 - ⇒ The whole mixing head unit has been vented

4.3 Machine maintenance before production start

- ✓ The machine is not in production mode
- 1. Check the cup in the filling or optional park or optional weighing position, change if necessary and in case of the park position, refill
- 2. Check the machine hoses for kinks, cracks, to make sure they are complete, etc.
- 3. Make sure the machine and workplace are clean
- 4. Check the pressure settings for the maintenance unit(s) and readjust if necessary
- 5. If dispensing machine: Check all dispensing components – e.g. mixing head(s), dispensing pump(s), dispensing valve(s) – to ensure there is free flow
- 6. If dispensing machine: Check the weighing-in
- 7. If dispensing machine: Check the material supply, refill if necessary or change the pail
- 8. If dispensing machine: replace optional mixing tube
- 9. If tempering or heating: Check the fill level of medium and check for contamination, if necessary add medium or replace
- 10. If cooling unit: Check refrigerant consumption: If the refrigerant is below the lower filling level mark, top up refrigerant up to max. the upper filling level mark
- 11. If needle check: Perform the needle check, if necessary insert a new needle and teach
- 12. If hot riveting system: Check the stamp for cleanliness, if necessary run a cleaning cycle
- 13. If plasma machine: Check the plasma nozzle for signs of wear; if necessary shut the system down and replace the nozzle
- 14. If transport device: Check the wear of the transport belt and replace if necessary
- 15. If transport device: check for obstacles and eliminate them
- 16. If dispensing needle cleaning: Check the sponges and if necessary replace if heavily soiled or worn
- 17. If discharge device for NOK parts: Empty the discharge chute

4.4 Mixing tube maintenance requirement

- ✓ Production was stopped
- ✓ MANUAL operation mode is active
- ✓ The output pressure of the dispensing pump has increased significantly
- 1. Execute blindshots until the entire mixing tube volume has been used once

2. If the pressure increase is still present during the next dispensing process: Increase the blindshot time if necessary to thoroughly flush the mixing tube
3. If the pressure increase is still present despite all measures having been performed: Replace the mixing tube

4.5 Dispensing pump maintenance requirements

- ✓ The mixing tube is dripping
 - ✓ The mixture is no longer hardening
 - ✓ Weighing error
 - ✓ Pump speed was increased for the same flow rate
1. Shut down the machine
 2. Disassemble the dispensing pump as per the maintenance instructions
 3. Clean the pump
 4. Replace the worn parts (e.g. seals, gear wheels, stators, etc.)
 5. Assemble the dispensing pump as per the maintenance instructions
 6. Bleed the dispensing pump
 7. Perform the weighing procedure (see also section "Weighing").
 8. Start a series of 50 weighing shots to be able to evaluate the stability of the weighing system

4.6 Maintenance requirement of emptying the condensate container



Illustration 157: Emptying the condensate cup

1. Place or hold the catch container under the outlet valve of the condensate collector
2. Unlock cup using blue detent

3. Unscrew cup counter-clockwise
4. Empty cup
5. Screw cup in clockwise
⇒ Cup engages in end position

**NOTICE****Disposal!**

Dispose of liquid according to applicable regulations!

4.7 Maintenance request - Blind shot - Replace or empty vessel

Blind shot bins take on the so-called blind shots that occur during pot life processing. These can be specified disposable cups from the lab resources or additional vessels whose presence and fill height are monitored. When a set fill level is reached, the system requests replacement (cup) or emptying (additional vessel).

The procedure is as follows:

1. If the system is in production mode: Press the *< Automatic on >* operating button, so that the ongoing process is ended
2. Press the *< Stop >* operating button
⇒ MANUAL operating mode is activated, access doors are unlocked
3. Open the access door(s)
4. If a cup: Replace the cup
5. If an additional vessel: Empty the vessel
6. Close the access door(s)

4.8 Maintenance requirement - replace weighing container

Weighing shot containers take on the so-called weighing shots that occur during the execution of the weighing sequence - e.g. when weighing in. There are specified disposable cups from the lab resources, the presence and filling weight of which is monitored. When a set weight limit is reached, the system requests a replacement (cups).

The procedure is as follows:

1. If the system is in production mode: Press the *< Automatic on >* operating button, so that the ongoing process is ended
2. Press the *< Stop >* operating button
⇒ MANUAL operating mode is activated, access doors are unlocked
3. Open the access doors
4. Replace the cup
5. Close the access doors

4.9 PPS maintenance requirement

- ✓ Target pressure can no longer be reached
 - ✓ The “HIGH” pre-warning limit has already been reached
1. Shut down the machine
 2. Disassemble the eccentric worm pump as per the maintenance instructions
 3. If the stator is worn: Replace the stator
 4. If the rotor is worn: Replace the rotor
 5. Assemble the eccentric worm pump

4.10 Maintenance prompt from EMERGENCY STOP command devices

- ✓ The machine has been cleaned (dynamic system) or moved to its home position (static and single-component system).
 - ✓ If automatic material transport takes place, the AUTOMATIC operation mode is active.
1. Press the EMERGENCY STOP command device.
 - ⇒ The control and load voltages are disconnected.
 - ⇒ All electrical power consumers are shut down.
 - ⇒ Air supply is disconnected (attention: in dynamic machines, some modules remain supplied with compressed air depending on the system requirements).
 - ⇒ EMERGENCY STOP operation mode is active.
 2. Unlock the EMERGENCY STOP command device again.
 - ⇒ The control and load voltages are switched to the machine.
- ⇒ If the access doors are closed: MANUAL operation mode is active; otherwise, SAFETY mode is active.



CAUTION

Residual energies

Depending on the machine configuration, residual energies may continue to be present immediately after the EMERGENCY STOP command device is triggered due to physical processes such as temperature control, plasma treatment or hot riveting, that are relieved only after a delay.

4.11 Maintenance requirements for the activation switch

- ✓ The access doors are closed
 - ✓ MANUAL operation mode is active
1. Turn the key switch into the 'Setup' position

2. Open the access doors
 - ⇒ SAFETY operation mode is active
 3. Push the operating button of the activation switch halfway in
 - ⇒ SETUP operation mode is active
 4. If required: The axes can be moved manually
 5. Release the operating button of the activation switch
 - ⇒ SAFETY operation mode is active
 - ⇒ Axes cannot be moved
 6. Push the operating button of the activation switch halfway in
 - ⇒ SETUP operation mode is active
 7. Push the operating button of the activation switch all the way in
 - ⇒ SAFETY operation mode is active
- ⇒ Axes cannot be moved

4.12 Maintenance requirements for the door safety switch

- ✓ The access doors are closed
 - ✓ MANUAL operation mode is active
1. Open all access points equipped with a door safety switch
 - ⇒ SAFETY operation mode is active
 2. Close all access points equipped with a door safety switch
 - ⇒ MANUAL operation mode is active
 - ⇒ Doors with a safety lock and doors without a safety lock can be opened
 3. Press the < *Process monitoring on* > operating button
 4. PROCESS MONITORING operation mode is active
 - ⇒ All door safety combinations equipped with door locks are also locked mechanically and cannot be opened
 5. Press the < *Automatic on* > operating button
 - ⇒ AUTOMATIC mode is active
 - ⇒ All door safety combinations equipped with door locks are also locked mechanically and cannot be opened
 6. Press the < *Stop* > operating button
 - ⇒ MANUAL operation mode is active
 - ⇒ All access points equipped with a door safety switch can be opened



NOTICE

Door safety switch with additional < *Unlock door* > button

In some systems, an additional < *Unlock door* > button is linked to the door safety circuit: In the MANUAL operating mode, secured access doors can be opened within 5 seconds of pressing this button.

7.

4.13 Light curtain maintenance requirement

- ✓ PROCESS MONITORING operation mode is active
- 1. Call up the “Axes” screen when checking via axes or the “Rotary table” screen when checking via rotary table
- 2. To check via the axis system: Press the symbol key for an axis move, e.g. “Set-up position”, and during the axis move, hold the black measuring tube (included in the system scope of delivery) in the detection range of the light barrier to check the light barriers
 - ⇒ Axis move must stop immediately
 - ⇒ When the measuring tube has been removed from the detection range of the light barrier, the axes are powered but do not move without a command
- 3. To check via a component feed, such as rotary table or slider: Start the rotary table manually, and during the rotary movement, hold the black measuring tube (included in the system scope of delivery) in the detection range of the light barrier to check the light barriers
 - ⇒ The rotary table stops immediately
 - ⇒ When the measuring tube has been removed, the rotary table rotates to the end position and remains there

4.14 Maintenance requirement - Festo cylinder



NOTICE

Lubrication of the piston rod

The piston rods of the Festo cylinder must always have a grease film. If there is no longer any film, this could lead to damage or severe wear to the seals.

Use the following grease for this purpose: LUB-E1.

4.15 Maintenance requirement - vacuum shut-off valve of follower plate

The vacuum shut-off valve is a double-acting cylinder mounted on the upper side of the follower plate. It actuates a sealed disc to open and close a bore that leads into the pail space. When the follower plate is lowered onto the pail, the bore is opened and a pneumatic vacuum hose draws out the remaining air. The bore is then sealed again. Inversely, when the follower plate is raised, e.g. for changing the pail, this pneumatic line is pressurised with compressed air to remove any material residues. The condition of the hose can be evaluated via a transparent hose section.

Depending on the material, volatile constituents such as oils, etc. can accumulate and solidify on the inner walls of the ventilation hose during the extraction process, possibly even blocking the hose. Furthermore, the seal on the disc is not resistant to every chemical constituent. Constituents of the material in the pail can then migrate via a porous seal into the hose.

Maintenance procedure:

For maintenance purposes, lift the follower plate out of the pail, ideally during a pail change, and inspect the condition of the seal of the plate valve disc on the underside of the follower plate and the transparent hose section that represents the start of the ventilation hose:

If the seal is porous or damaged in any other way, replace the complete vacuum shut-off valve.

If deposits are discovered in the transparent section of the ventilation hose, replace this hose section.



NOTICE

Seal material

If the disc seal shows regular signs of wear, the cause may also lie in an incompatibility of constituents of the material in the pail with the seal materials. If in doubt, contact bdtronic Applications Engineering for advice on how to proceed.



NOTICE

Pail change

When changing the old pail, a fleece made from a compatible material must be placed on the middle of the pail surface. The corresponding order number can be found in the machine pass of the system. Please use only this fleece type!

4.16 Maintenance prompt – Internal calibration of Mettler scales

The visualisation system function "Internal Calibration" is used when scales are put into operation for the first time and after a change of location. This means that every machine that is put into operation on the customer's premises, for example, must first go through this procedure.

- ✓ Scales of type 'WKC' are installed
- ✓ Serial interface via the BECKHOFF terminal is active: the serial number of the scales module appears on the visualisation system
- ✓ This function was enabled in the machine visualisation system during commissioning
- ✓ User is logged in with the login right 'Level 4'

1. Press the < *Internal calibration* > short-cut key





NOTICE

Key status

If an internal calibration has already been carried out beforehand, the < *Internal calibration* > short-cut key lights up green, otherwise grey or red in the event of a fault.

- ⇒ The following message appears: "Calibration started"
- ⇒ If calibration is OK: < *Internal calibration* > short-cut key lights up green
- ⇒ If the calibration has failed: < *Internal calibration* > short-cut key lights up red

4.17 Maintenance of mixing and dispensing components

4.17.1 Maintenance work on the static Duplex mixing head

A static dispensing system usually works without shut off valves. With this system type, the specific potting materials are routed through the corresponding channels into a mixing tube by means of dispensing pumps. Here, the mixing tube has the task of homogeneously mixing the two different components with each other by means of the mixing-spiral geometry.

During the ongoing dispensing process, a so-called working pressure forms between the individual modules of the dispensing pump and the outlet nozzle. This working pressure arises dependent on various factors, for example: dispensing output, mixing tube length, mixing tube diameter, material viscosity.

In the event of a dispensing stop, the existing overpressure equalises to the atmospheric pressure. This equalisation is time-dependent on the previously mentioned factors. As a result, during the drop in pressure the mixing tube is relieved, so to speak. Volume is displaced due to this relief, which can in turn lead to a certain run-on effect of the dispensing material.

If necessary, this response can be compensated by briefly turning back the dispensing pumps after dispensing stop, and can be influenced by the parameters "Switch-off delay" and "Outlet speed" in the dialog "Pumps A/B Dialog".

Relevant are different factors, for example: wear of machine components, leaks, air bubbles in the system, incorrect parameters, etc., which could still lead to running-on or dripping.

Running-on and dripping:

The terms running-on and dripping are used if, after the dispensing process, the dispensing material continues to flow steadily or in drip form from the mixing tube.

4.17.1.1 Analysis of the material flow after the dispensing



NOTICE

Maintenance program!

With respect to various maintenance operations such as, for example, bleeding of the dispensing system, the set monitoring limits are often violated. For this it is recommended to create a copy of the previously set-up dispensing program and to name it accordingly "Maintenance" and save it. In this copy you can change all process-relevant parameters to meet your specific maintenance needs.



NOTICE

For machine run-in!

If relevant dispensing components are replaced during the course of a maintenance, such as: dispensing pumps or mixing head elements, these components must be previously filled without a mixing tube. Fill them on the system side with the corresponding material component until material is discharged from the mixing tube. In this case, a mounted mixing tube, could serve as a connecting bridge for the components A and B and thus press dispensing material in the unfilled dispensing channel side of the component on the opposite side.

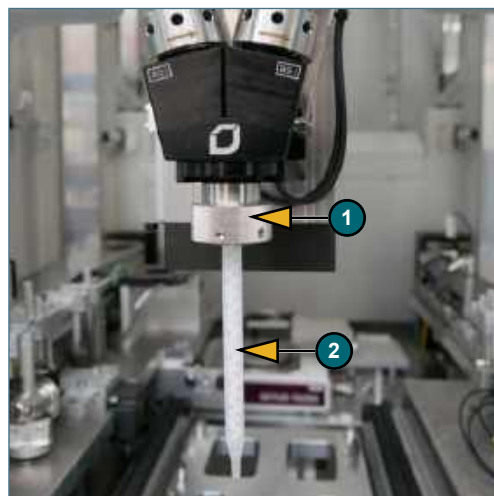


NOTICE

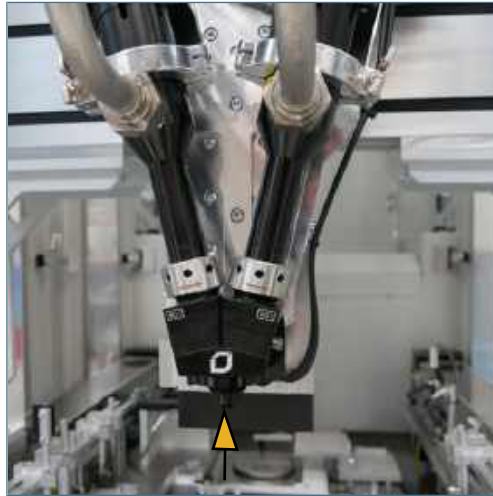
Procedure for bleeding!

Bleed each side of the system sequentially; this allows mixing of different dispensing materials in the respective dispensing chambers.

1. Loosen and take off the cap nut with a suitable hook spanner and remove the mixing tube of the static mixing head.



2. During the dispensing, inspect the individual components and observe whether there is a continuous material flow from the respective outlet holes.



Case representation of disruptions during the dispensing or after dispensing of the material flow.

Case 1:

In the system visualization, check the process-relevant parameters (initial values can be found in the original MFU setting data) for their process capability, e.g. "Dispensing speed" and "Switch-off speed".

Case 2:

Not all material flows through the intended outlet holes of the respective component. -> Refer to the item: "**Dismantling the mixer holder**".

Case 3:

No abnormalities in the material flow can be detected. -> Refer to the item: "**Bleeding operation at the pump holder**".

Case 4:

Fracturing or air inclusions during the flow of material can be detected. -> Refer to the item: "**Bleeding operation before the dispensing pump**".

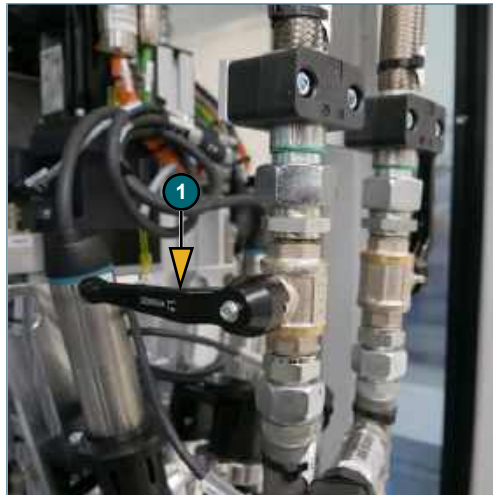
Case 5:

After the pumping has stopped, continued running-on of the dispensing material. -> Refer to the items: "**Dismantling the eccentric worm pump from the mixing head**" "**Replacing a rotor/stator on an eccentric worm pump**".

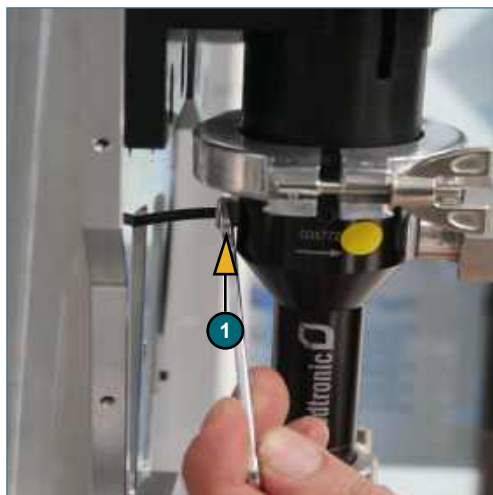
4.17.1.2 Bleeding operation before the dispensing pump

1. In the system visualization, put the material processing in the "MANUAL" mode.

2. Turn the ball valve of the system side to be bled to the "Close" position.



3. Attach the ventilation hose to the adjusting screw for bleeding and place a suitable catch container underneath.
4. Unscrew the adjusting screw about 1-1.5 turns to bleed.
⇒ The material flows.



5. Turn the ball valve to the "Off" position.



6. Put the material preparation in the system visualization in the "Automatic" mode.
7. Pump the dispensing material until bubble-free material flows.
8. Switch off material preparation.



NOTICE

No bubble-free dispensing material

No bubble-free dispensing material is attained. Bleed the material supply lines of the material preparation.

4.17.1.3 Bleeding operation at the pump holder

1. Unscrew the adjusting screw for bleeding about 1-1.5 turns on the pump housing with a 8-size open-ended spanner.

⇒ The material flows.



2. Continue dispensing until a bubble-free dispensing material flows from the adjusting screw for bleeding and then close the adjusting screw again.

4.17.1.4 Dismantling the mixer holder

1. Loosen the Allen screws on the mixer holder with an Allen key and remove them.



2. Remove the mixer holder.



NOTICE

Cleaning the underside of the pump housing

After dismantling the mixer holder, clean the underside of the pump housing, but only from the back of the machine in the direction of the machine access side (loading door). The reason for this is that, with this kind of cleaning, no components get mixed together.



NOTICE

When dismantling the mixer holder

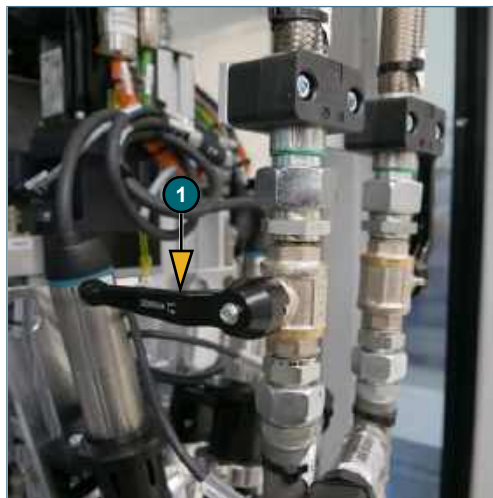
When dismantling the mixer holder, ensure that none of the installed small parts are lost. The following small parts are required: 3 x centring sleeves, 2 x seals

**NOTICE****Cleaning residual material from the mixer holder**

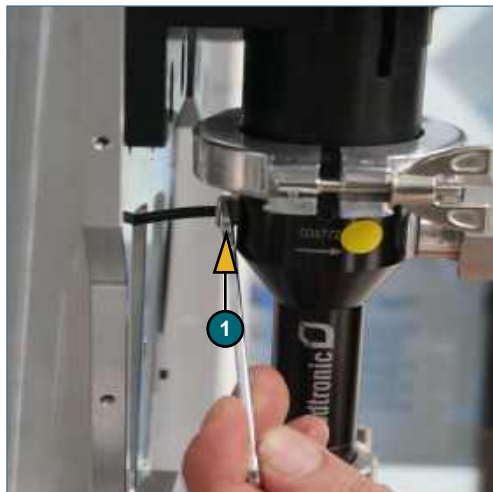
After dismantling the mixer holder, properly remove all O-rings in case of any wear and tear (cracks, pinched areas, etc.) is detected; replace them if necessary with new O-rings. Then, clean any residual material from the mixer holder.

4.17.1.5 Dismantle the eccentric worm pump from the mixing head

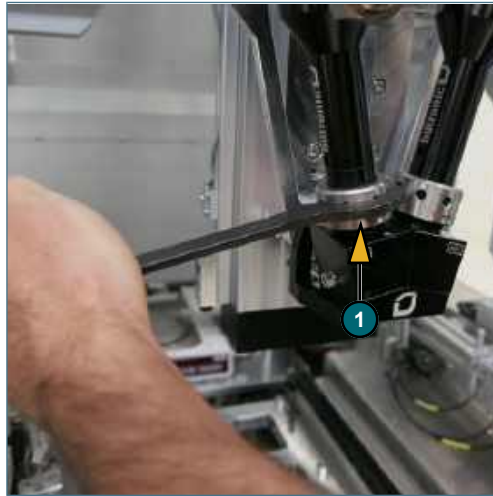
1. In the system visualization, put the material processing in the "MANUAL" mode.
2. Turn the ball valve of the respective system side to the "Close" position.



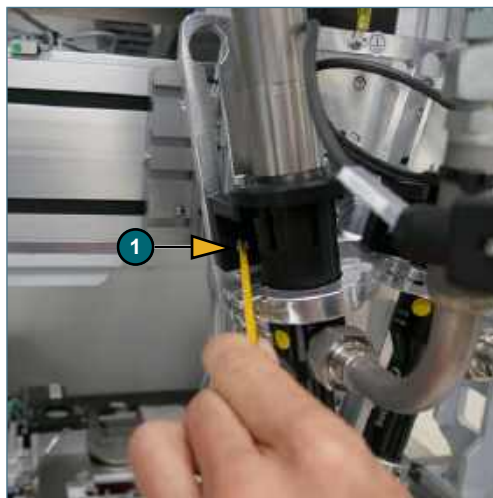
3. Put a suitable catch container in place and relieve the remaining pressure with the adjusting screw for bleeding, located above the dispensing pump.



4. Loosen the fastening nuts of the dispensing pump with suitable hook spanner opposite to the direction indicated by the arrow and completely screw on (left-hand thread).



5. Loosen the Allen screws on the dispensing pump bracket.



6. Pull the dispensing pump upward, then turn it to the side and secure it with the Allen screws.



7. Loosen the material supply lines on the dispensing pump and remove them.



8. Open the fastening clamps and remove them.



9. Pull the dispensing pump downward.



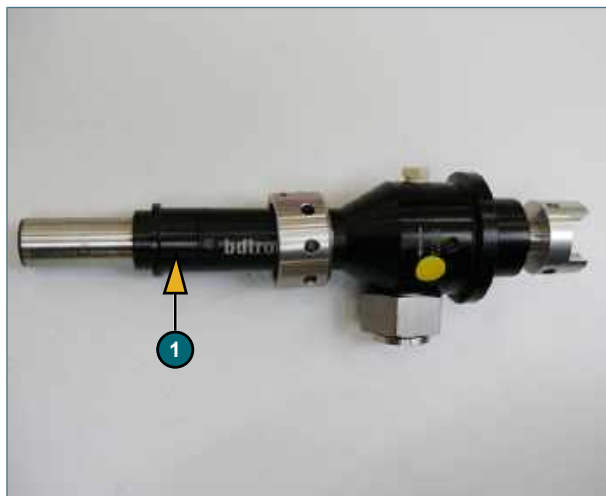
NOTICE! Further procedure for the dismantling of the dispensing pump is described in the dialog -> **"Replacing a rotor/stator on an eccentric worm pump"**.

4.17.1.6 Replacing a rotor/stator on an eccentric worm pump

- ✓ A removed bdtronic eccentric worm pump.



1. Loosen and remove the housing/adapter.



2. Removing the stator from the rotor shaft

- ⇒ Pull the stator downward out of the axis lock. **NOTICE! If this is not possible, hold the stator firmly and turn the drive shaft at the rotor head end in the direction of the arrow marking on the dispensing pump.**



3. Loosen the Allen screw on the pump coupling with an Allen key and remove it.



4. Secure the drive shaft end against twisting using an appropriate tool, loosen and remove the Allen screw of the rotor, then downwardly pull out the rotor.



5. Clean the remaining material from the dispensing pump.

4.17.1.7 Mounting a bdtronic eccentric worm pump

1. Check the rotor for signs of wear and tear, e.g.: scoring, scratches, etc.



NOTICE

New rotor

If a new rotor is required due to wear and tear, then it is always delivered by bdtronic GmbH with a new Allen screw and seal (O-ring). This new Allen screw has a screw lock and therefore it may only be used once for safety reasons.

2. Completely moisten the O-ring on the rotor from the **inside** and the **outside** with Vaseline.



3. Carefully insert the rotor into the closed guide groove until it is exactly in position, then exert gentle pressure on the rotor until a slight click sound is heard.



NOTICE

Check the seating of the O-ring!

The seating of the O-ring can be examined by glancing over the pump inlet. If the O-ring is seated correctly in the groove, then it is completely hidden in the guide groove and is no longer visible from the outside.

4. Assemble the rotor with the new supplied parts (**Allen screw and o-ring**) and fasten securely with the exact torque of 3 Nm.



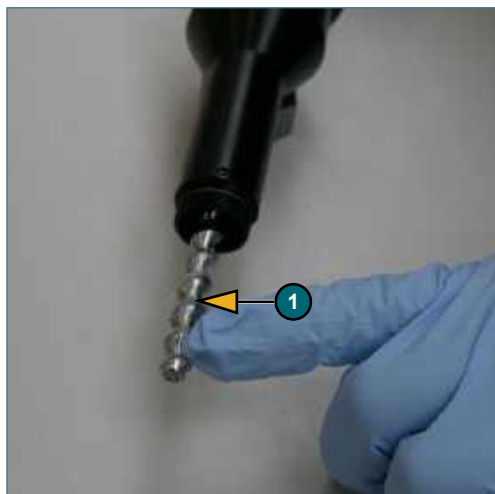
5. Moisten the rotor with potting compound.



NOTICE

Putting the rotor and stator together

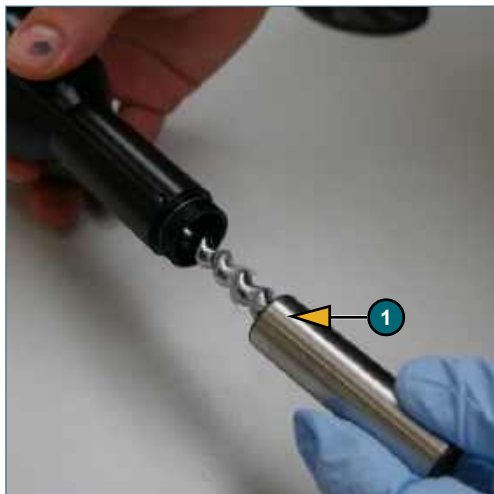
The rotor and the stator must never assembled together dry.



6. Screw on the stator with the axis lock upward and in clockwise direction until just before contact with the locking screw.

**NOTICE****Axis lock on the stator**

Turn in the axis lock until the groove is still easily recognizable. The groove is located on the same position line as the marking on the stator.

**CAUTION****Grub screw for locking**

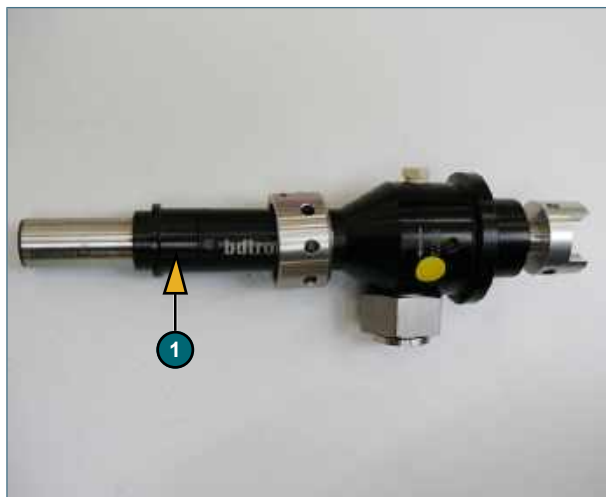
The grub screw must **not be** tightened or loosened with an Allen wrench as it is **set and glued in place**.

7. The stator is pulled into its desired position by means of the rotor shaft. To the stator must be aligned so that the axis lock on the stator is in line with the locking screw in the pump housing. Using the supplied special spanner, turn at the end of the rotor until the stator is properly seated in the axis lock.

**NOTICE****Rotor/stator control check**

Upon successful installation of the rotor and stator, the rotor is positioned flush vis-à-vis the stator or slightly below it.

8. Push on the cap nut and screw on the housing adapter by hand.



9. Push on the coupling part with the cylinder pin until the cylinder pin sits at the end of the rotor shaft recess, then tighten the Allen screws with exactly 3 Nm torque.

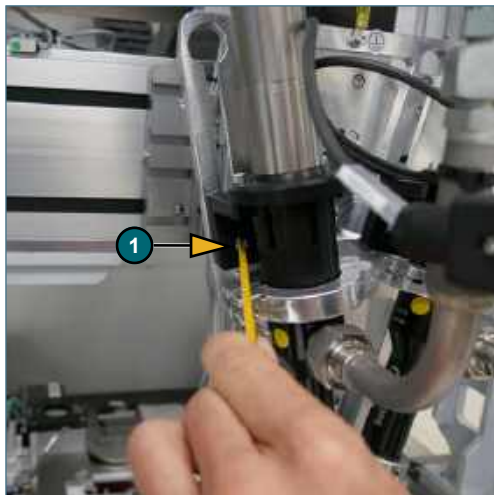


4.17.1.8 Replacing the Wittenstein motor

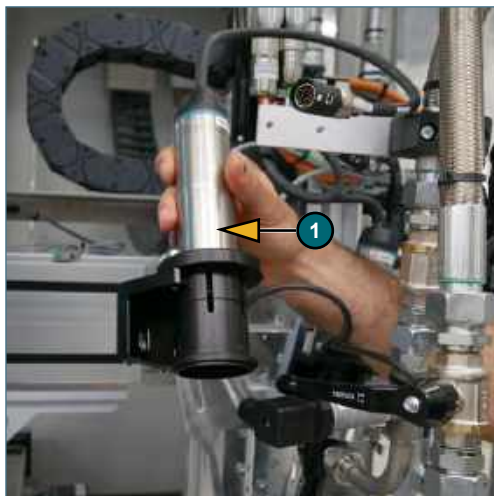
1. Unlock and disconnect the electrical plug connector.



2. Loosen 2x Allen screws on the motor mounting plate and remove the fastening clamp.



3. Upwardly pull away the Wittenstein motor and remove it.



4. Loosen 2x Allen screws from the bracket to the Wittenstein motor and remove.



5. Loosen the grub screws and remove the coupling.



6. Loosen 4x Allen screws in the coupling housing and remove.



7. Pull off the coupling housing.

4.17.2 Maintenance work on bdtronic progressing cavity pump PPS

Application

The progressing cavity pump for use in bdtronic PPS pail preparation systems comes in two designs: one for vacuum chamber (depicted here) and one for use without vacuum chamber. The following description of the maintenance work can be used for both types.

4.17.2.1 Changing the wiping ring

1. Loosen screws of the holder of the follow-up plate (3x pos. 1)

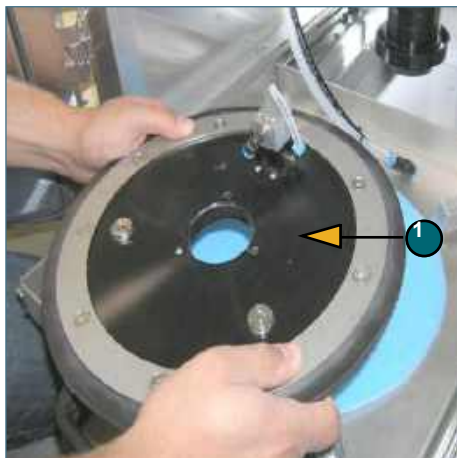


2. Disconnect pneumatic hoses (pos. 1)

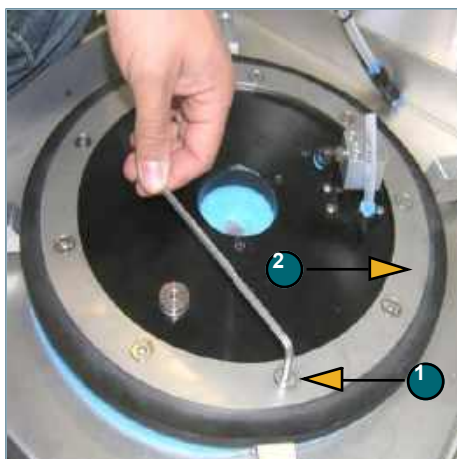


3. Press button ("MANUAL") (MANUAL operating mode is active)

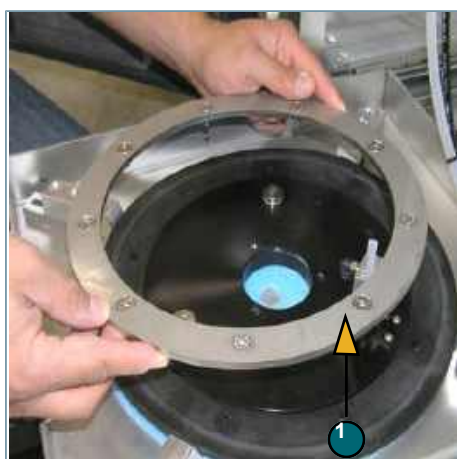
4. Press Arrow up button ("Manual pail") until you can pull the follow-up plate (pos. 1) from the pump body



5. Loosen holder screws (8x pos. 1) on support ring (pos. 2)



6. Remove support ring (pos. 1)



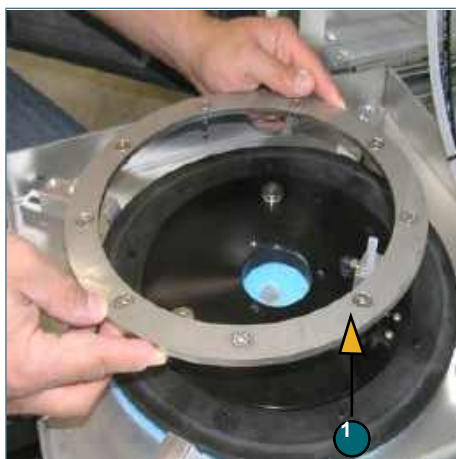
7. Wipe off holder plate of wiping ring (pos. 1)



8. Change wiping ring (pos. 1) if worn



9. Attach support ring (pos. 1)



10. Screw support ring to retaining plate

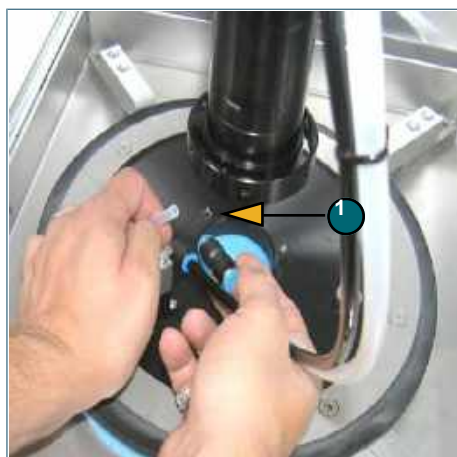


4.17.2.2 Changing the stator

1. Loosen screws of the holder of the follow-up plate (3x pos. 1)

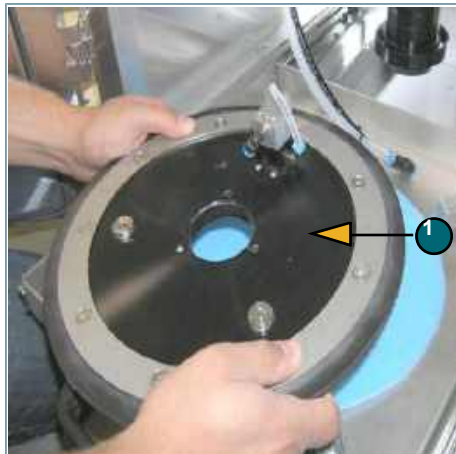


2. Disconnect pneumatic hoses (pos. 1)

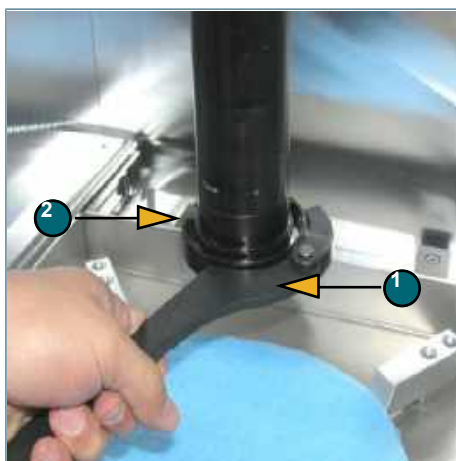


3. Press button ("MANUAL") (MANUAL operating mode is active)
4. Press Arrow up button ("Manual pail lift") until the follow-up plate (pos. 1) is in the upper end position

5. Pull follow-up plate off pump body



6. Loosen end supports (pos. 2) with hook spanner (pos. 1)



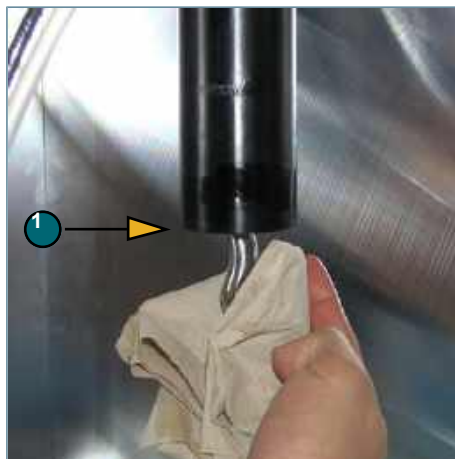
7. Unscrew end supports (pos. 1)



8. Push stator (pos. 1) downward while turning slightly

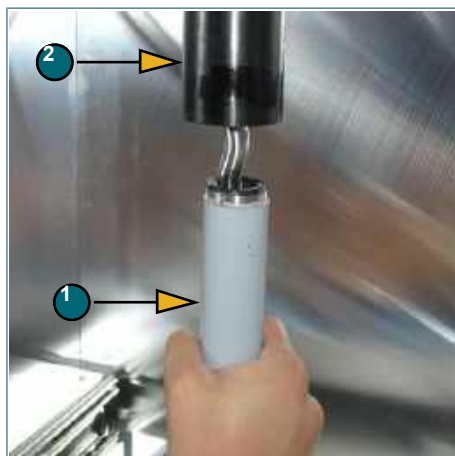


9. Clean rotor (pos. 1) with cloth and spatula



10. Grease rotor slightly with dispensing material due to friction

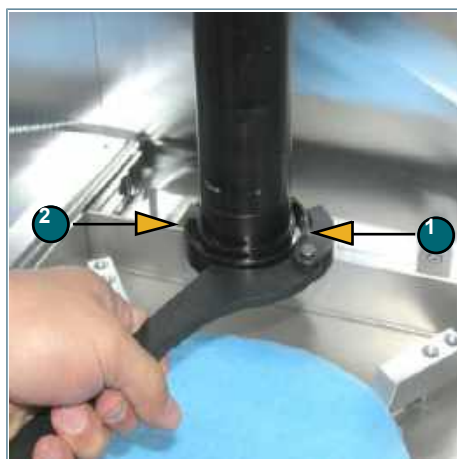
11. Screw the new stator (pos. 1) onto the rotor (pos. 2) so that the stator side points upward with the notch on the material side



12. Screw end supports (pos. 1) to pump housing



13. Fix end supports (pos. 2) with hook spanner (pos. 1)



14. Connect pneumatic hoses (pos. 1)



15. Fix follow-up plate (3x pos. 1)

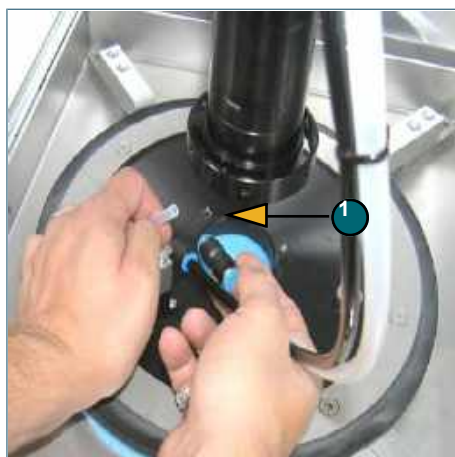


4.17.2.3 Changing the rotor

1. Loosen screws of the holder of the follow-up plate (3x pos. 1)

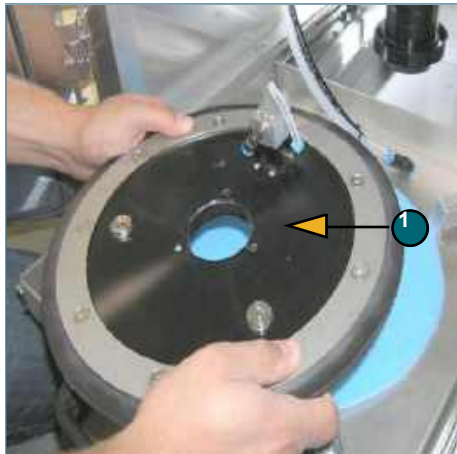


2. Disconnect pneumatic hoses (pos. 1)

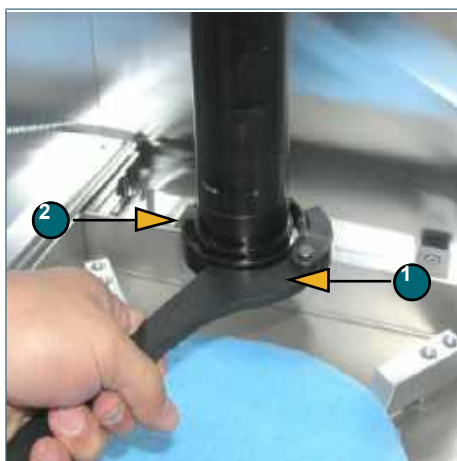


3. Press button ("MANUAL") (MANUAL operating mode is active)
4. Press Arrow up button ("Manual pail lift") until the follow-up plate (pos. 1) is in the upper end position

5. Pull follow-up plate off pump body



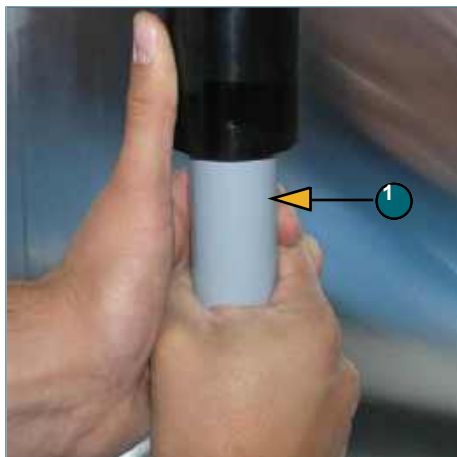
6. Loosen end supports (pos. 2) with hook spanner (pos. 1)



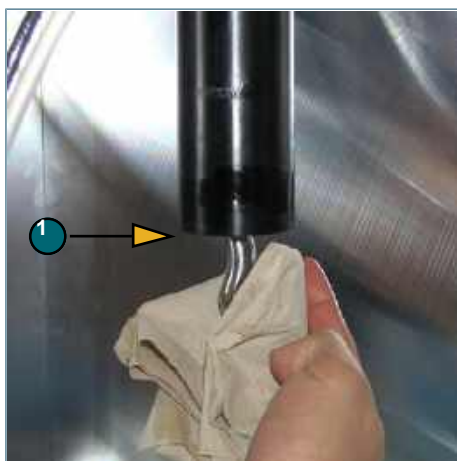
7. Remove end supports (pos. 1)



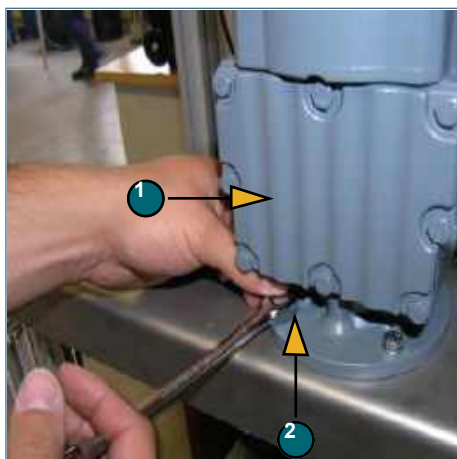
8. Pull stator (pos. 1) downward and out



9. Clean rotor (pos. 1) with cloth and spatula



10. Unscrew holder screws (4x pos. 2) of the motor (pos. 1)



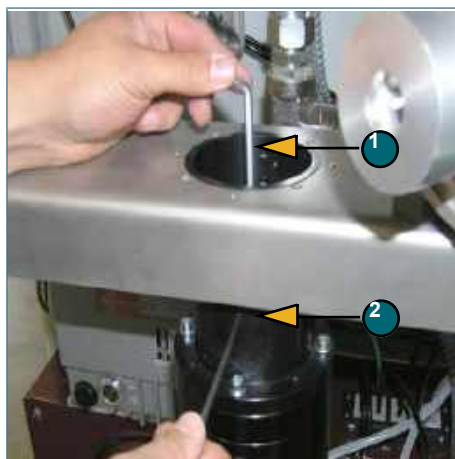
11. Remove motor



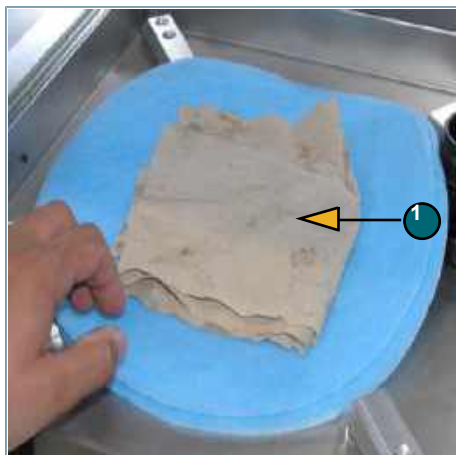
12. Put motor aside and secure against falling



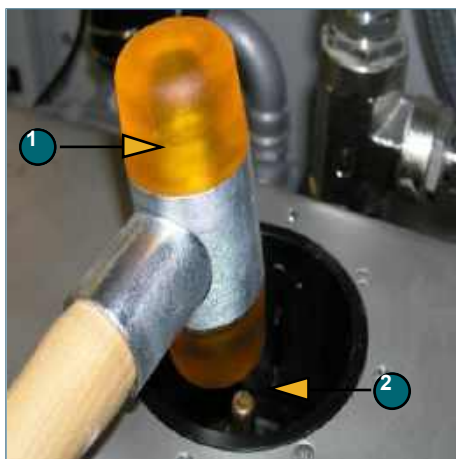
13. Loosen rotor locking screw with Allen spanner (pos. 1) while blocking shaft with second Allen spanner (pos. 2)



14. Pad pail seat (pos. 1)

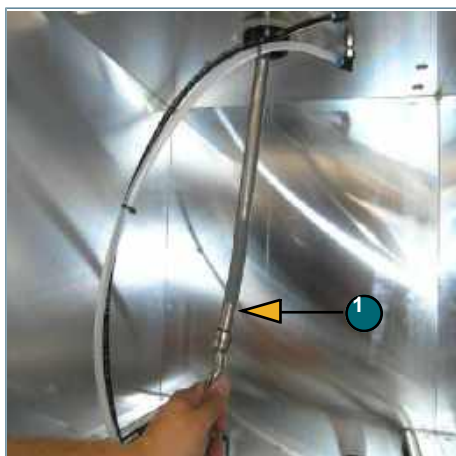


15. Knock rotor carefully out downward with rubber mallet (pos. 1) and puncture (pos. 2)

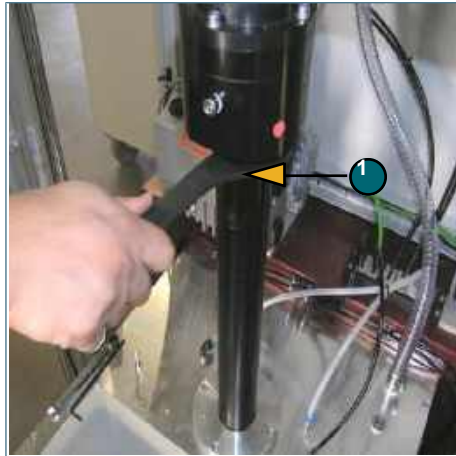


16. Pull drawer out

17. Pull out rotor (pos. 1)



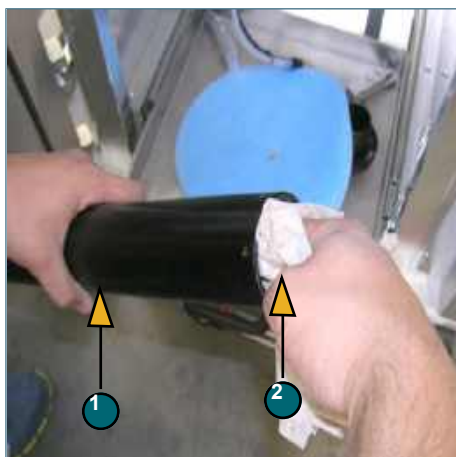
18. Open threaded pump body-upper part of pump with hook spanner (pos. 1)



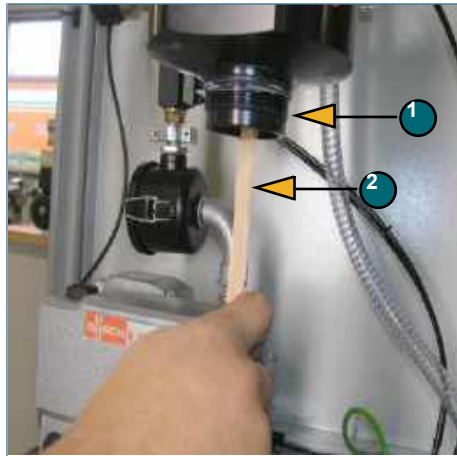
19. Open threaded pump body-upper part of pump with hook spanner (pos. 1)



20. Clean pump body (pos. 1) with spatula and cloth (pos. 2)



21. Clean upper part of pump (pos. 1) with spatula (pos. 2) and cloth



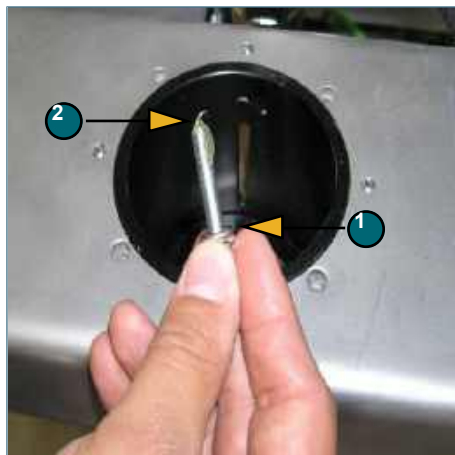
22. Grease seal on plug-in coupling of the new rotor



23. Insert rotor, turning it, until it snaps in



24. Place USIT-ring (pos. 1) on rotor locking screw (pos. 2)



25. Insert rotor locking screw and tighten, while blocking rotor movement .B. with an Allen spanner



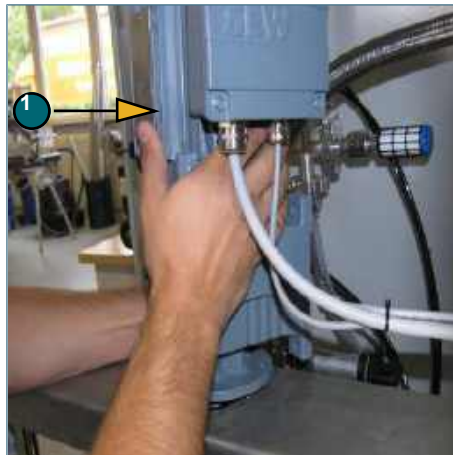
26. Insert lower part of coupling (pos. 1)



27. Screw lower part of coupling (pos. 1) on motor shaft so that shaft end ends approx. 3 mm before end of upper part of coupling



28. Attach motor (pos. 1) so that both coupling parts grip one another



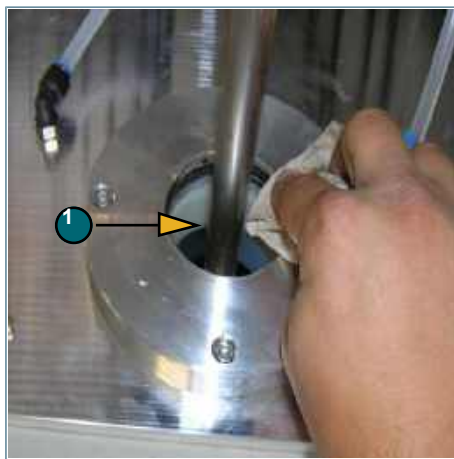
29. Tighten motor fastening screws (4x pos. 1) crosswise



30. Grease seal (pos. 1) on the attachment of the upper part of pump

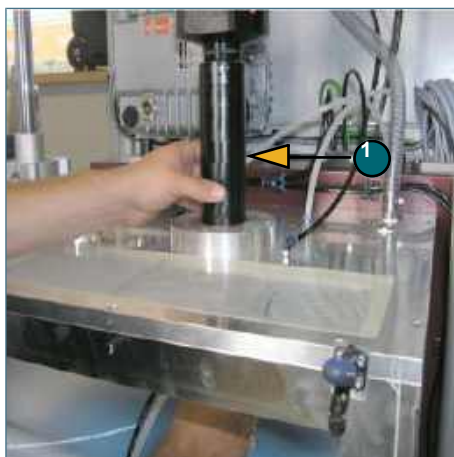


31. Clean lip seal (pos. 1) if necessary; replace if damaged

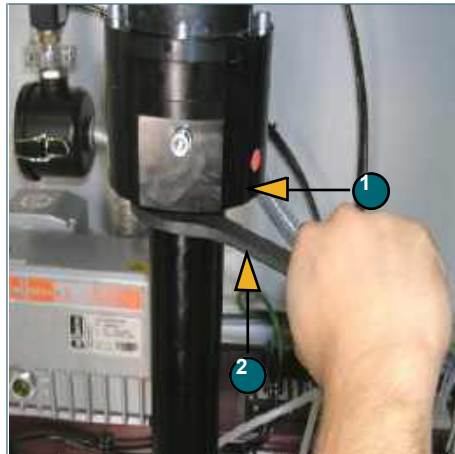


32. Insert pump body (pos. 1) from below through the chamber ceiling

33. Screw pump body to upper part of pump



34. Fasten pump body (pos. 1) with hook spanner(pos. 2)

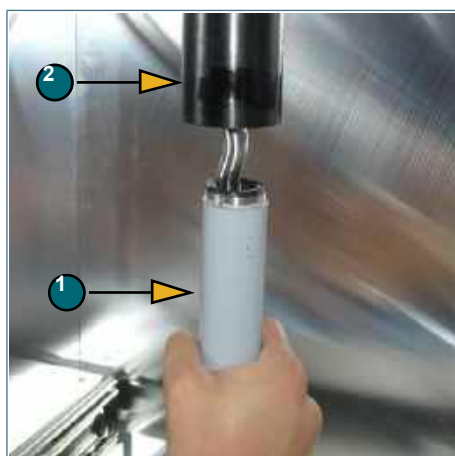


35. Remove rotor guard (pos. 1)

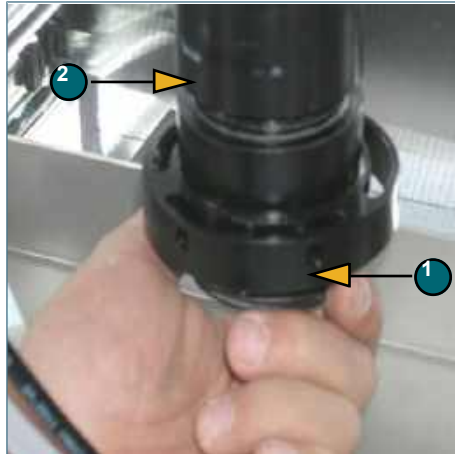


36. Grease rotor slightly with dispensing material due to friction

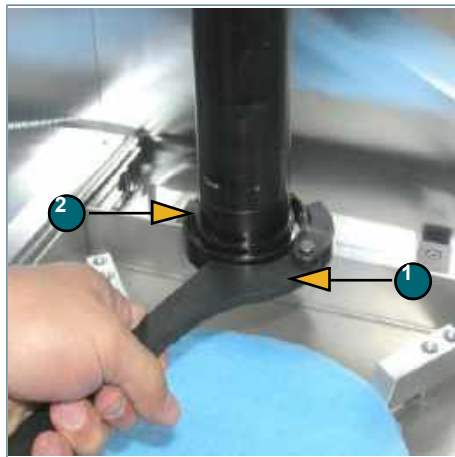
37. Turn stator (pos. 1) onto rotor (pos. 2)



38. Screw end supports (pos. 1) onto pump housing (pos. 2)



39. Fix end supports (pos. 2) with hook spanner (pos. 1)



40. Connect pneumatic hoses (pos. 1)



41. Fix follow-up plate with screws (3x pos. 1)



4.18 Maintenance of special components

4.18.1 Oil change vacuum pump

- ✓ Oil is contaminated
 - ✓ The machine has been shut down
 - ✓ Vacuum pump has cooled off
 - ✓ Protective clothing is worn
1. Place empty oil catch container with sufficient capacity under oil drain screw
 2. Open oil drain screw
 - ⇒ Oil runs into catch container



NOTICE

Disposal of old oil

Dispose of old oil as special waste according to the applicable provisions!

3. Close the oil drain screw again
4. Change the oil separator according to the manufacturer's instructions



NOTICE

Original replacement part

Only use original replacement parts during maintenance!

5. Change filter cartridge according to manufacturer's instructions

**NOTICE****Original replacement part**

Only use original replacement parts during maintenance!

6. Fill vacuum pump oil

**NOTICE****Type of oil for vacuum pump**

Only use oil recommended by the manufacturer to refill (see pump nameplate!) : Incorrect oil properties damage vacuum pumps!

7. Close oil filling screw
8. Blow ventilator cover, ventilator wheels, ventilator grille and cooling ribs dry
9. Re-start machine

4.18.2 Oil level vacuum pump

1. Check filling level through the inspection glass: Level must be within upper and lower setting mark
2. If filling level is outside the permitted range:
 1. Stop the machine immediately
 2. Check oil type for refilling
 3. Open the oil inlet screw
 4. Only refill permitted oil type until upper setting mark at most
 5. Close oil inlet
 6. Start the machine

**NOTICE****Oil type for refilling use**

Only use recommended oil type for refilling! Wrong oil specification irreversibly could damage pump!

4.19 Lubricating LineTech axles

- ✓ The machine has been shut down

1. Press the lubricant into the lubrication nipples of all axes with a grease press

**NOTICE****Type of grease**

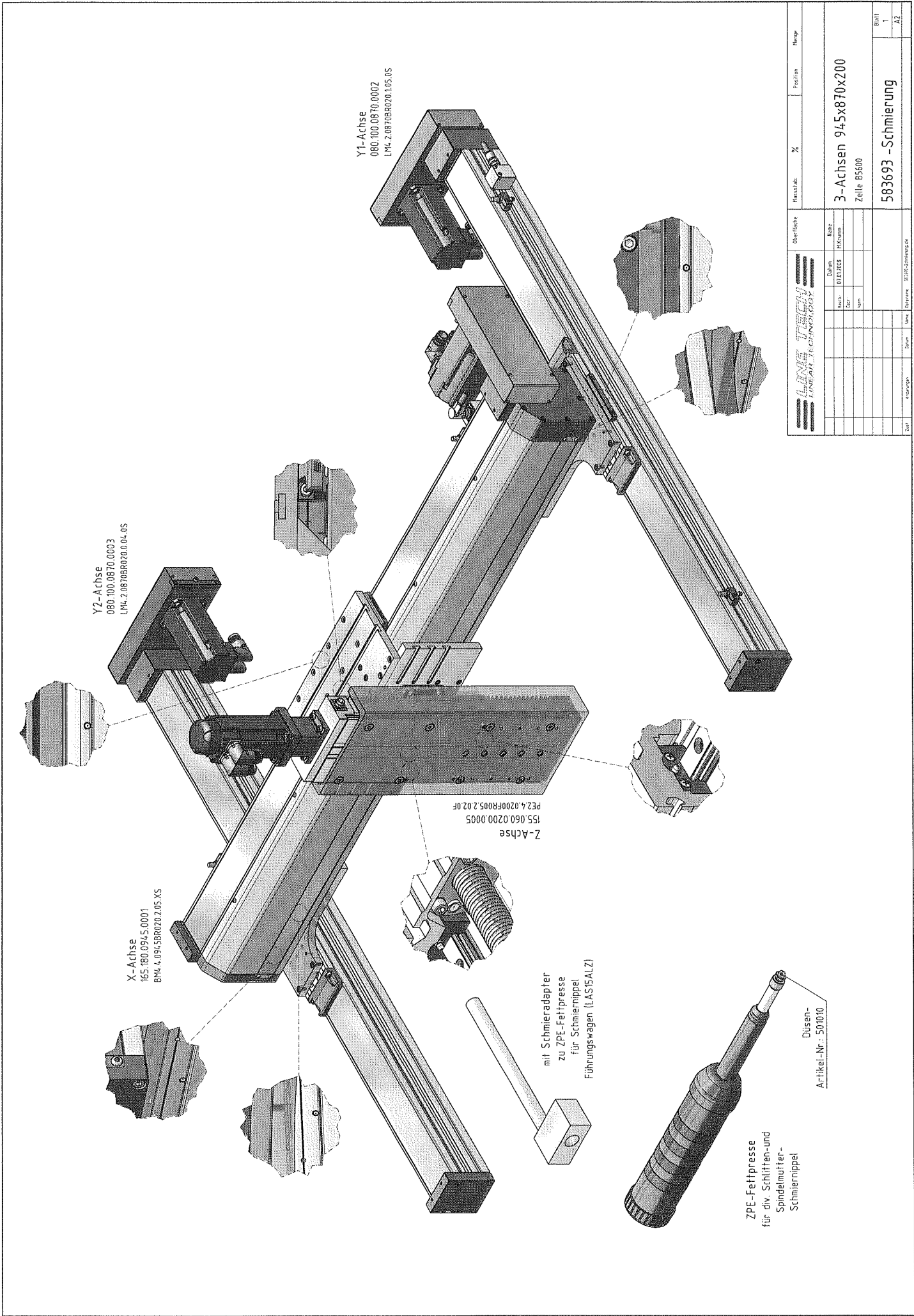
Only use the grease recommended by the manufacturer, e.g. "MICROL-UBE 131" for LineTech axes.

2. See the manufacturer's lubrication plan:

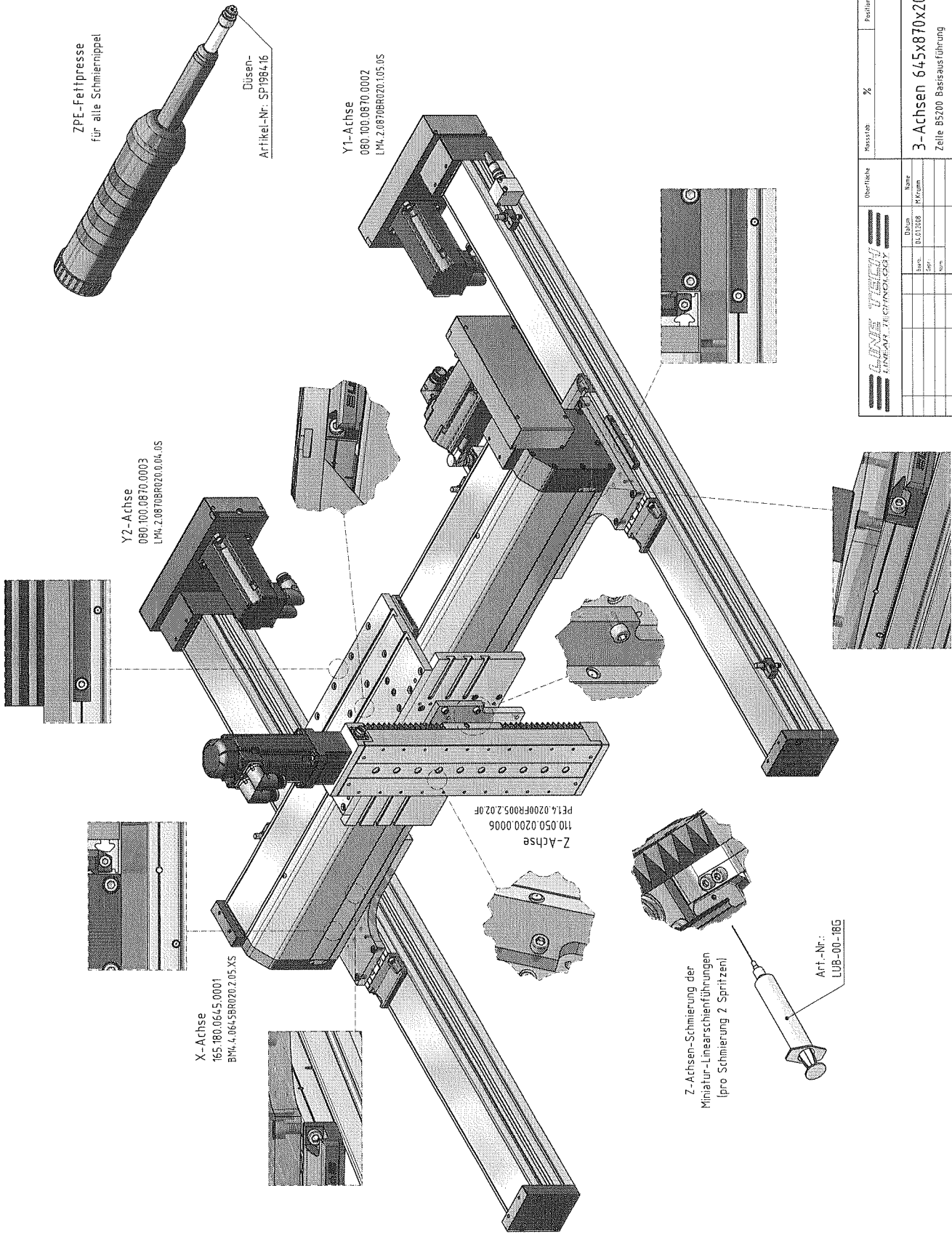
Also see about this

 Lubricating LineTech axles [► 70]

 Lubricating LineTech axles [► 79]



Maschinen		Modell	Position	Reihe
3-Achsen 945x870x200				
Zelle 85500				
583693 -Schmierung				
Bauteil				
1				
A2				



Oberfläche		Profil	Reihe
Material	%		
3-Achsen 645x870x200			
Zelle 85200 Basisausführung			
583691 -Schmierung			
Blatt		1	
Zell		A2	



Handhabungs-Hinweise - 2 -
Positioniereinheiten

Handling Instructions - 3 -
Positioning Unites

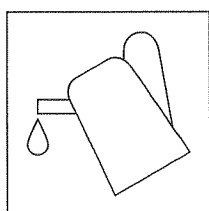
Conseils de manipulation - 4 -
Unités de positionnement

Istruzioni per l'uso - 5 -
Unità di posizionamento

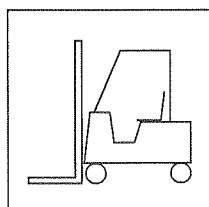
Instrucciones de manejo - 6 -
Unidades de posicionamiento

Handhabungshinweise LineTech-Positioniereinheiten

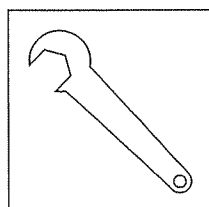
Bitte lesen Sie folgende Handhabungshinweise sorgfältig durch. Um einen optimalen Einsatz und eine lange Lebensdauer der LineTech-Positioniereinheiten zu garantieren, müssen sämtliche Punkte genau befolgt werden. Bei Unklarheiten wenden Sie sich bitte an Ihr LineTech-Team.



Schmierung → LineTech-Positioniereinheiten sind ab Werk mit Microlube GBU Y 131 geschmiert: ca. alle 500 Betriebsstunden nachschmieren. Bei Verwendung von Fremdfabrikaten die Verträglichkeit mit dem Standard-Schmierfett prüfen, falls nötig auswaschen. Linearschienen und Spindeln über Schmiernippel genügend schmieren. Zahnriemen und Lager sind lebensdauergefettet.



Transport → Achsen mit mehr als 700 mm Länge beim Transport immer unterstützen.



Montage → Auflagefläche

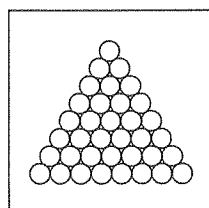
	0.05
--	------

Einzelachsen

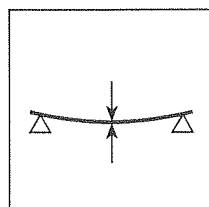
	0.05	1000
--	------	------

Doppelachsen

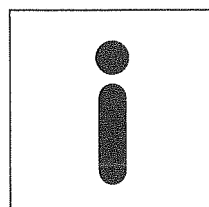
	0.02
--	------



Lagerung → eben und trocken lagern.



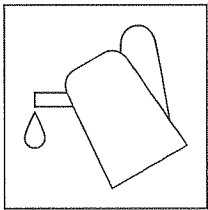
Durchbiegung → Bei freitragenden Positioniereinheiten darf die Neigung nicht grösser als 5' werden.



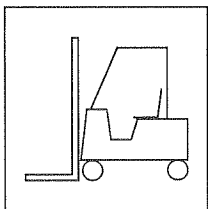
Achtung → Standardpositioniereinheiten werden mechanisch betriebsbereit geliefert. Nachjustierungen sind keine mehr nötig.

Handling Instructions LineTech Positioning Units

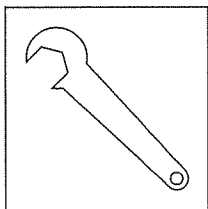
Please read the following handling instructions very carefully. To ensure the best possible performance and a long life with LineTech positioning units all instructions must be strictly observed. If you have any queries please contact your LineTech team.



Lubrication → LineTech positioning units are lubricated at the works with Microlube GBU Y 131 grease: regrease after approx. every 500 operating hours. If using other lubricants, test compatibility with the standard grease, and if necessary wash out. Grease the linear ways and screws adequately via the lubricating nipples. Toothed belts and bearings are lubricated for life.



Transport → During transport always support axles with lengths of more than 700 mm.



Mounting → Bearing surface

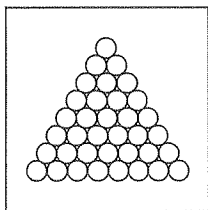
 0.05

Single axles

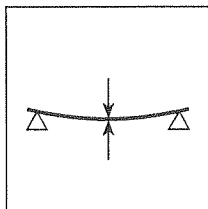
 0.05 1000

Double axles

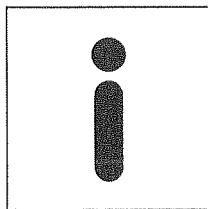
 0.02



Storage → Store level and dry.



Deflection → The deflection of unsupported positioning units is not to exceed 5'.

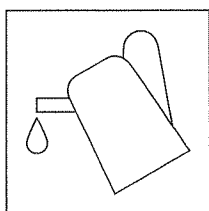


Please not → Standard positioning units are supplied mechanically ready for use. No readjustments are required.

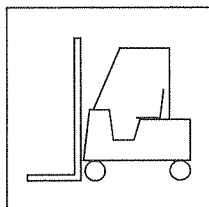
Conseils de manipulation

Unités de positionnement LineTech

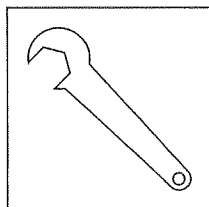
Lire attentivement les directives suivantes. Pour garantir une utilisation optimale des unités de positionnement LineTech, respecter rigoureusement tous les détails indiqués. En cas de doute, n'hésitez pas à contacter votre équipe LineTech.



Lubrification → Les unités de positionnement LineTech sont lubrifiées départ usine avec de la graisse Microlube GBU Y 131: Renouveler le graissage env. tous les 500 heures de marche. En cas d'emploi de produits étrangers, vérifier leur compatibilité avec la graissage standard. Laver à fond, si nécessaire. Lubrifier suffisamment les guidages linéaires et les broches par les graisseurs. Les courroies crantées et les paliers sont graissés à vie.



Transport → Au transport, toujours étayer les axes plus de 700 mm de longueur.



Montage → Surface d'appui

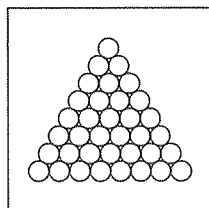
 0.05

Axes simples

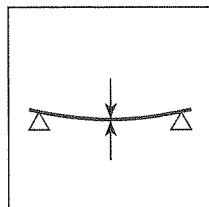
 0.05 1000

Axes doubles

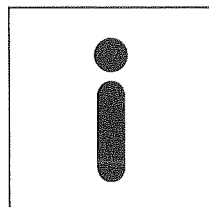
 0.02



Stockage → A plat et au sec.



Flexion → Pour les unités de positionnement non soutenues, la flèche ne doit pas dépasser 5'.

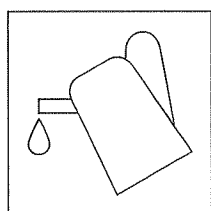


Attention → Les unités de positionnement standard sont livrées mécaniquement prêtes à fonctionner. Plus nécessaire de les réajuster.

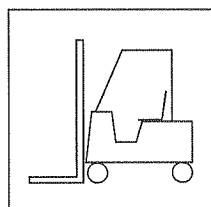
Istruzioni per l'uso

Unità di posizionamento LineTech

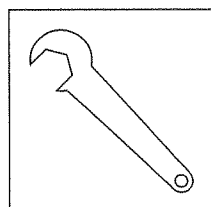
Si prega di leggere attentamente le seguenti istruzioni per l'uso. Per garantire un impiego ottimale ed una lunga durata delle unità di posizionamento LineTech, bisogna osservare scrupolosamente ogni punto. Nel caso in cui sussistano dei dubbi, siete pregati di rivolgervi al vostro team LineTech.



Lubrificazione → Le unità di posizionamento LineTech vengono lubrificate in fabbrica con grasso Microlube GBU Y 131: ripetere la lubrificazione dopo circa 500 ore di funzionamento. Nel caso in cui si utilizzino altri prodotti, bisogna verificarne la compatibilità con il grasso lubrificante standard; se necessario toglierlo. Lubrificare in misura sufficiente rotaie lineari e mandrini attraverso il raccordo di lubrificazione. Cinghie dentate e cuscinetti sono ingrassati a vita.



Trasporto → Durante il trasporto, appoggiare sempre gli assi di una lunghezza maggiore a 700 mm.



Montaggio → Superficie di appoggio

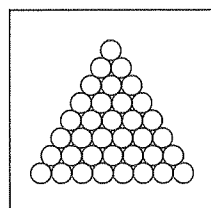
	0.05
--	------

Asse individuale

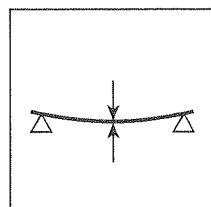
	0.05	1000
--	------	------

Asse doppia

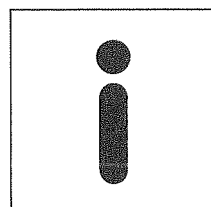
	0.02
--	------



Magazzinaggio → Immagazzinare in piano e all'asciutto.



Inflessione → In case di unità di posizionamento a sbalzo, l'inclinazione non deve essere superiore a 5'.

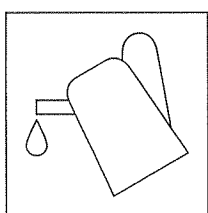


Attenzione → Per quanto concerne la meccanica, le unità di posizionamento standard vengono fornite pronte per l'uso. Messe a punto ulteriori non sono più necessarie.

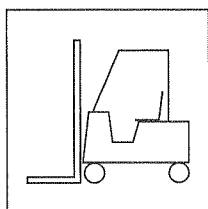
Instrucciones de manejo

Unidades de posicionamiento LineTech

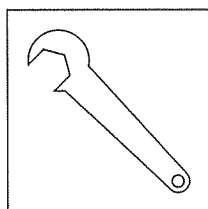
Por favor, lea con atención las siguientes instrucciones. Para garantizar una aplicación óptima y una larga duración de las unidades de posicionamiento LineTech, han de tenerse en cuenta los siguientes puntos. En caso de dudas diríjase a su distribuidor LineTech.



Lubricación → Las unidades de posicionamiento LineTech vienen lubricadas de fábrica con Microlube grasa GBU Y 131: aproximadamente cada 500 horas de servicio deben lubricarse de nuevo. Si se emplearan otras marcas de grasa, verificar previamente la compatibilidad de las mismas con la grasa estándar y en caso necesario lavar. Barras lineales y usillos deben ser lubricados suficientemente por el tracor de engrase. Correas trapezoidales dentadas y cojinetes llevan engrase perpetuo.



Transporte → Para transportar husillos de una longitud superior a 700 mm, deberán tener siempre una base de apoyo.



Montaje → Superficie de apoyo

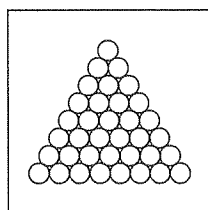
	0.05
--	------

Eje unitario

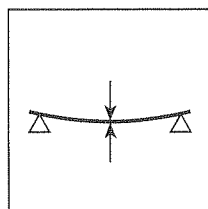
	0.05	1000
--	------	------

Doble eje

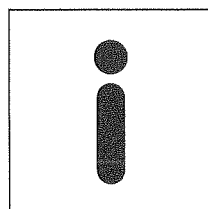
	0.02
--	------



Almacenamiento → Almacenar en lugar plano y seco.



Flexión → Para unidades de posicionamiento en voladizo, la inclinación del eje no ha de ser superior a 5'.



Atención → Las unidades de posicionamiento estándar se suministran mecánicamente dispuestas par el funcionamiento. No son necesarios reajustes posteriores.



17. Maintenance - Service - Upkeep

17.1 Lubricant Quantities, Lubrication Points, Lubrication Interval



The Linear Modules may only be lubricated with grease.

The initial greasing is performed by LINE TECH AG. The bearings used are usually maintenance-free and do not have to be re-lubricated under normal conditions.

The Linear Modules may only be lubricated with grease. LINE TECH AG recommends MICROLUBE GBU-Y 131 as a standard grease. If using other greases (other brands), the compatibility with the standard grease must be checked first. Observe the safety data sheet.

In the event of special operating conditions, inquire about the lubricant quantities. Values and information about lubricant quantities can be found in the adjacent table, corresponding to use under normal operating conditions

There are grease nipples on both sides of the carriage parts. It is sufficient if the amount of grease is pressed in from one side only.

Type	BSD d x P (mm)	Grease	Grease relubrication		
			Ball screw (cm³)	Runner blocks (cm³) Grease point ¹⁾	Grease point ²⁾
LM3	16 x 5	Klüber Microlube GBU-Y 131	0.60	3.50	1.75 (per carriage)
	16 x 10		0.85		
	16 x 16		1.15		
LM4	20 x 5		0.90	3.50	1.75 (per carriage)
	20 x 20		2.30		
	32 x 5		2.10		
LM5	32 x 10		2.95	4.40	2.20 (per carriage)
	32 x 32		5.15		



Depending on which value is reached first, the linear guides and the ball screw are to be relubricated according to the adjacent table

Type	BSD d x P (mm)	Operating interval (h)	Time interval (months)	Running interval (km)
LM3	16 x 5	500	3 - 6	250
	16 x 10			500
	16 x 16			800
LM4	20 x 5	500	3 - 6	250
	20 x 20			1 000
	32 x 5			250
LM5	32 x 10	500	3 - 6	500
	32 x 32			1 600





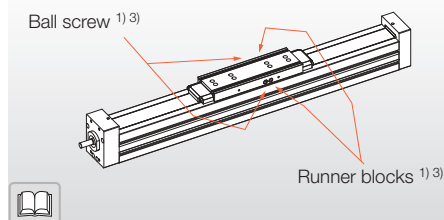
Various grease nipples exist on the Linear Modules.

- ¹⁾ Grease nipple according to DIN 3405; The carriage positions for the purposes of lubrication are independent of the stroke.
- ²⁾ Grease nipple according to DIN 71412; The carriage positions for the purposes of lubrication are independent of the stroke.
- ³⁾ Lubrication is possible either from the left or right.

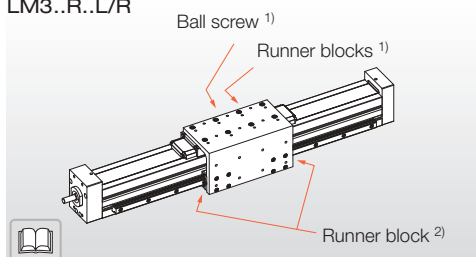
For pressing in the lubricant, we recommend the grease gun with the item number: ZPE.GREASEGUN-03

- Lubricate the table part(s)
- Calculate the amount of grease per stroke when lubricating with a hand press
- After the lubrication, remove the excess grease with a clean cloth.

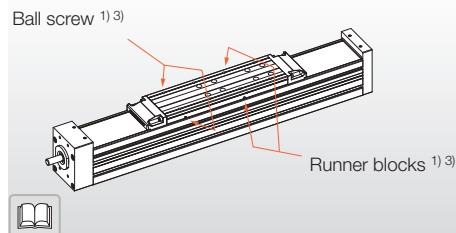
LM3..R..N



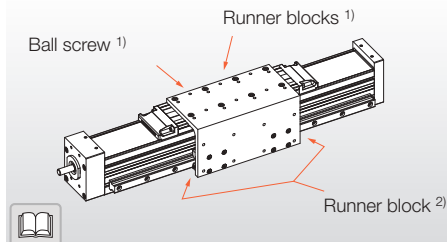
LM3..R..L/R



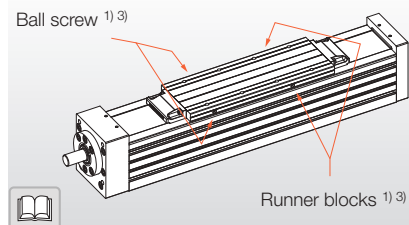
LM4..R..N



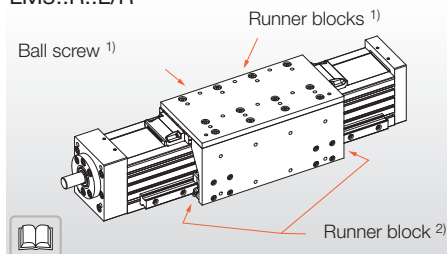
LM4..R..L/R



LM5..R..N



LM5..R..L/R





19. Maintenance - Service - Upkeep

19.1 Lubricant Quantities, Lubrication Points, Lubrication Interval



The Compact Units may only be lubricated with grease.

The initial greasing is performed by LINE TECH AG. The bearings used are usually maintenance-free and do not have to be re-lubricated under normal conditions.

The Compact Units may only be lubricated with grease. LINE TECH AG recommends MICROLUBE GBU-Y 131 as a standard grease. If using other greases (other brands), the compatibility with the standard grease must be checked first. Observe the safety data sheet.

In the event of special operating conditions, inquire about the lubricant quantities. Values and information about lubricant quantities can be found in the adjacent table, corresponding to use under normal operating conditions

There are grease nipples on both sides of the carriage parts. It is sufficient if the amount of grease is pressed in from one side only.

Type	BSD d x P (mm)	Grease	Grease relubrication	
			Carriage A (cm³)	Carriage B (cm³)
KE1	12 x 5	Klüber Microlube GBU-Y 131	0.7	0.4
	12 x 10		0.7	
KE2	16 x 5		1.6	1.0
	16 x 10		1.8	
	16 x 16		1.9	
KE3	20 x 5		2.5	1.5
	20 x 10		2.9	
	20 x 20		3.2	



Depending on which value is reached first, the linear guides and the ball screw are to be relubricated according to the adjacent table

Type	BSD d x P (mm)	Operating interval (h)	Time interval (months)	Running inter- val (km)
KE1	12 x 5	500	3 - 6	250
	12 x 10			500
KE2	16 x 5	500	3 - 6	250
	16 x 10			500
	16 x 16			800
KE3	20 x 5	500	3 - 6	250
	20 x 10			500
	20 x 20			1000





Different grease nipples exist on the Compact Units.

- 1) Grease nipples according to DIN 3405; the cover caps must be removed for lubrication. The carriage positions can be found in the table below for lubrication purposes.
- 2) The connection is according to the interface for customer add-on and can be found on the following page. The carriage positions for the purposes of lubrication are independent of the stroke.
- 3) Grease nipple according to DIN 3405; The carriage positions for the purposes of lubrication are independent of the stroke.

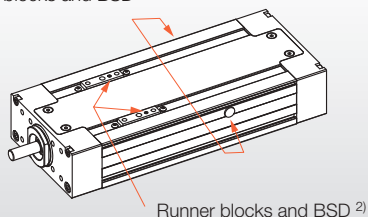
For pressing in the lubricant, we recommend the grease gun with the item number: ZPE.GREASEGUN-04

Procedure:

- Remove the cover caps
- Lubricate the table part(s)
- Calculate the amount of grease per stroke when lubricating with a hand press
- After lubricating with the hand press, close the nipple with a cover cap and remove excess grease with a clean cloth.

KE...2N...

Runner blocks and BSD ¹⁾

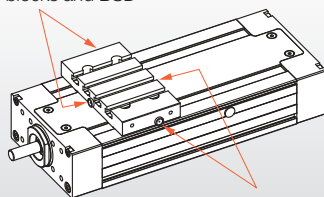


Runner blocks and BSD ²⁾



KE...2V...

Runner blocks and BSD ³⁾

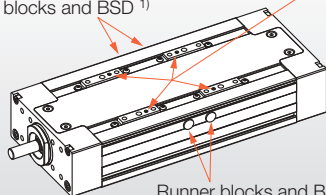


Runner blocks and BSD ³⁾



KE...4N...

Runner blocks and BSD ¹⁾

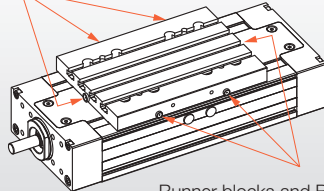


Runner blocks and BSD ¹⁾



KE...4V...

Runner blocks and BSD ³⁾



Runner blocks and BSD ³⁾



4.20 How to read the maintenance table

General

The maintenance table collects all of the information on the maintenance of modules that were declared as worthy of maintenance when the various text modules of the operating manual were created. The entries in the first column are linked with the descriptions relating to the entry. When the book is generated, the table is updated once with the filtered information.

Structure

The maintenance information is arranged in a table such that:

1. The first column of the maintenance requirement describes the module to which the maintenance pertains.
2. The second and any other columns indicate the maintenance intervals.
3. The last column may provide additional information.
4. Each line, which is coded with an ID, indicates the maintenance personnel required.
5. Possible entries for maintenance cases:

"Daily", "weekly", "monthly", "quarterly", "annually", "regular visual inspection", "every 2 years", "machine requirement", "prior to beginning of shift".

The entries can also appear in combination, e.g. "visual inspection" and "daily". The table then lists these same entries in duplicate or multiple times.

1. Possible entries for personnel:

Codes "1" or "2" or "3" replace the entries for "floor personnel", "maintenance personnel" and "service personnel" in the table.

4.21 Maintenance table

		daily	monthly	quarterly	annually	regular visual inspection	Machine requirement	prior to beginning of shift	Pail change	After program parameter changes	Additional information
4.4	Mixing tube maintenance requirement	2									
4.10	Maintenance prompt from EMERGENCY STOP command devices		2								
4.13	Light curtain maintenance requirement			2							
4.11	Maintenance requirements for the activation switch				2						
4.12	Maintenance requirements for the door safety switch				2						
4.16	Maintenance prompt – Internal calibration of Mettler scales				2						
4.18.1	Oil change vacuum pump				2						
4.19	Lubricating LineTech axles				2						
4.5	Dispensing pump maintenance requirements					2					
4.6	Maintenance requirement of emptying the condensate container					2					
4.9	PPS maintenance requirement					2					
4.14	Maintenance requirement - Festo cylinder					2					

		daily	monthly	quarterly	annually	regular visual inspection	Machine requirement	prior to beginning of shift	Pail change	After program parameter changes	Additional information
4.15	Maintenance requirement - vacuum shut-off valve of follower plate					2					
4.18. 2	Oil level vacuum pump					2					
4.7	Maintenance request - Blind shot - Replace or empty vessel						1				
4.8	Maintenance requirement - replace weighing container						1				
4.17. 2	Maintenance work on bdtronic progressing cavity pump PPS					2					
4.3	Machine maintenance before production start						2				
4.15	Maintenance requirement - vacuum shut-off valve of follower plate							2			
4.1	BDT folder backup								2		

1 - Operating personnel; 2 - maintenance personnel

5 Overview of faults

Faults and troubleshooting

Faults arise especially due to:

- Change of the properties of the material to be processed or of the components to be processed
- Incorrect or changed settings
- Wear, abrasion
- Jams, accumulations
- Leaks in the system
- Faults caused electrically or electronically



NOTICE

Troubleshooting!

Before reaching into moving machine parts:

Press the "Automatic" operating button: Component will be completed

Turn the key switch into the "Setup" position

Press the "Machine off" button: The power supply to the machine part is interrupted, all movements are stopped

Open the access doors

Eliminate the cause of the fault

Or immediate stop:

Press the "Stop" operating button: The door is unlocked

Let the components cool off if heated

Block the compressed air supply on the maintenance unit

Before reaching into pressurized components (e.g. mixing head, valves, vessels) and machine parts:

Press the "Automatic on" operating button: Component will be completed

Depressurise the component (e.g. release the pressure on the vessel or clean the mixing head, block the material feed to the dispensing pumps and dispense the dispensing valve or pump manually until it is empty -> the mixing chamber must be empty)

Press the "Stop" operating button: The door is unlocked

Let the components cool off if heated

Block the compressed air supply on the maintenance unit



CAUTION

Climbing into the machine

Before climbing into the machine, it is essential to ensure that:

- The main switch is in the "OFF" position.
- Let the components cool off if heated.
- Block the compressed air supply on the maintenance unit.

When working with the machine where there may be contact with dispensing materials and detergents, it is essential to:



NOTICE

Wear eye protection!

Wear suitable eye protection for every dangerous situation, e.g. laser, infrared, etc.



NOTICE

Wear protective gloves!

Wear suitable protective gloves for every dangerous situation.



NOTICE

Fault during dispensing

In case of faults during the process (pressure faults, etc.), the tool moves (e.g. mixing head, hot rivet head, etc.) into the home position if the "PROCESS MONITORING" or "AUTOMATIC" operation mode is active. The specified message light lights up and optional horn sounds a warning tone. For hot riveting machines, an emergency cooling cycle starts if the machine has been programmed this way. Due to the relatively high heat capacity of components of the hot rivet head, a wait time must elapse before these parts may be touched.

List of error messages and faults

The following list of possible faults and errors is organised according to topics:



NOTICE

General faults

Not all faults can be assigned to concrete events on particular components: an error state is frequently caused by several events, which in the course of the PLC algorithm are set into temporal relation by a type of parallel processing of various task modules and are thus made known as an error or fault at the end of a chain of events. Troubleshooting also has to do with the experience of the operating and service personnel working on machines of this type.



NOTICE

Faults - "on our own behalf"

Naturally, nobody can foresee all faults and simulate them in the creation and acceptance process at the manufacturer. Therefore, a fault list is also only a reflection of current experience which is not just created deterministically, but also empirically. The manufacturer depends on insights "from the field on site" to incorporate them in the documentation. Not just the actual fault cause and effect mechanisms that the bdtronic Service Team incorporates, but also those coming from customers to the editorial team in order to incorporate them into other machines of this type are desired and indispensable for product optimisation.



NOTICE

Consumption of replacement and wearing parts

The order numbers for replacement and wearing parts are in the machine log or on the electrical and pneumatic function plans or, if requested, they are available on a specially-created list. For replacement or wearing parts for which there is no analogue in the current version of your documentation, contact with the "Central Processing" or "Service" departments can help in order to initiate a replacement parts offer or actual delivery in timely fashion. For the appropriate contact data, see the "Service-legal notices" section of these operating instructions.

5.1 Faults on compressed air-driven units and components

5.1.1 Pneumatic cylinder shut-off valve SRU

Message text and effect on the component or machine:

"Fault on cleanser-recirculation shut-off valve"

Possible causes	Recommended troubleshooting
Cylinder end position not reached	Check cylinder end switch
Cylinder moves sluggishly	Clean and lubricate cylinder

Table 2: Table: Pneumatic cylinder shut-off valve SRU

5.1.2 Pneumatic cylinder end position inlet

Message text and effect on the component or machine:

"Inlet error vacuum chamber"

Possible causes	Recommended troubleshooting
Cylinder end position not reached	Check cylinder end switch

Possible causes	Recommended troubleshooting
Cylinder moves sluggishly	Clean and lubricate cylinder

Table 3: Table: Pneumatic cylinder end position inlet

5.1.3 Pneumatic cylinder end position

Message text and effect on the component or machine:

"Outlet error vacuum chamber"

Possible causes	Recommended troubleshooting
Cylinder end position not reached	Check cylinder end switch
Cylinder moves sluggishly	Clean and lubricate cylinder

Table 4: Table: Pneumatic cylinder end position

5.1.4 Power supply - Compressed air

Message text and effect on the component or machine:

"Compressed air fault"

Possible causes	Recommended troubleshooting
Hand switch-on valve of the compressed air inlet valve closed	Open hand switch-on valve
Compressed air supply missing	Check compressed air supply
Compressed air supply not stable	Check compressed air supply

Table 5: Table: Power supply - Compressed air fault

5.2 Faults on sensor units and components

5.2.1 Sensors - scanner - read error

Message text and effect on the component or machine:

"Scanner read error"

Possible causes	Recommended troubleshooting
Power supply to scanner lacking	Switch on scanner supply
Scanner defective	Change scanner

Table 6: Table: Sensors - scanner - read error

5.2.2 Sensors - balance - cup missing

Message text and effect on the component or machine:

"No cup on balance"

Possible causes	Recommended troubleshooting
There is no cup on the balance when it is switched on	Place cup on balance

Table 7: Table: Sensors - balance - cup missing

5.2.3 Sensors - balance - cup full

Message text and effect on the component or machine:

"Cup on balance full"

Possible causes	Recommended troubleshooting
Cup is full of material above tolerance limit	Change cup

Table 8: Table: Sensors - balance - cup full

5.2.4 Sensors - balance - communication error

Message text and effect on assembly or machine:

"Scale communication error"

Possible causes	Recommended troubleshooting
Connection to scale interrupted	Establish connection
Scale power pack is not plugged into 220 V outlet	Plug power pack into 220 V

Table 9: Table: Sensors - scale - communication error

5.2.5 Sensors - balance - taring

Message text and effect on the component or machine:

"Tare balance"

Possible causes	Recommended troubleshooting
Cup is on balance when switched on	- Remove cup from the balance for 2 seconds - Balance is tared - Place new cup on balance if previous one was full

Table 10: Table: Sensors - balance - taring

5.2.6 Sensors - needle check - outside of tolerance

Message text and effect on the component or machine:

"Outside of tolerance"

Possible causes	Recommended troubleshooting
Needle bent	• Press Stop button • Align needle • Start needle check if needle defective • Change needle • Teach new needle
Needle soiled	• Clean needle with cloth or brush • Perform needle check

Table 11: Table: Sensors - needle check - outside of tolerance

5.2.7 Sensors - needle check needle check error

Message text and effect on the component or machine:

"Needle check error"

Possible causes	Recommended troubleshooting
Needle outside of tolerance	• Press Stop button • Open door • Bend needle straight • Close doors

Table 12: Table: Sensors - needle check needle check error

5.2.8 Sensors - needle check position calculation incorrect

Message text and effect on the component or machine:

"Position calculation incorrect"

Possible causes	Recommended troubleshooting
Needle deviates too much	• Press Stop button • Check needle visually • Align or change needle • Perform needle check or teach new needle

Table 13: Table: Sensors - needle check position calculation incorrect

5.2.9 Sensors - needle check sensor incorrect

Message text and effect on the component or machine:

"Sensor incorrect"

Possible causes	Recommended troubleshooting
Sensor(s) soiled	• Press Stop button • Needle check • Check procedure in the graphic on "SETUP MOVEMENT" screen • Clean sensors if still steamed up
Sensor(s) defective	• Check sensor activity in the TWINCAT SystemManager • Replace sensor(s) if defective

Table 14: Table: Sensors - needle check - sensor incorrect

5.2.10 Sensors – Light barrier

	NOTICE Appearance of the error message These error messages are shown on the display of the light barrier controller.
---	--

If the GL-R is in error state.

Display	Error description	Cause	Check and remedy
E -> 2	Wiring error	The end cover of the receiver is not connected.	Check whether the end cover is correctly installed on the receiver.
		The connection cable is connected to the upper part of the GL-R.	Connect the connection cable to the lower part of the GL-R.
		Transmitter and receiver are different models.	Check whether all transmitter and receiver models have been combined correctly.
		If the GL-R is operated with cable synchronisation:	
		The synchronisation cable is not connected correctly or is disconnected.	Check the wiring of the synchronisation cable.
		If the GL-R is operated as a one-line system:	
		The connection cable is connected to the transmitter.	Connect the connection cable to the receiver. Do not connect the connection cable to the transmitter.
		The series connection cable is connected to the upper part of the GL-R.	Connect the series connection cable to the lower part of the transmitter and receiver.
		If GL-R is operated in a series circuit	

Display	Error description	Cause	Check and remedy
		The expansion unit is not connected correctly.	Check the direction of installation of the expansion unit. Check that the receiver of the expansion unit is connected to the transmitter of the main device. Check whether the total number of beam axes is more than 240.
E -> 2		The expansion unit is defective.	Check whether the extension unit functions correctly when it is not connected in series.
		The synchronisation cable is being affected by external interference.	Check whether there are sources of interference (inverters, servo motors, etc.) in the vicinity of the installation location and cables of the GL-R
E -> 4	Dip switch error	The dip switch configuration does not meet the specifications.	Check the dip switch configuration.
		The dip switch is not set to channel 0 with cable synchronisation.	

Display	Error description	Cause	Check and remedy
E -> 5	Configuration error	The dip switch configuration overlaps the setting uploaded from the configuration software.	Change the configuration of the dip switch, or change the configuration with the configuration software.
		The device configuration does not match the configuration uploaded from the software.	Check the model name of the main device and expansion unit, and check that the device configuration from the software matches the actual device configuration.
		Uploading of the configuration was not completed correctly when using the configuration software.	Upload the configuration again.

Display	Error description	Cause	Check and remedy
E -> 7	Interlock error	Interlock mode selection input or reset input are wired incorrectly.	Wire the Interlock mode selection input or reset input correctly.
E -> 8	EDM error	EDM input is not connected automatically.	If the EDM function is needed, wire the EDM correctly. If the EDM function is not needed, wire the EDM and AUX correctly or deactivate the EDM function with the configuration software.
		There is a defective welding contact on the external device.	Replace the external device.
E -> 1 -> 0	Receiver error	The receiver is affected by ambient light.	Shield the receiver from ambient light.
E -> 1 -> 0	Receiver error	The receiver is affected by ambient light.	Shield the receiver from ambient light.
E -> 1 -> 2	Transmitter error	The transmitter is defective.	Replace the transmitter.
		If the error is rectified by restarting the GL-R:	
		The transmitter is being affected by external interference.	Check whether there are sources of interference (inverters, servo motors, etc.) in the vicinity of the installation location and cables of the GL-R.

Display	Error description	Cause	Check and remedy
E -> 1 -> 4 E -> 1 -> 5	OSSD1 error OSSD2 error	If the error is rectified by restarting the GL-R:	
		The OSSD is being affected by external interference.	Check whether there are sources of interference (inverters, servo motors, etc.) in the vicinity of the installation location and cables of the GL-R.
		There is a surge voltage that affects the OSSD with its inductive load.	If it is an inductive load, e.g. relay, use a load with absorbing function.
		The voltage of the power supply unit has continuously decreased or suddenly dropped.	Rectify the problem by replacing the power supply unit, increasing the power or specifically assigning the power supply unit to the GL-R.
		If the error is not rectified by restarting the GL-R:	
		OSSD is shorted at 0 V or 24 V of the power supply unit. OSSDs are shorted to each other. OSSD is shorted with other cables.	Wire the OSSD correctly.
		OSSD is defective.	Replace the receiver.

Display	Error description	Cause	Check and remedy
E -> 1 -> 8	Error in the expansion unit.	The sub-unit is defective.	Replace the sub-unit.
		If the error is rectified by restarting the GL-R:	
		The expansion unit cable or series connection cable is being affected by external interference.	Check whether there are sources of interference (inverters, servo motors, etc.) in the vicinity of the installation location and cables of the GL-R.
E -> 2 -> 0	Communication error*1	If the error is rectified by restarting the GL-R:	
		The synchronisation cable is not connected correctly or is disconnected.	Check the connection of the synchronisation cable.
		The synchronisation leads in the cable are being affected by external interference.	Check whether there are sources of interference (inverters, servo motors, etc.) in the vicinity of the installation location and cables of the GL-R.
		The voltage of the power supply unit has continuously decreased or suddenly dropped.	Rectify the problem by replacing the power supply unit, increasing the power or specifically assigning the power supply unit to the GL-R.
		If the error is not rectified by re-starting the GL-R:	
		The synchronisation cable or series connection cable is not connected correctly or is disconnected	Check the connection of the synchronisation cable.

Display	Error description	Cause	Check and remedy
E -> 2 -> 4	Deactivation lamp Disconnection error *2	The deactivation lamp is not connected correctly.	Check the connections.
		The deactivation lamp is defective.	Replace the deactivation lamp.
E -> 2 -> 5	Deactivation lamp Overcurrent error *2	The deactivation lamp is not connected correctly.	Check the connections.
		The deactivation lamp consumes too much current.	Check that the deactivation lamp is not consuming more current than the GL-R can handle.
		The deactivation lamp is defective.	Replace the deactivation lamp.
E -> 2 -> 7	Synchronisation error	If the GL-R is operated with optical synchronisation, the fixed blanking function or the deactivation function are applied to both the upper and lower beam axes (for synchronisation control).	Apply the fixed blanking function or the deactivation function to either the upper or the lower axis. Operate the GL-R as a one-line system or with cable synchronisation.
E -> 4 -> 0 : E -> 5 -> 9	System error	The voltage of the power supply unit has continuously decreased or suddenly dropped.	Rectify the problem by replacing the power supply unit, increasing the power or specifically assigning the power supply unit to the GL-R.
		The GL-R is defective.	Replace the GL-R.

If the GL-R is not in an error state:

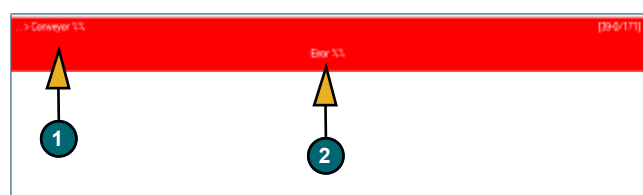
Error designation	Cause	Remedial measure
All displays are OFF.	The device is not switched on, or the voltage of the power supply is insufficient.	Adjust the voltage of the power supply according to the specifications. Wire the power supply correctly.
	Cables are disconnected or not correctly connected.	Check the connection and reconnect the cables if necessary.
The OSSD does not switch on. (Central display lights up red or does not switch on)	Are the beam axes correctly aligned?	Perform an optical alignment.
	Are there any objects in the detection range?	Remove all objects from the detection range. Ensure the distance between transmitter and receiver is within the specified range.
	Is the distance between transmitter and receiver within the specified range?	
	Is there dust or other particles on the detection surface blocking the beam axes?	Clean the surface. Carefully wipe off the dirt with a cloth moistened with a mild detergent.
The OSSD does not switch on. (Central display lights up green)	If the function indicator for Interlock is lit, the GL-R is in Interlock mode.	In manual start/manual reset mode, the OSSD is only switched on when light is received from all beam axes and the reset input is activated.
	If "U" is shown on the 7-segment display, the Wait input is active.	Connect the Wait input correctly.
	With series connection, are the beam axes for the other GL-R devices aligned?	Check the status of the other GL-R devices connected in series.

Error designation	Cause	Remedial measure
The beam axes are aligned, but sometimes the OSSD switches OFF.	The receiver is affected by ambient light or light from other photoelectric devices.	Shield the receiver from ambient light.
	The synchronisation leads in the transmission cable are being affected by external interference.	Check the environment of the cables for interference.
The device connected to the OSSD is switched on and off very quickly.	The self-diagnostic function switches off the OSSD periodically so that the device may detect this short OFF signal.	Connect a device that does not detect the regular OSSD OFF signal.
All transmitter displays except "POWER" are switched off.	The GL-R is operated with optical synchronisation.	If you wish to use optical synchronisation, the GL-R will operate correctly and you do not need to do anything. If you do not want to use optical synchronisation, check the connection of synchronisation wires 1 and 2 in the cable.

Error designation	Cause	Remedial measure
The GL-R does not switch to the deactivation state when the deactivation inputs are activated.	The 7-segment display and the function displays do not change when the deactivation inputs are activated.	
	Wires are not connected correctly.	Check the connections of the wires.
	An external device connected to the deactivation inputs is defective.	Connect another external device to the deactivation inputs.
	Wires of the deactivation inputs are interrupted.	Connect a different connection cable.
	Deactivation inputs are defective.	Connect a different GL-R device.
	The 7-segment display and the function displays change when the deactivation inputs are activated.	
	The conditions for triggering a deactivation situation are not satisfied.	Check the deactivation states.
	The combination of deactivation group inputs is not correct.	Check the deactivation group inputs when the deactivation group function is used.

5.2.11 Fault - Drive motor - SEW

Faults caused by an SEW drive in the machine are described below.



1	"Site of error source" description	2	"Error code from SEW error list" description
---	------------------------------------	---	--

"Site of error source" description (1):

A description is shown here of the site in the machine visualisation containing the section (e.g. trickling station, preheating, curing, etc.) that is the source of the error.

"Error code from SEW error list" description (2):

The error code from the SEW drive is shown here. This error code is found chapter "Service/error list" in the manufacturer documentation.

No.	Designation	Reaction	Possible cause	Measure
00	No error	-	-	-
01	Overcurrent	Immediate shutdown with lock	Short circuit on output	Remove short circuit
			Switching on output	Switching only with blocked end stage
			Motor too large	Connect smaller motor
			Defective end stage	If error cannot be reset, consult SEW-Service
03	Earth fault	Immediate shutdown with lock	Earth fault in motor	Replace motor
			Earth fault in converter	Replace MOVITRAC® B
			Earth fault in motor supply cable	Repair earth fault
			Overcurrent (see F01)	See F01

No.	Designation	Reaction	Possible cause	Measure
04	Brake chopper	Immediate shutdown with lock	Generator-based power too great	Delay ramps extension
			Brake resistor circuit interrupted	Check brake resistor supply cable
			Short circuit in brake resistor circuit	Remove short circuit
			Brake resistor too highly resistive	Check technical data of the brake resistor
			Brake chopper defective	Replace MOVITRAC® B
			Earth fault	Repair earth fault

No.	Designation	Reaction	Possible cause	Measure
06	Mains phase failure	Immediate shutdown with lock (for 3-phase converter only)	Phase failure	Check mains cable
			Mains voltage too low	Check mains voltage
07	Intermediate circuit over-voltage	Immediate shutdown with lock	Intermediate circuit voltage too high	Delay ramps extension
				Check brake resistor supply cable
				Check technical data of the brake resistor
			Earth fault	Repair earth fault
08	Speed monitoring	Immediate shutdown with lock	Current regulator works on the control limit due to:	-
			Mechanical overload	Reduce load
				Check current limit
				Delay ramps extension
				Increase set delay time P501 >
			Phase failure on the mains supply	Check mains phases
			Phase failure on the motor	Check motor supply cable and motor
			Maximum speed for VFC operation setpoints exceeded	Reduce maximum speed

No.	Designation	Reaction	Possible cause	Measure
09	Commissioning	Immediate shutdown with lock	Converter not yet put into operation	Put converter into operation
			Unknown motor selected	Select another motor
			VFC & lifting equipment selected and P207 not set	Set P207
10	IPOS-ILLOP	Stop with lock With IPOS only	Faulty command when running program	Validate program
			Faulty conditions when running program	Validate program sequence
			Function in converter not present / implemented	Use other functions
11	Overtemperature	Stop with lock	Thermal overload of the converter	Reduce load and / or ensure sufficient cooling
				If brake resistor is integrated in the cooling element: Mount brake resistor externally
17-24	System fault	Immediate shutdown with lock	Converter electronics faulty, potentially via EMC effect	Earth connections and check shields and repair if necessary
				Obtain advice from SEW-Service on repeated occurrence

No.	Designation	Reaction	Possible cause	Measure
25	EEPROM	Stop with lock	Error when accessing EE-PROM	Call up factory setting, perform reset and re-parameterise
				In case of recurrence Obtain advice from SEW-Service

No.	Designation	Reaction	Possible cause	Measure
25	EEPROM	Stop with lock	Error when accessing EEPROM	Call up factory setting, perform reset and re-parameterise
				In case of recurrence Obtain advice from SEW-Service
26	External clamp	Programmable	External fault signal to be imported via programmable input	Eliminate the respective cause of the error, reprogram the clamp if necessary
31	TF/TH trigger	Stop without lock "Ready for operation maintained" message	Motor too hot, TF has been triggered	Allow motor to cool down and reset error
			TF of the motor not connected or not correctly connected	Check connections / joints between MOVITRAC® B and TF
			MOVITRAC® B and TF connected on the motor interrupted	
32	!POS-Index overflow	Stop with lock	Programming principles violated, resulting in internal stack overflow	Check and correct user program

No.	Designation	Reaction	Possible cause	Measure
34	Ramp timeout	Immediate shutdown with lock	Overwriting of the set ramp time	Extend ramp time
			If you remove the clearance and the drive exceeds the stop ramp time t13 by a certain time, the converter reports error message F34	Extend stop ramp time

No.	Designation	Reaction	Possible cause	Measure
35	Ex-e-protected mode	Immediate shutdown with lock	Incorrect operation mode selected	Approved procedure: U/f, VFC, VFC lifting gear
				Unapproved procedure: Snap function DC brake Group operation
			Non-permitted parameter set	Only use parameter set 1
			Do not put Ex protected motor into operation	Put Ex protected motor into operation
			Incorrect parameterisation of the frequency points	Frequency A < Frequency B Frequency B < Frequency C
			Incorrect parameterisation of the current limits	Current limit A < current limit B Current limit B < current limit C

No.	Designation	Reaction	Possible cause	Measure
36	Option missing	Immediate shutdown with lock	Option card type not permitted	Insert correct option card
			Set value source, control source or operation mode for this option card is not permitted	Set correct set value source
				Set correct control source
				Set correct operation mode
				Check parameters P120 and P121
			Required option missing	Check the following parameters:
				P121 for FBG11B
				P120 and P642 for FIO12B
			Front module FIO21 B not being supplied	Set P808 to "On" or supply basic device with external 24 V

No.	Designation	Reaction	Possible cause	Measure
37	System watchdog	Immediate shutdown with lock	Error in the System-• Software sequence	Earth connections and check shields and repair if necessary
				Obtain advice from SEW-Service on repeated occurrence
38	System software	Immediate shutdown with lock	System fault	Earth connections and check shields and repair if necessary
				Obtain advice from SEW-Service on repeated occurrence
43	RS485 Timeout	Stop without lock ²	Communication between converter and PC disrupted	Check connection between converter and PC
			Communication with FSE24B disrupted	Check power supply
				Check parameter P808
44	Device utilisation	Immediate shutdown with lock	Device utilisation (1 x t-value) • too great	Reduce power output
				Extend ramp
				If named points not possible: Insert larger converter
45	Initialisation	Immediate shutdown with lock	Error during initialisation	Obtain advice from SEW-Service

No.	Designation	Reaction	Possible cause	Measure
47	System bus 1 timeout	Stop without lock ²	Error with communication via the system bus	Check system bus connection
				Check P808
				Check power supply of the FSE24B
				With FSE24B plugged in, check Ether-CAT communication

No.	Designation	Reaction	Possible cause	Measure
77	IPOS control word	Stop with lock	System fault	Obtain advice from SEW-Service
80	RAM test	Immediate shutdown	Internal device error, Random Access Memory defective	Obtain advice from SEW-Service
81	Start condition	Immediate shutdown with lock	Only in "VFC lifting gear" mode: The converter could not impress the required current into the motor during the pre-magnetisation time:	
			Motor rated power too low in relation to converter rated power	Check converter and motor connection
				Check commissioning data and, if necessary, re-commission
			Cross section of motor supply line too small	Check cross section of motor supply line and, if necessary, increase
82	Output open	Immediate shutdown with lock	Only in "VFC lifting gear" mode:	
			2 or all output phases interrupted	Test converter and motor connection
			Motor rated power too low in relation to converter rated power	Check commissioning data and, if necessary, re-commission

No.	Designation	Reaction	Possible cause	Measure
84	Motor protection	Stop with lock	Motor load too high	Check P345 I P346 IN-UL monitoring
				Reduce load
				Extend ramp
				Longer pause times
94	EEPROM checksum	Immediate shutdown with lock	EEPROM defective	Obtain advice from SEW-Service
97	Copy error	Immediate shutdown with lock	Remove parameter mode during copying	Before error acknowledgment: Factory setting or load entire data set from parameter module
			Switch on/off during copying	
98	CRC Error Flash	Immediate shutdown	Internal device error, Flash memory defective	Send in device for repair
100	Vibration/warning	Show error	Vibration sensor warning (see operating instructions for "Diagnostic unit DUV10A")	Determine the cause of vibration, operation still possible until F101 occurs
101	Vibration error	Rapid stop	Vibration sensor reports error	SEW-EURODRIVE recommends remedying the cause of the vibration immediately
102	Oil ageing/warning	Show error	Oil ageing sensor warning	Schedule oil change

No.	Designation	Reaction	Possible cause	Measure
103	Oil ageing/error	Show error	Oil ageing sensor reports error	SEW-EURODRIVE recommends replacing the device oil immediately
104	Oil ageing/overtemperature	Show error	Oil ageing sensor reports overtemperature	Allow oil to cool
				Check correct gearbox cooling
105	Oil ageing/ready for operation message	Show error	Oil ageing sensor is not operationally ready	Check power supply of the oil ageing sensor
				Check oil ageing sensor, replace if necessary

No.	Designation	Reaction	Possible cause	Measure
106	Brake wear	Show error	Brake lining worn	Replace brake lining (see "Motors" operating instructions)
110	Ex-e-protection	Emergency stop	Time period of operation below 5 Hz exceeded	Check project management
				Reduce time period of operation below 5 Hz
111	System bus (SBus) error	This error number indicates to the EtherCAT or Fieldbus-Master that communication between the FSE24B and MOVITRAC® B is disrupted. The MOVITRAC® B device itself would detect error 4 7.	Check plug connector FSE24B	
113	Wire break Analogue input	Program-mable	Wire break on analogue input A1	Check wiring
116	Application module error Sub-error: 14: Device error 29: Approach limit switch 42: Tracking error 78: Software limit switch approached			

5.3 Faults on transport and movement components

5.3.1 Transport - lift-turn unit - turning working position NOK

Message text and effect on the component or machine:

"Turning working position not reached"

Possible causes	Recommended troubleshooting
Workpiece carrier jammed	Eliminate mechanical blockage
Cylinder jammed	• Check initiators • Reposition initiators • Eliminate mechanical interlocking
Compressed air supply fault	Check compressed air supply

Table 15: Table: Transport - lift-turn unit - turning working position NOK

5.3.2 Transport - lift-turn unit - turning home position NOK

Message text and effect on the component or machine:

"Turning home position not reached"

Possible causes	Recommended troubleshooting
Workpiece carrier jammed	Eliminate mechanical blockage
Cylinder jammed	• Check initiators • Reposition initiators • Eliminate mechanical interlocking
Compressed air supply fault	Check compressed air supply

Table 16: Table: Transport - lift-turn unit - turning home position NOK

5.3.3 Transport - lift-turn unit - home position NOK

Message text and effect on the component or machine:

"Lift-turn unit: "Lifting home position not reached"

Possible causes	Recommended troubleshooting
Workpiece carrier jammed	Eliminate mechanical blockage
Cylinder jammed	• Check initiators • Reposition initiators • Eliminate mechanical interlocking
Compressed air supply fault	Check compressed air supply

Table 17: Table: Transport - lift-turn unit - home position NOK

5.3.4 Transport - WC

Message text and effect on the component or machine:

"WC conveyor belt"

Possible causes	Recommended troubleshooting
Workpiece carrier jammed	Eliminate mechanical blockage
Stopper jammed	• Check initiators • Reposition initiators
Drive fault	Reactivate motor protection switch when triggering error eliminated

Table 18: Table: Transport - WC

5.3.5 Transport - pallet holder not open

Message text and effect on the component or machine:

"Cover holder sensor open" not programmed

Possible causes	Recommended troubleshooting
Cover holder is in incorrect position (e.g. is still on housing)	• Open cover holder • If necessary, change old component

Table 19: Table: Transport - pallet - holder not open

5.3.6 Drive technology - error in axis system

Message text and effect on assembly or machine:

Various error messages.

Possible causes	Recommended troubleshooting
Fault is drive train, e.g. In drive amplifier	• Press the < Restart controller > key on the "Axes" mask: The drive amplifier is reset and starts again with the previous parameter set

Table 20: Table: - Drive technology – error in axis system

5.3.7 Drive technology - lift-turn unit - mechanical system

Message text and effect on the component or machine:

"LRU place: rotator mechanical error"

Possible causes	Recommended troubleshooting
Pallet jammed	• Open cover on LRU • Put pallet in correct position

Table 21: Table: Drive technology - lift-turn unit - mechanical system

5.3.8 Drive technology - lift-turn unit - working position

Message text and effect on the component or machine:

"Working position NOK"

Possible causes	Recommended troubleshooting
Compressed air lacking	Check compressed air supply
Pallet jammed	• Open cover • Put pallet in correct position

Table 22: Table: Drive technology - lift-turn unit - working position

5.3.9 Drive technology - lift-turn unit - home position

Message text and effect on the component or machine:

"Home position NOK"

Possible causes	Recommended troubleshooting
Compressed air lacking	Check compressed air supply
Pallet jammed	• Open cover • Put pallet in correct position

Table 23: Table: Drive technology - lift-turn unit - home position

5.3.10 Drive technology - Reference run

Message text and effect on the component or machine:

"Reference of the axes NOK"

Possible causes	Recommended troubleshooting
Error in reference run	Start reference run on SETUP MOTION screen

Table 24: Table: Drive technology - Reference run

5.3.11 Drive technology - Transport belt

Message text and effect on the component or machine:

"Conveyor belt error"

Possible causes	Recommended troubleshooting
Belt motor	• Check actuation • Check belt motor • If necessary, replace motor if defective

Table 25: Table: Drive technology - Reference run

5.4 Faults caused by processes

5.4.1 Dispensing technology - filling prohibited

Message text and effect on the component or machine:

"Filling prohibited"

Possible causes	Recommended troubleshooting
Blindshot container is full	Change container

Table 26: Table: Dispensing technology - filling prohibited

1. Press < *Stop* > operating button
 - ⇒ Component processing stops
 - ⇒ Machine is in HAND
 - ⇒ Doors are no longer locked
1. Open access doors
 - ⇒ SAFETY operating mode is active
2. Remove container
3. Place new container in position
1. Close access doors
 - ⇒ Safety is ensured
 - ⇒ HAND operating mode is active
2. Press < *AUTOMATIC* > operating button
 - ⇒ Machine goes into AUTOMATIC operating mode

5.4.2 Dispensing technology - process dispensing pressure

Message text and effect on the component or machine:

"Pump speed above max."

Possible causes	Recommended troubleshooting
Dispensing valve blocked	Take apart dispensing valve and clean

Possible causes	Recommended troubleshooting
Outlet nozzle of the mixing chamber blocked	- Block material feed to pump - Depressurize system - Clean or replace hose - Clean actuation of dispensing valve
Dispensing valve does not switch	- Clean actuation of dispensing valve - Clean dispensing valve
Mixing head is hardened	- Depressurize system - Disassembling the mixing head

Table 27: Table: Process dispensing pressure

5.4.3 Dispensing technology - blindshot container

Message text and effect on the component or machine:

"Blindshot container missing"

Possible causes	Recommended troubleshooting
No container in position	Place empty container in position

Table 28: Table: Dispensing technology - blindshot container

5.4.4 Dispensing technology - blindshot container full

Message text and effect on the component or machine:

"Blindshot container too full, please empty"

Possible causes	Recommended troubleshooting
Container above maximum fill level	Place empty container in position

Table 29: Table: Dispensing technology - blindshot container - full

5.4.5 Dispensing technology - dispensing pumps - pressure low

Message text and effect on the component or machine:

"Dispensing pressure below low limit"

Possible causes	Recommended troubleshooting
Pump wear	- Check pump according to manufacturer's instructions - Maintain pump according to manufacturer's instructions

Table 30: Table: Dispensing technology - dispensing pumps - pressure low

5.4.6 Dispensing technology - material preparation - material NOK

Message text and effect on the component or machine:

"Material NOK"

Possible causes	Recommended troubleshooting
Automatic material preparation still running	Wait until Material OK is sent to PLC

Table 31: Table: Dispensing technology - material preparation - material NOK

5.4.7 Dispensing technology - pump speed above max.

Message text and effect on the component or machine:

"Speed above max..."

Possible causes	Recommended troubleshooting
Actuation pump NOK	Check actuation card for dispensing pump
Tacho monitoring	Check tacho cable

Table 32: Table: Dispensing technology - pump - speed above max.

5.4.8 Dispensing technology - pump speed under min.

Message text and effect on the component or machine:

"Speed under min..."

Possible causes	Recommended troubleshooting
Actuation pump NOK	Check actuation card for dispensing pump
Pump does not run smoothly	Take pump apart and clean

Table 33: Table: Dispensing technology - pump - speed under min.

5.4.9 Process - dispense - chamber not ready

Message text and effect on the component or machine:

"Chamber not ready"

Possible causes	Recommended troubleshooting
Dispense without previous filling	Press fill button on "SETUP DISPENSING" screen

Table 34: Table: Process - dispense - chamber not ready

5.4.10 Process - dispense - chamber error after cleaning

Message text and effect on the component or machine:

"Chamber error after cleaning"

Possible causes	Recommended troubleshooting
Cleaning was interrupted	Press Clean button again in PROCESS MONITORING operating mode

Table 35: Table: Process - dispense - chamber error after cleaning

5.4.11 Process - dispense - undef. Chamber status after recipe loading

Message text and effect on the component or machine:

"Undef. Chamber status after recipe loading"

Possible causes	Recommended troubleshooting
Change dispensing program	• Clean system (only for dynamic system) • Put system into home position (only for 1K or static) • Select recipe

Table 36: Table: Process - dispense - undef. Chamber status after recipe loading

5.4.12 Software - Drive program

Message text and effect on the component or machine:

"CNC error"

Possible causes	Recommended troubleshooting
Error in drive program	Check drive program

Table 37: Table: Software - Drive program

5.4.13 System - vacuum - not filled

Message text and effect on the component or machine:

"Vacuum chamber not ventilated"

Possible causes	Recommended troubleshooting
Ventilation valve ball cock jammed	• Check actuation ventilation ball cock • Check ball cock

Possible causes	Recommended troubleshooting
Ventilation valve stuck	Check valve

Table 38: Table: System - vacuum - not filled

5.4.14 System - vacuum error

Message text and effect on the component or machine:

"Error in vacuum"

Possible causes	Recommended troubleshooting
Vacuum filter time too short	Change setting on screen
Container leaky	• Use leakage search fluid to search for leak • Seal up spot • Attach hoses properly
Vacuum pump defective	• Check device according to manufacturer's operating instructions • Replace device if defective

Table 39: Table: System - vacuum

5.4.15 System - vacuum - vacuum pump

Message text and effect on the component or machine:

"Vacuum pump error"

Possible causes	Recommended troubleshooting
Motor protection switch triggered	• Check vacuum pump according to manufacturer's instructions • Switch motor protection switch on again when cause of error eliminated

Table 40: Table: System - vacuum - vacuum pump

5.4.16 System - vacuum - vacuum value

Message text and effect on the component or machine:

"Vacuum value error"

Possible causes	Recommended troubleshooting
Leak in system	Search for leak in container, hoses, etc.
Vacuum regulator valve defective	Replace valve

Possible causes	Recommended troubleshooting
Pump defective	Replace pump

Table 41: Table: System - vacuum - vacuum value

5.4.17 System - vacuum chamber - not filled

Message text and effect on the component or machine:

"Vacuum chamber not ventilated"

Possible causes	Recommended troubleshooting
Vacuum chamber still under vacuum	Ventilate vacuum chamber

Table 42: Table: System - vacuum chamber - not ventilated

5.4.18 System - vacuum chamber - process still active

Message text and effect on the component or machine:

"Vacuum regulation still active"

Possible causes	Recommended troubleshooting
Vacuum regulation is still switched on	Switch off vacuum regulation

Table 43: Table: System - vacuum chamber - process still active

5.4.19 System - vacuum - vacuum not reached

Message text and effect on the component or machine:

"Vacuum not reached"

Possible causes	Recommended troubleshooting
Suction line not connected	Connect suction line (only for ECU)
Suction cup defective	Change suction cup
Cover too bent	Replace cover

Table 44: Table: System - vacuum - vacuum not reached

5.5 Material-conditioned faults

5.5.1 Material preparation - container - fill level max.

Message text and effect on the component or machine:

"A1 over max."

Possible causes	Recommended troubleshooting
"HIGH" and "Max." limits are selected too close on container dialog	Reassign limits on container dialog (screen: "MATERIAL A")
Fill level electrode may be defective	• Check limits on fill level electrode • Replace fill level electrode if defective
Ball cock jammed in front of container inlet	• Take ball cock apart and clean • Replace ball cock if necessary

Table 45: Table: Material preparation - container - fill level max.

5.5.2 Material preparation - container - fill level min.

Message text and effect on the component or machine:

"Container under min."

Possible causes	Recommended troubleshooting
Barrel feed interrupted	• Check barrel fill level • Check recirculation path • Press "Change mode" button if still in barrel change mode

Table 46: Table: Material preparation - container - fill level min.

5.5.3 Material preparation - container - stirrer

Message text and effect on the component or machine:

"Stirrer: error..."

Possible causes	Recommended troubleshooting
Motor protection failed	• Eliminate cause of fault • Switch motor protection on again
Overtemperature switch in winding has triggered	• Let motor cool off • Switch off material preparation • Check pail

Table 47: Table: Material preparation - container - stirrer

5.5.4 Material preparation - container - sensors - safety switch

Message text and effect on the component or machine:

"Safety switch container NOK"

Possible causes	Recommended troubleshooting
Cable break on safety switch	• Change cable • Replace safety switch if defective
Container overfilled	• Ventilate container • Unscrew container cover • Remove material • Close container cover again
Fill level electrode incorrectly adjusted	Readjust fill level electrode according to manufacturer's instructions

Table 48: Table: Material preparation - container - sensors - safety switch

5.5.5 Material preparation - container - prepressure low

Message text and effect on the component or machine:

"Prepressure material container below low limit"

Possible causes	Recommended troubleshooting
Material feed too low	• Check material supply • Check ball cocks for blockage • Check recirculation pump

Table 49: Table: Material preparation - container - prepressure low

5.5.6 Material preparation - container - prepressure max.

Message text and effect on the component or machine:

"Prepressure over max. for container"

Possible causes	Recommended troubleshooting
Pressure reducer valve pushes through	• Trigger dispensing shot manually • Readjust setting on pressure reducer valve

Table 50: Table: Material preparation - container - prepressure max.

5.5.7 Material preparation - ball cock - end position

Message text and effect on the component or machine:

"End position closed ball cock not reached"

Possible causes	Recommended troubleshooting
Ball cock could not be closed in container inlet	• Maintain ball cock according to manufacturer's instructions • Check switch box of the ball cock

Table 51: Table: Material preparation - ball cock - end position

5.5.8 Material preparation - ball cock - message box

Message text and effect on the component or machine:

"Error in message box of the ball cock, repump"

Possible causes	Recommended troubleshooting
End switch blocked on message box	Set cam switch of end switch to largest possible switch distance

Table 52: Table: Material preparation - ball cock - message box

5.6 Safety faults

5.6.1 Safety - E-STOP pressed

Message text and effect on the component or machine:

"E-stop button pressed"

Possible causes	Recommended troubleshooting
E-stop button pressed	• Check cause • Unlock e-stop button; if causes of error eliminated

Table 53: Table: Safety - E-STOP pressed

5.6.2 Safety - Door not locked

Message text and effect on the component or machine:

"Safety door not closed"

Possible causes	Recommended troubleshooting
Access doors not locked	Close and lock doors

Table 54: Table: Safety - Door not locked

5.7 Fault due to pump wear

In classic mixing and dispensing technology, the task of pumps is to move various liquid and also pasty, low- to high-viscosity, potted and unpotted materials. Due to the direct contact with the conveying medium, parts of the pumps such as gears, rotors, stators, etc. are subject to material and pro-

cess-induced abrasion that has a negative effect on the conveying power and behaviour of the pumps such that - despite pump shutdown - material leaves the pump outlet or with controlled pump operation unstable material volumes are conveyed. Therefore, it is important to recognise characteristics that allow you to make conclusions about the wear situation of the pumps.

A typical characteristic is the escaping of dispensing mass without speed specification: this could indicate wear of the gears in gear pumps or stator abrasion in eccentric worm pumps.



NOTICE

Slippage

Internal and external gear pumps and eccentric worm pumps have natural slippage due to their construction. For unused or new pumps, this is within the precision limits of the conveying power.

A functional check can be conducted as follows:

1. Put machine into HAND operating mode
2. Call up pump tab for this pump
3. Press the < *Start* > button
 - ⇒ Pump starts, whereby downstream dispensing valve remains closed
 - ⇒ With sufficient supply of the pump, for pumps without a lot of wear there should be a significant pressure increase at the pump outlet (e.g. to > 15 bar)
 - ⇒ Pump running is generally stopped by an error message since the set maximum pressure limit is exceeded
4. Stop pump with < *Stop* > button



NOTICE

Running dry of pump

For this check, especially the gear pumps may not run dry in order to prevent damage to the internal parts!

5. If no significant pressure increase: Shut down machine, take apart pump and check gears and stator-rotor system for signs of wear (traces of abrasion, score formation, etc.), replace damaged parts.
6. If there are no signs of wear: Ventilate dispensing circuit step by step; starting with the dispensing pump to the mixing unit; check other places for possible leaks such as dispensing valve, sealing element, etc. Another error source for unstable conveying volumes is insufficient material preparation, which transports included air into the following dispensing circuit. In case of doubt, evacuate material.

Error pattern: Material escapes permanently

Possible cause	Recommended troubleshooting
Contamination between plates	Take apart pump according to maintenance instructions and clean

Seals defective	1. Replace seals of the seal package (2nd gen.) 2. Change O-ring between plates (only 4th gen.) 3. Tighten plates properly (7-8 Nm)
Plates run-in	Return pump to bdtronic to have the plates ground

Table 55: Error pattern: External gear pump leaks

Error pattern: Material escapes permanently between housing cover and pump housing	
Possible cause	Recommended troubleshooting
O-ring defective	Change O-ring
Rod seals defective	Change rod seals

Table 56: Error pattern: Internal gear pump leaks

Error pattern: After dispensing stop, pump drips	
Possible cause	Recommended troubleshooting
Air inclusions in the pump	Ventilate pump at ventilation connection
Stator/rotor worn	Dismantle the pump as described in the maintenance instructions and install new stator/rotor, if necessary.

Table 57: Error pattern: Dripping of eccentric worm pump

6 Layout

The following layout provides the exterior dimensions of the entire machine and the most important components, optional drive and transport paths and the position of special machine positions.

7 Setting data

The settings data sheet lists a sequence of weighing measurements that were performed under customer specifications using defined parameter settings such as pump capacity, etc. on this machine. A Cmk value determined through calculation provides a starting point for the quality of the dispensing pump.

8 Machine log

The usually multi-page machine log is a listing of important process modules of the machine such as dispensing modules, hot riveting modules, depending on the machine type. In the 'Item no.' column, this part or module can be ordered as a replacement part from the 'Central Processing' department of bdtronic GmbH (for contact data, see 'Service legal notices'). For precise identification of module details, see "Spare part documents" in the appendix.

9 Replacement part documents

Below is the replacement part list for the components specified in the machine log.

10 Electrical schematic

The electrical circuit diagram is split into the following areas:

1. Side tree
2. Side list
3. Equipment tree
4. Equipment list



NOTICE

Electrical circuit diagram

For reasons of environmental protection, bdtronic GmbH has decided to provide the comprehensive electrical circuit diagram in digital form only. It is supplied on the documentation CD together with the operating instructions. On request, a PDF can be sent directly to the customer by e-mail. The information on the type plate is also binding for the connection of the machine!

11 Supplier documentation

In this additional section you will find optional certificates and licences from our suppliers for your machine components together with supplier-specific CDs and a CD with a selection of the most important documentation from our suppliers.

The provision of supplier documentation in paper form can therefore be essentially avoided.